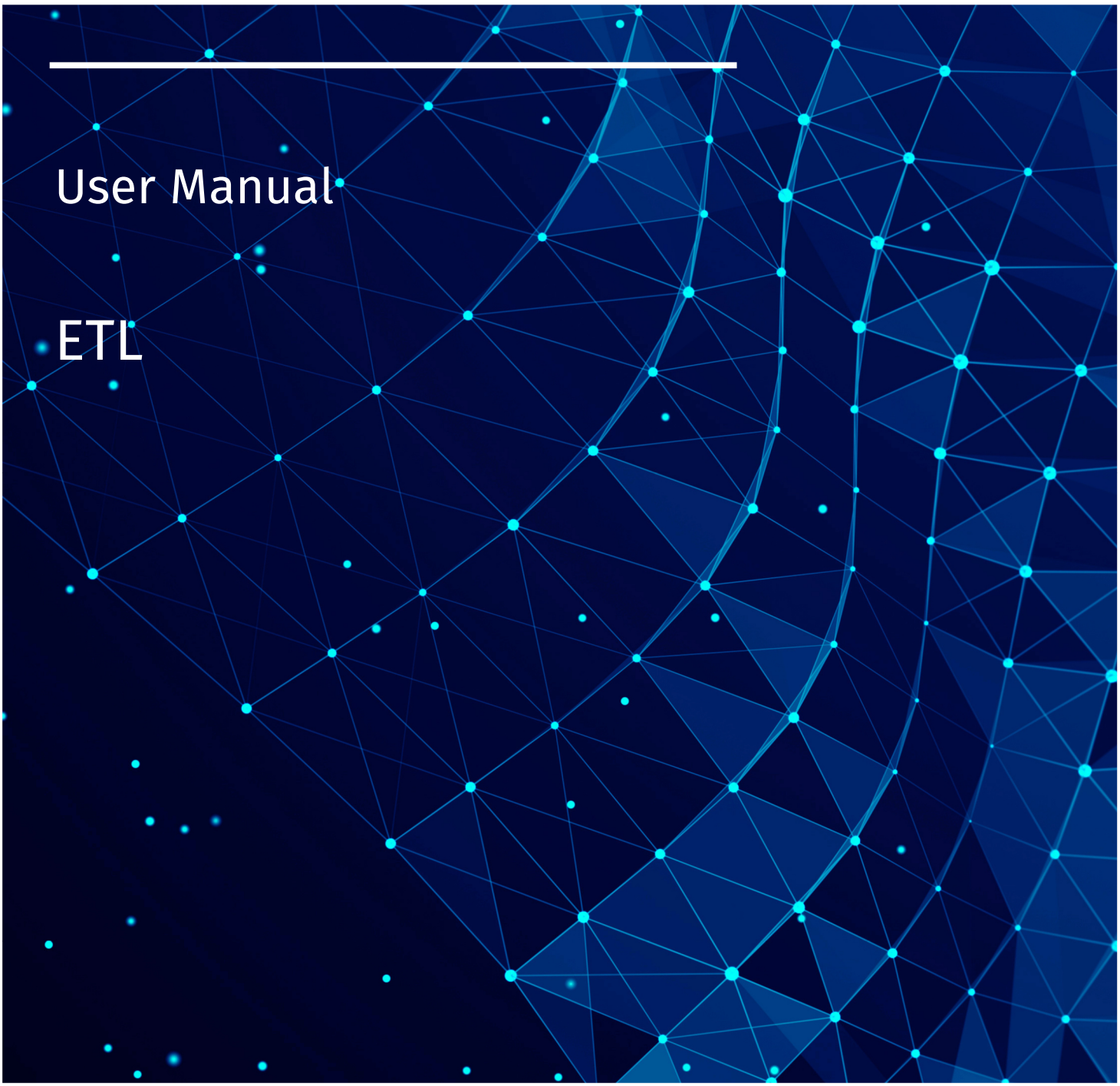


User Manual

ETL

This manual provides an overview of the ETL engine ("Extraction, Trasformation & Loading") which involves all processes and is fully configurable .

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Wolters Kluwer

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Introduction

Objectives of the manual

This manual provides an overview of the ETL engine (**"Extraction, Trasformation & Loading"**) which involves all processes, is fully configurable and is useful to automatize:

- the extraction of data from any external source, such as excel files, text files, data warehouse, ERP systems (SAP, Oracle or Microsoft) or from a database of Tagetik;
- the processing of extracted data with possible aggregation or enrichment;
- the loading of processed data to different destination types, both internal (Tagetik's database) or external (generic DB, excel or text file).

Contents

Below is a list of the topics covered in this manual.

Section	Content
Introduction	Introduction to the ETL process and to the concepts of ETL rule and structure
Configuration	Describes the general information, used by the whole application, the administrator user must set (domains, dimensions, dictionaries, sql models and parameters)
Routine	Introduction to the concept of standard routine and custom routine. Detailed description of the routine definition steps
Formats	Introduction to the concept and to the management of • Data Source - the structure containing the information the system needs to extract data from an external source, manage extracted data and save them in the staging table • Lookup table - transformation rules needed to allow the loading of data extracted and copied in the staging table • Data target - the structure which indicates *how and where* the system must load extracted and transformed data. Data can be loaded in a table of the Tagetik application database. In this case it is necessary to define a "Data Target: Tagetik" or an external target (generic DB, excel or text file, another application DB of Tagetik different from the source DB). In this case it is necessary to define a "Data Target: Other".

Section	Content
Run a routine	Detailed description of the execution of • a standard routine • a custom routine
ETL Job	Provides a detailed description of the steps to follow to define and run an ETL Job
Data check	Describes the tools supplied to the user to keep track of extracted and transformed data
Utility	Describes the copy / deletion / import - export utilities for the management of the ETL parametrization

ETL Domains

An ETL domain includes some settings which allow the user to define a series of rules to:

- extract data from various sources,
- transform data by configurable logics,
- load data into CCH Tagetik as well as into other target.

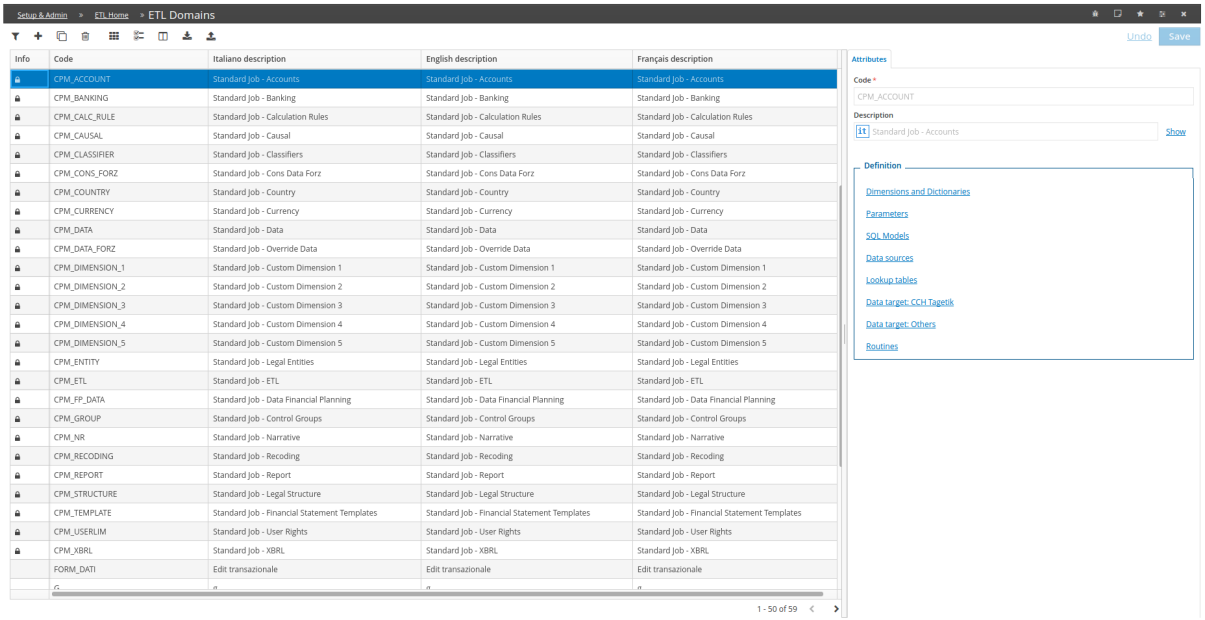
CCH Tagetik includes by default the **Standard ETL Domains** which allow for the loading of "structural" data (elements of dimensions, hierarchies and groupings) and "values" (amounts, IC entries, journals and FX rates).

New and entirely configurable **Custom ETL Domain** can be configured in order to meet the specific business needs.

Definition

The definition of the ETL domains can be accessed from the ETL tile displayed on the admin user's home page.

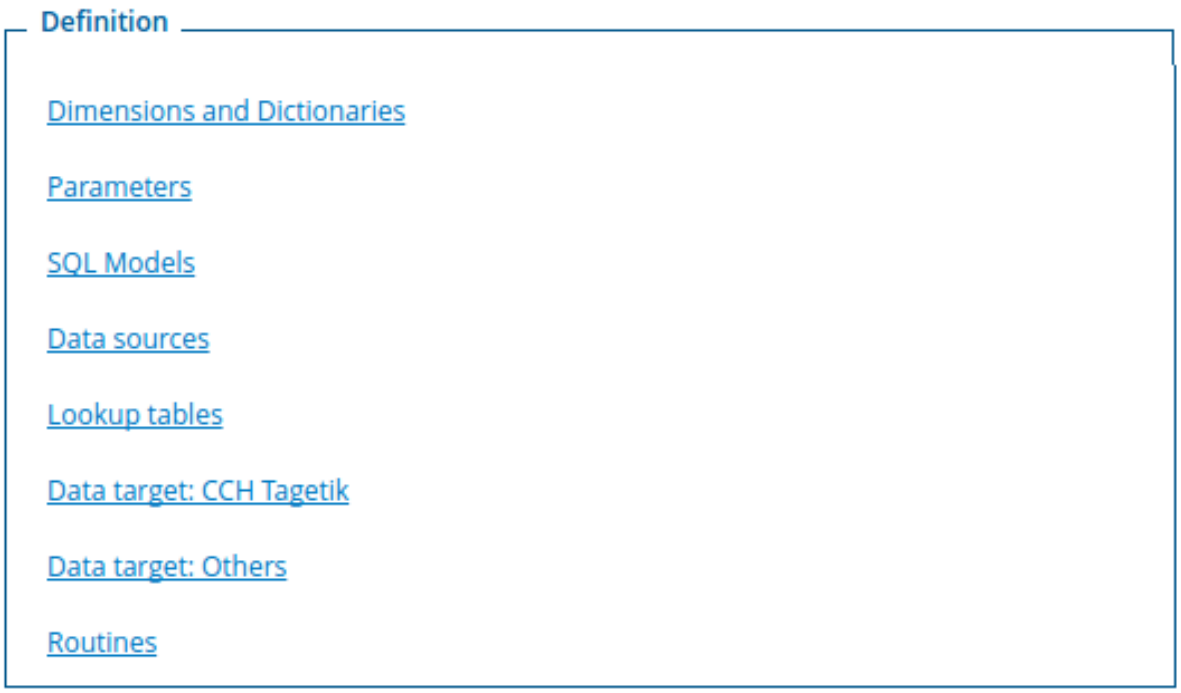
The standard ETL domains definition is locked. For these records, the icon is displayed in the first column of the grid.



Each ETL Domain is identified by a **Code** and **Descriptions**.

Attributes

Besides the **Code** and the **Descriptions**, the Attributes details include a list of links to the parametrization details. Those links are enabled only if one single ETL Domain is selected.



Relationships

From this tab it is possible to display and modify the nodes of the grouping the selected ETL Domain is related to.

Each new ETL Domain will be automatically related to the default nodes of the grouping.



User rights

User visibility Rights apply to the definition of the ETL Domains.

If user restrictions apply, the window will show only the visible ETL Domains and it won't be possible to add or clone new records.

User rights can be defined by accessing the old web interface of the admin user by the path *Setup & Admin > User rights > User rights* (for further details on the user rights management, please see the "User rights on data" section in the user manual).

Utility

The Actions menu in the window includes some ETL Domain copy, deletion, import and export utilities. These utilities can be used only if one single record is selected.

Reset

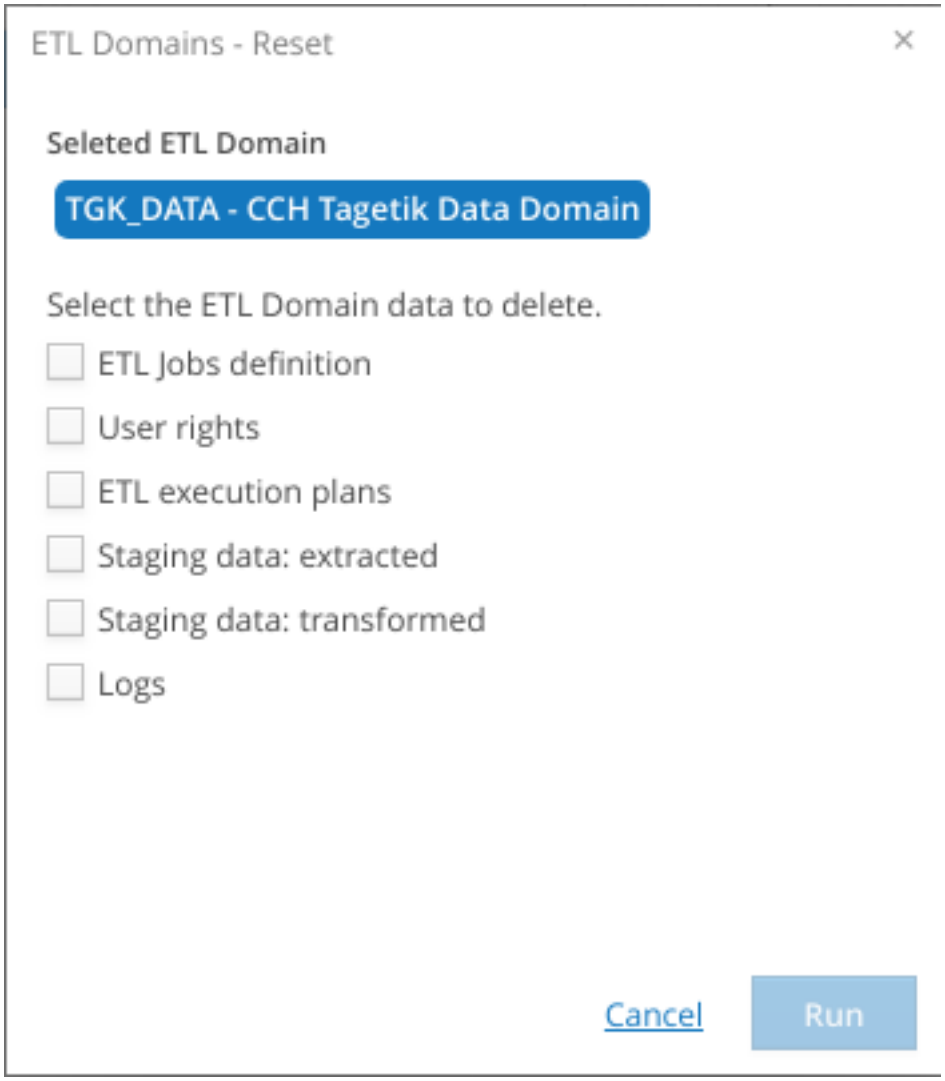
The *Reset* menu allows you to reset all uses of the ETL Domain and of its parametrization:

- in the definition of all ETL Jobs,
- in the users rights,
- in the ETL execution plans of all processes

as well as the execution data related to its Routines, namely:

- the staging table of extracted data,
- the staging table of transformed data,
- the execution logs.

In order to run the operation, it is necessary to select at least one element to delete.

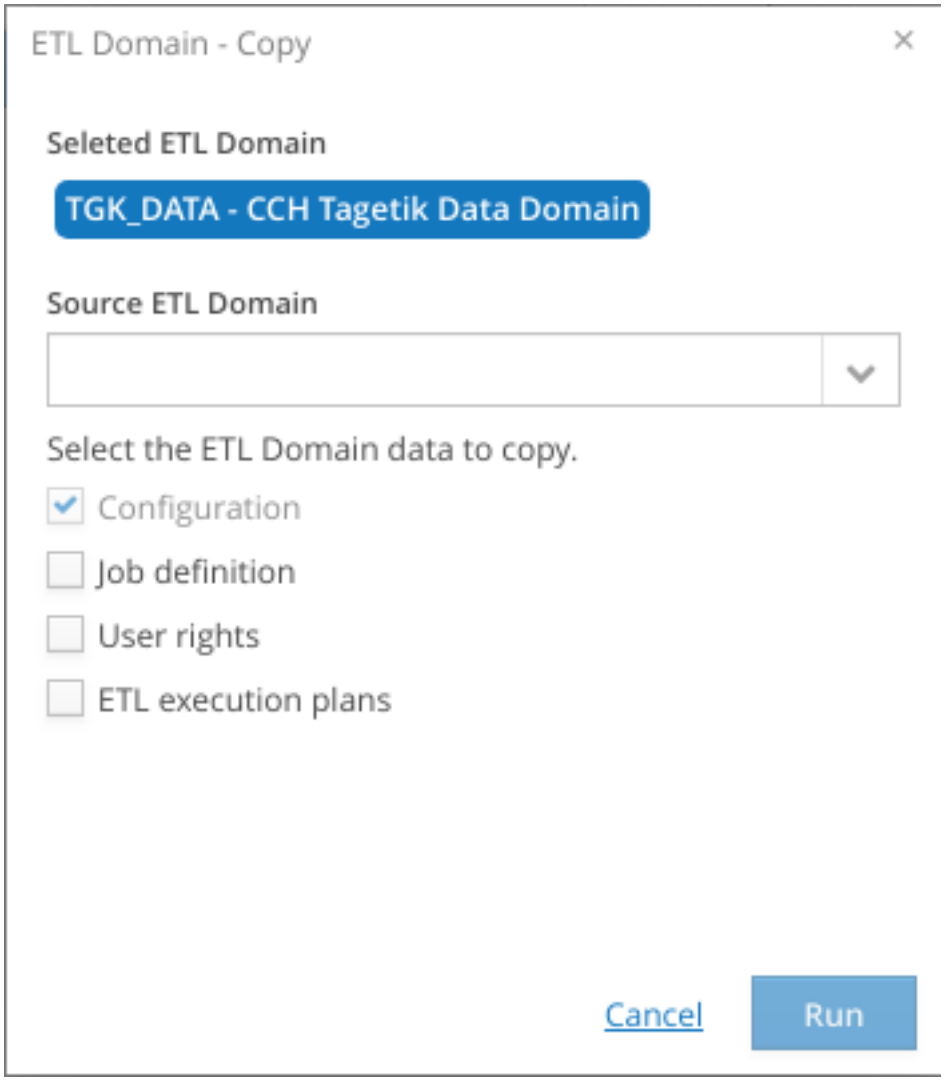


Copy from

The *Copy from* menu enables the copying (by overwriting) on the selected ETL Domain of the entire default configuration and of the following uses of the ETL Domain specified in the copy window:

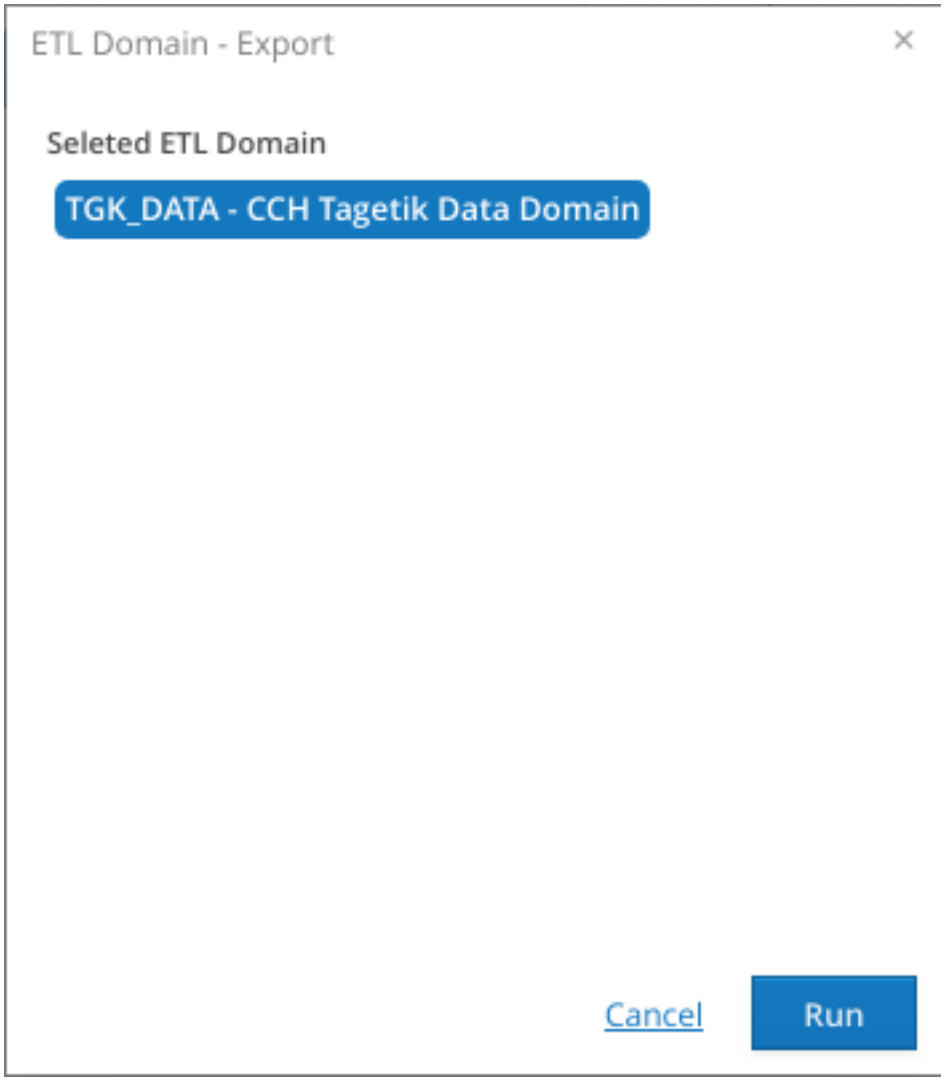
- definition of ETL Jobs,
- user rights,
- ETL execution plans of all processes

In order to run this operation, it is necessary to specify the source ETL Domain.



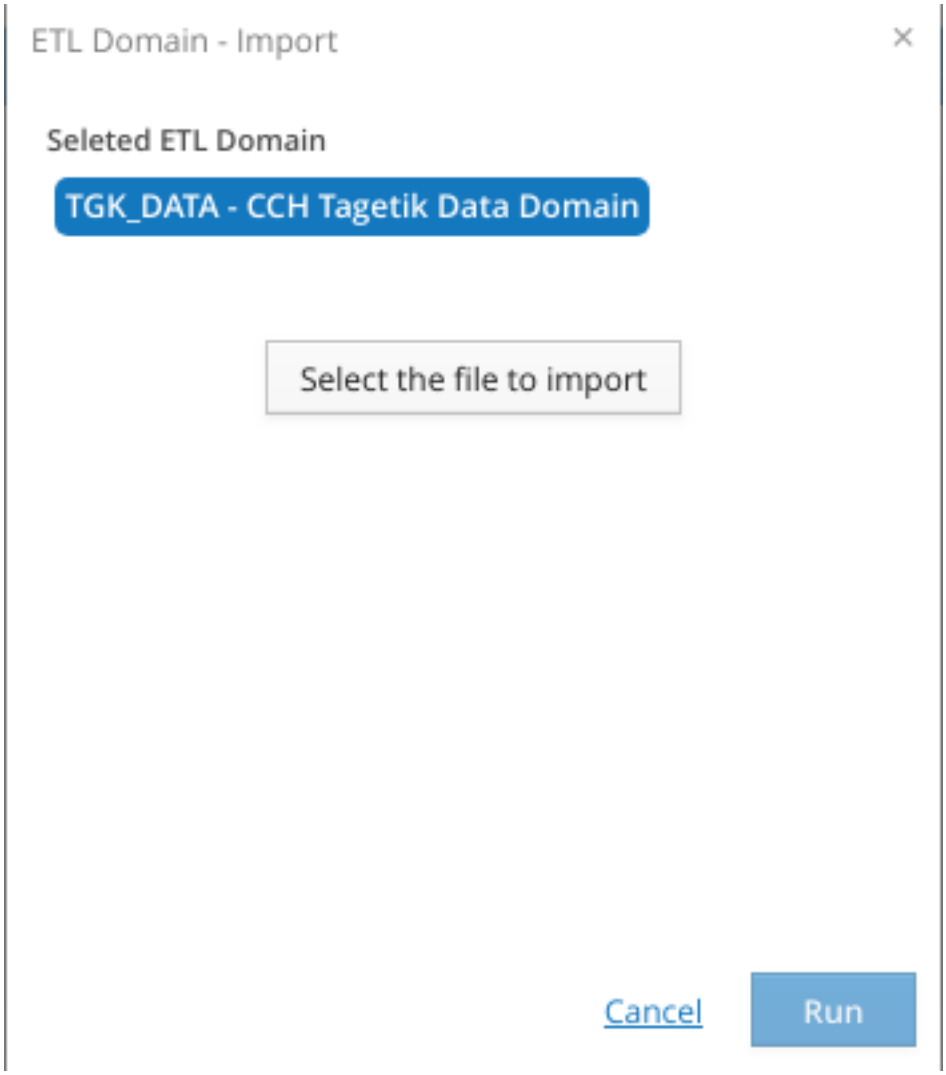
Export

The *Export* menu allows you to export to a file (with .dat extension) the entire configuration of the selected ETL Domain and of its Routines, so that it can be duplicated on another existing ETL Domain (in the same domain or in another one) by the *Import* menu.



Import

The *Import* menu allows you to import a file (with .dat extension) that was previously exported by the *Export* menu, in order to duplicate on the selected ETL Domain the entire configuration of the exported ETL Domain and of its Routines. The target ETL Domain will be entirely deleted and then populated with the configuration contained in the file.


ETL Domain - Import
Seleted ETL Domain
TGK_DATA - CCH Tagetik Data Domain
Select the file to import
Cancel
Run

Dimensions and dictionaries

When running, the ETL populates a temporary table ("*Staging*" table) by extracting data from one or more external sources, transforms them by applying a series of rules and then loads them into CCH Tagetik or into other custom dimensions, such as files or custom tables. This is why it is necessary to set a correspondence between:

- the structure of data and the structure of the "*Staging*" table;
- the structure of the "*Staging*" table and the structure of the target table or file.

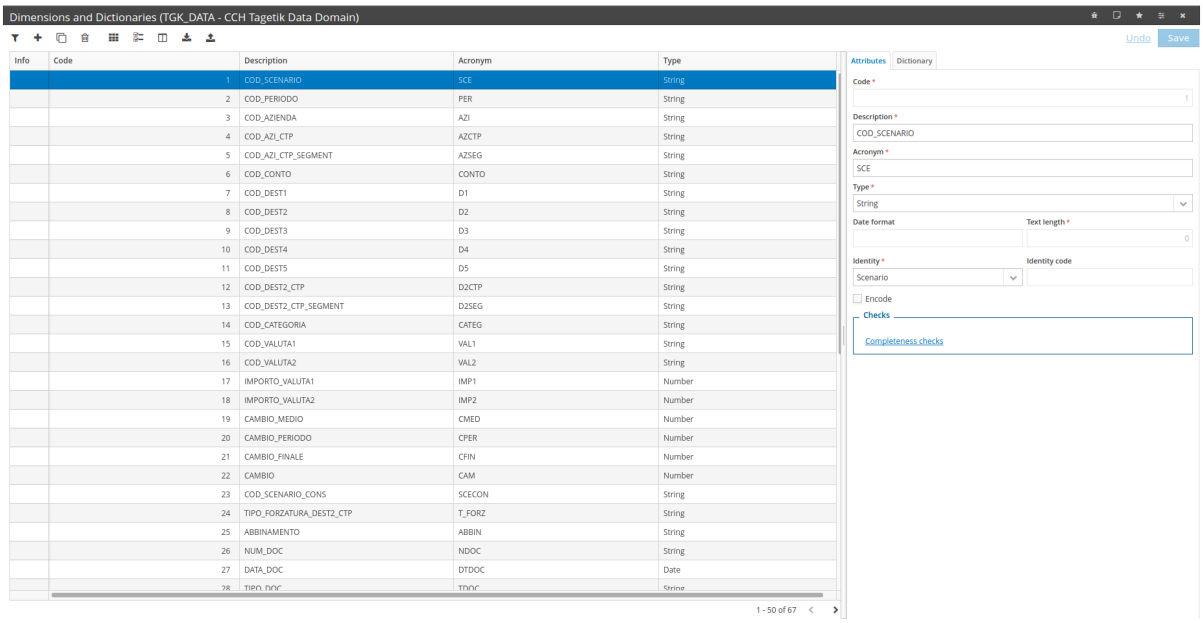
The Dimensions represent the columns of the "*Staging*" table (the names of the fields are made up of "CAMPO" + dimension **Code**). It is possible to manage up to 200 dimensions.

It is good practice to define a**Description** and an **Acronym**that clearly identify the content of the dimension.

Definition

The Definition of the Dimensions and Dictionaries can be accessed by the *Dimensions and Dictionaries* link in the **ETL Domaines** window.

If the ETL Domain is locked, then the window will be in read only mode.



Each Dimension is identified by:

- **Code**: a number between 1 and 200. This code corresponds to one of the columns of the staging table as described above;
  - dimensions from 1 to 89 are *"varchar"* type and 50 characters long;
  - dimensions from 90 to 99 are *"varchar"* type and 255 characters long;
  - dimensions from 100 to 130 are *"varchar"* type and their length is defined in the **Text length** field;
  - dimensions from 131 to 200 are *"clob"* type

- **Description**;
- **Acronym**: short description of the dimension to use in the dynamic syntaxes of SQL filters and text of the ETL;
- **Type**: specifies the type of data that will populate the dimension. Managed types:
  - *String*: suitable only for dimensions with code from 1 to 99;
  - *Number*: suitable only for dimensions with code from 1 to 99;
  - *Date*: suitable only for dimensions with code from 1 to 99;
  - *Text*: suitable only for dimensions with code from 100 to 130;
  - *Clob* : suitable only for dimensions with code from 131 to 200.

 Normally, the *Clob* type dimensions for the management of large amounts of text can be used in the whole ETL parametrization but cannot be stored, neither in extracted data nor in transformed data. Moreover, for the ETL domains that use these dimensions, it is not possible to run the Routines in partial mode.

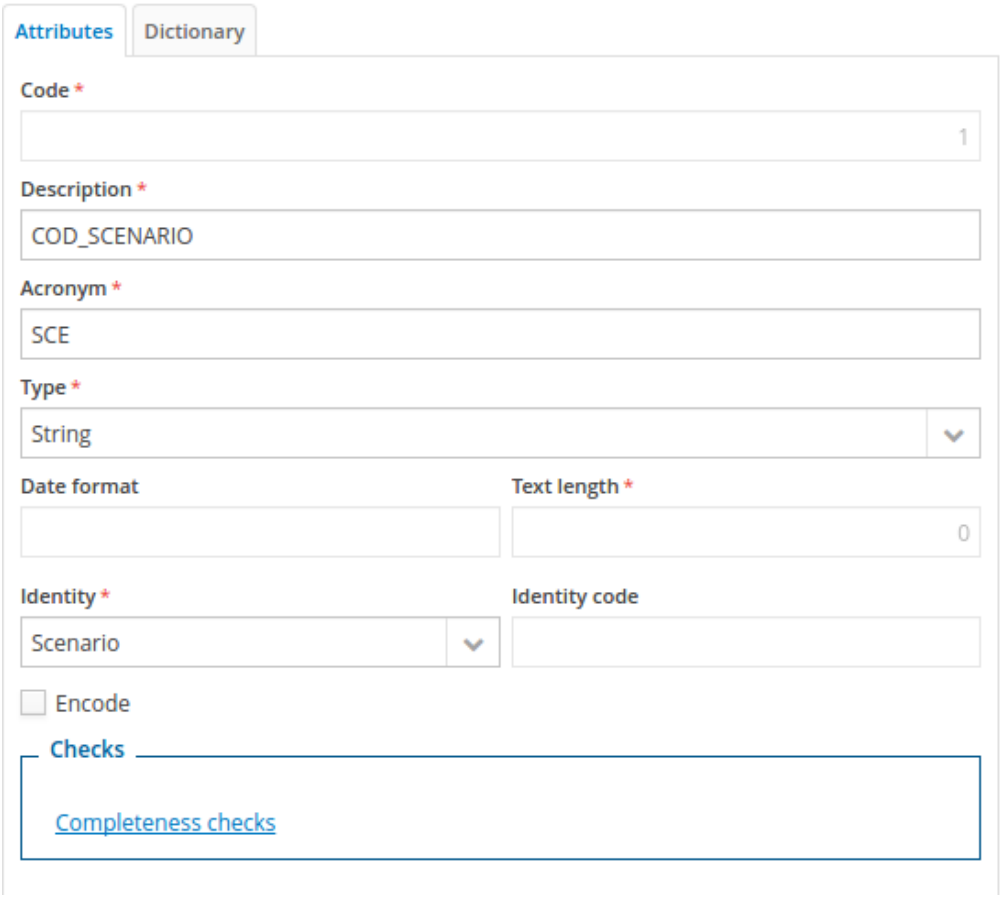
 Since in the staging table all the columns are of *"varchar"* type, the **Type** specified for each dimension is used by the system only in the following cases:

- to format data to display after a query on the staging table or on the log table. Numeric data are formatted based on the user's nationality;
- to automatically convert the value of a dimension's element whenever the **Acronym** is inserted in a dynamic syntax in an ETL filter, a SQL model or a SQL query.

  For example, if we have a numeric ETL dimension (e.g. CAMPO10) with acronym *"IMP"* and we want to multiply the value by -1, we just have to write in the query {IMP} * -1 and the system will automatically convert the value into a number before multiplying it by-1.

Attributes

In addition to **Code**, **Description**, **Acronym** and **Type**, in the attributes detail it is possible to define some additional information.

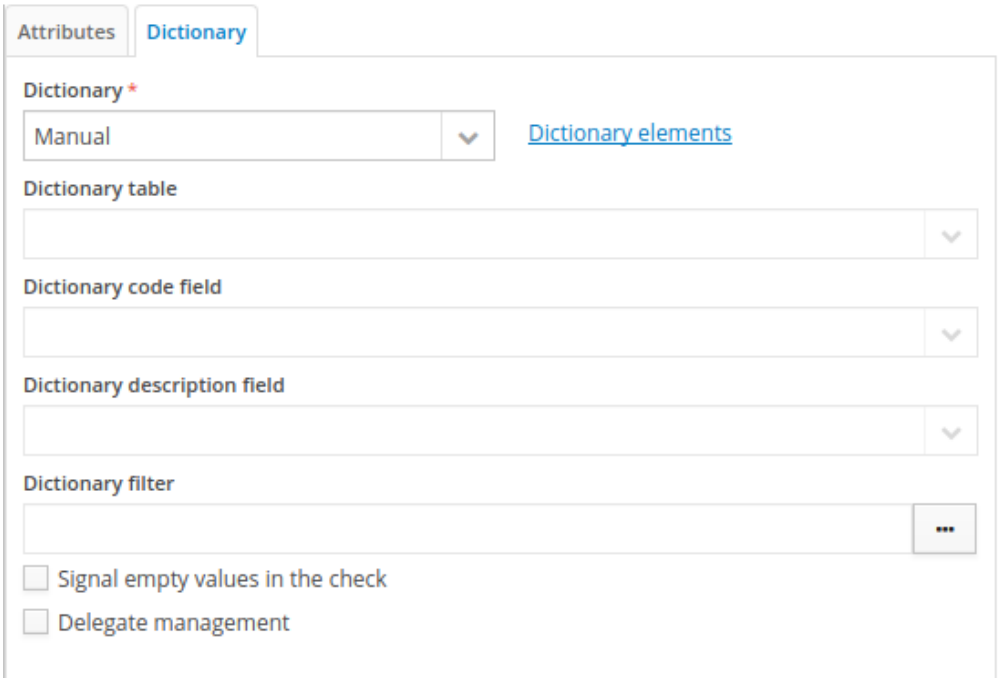


- **Date format**: is mandatory for the *"Date"* **type** dimension and specifies the date format of data to import.
- **Text length**: is mandatory for *"Text"* **type** dimensions and specifies the number of characters the dimension can have.
- **Identity**: the values in the dimension belong to the corresponding logical dimension of CCH Tagetik. The identities are used to limit the possible values in the selection of the parameters related to such dimensions and in some transformation or loading tasks. The identity cannot be set for the *"Clob"* **type** dimensions. The identity can be:
    - *None*
    - *Entity*
    - *Scenario*
    - *Period*
    - *Event scenario*
    - *Account*
    - *Target*
    - *Classifier*
    - *Counterparty zone*
    - *Counter*
    - *Category*
    - *Currency*
    - *Original Currency*
    - *Record num.*

- **Identity code**: mandatory if the **Identity** is *Target* or *Classifier* and allows specifying respectively one of the managed custom dimensions or a classifier. This field's value must be unique for the ETL Domain.
- **Encode**: specifies that the values that are inserted in this dimension during extraction must be encoded.
- *Checks* box containing a link to the Completeness checks

Dictionaries

For each *"String"* or *"Text"* **type** dimension, it is possible to specify a set of accepted values for the dimension in the Dictionary details.



Using a Dictionary does not imply that the user cannot use some specific values for a given dimension, but that if a transformation task of *Dictionary check* is defined within an ETL Routine, any processed data having invalid values will be reported to the user as a warning. When the **Signal empty values in the check** option is enabled, empty fields are reported as warnings as well.

Available dictionaries:

- *No dictionary*: no dictionary has been defined for this dimension.
- *Manual*: for this dimension, the dictionary elements will be inserted manually by a specific management accessible by the Dictionary elements link. Generally, the Dictionary Elements are managed and updated by the admin user who manages the parametrization of the ETL Domain. Nevertheless, it is possible to task the End User with this activity by enabling the **Delegate management** option.
- *From Tagetik*: for this dimension, the dictionary elements are represented by the values of a given field on a CCH Tagetik table. For example, if you want to use as dictionary elements the entity codes in the "AZIENDA" table, you have to set the **Tagetik Table** attribute to AZIENDA, **Dictionary code field** to COD_AZIENDA and **Dictionary description field** to DESC_

AZIENDA0. It is possible to limit the dictionary elements by setting a filter condition in the **Dictionary filter** attribute. For example, even if all entity codes available in the AZIENDA table have been inserted in the dictionary, it is possible to consider only those whose code starts with A0 (in this case, the sql condition to use is COD_AZIENDA LIKE 'A0%').

Dictionary Elements

The Dictionary Elements specify the set of accepted values for the dimensions having a *Manual* **Dictionary**.

Definition

The Dictionaries definition can be accessed from:

- *Dictionary Elements* link in the **Dimensions** window
- *Dictionary* workflow task of the Data Integration group
- *Dictionary - elements* tile in the ETL home page accessible from the End User home page by selecting an ETL Domain and one of the dimensions for which a *Manual* **Dictionary** is defined.

All three entry points open the same window.

Generally, the Dictionary Elements are managed and updated by the admin user who manages the parametrization of the ETL Domain. Nevertheless, it is possible to delegate this to the End User by enabling the **Delegate management** option in the definition of the dimension concerned; in this case, the window accessed from the End User's home and from the Workflow task will be editable.

If the ETL Domain is locked, then the window will be in read only mode.

Each Dictionary Element is identified by a **Code** and some **Descriptions**.

Completeness check

The Completeness Checks allows you to verify that, in presence of some values of the dimension in question, the values of the other dimensions are mandatory, optional or forbidden.

Example:

Let us assume that we have enabled the management of two custom dimensions, "Function" and "Segment", and that we have the "1000 - Salaries" account that needs to be detailed by Function but not by Segment. To verify the consistency of extracted data, it is possible to define a rule specifying that for the "Account" dimension in presence of the element 1000, the "Function" dimension has to be mandatory, whereas the "Function" dimension has to be empty.

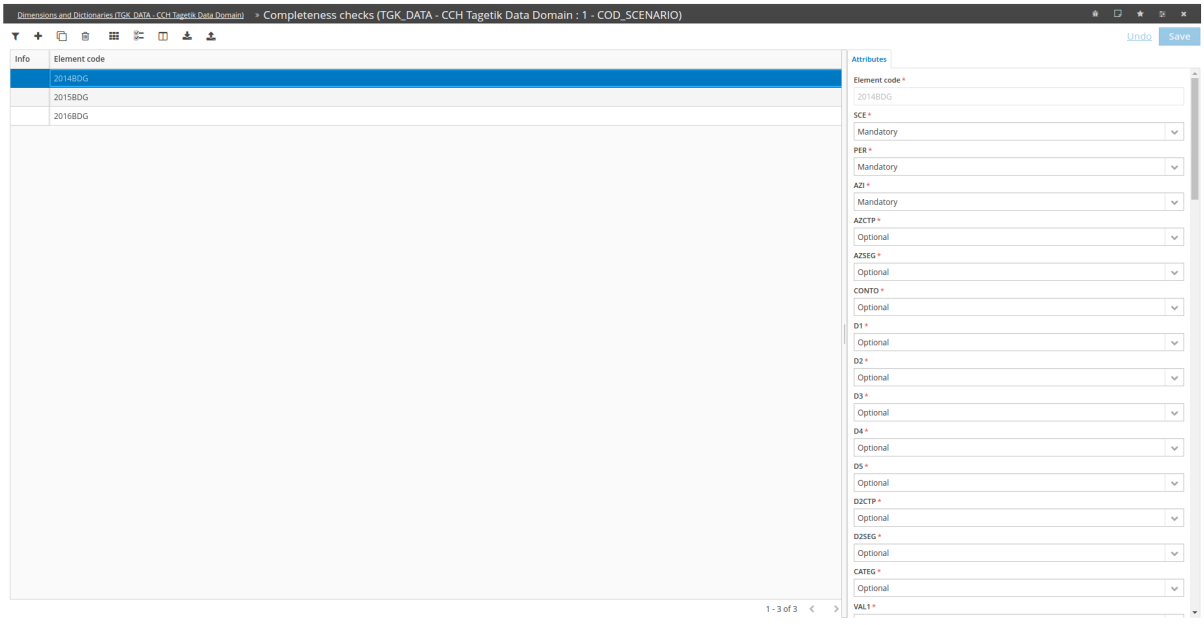
For the checks to be done, it is necessary to define a transformation task of *"Completeness check"* type inside an ETL routine. In this way, at runtime any processed data that do not comply with the rule will be reported to the user as warnings.

Definition

The definition of the Completeness Check can be accessed from the *Completeness Check* actions menu of the **Dimensions**

window.

If an ETL Domain is locked, then the window will be read-only.



Each Completeness Check is identified by:

- an **Element code** containing the value that the dimension in question can assume at runtime;
    - a list of options related to one of the other dimensions of the ETL Domain that, if the value of the dimension is the one defined in the Element Code field, allow you to

indicate if each one has to be:

- **Mandatory** if the definition of the dimension related to the option mandatory;
- **Optional** if the dimension related to the option is optional;
- **Forbidden** if the dimension related to the option has to be left empty.

Parameters

The Parameters allows for the dynamic selection of data to be processed when running an ETL.

They can be used:

- in the definition of the Routine for filtering the log of extracted and transformed data,
- in any extraction tasks to limit the amount of data to extract from the source,
- in a *Parameter choice* transformation task to limit the amount of data to transform or to assign a Dimension a fixed value,
- in any loading tasks to limit the amount of data to delete, update or insert from or into the data target.

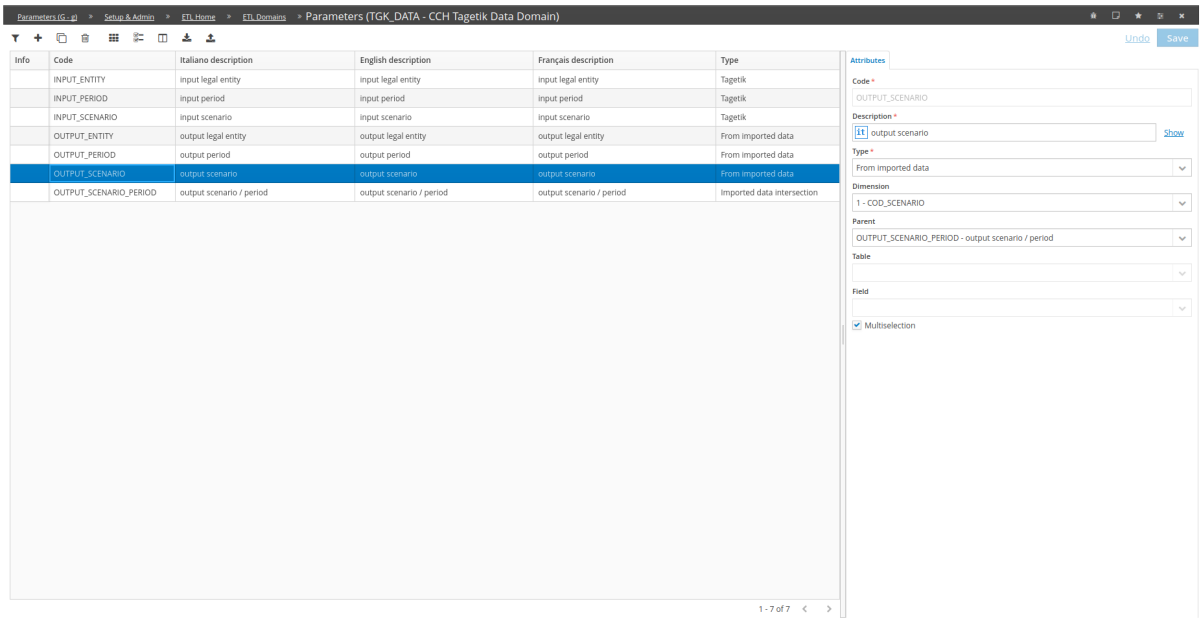
Each one is related to a Dimension and:

- if used during the extraction phase, it filters the data source field linked to the Dimension in the Definition of the Data Source fields
- if used in the extracted and transformed data log or in the *Parameter choice* transformation task, it deletes all records having a null value in the field of the *staging* table related to the Dimension,
- if used during the loading phase, it filters the field of the *staging* table related to the Dimension in the data entry operations and the data target field linked to the Dimension in the Definition of the Data Target fields during deletion.

Definition

The Parameters definition can be accessed from the *Parameters* link of the ETL Domains.

If the ETL Domain is locked, then the window will be in read-only mode.



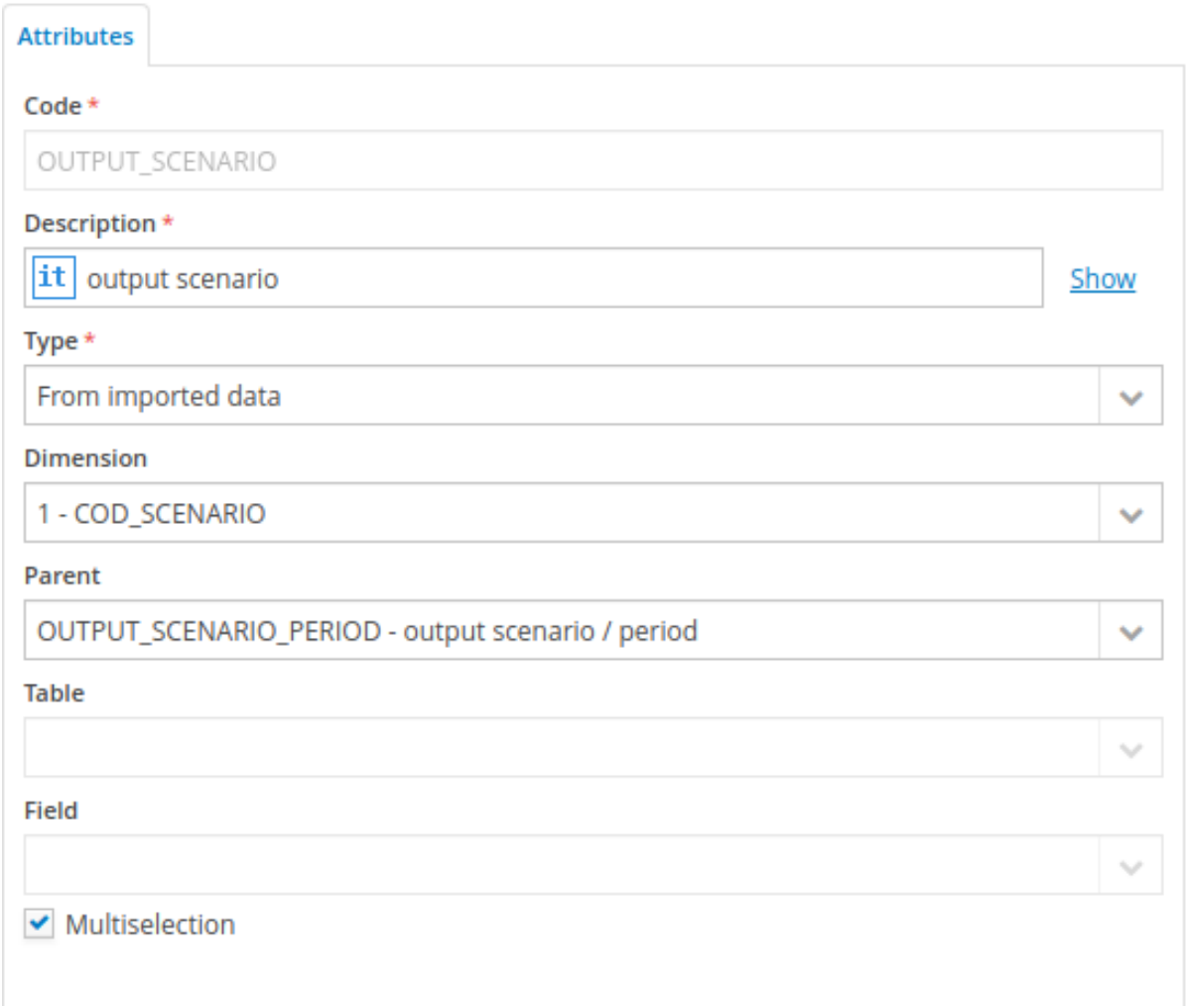
Each Parameter is identified by:

- a **Code**;
- **Descriptions**;
- a **Type**: specifies the Parameter type; the values to be selected are loaded in a different way for each type. Possible values:
  - *Free*. The parameter value is specified manually by the user. In case of **Multi-selection**, values must be separated by a *comma*;
  - *From imported data*. The range of possible values is given by the values contained in the **Dimension** specified when setting the Parameter. This type of Parameter can be used only after the *Extraction* step.
  - *From dictionary dimension*. The range of possible values is given by the values contained in the Dictionary defined for the specified **Dimension**;
  - *From Tagetik*. The range of possible values is given by the values available in a field of a CCH Tagetik table. For example, it is possible to create Parameters having as values the entity codes contained in the Entity table. In this case, it is necessary to set the **Table** attribute to AZIENDA and **Field** to COD_AZIENDA;
  - *Imported data intersection*. The range of possible values is given by the combinations of values of other **Dimensions** defined on other Parameters of *From imported data* type for which the usual Parameter **Code** is defined in the **Parent** field. For example, in order to filter by Scenario/Period, we can:
    - define one "Scenario" and one "Period" of *From imported data* **Type**;
    - define one "Scenario/Periodo" parameter of *Imported data intersection* **Type**;
    - link the "Scenario/Period" parameter to the "Scenario" and "Period" parameters as **Parent** parameter

    so the "Scenario/Period" parameter has as values the possible combinations of the "Scenario" and "Period" parameters values.

Attributes

Besides the **Code**, the **Descriptions** and the **Type**, among the Attributes details it is possible to define further pieces of information.



- **Dimension**: this is mandatory for all Parameters without *Imported data intersection* **Type** and specifies the Dimension to which the filter must be applied or that must be set to the value or the Parameter.
- **Padre**: only for Parameters of *From imported data* type, it specifies the parameter of *Imported data intersection* type that associates it with other Parameters with the usual **Parent** code.
- **Table**: it specifies the source table and is mandatory for the Parameters of *From Tagetik***type** .
- **Field**: it specifies the source field and is mandatory for the Parameters of *From Tagetik***type**.
- **Multi-selection**: specifies if it is possible to select more than one value for the Parameter.

In order to limit the selectable values of a Parameter to those compatible with the selected Process and with the visibility rights of the current user, it is possible to use the **Identity** of a Dimension. If the **Dimension** related to a Parameter has the **Identity** field set to *Entity*, *Scenario*, *Period*, *Event Scenario*, *Category*, *Custom Dimensions*, then the system will not display among the selectable values those that are not defined in the Process and those the current user cannot display.

Example: Let us assume that we have a user with visibility rights only on the Entities A01, A02 and A03. In order that the user can select only the entities he/she is allowed to display, it is necessary to define a parameter of *From Tagetik* type with the **Table** field set to AZIENDA, and **Field** set to COD_AZIENDA and then relate it to a **Dimension** with *Entity* **Identity**.

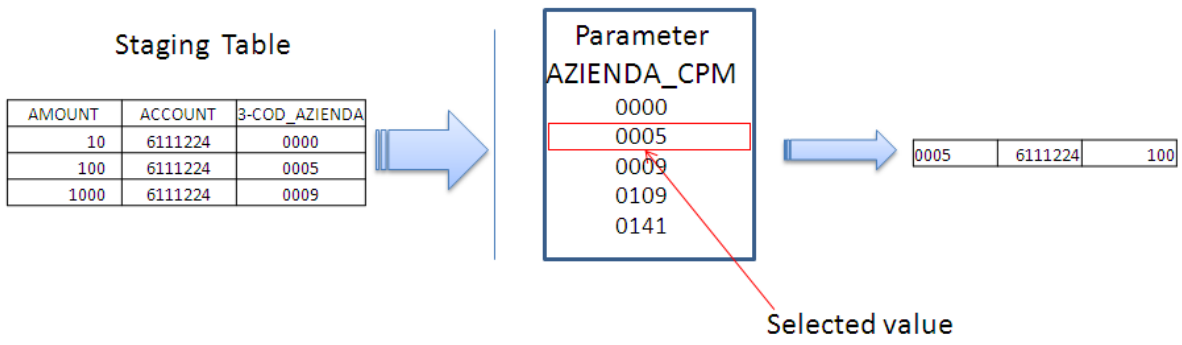
All Parameters are used to filter (exclude / remove) records according to where they are used except those of *Free*, *From dimension dictionary* and *From Tagetik* type with the **Multi-selection** option disabled that are used in a transformation task of *Parameter choice* type; in this case, the selected value is not used to delete the records with a different value but to override the value setting of the field related to the Dimension related with the Parameter for all the records currently available in the *staging* table.

Examples

Example 1: Let us assume that we want to filter the Dimension "3 - Entity Code" by selecting one or more Entity values from the CCH Tagetik table. We just have to define:

- **Code**: AZIENDA_CPM
- **Description**: Entity from CCH Tagetik
- **Type**: *From Tagetik*
- **Dimension**: 3 - Entity Code
- **Table**: AZIENDA
- **Field**: COD_AZIENDA

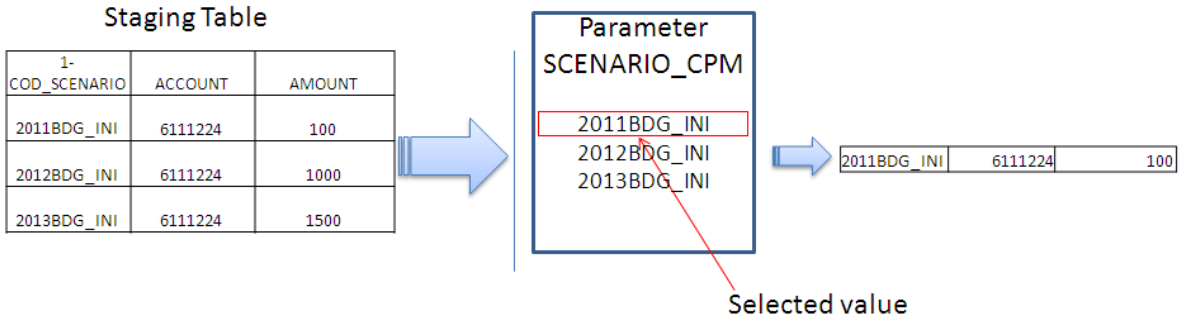
So, if we select the value 0005 when running an ETL when the Parameter needs to be populated, the system will filter all the parameters that do not meet the criteria (see image below).



Example 2: Let us assume that we want to filter the Dimension "1 - Scenario Code" by selecting one or more values available on this Dimension after the data extraction. We just have to define:

- **Code**: SCENARIO_ETL
- **Description**: Scenario from Imported Data
- **Type**: *From imported data*
- **Dimension**: 1 - Scenario Code

So, if we select the value 20011BDG_INI iwhen running an ETL when the Parameter needs to be populated, the system will filter all the parameters that do not meet the criteria (see image below).



SQL Models

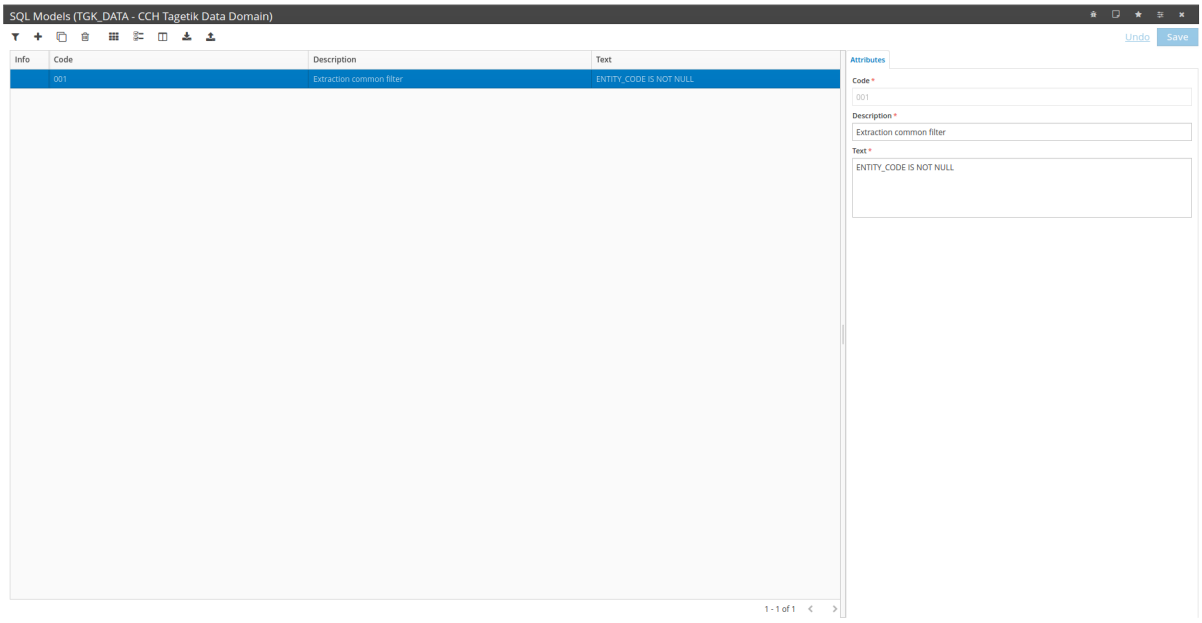
The SQL Models allows defining SQL syntaxes that can be used several times as text or as filter in the tasks of an ETL Routine.

The centralised definition of the SQL syntaxes allows an easier maintenance of the parametrizations: any modifications to a SQL Model will affect all the tasks in which it is used.

Definition

The definition of the SQL Model can be accessed by the *SQL Models* link of the **ETL Domains.**

If the ETL Domain is locked, the window will be in read-only mode.



Each SQL Model is identified by:

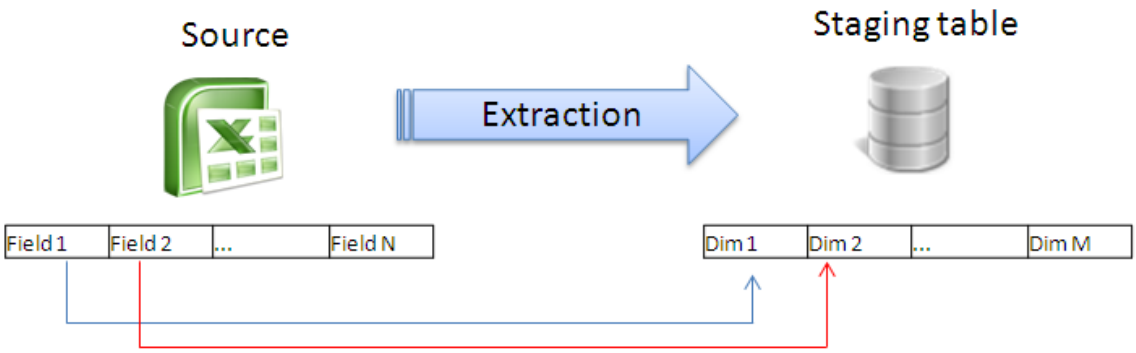
- a **Code**;
- a **Description**;
- a **Text**. It contains the SQL syntax. The dynamic syntax of the ETL can be used (see **Filter Strings Syntax or SQL Query**

Data Source

Data sources are used to define the data source from which the values are extracted. Defining a Data source is not enough to extract data; it should be related to an Extraction task of a Routine. At runtime, the system will as you to select the data source type to use.

After setting the data source types, it is necessary to access the Data source fields definition to define the relationships between the information and the Dimensions of the ETL Domain.

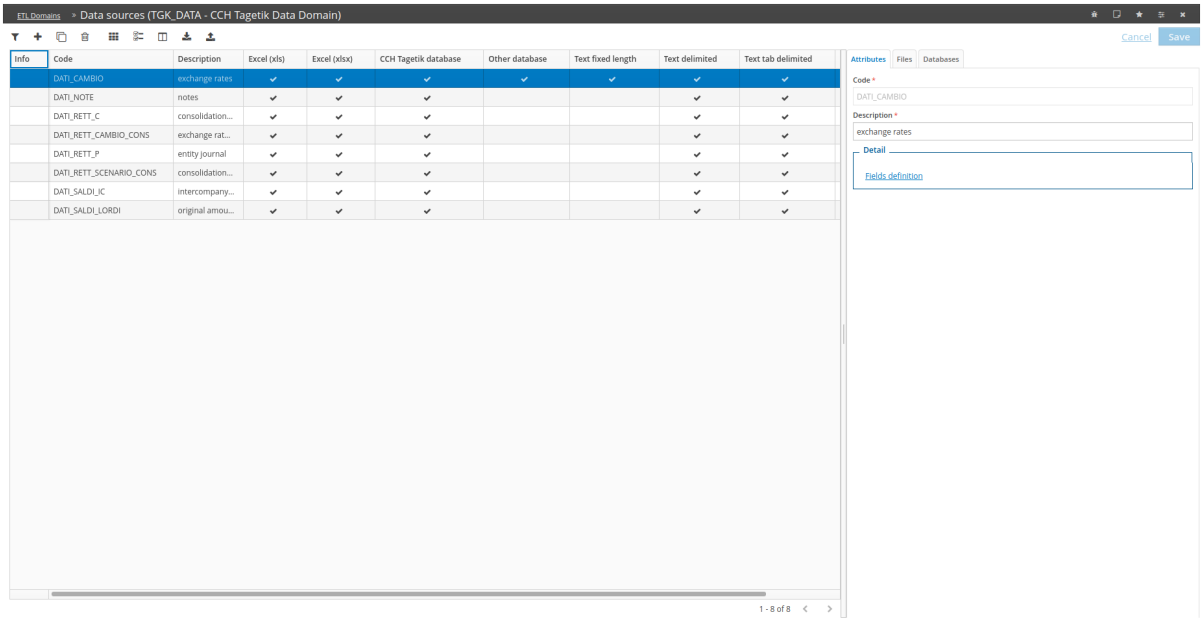
Several data source types can be enabled within the same Data source only if the pieces of information are shown in the same way.



Definition

The definition of the Data sources can be accessed from the *Data sources* link of the ETL Domain window.

If the ETL Domain is locked, then the window will be in read-only mode.

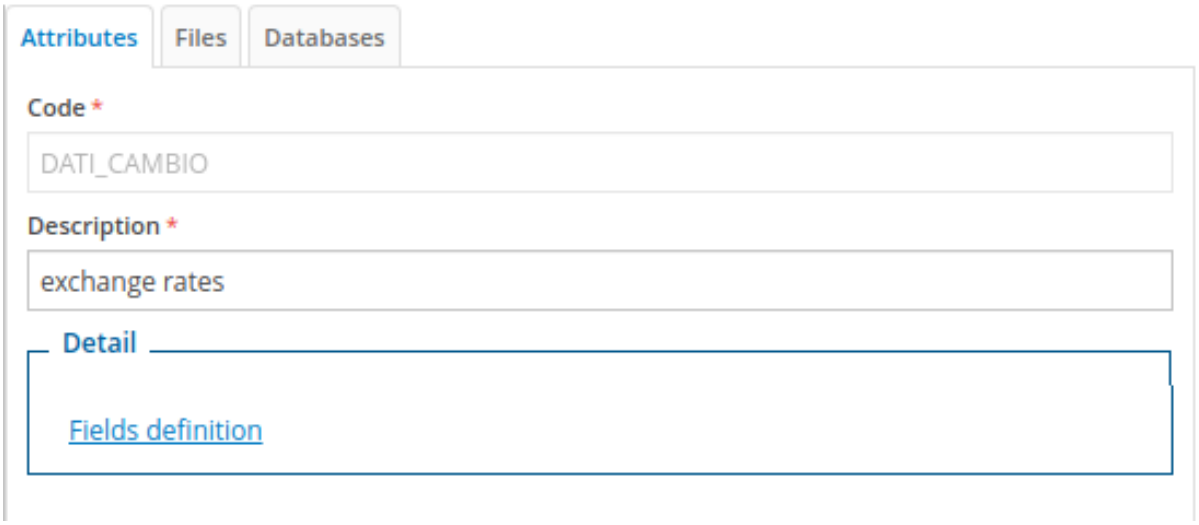


Each data source is identified by:

- **Code**: code of the Data source to be used in the extraction tasks;
- **Description** description of the Data source;

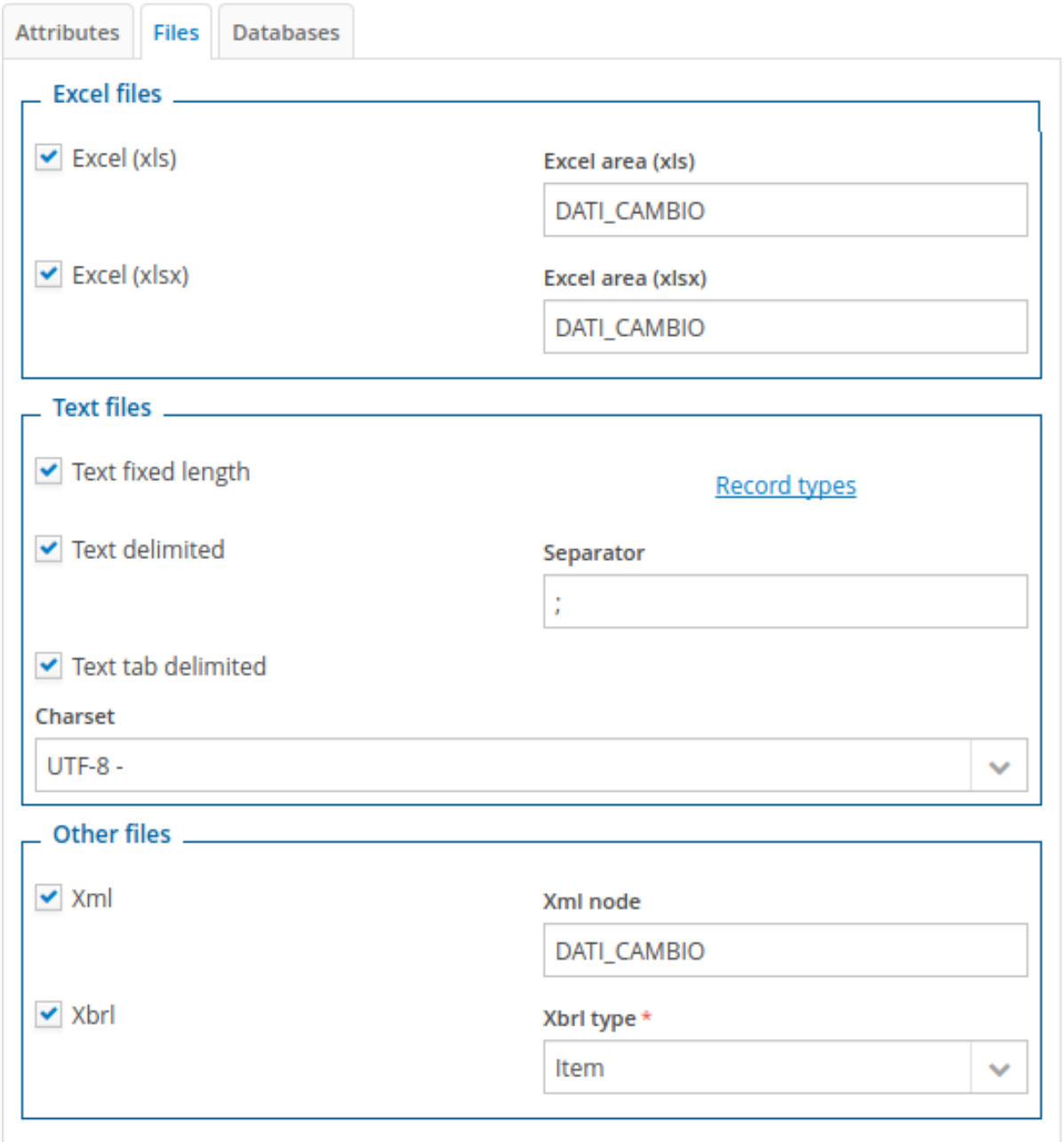
Attributes

Besides the **Code** and the **Description**, the Attributes include the *Fields definition* link to the **Data source fields definition** window where it is possible to link the dimensions of the ETL Domain with the values present in the data sources defined in the selected Data source.



Files

In this tab it is possible to define the data sources of File type for the selected Data source. Each type can be enabled by the related option and may need additional information.



Excel files

- When the **Excel (xls)** option is enabled, it is possible to extract data from an Excel file (xls). This file must have a data area including a header record and values records. The name of the data area must be displayed with the same case in the **Excel Area (xls)** field.

- When the **Excel (xlsx)** option is enabled, it is possible to extract data from an Excel file (xlsx). This file must have a data area including a header record and values records. The name of the data area must be displayed with the same case in the **Excel Area (xlsx)** field.



	A	B	C	D	E	F	G	H	I
1	COD_SCENARIO	COD_PERIODO	COD_AZIENDA	COD_CONTO	COD_DEST1	COD_DEST2	IMPORTO	VALUTA	
2	2011C	12	A00	COAN50010	G01	A01	-130.000	EUR	
3	2011C	12	A00	COAN50010	G01	A02	-40.000	EUR	
4	2011C	12	A00	COAN50010	G01	A03	-42.000	EUR	
5	2011C	12	A00	COAN50020	G01	G01	-20.000	EUR	
6	2011C	12	A00	COAN50110	G01	A01	65.000	EUR	
7									

Text files

- When the **Text fixed length** option is enabled, it is possible to extract data from a text fixed length file. In this text file type, each record must have always the same number of characters, each information must start from the same position (index of the character inside the record) and must have the same length; if the value's length in characters does not equal the number of characters of the information, it is necessary to add some *space* characters (before or after, depending on whether you want to set the value to a numeric format) in order that the following information starts in the prearranged position. If this type is enabled, the *Record types* link will be enabled as well and will allows defining the Record types if the source file includes records that are not related to the values to be extracted but that contain information that must be reported for each record upon extraction.

```
2018BDG     01A0     P1     $AMOUNT     100
2018BDG     02A0     P1     $AMOUNT     100
2018BDG     03A0     P1     $AMOUNT     100
2018BDG     04A0     P1     $AMOUNT     100
2018BDG     05A0     P1     $AMOUNT     100
2018BDG     06A0     P1     $AMOUNT     100
2018BDG     07A0     P1     $AMOUNT     100
2018BDG     08A0     P1     $AMOUNT     100
2018BDG     09A0     P1     $AMOUNT     100
2018BDG     10A0     P1     $AMOUNT     100
2018BDG     11A0     P1     $AMOUNT     100
2018BDG     12A0     P1     $AMOUNT     100
```

- When the **Text delimited** option is enabled, it is possible to extract data from a delimited text file. In this type of text file, the pieces of information for each record can have different length but must be separated by a specific character that must be defined in the **Separator** field.

```
2018BDG;01;A0;P1;$AMOUNT;100
2018BDG;02;A0;P1;$AMOUNT;100
2018BDG;03;A0;P1;$AMOUNT;100
2018BDG;04;A0;P1;$AMOUNT;100
2018BDG;05;A0;P1;$AMOUNT;100
2018BDG;06;A0;P1;$AMOUNT;100
2018BDG;07;A0;P1;$AMOUNT;100
2018BDG;08;A0;P1;$AMOUNT;100
2018BDG;09;A0;P1;$AMOUNT;100
2018BDG;10;A0;P1;$AMOUNT;100
2018BDG;11;A0;P1;$AMOUNT;100
2018BDG;12;A0;P1;$AMOUNT;100
```

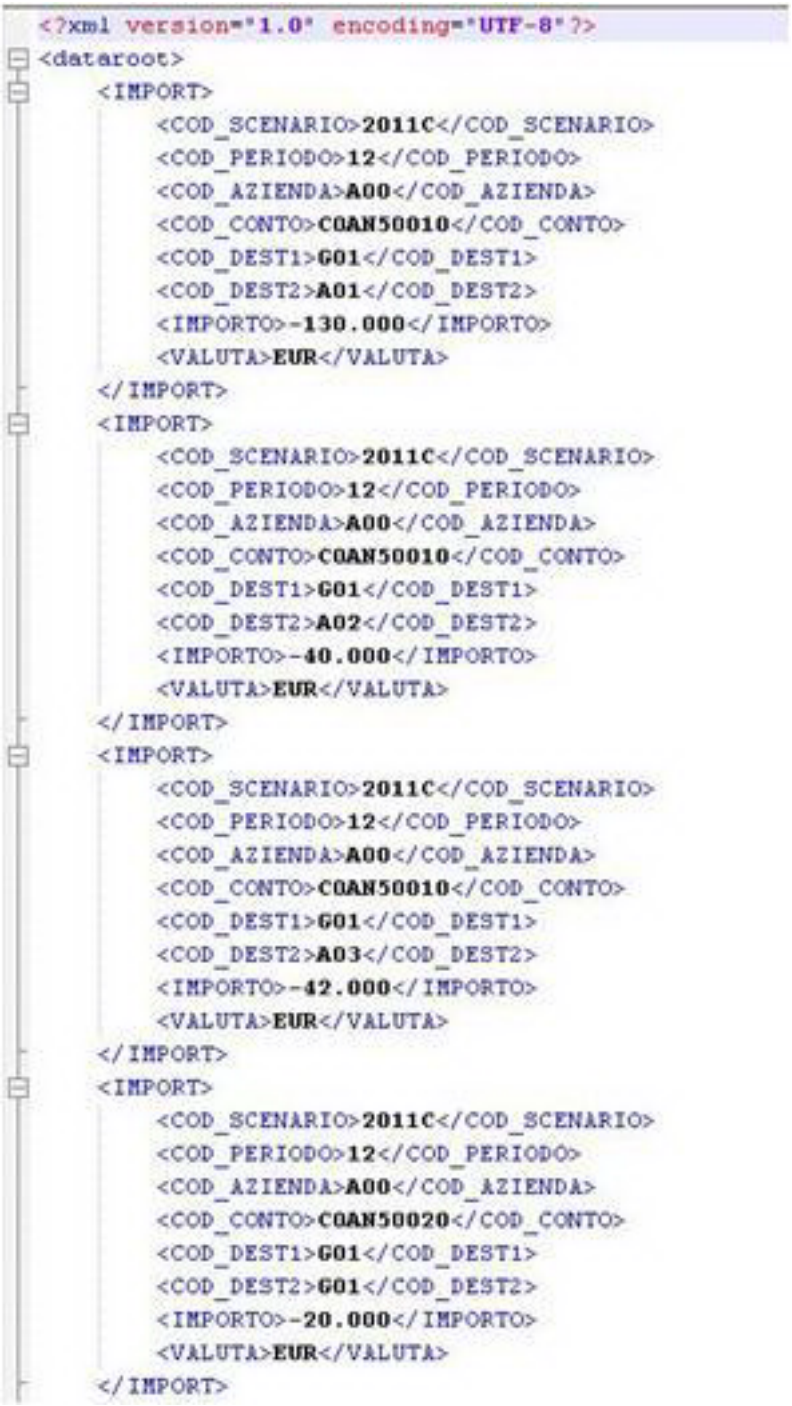
- When the **Text tab delimited** option is enabled, it is possible to extract data from a delimited text file but in this case the separator is the *tab* character.

```
2018BDG 01      A0      P1      $AMOUNT 100
2018BDG 02      A0      P1      $AMOUNT 100
2018BDG 03      A0      P1      $AMOUNT 100
2018BDG 04      A0      P1      $AMOUNT 100
2018BDG 05      A0      P1      $AMOUNT 100
2018BDG 06      A0      P1      $AMOUNT 100
2018BDG 07      A0      P1      $AMOUNT 100
2018BDG 08      A0      P1      $AMOUNT 100
2018BDG 09      A0      P1      $AMOUNT 100
2018BDG 10      A0      P1      $AMOUNT 100
2018BDG 11      A0      P1      $AMOUNT 100
2018BDG 12      A0      P1      $AMOUNT 100
```

For the text files, it is possible to specify in the **Charset** field the character encoding by which it was created; this information will be used by the system during the extraction phase to understand the processed characters. If the **Charset** is not defined, during the extraction stage, the system will try to recognise it automatically.

Other files

- When the **Xml** option is enabled, it is possible to extract data from a Xml file (eXtensible Markup Language) with a predefined structure (as sown in the image below). This file must comprise a *root* and as many *tags*, with the same name, as records to be extracted; the name of the tag must be defined in the **Xml Node** field. Each tag must contain as many *sub tags* (always the same) as pieces of information to be extracted; then the names of these sub-tasks must be related to the dimensions in the Data source fields definition.

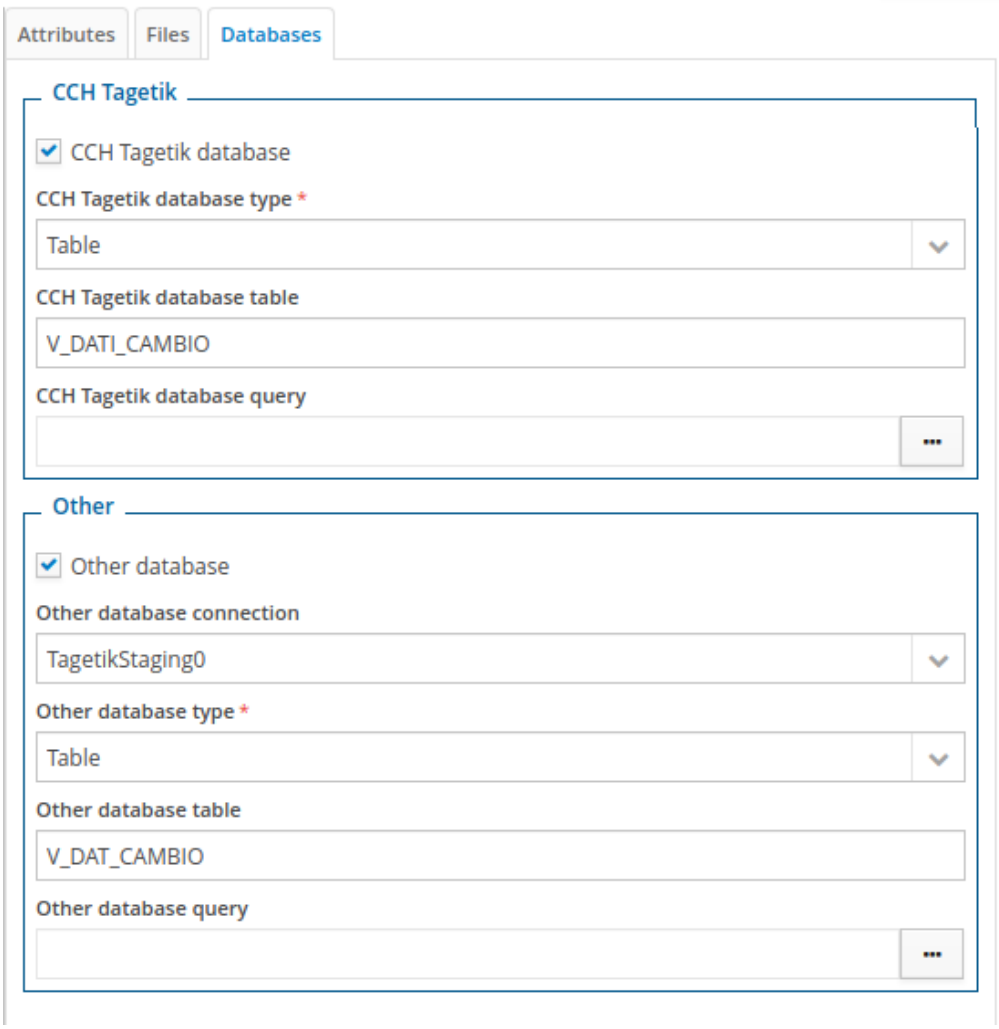


- When the **Xbrl** option is enabled, it is possible to extract data from a standard zip file containing the information related to the Xbrl. It is possible to tell the system what information is contained in the file by selecting the *Item*, *Context* and *Instance* values for the **Xbrl Type** field.

Databases

In this tab, it is possible to defined the data sources of Database type for the selected Data source. As with the tab related to the data sources of File type, each type can be enabled by the related option and may require additional information.

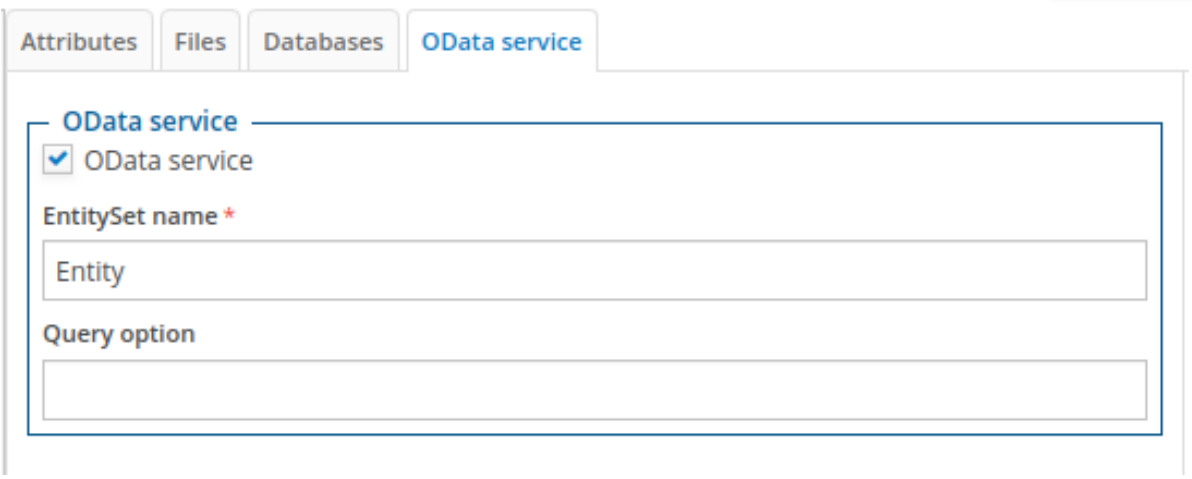
- When the **CCH Tagetik database type** option is enabled, it is possible to extract data from the database of CCH Tagetik. By selecting a value in the **CCH Tagetik database type** field, it is possible to specify:
  - *Table*: data are extracted from Table or a Window specified in the **CCH Tagetik database table** field.
  - *Query*: data are extracted by a query specified in the **Query database CCH Tagetik** field. In the query's journal it is possible to use the dynamic syntax of the ETL (see **Filter Strings Syntax or SQL Query**).
- When the **Other database** option is enabled, it is possible to extract data from another database that can be reached by a JDBC connection defined in the **Other database connection** field. Data are extracted in the same way as with the **CCH Tagetik database** type, but in this case the **Other database type**, **Other database table** and **Other database query** fields are used for the definition.



OData service

In this tab you can define the OData type data sources for the selected Data source. As with the File and Database type data sources tabs, the type can be enabled by the relevant option and requires some additional information.

- Enabling the **OData service** option activates the extraction of data from an OData service by setting up:
  - *EntitySet name*: name of the entity exposed by the OData service
  - *Query option*: query options to be specified with the standard OData syntax



At runtime or when defining the ETL Job Extractions, it is possible to select the OData Endpoint on which the root of the service and the possible authentication method are defined.

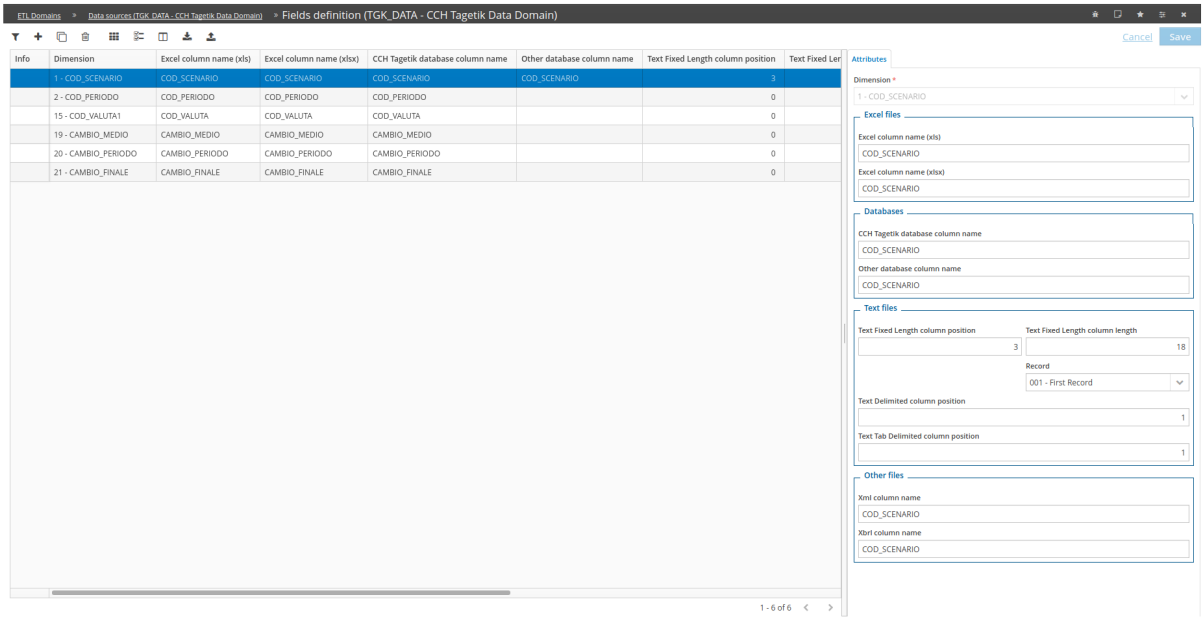
Data Source - fields definition

For each Data source and for each data source type managed, it is necessary to define the correspondence between the information available in the data source and the dimensions of the ETL Domain, in order to control the insertion of extracted data into the related columns of the ETL *Staging* table when running the extraction task that uses such Data source.

Definition

The Fields Definition of the Data source can be accessed from the *Fields definition* link in the Data source window.

If the ETL Domain is locked, then the window will be in read-only mode.



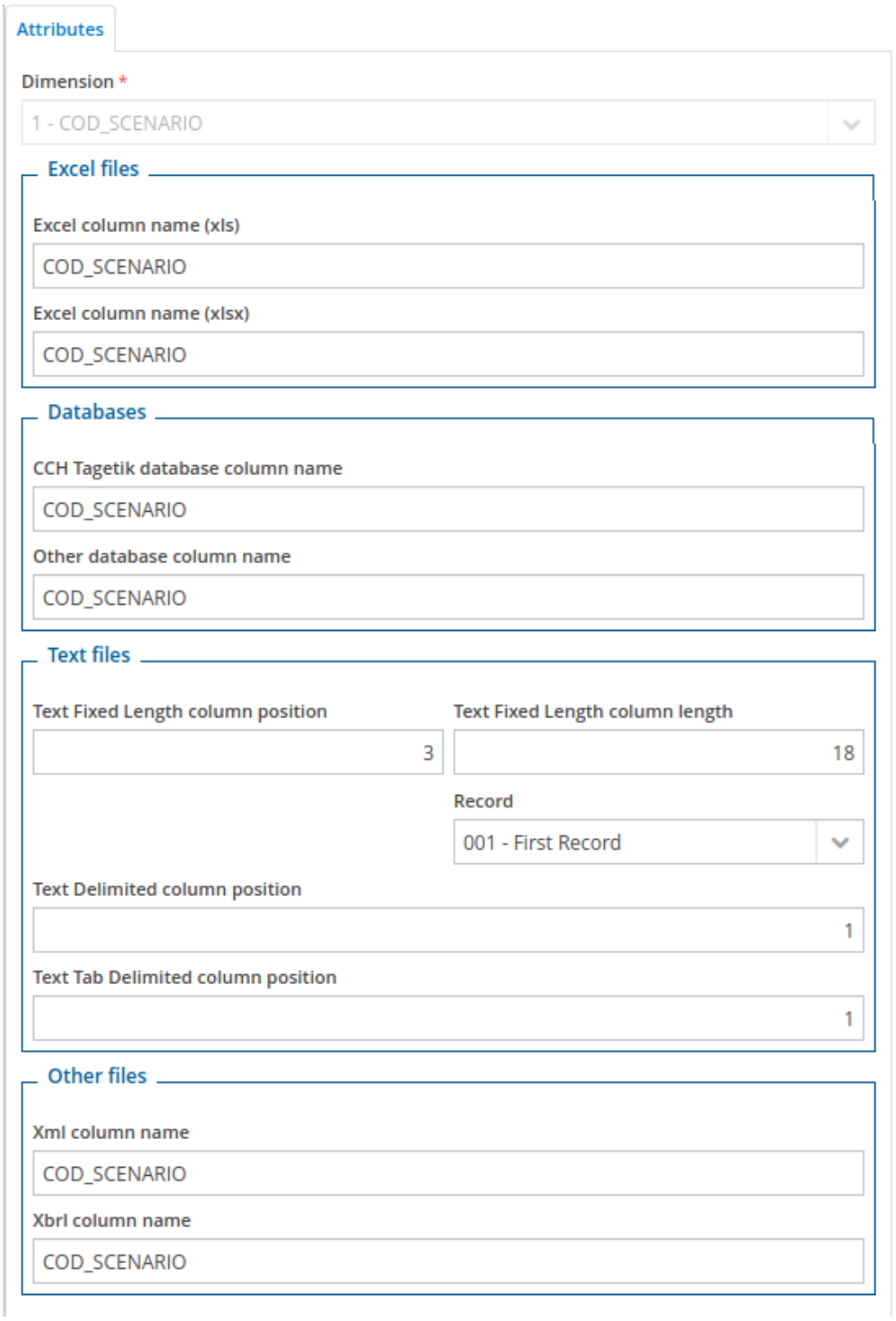
Each record of the Data Source's Fields Definition is identified by:

- **Dimension**: the dimension of the ETL Domain containing the field of the *Staging* table in which the value extracted from the data source will be saved.

  Depending on the type managed in the **Data source**, the system will display different fields for the definition of each correspondence between the **Dimension** and the values present in the various data sources:

  - if the extraction from *Excel (xls)* is enabled, the system will display the **Excel column name (xls)** field which allows specifying the values present on the xls file in one of the cells of the header record in order to determine the source column.
  - if the extraction from *Excel (xlsx)* is enabled, the **Excel column name (xlsx)** field will be used to specify the value present on the xlsx file in one of the cells of the header record in order to determine the source column.
  - if the extraction from *Database CCH Tagetik* is enabled, the system will display the **CCH Tagetik database column name** field which allows specifying the name of the source column of the table / window or of the query run on the CCH Tagetik database.
  - if the extraction from *Other database* is enabled, the system will display the **Other database column name** field which allows specifying the name of the source column of the table / window or of the query run on another database.
  - if the extraction from *Text Fixed Length* is enabled, the system will display:
    - the **Text Fixed Length column position** field which allows specifying the index of the first character of the value to be extracted from each record of the file;
    - the **Text Fixed Length column length** field which allows specifying the length (in characters) of the value to be extracted from each record of the file;
    - the **Record** field which allows specifying a **Record type** for the extraction of the value from a specific record of the file.
  - if the extraction from *Text Delimited* is enabled, the system will display the **Text Delimited Column Position** field which allows specifying the number of the file's source column.

- if the extraction from *Text Tab Delimited* is enabled, the system will display the **Text Tab Delimited Column position** field which allows specifying the number of the file's source column.
- if the extraction from *Xml* is enabled, the system will display the **Xml Column name** field which allows specifying the name of the source tag.
- if the extraction from *Xbrl* is enabled, the system will display the **Xbrl column name** field which allows specifying the name of the information provided by the parser of the source Xbrl file.
- if the extraction from OData service is enabled, the system displays the **OData service column name** field where you can specify the name of the property of the entity exposed by the service from which the values have to be extracted



If new data source types are added into the Data Source, it is necessary to access this window in order to complete the parametrization; otherwise, an error will occur when running the extraction task that uses this Data source.

It is not possible to relate those **Dimensions** that:

- are used in the **Dimension1..9** fields of the extraction tasks
- have the **Identity** set to *Record num.* that automatically defines the field of the *Staging* table related to the dimension with a progressive number for each extracted record.

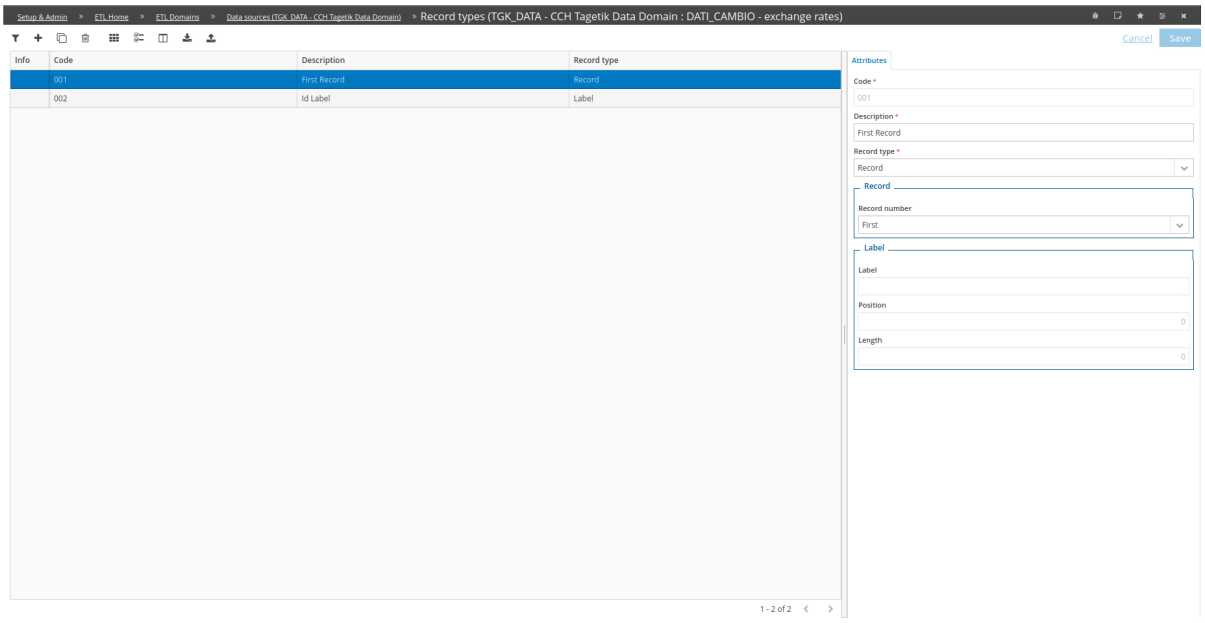
Data Source - Record Types

For those Data sources in which the extraction from **Text Fixed Length** file is managed, it is possible to indicate the presence of some records that do not belong to data to extract but contain information to be used for the automatic definition of some fields of the *Staging* table for each records extracted from the file. The record types identify these specific records and can be used, once related to the dimensions managed in the Data source fields definition, to enable their automatic definition with the information extracted from those records.

Definition

The definition of the record types can be accessed from the *Record types* link in the Data sources window.

If the ETL Domain is locked, then the window will be in read-only mode.



Each Record type is identified by:

- **Code**: record type id to be used in the **Record** field of the Data source fields definition;
- **Description**: Record type description;
- **Record type**: specifies the type of record. Available types:
  - *Record*: allows you to specify one of the first 5 or one of the last 5 records present in the file.

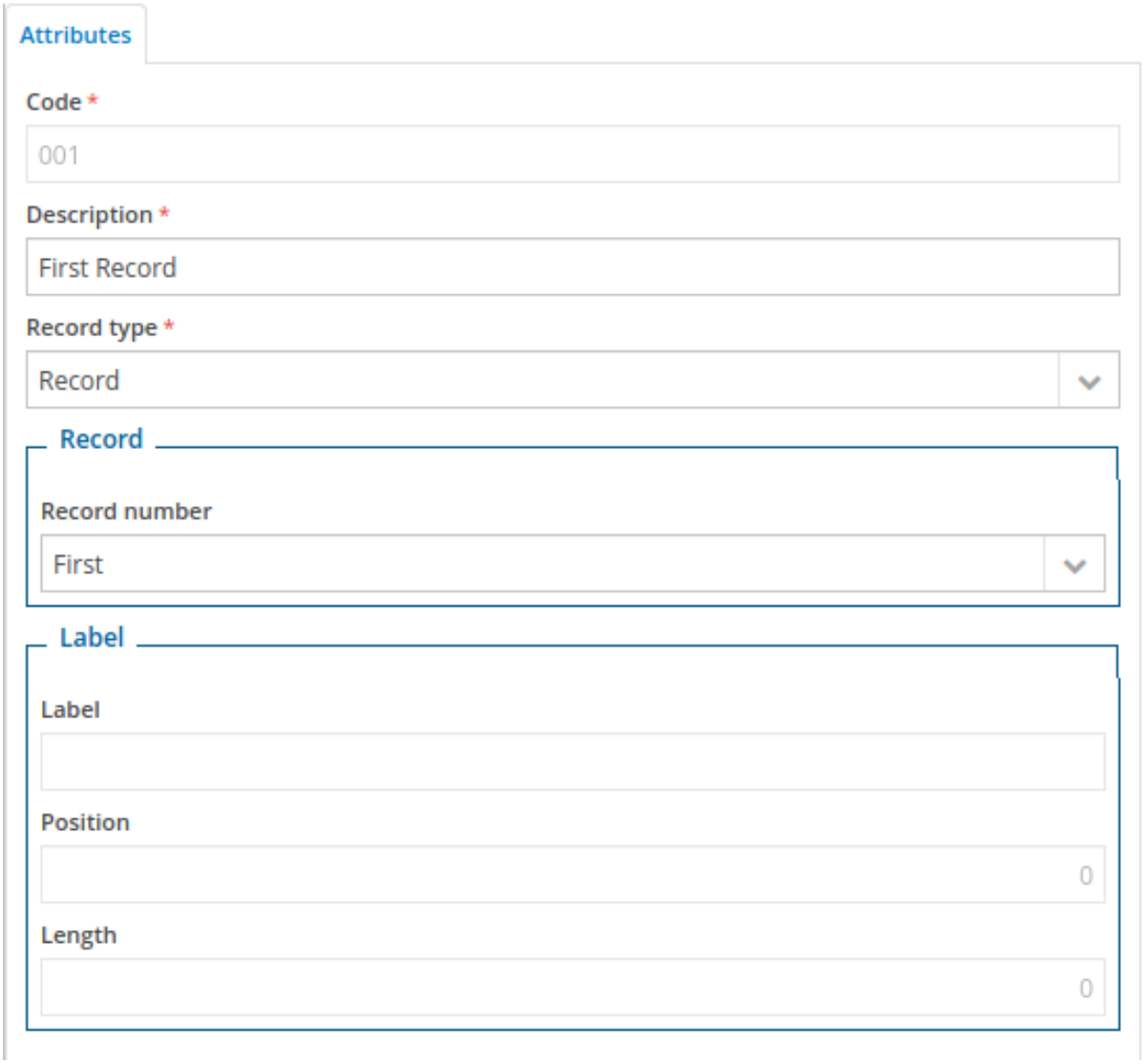
- *Label*: allows you to specify a record containing a specific id defined in the **Label** field that starts with the character defined in the **Position** field and whose length is defined in the **Length** field.

Attributes

Besides the **Code**, the **Description** and the **Record type**, in the Attributes tab it is possible to define other pieces of information on the value of the selected **Record type** field.

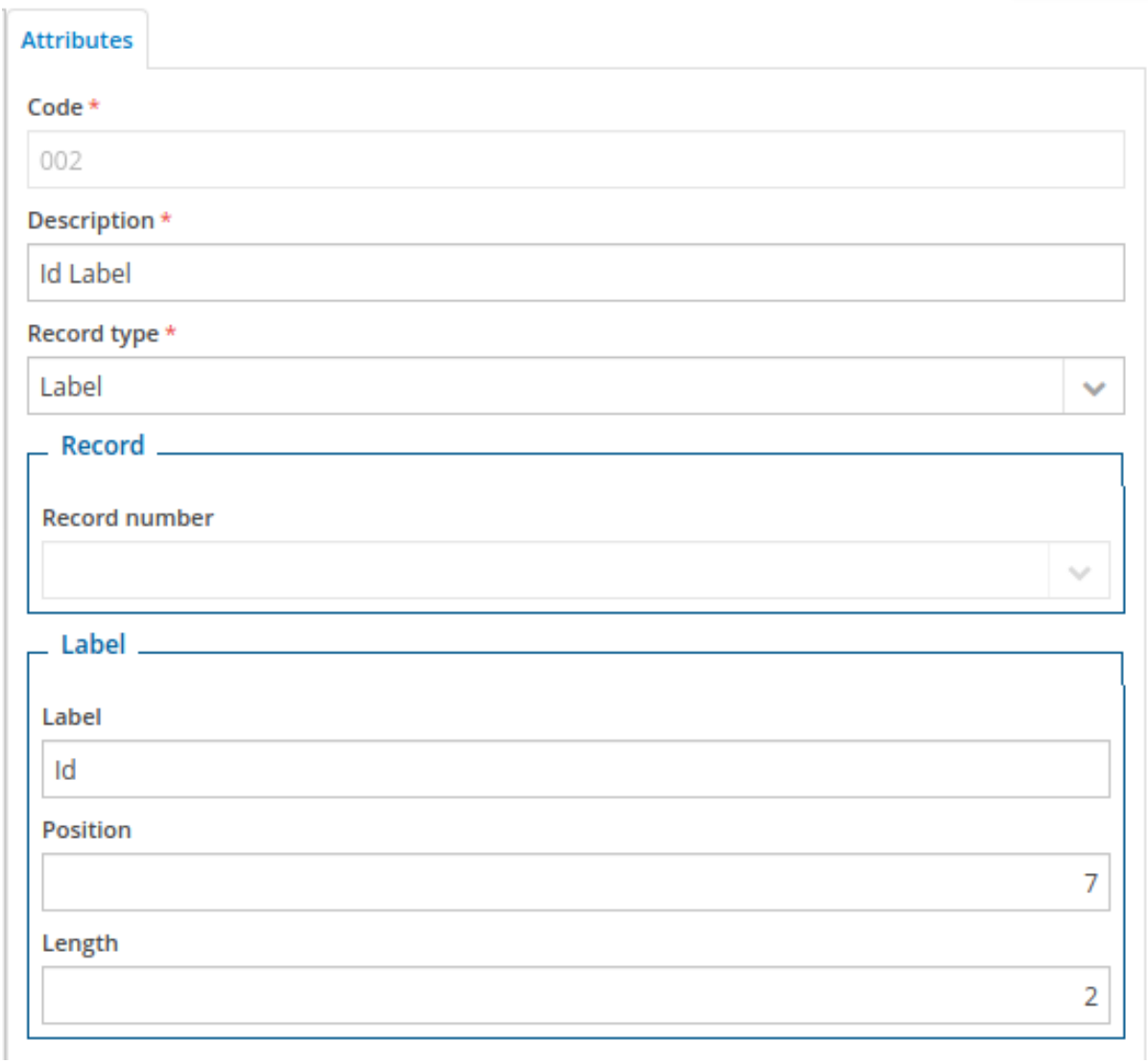
If *Record* has been selected, it must be specified in the **Record** field. Possible values:

- *F1*: first record
- *F2*: second record
- *F3*: third record
- *F4*: fourth record
- *F5*: fifth record
- *L5*: fifth from last record
- *L4*: fourth from last record
- *L3*: third from last record
- *L2*: penultimate record
- *L1*: last record



If *Label* has been specified, the following must be defined:

- **Label**: id to search for in each row of the file to identify the record;
- **Position**: index of the character used to verify the presence of the **Label** for each row of the file;
- **Length**: characters in length of the **Label**.



The records of the file identified by this parametrization are removed only if the Record types are used in the Data source fields definition.

Lookup Table

A lookup table defines a logical structure that establishes a correspondence (mapping) between one or more input values (Input ETL Dimensions) and one or more output values (Output ETL Dimensions). These ETL Dimensions are related to some fields present on the *staging* table.

Input1	Input2	...	InputN	Output1	Output2	...	OutputN
In_1	In_2	...	In_N	Out_1	Out_2	...	Out_N
In_3	In_4	...	In_X	Out_3	Out_4	...	Out_X

These rules are used by some transformation tasks of the Routine when it is necessary to transcode extracted data, in order to conform them to the loading step.

For example, the lookup tables are required in the following cases:

- the amounts present on different accounts of the general accounting must be aggregated in a single account of the chart of accounts of CCH Tagetik

General Ledger Account (in)	Tagetik Account (out)
Revenues 1	Product Revenues
Revenues 2	Product Revenues
Revenues 3	Product Revenues
Revenues 4	Other Revenues
Revenues 5	Other Revenues

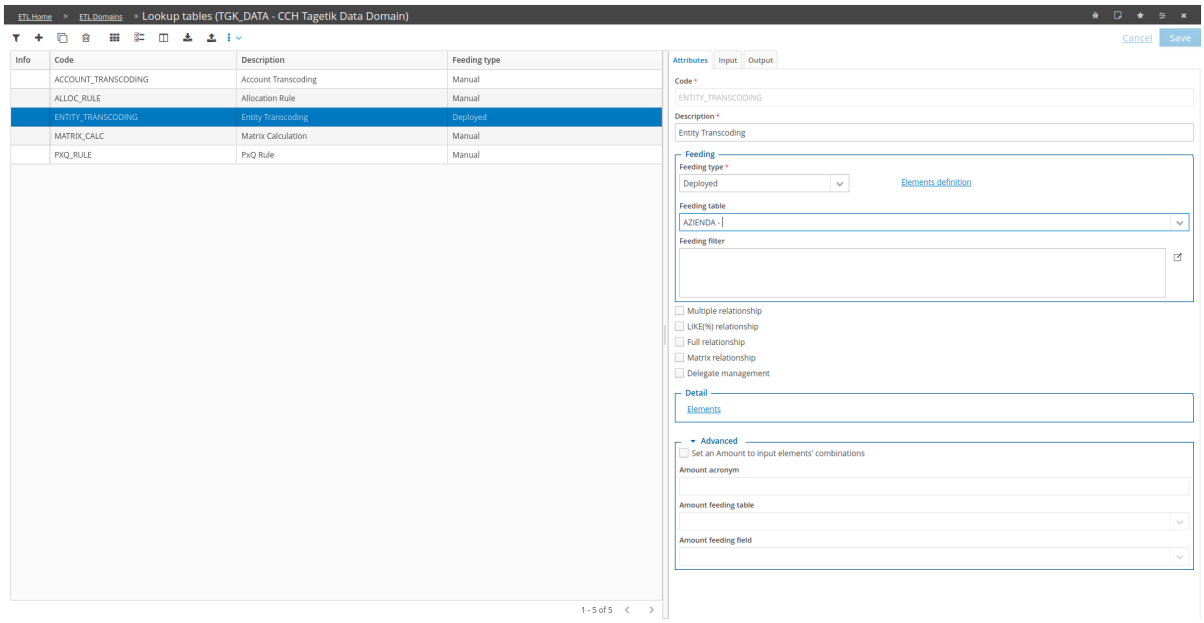
- a client must be transcoded into another CCH Tagetik client based on the combinations with the Cost centres (for example Premiere or Generic customers). The pair (Customer 2, Sales) is transformed into Premiere while the pair (Customer 2, IT) is transformed into Generic customer.

Customer (in)	Cost Center (in)	Tagetik Customer (out)
Customer 1	Sales	Premiere
Customer 2	Sales	Premiere
Customer 3	Sales	Generic
Customer 1	IT	Premiere
Customer 2	IT	Generic

Definition

The definition of the Lookup tables can be accessed by the *Lookupt table* link in the ETL domains window.

If the ETL Domain is locked, then the window will be in read-only mode.



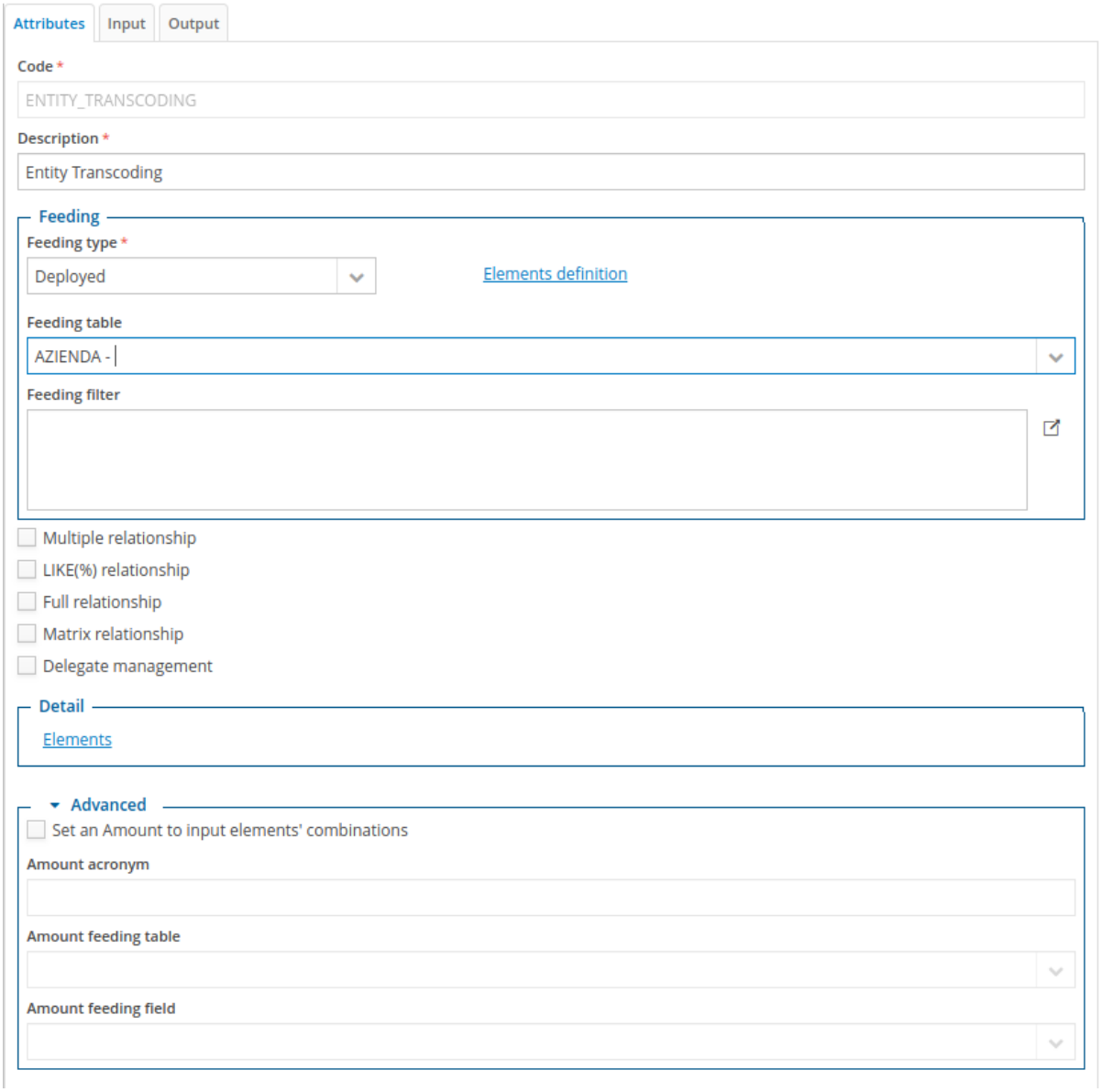
Each Lookup table is identified by:

- **Code**: code of the Lookup table to use in the transformation tasks;
- **Description**: description of the Lookup table;
- **Feeding type**: specifies how the Lookup table elements must be generated. It can be set to:
  - **Manual**: the user defines the elements manually through the related management;
  - **Tagetik**: the elements are generated automatically by the system by the Automatic feeding data processing according to the Lookup table definition. In a second step, through the management of the elements, it is possible to edit the automatically generated elements in order to meet any specific needs;
  - **Deployed**: the elements are generated automatically by the system by the Automatic feeding data processing according to the definition of the Lookup table and to the

definition of the Lookup table's elements that allows for a synthetic definition of the rules for the creation of the actual elements. In a second step, through the elements management it is also possible to edit the automatically generated elements in order to meet any specific needs.

Attributes

In addition to **Code**, **Description** and **Feeding type**, in the Attributes detail it is possible to define some additional information:



Feeding

- *Elements definition*link: if the **Feeding type** is *Deployed*, from this link it is possible to access the Lookup table elements definition window and define - also synthetically - the rules for the creation of the actual elements that will be generated by the Automatic feeding data processing;
- **Feeding table**: if the **Feeding type** is *Tagetik* or *Deployed*, it specifies the table of the CCH Tagetik database from which it is possible to generate the Lookup table's elements by the Automatic feeding data processing according to the definition of the input and output dimensions;
- **Feeding filter**: if the **Feeding type** is *Tagetik* or *Deployed*, it specifies the filter to apply to the creation of the Lookup table elements by the Automatic feeding data processing.

Options

The options described in this paragraph define the relationship between the input elements and the output elements of the Lookup table.

- **Multiple relationship**: if this option is enabled, the same combination of input elements can be connected with the different combinations of output elements;

Input 1	Output 1
AA000	D00
AB000	D00
AC000	D00
AA000	D01

- **LIKE(%) relationship**: if this option is enabled, each input element can be defined *synthetically* by *%*.

  In a definition of the elements in which the value D00 is associated with three different input values starting with the character A,

Input 1	Output 1
AA000	D00
AB000	D00
AC000	D00

it is possible to define the elements as shown below:

Input 1	Output 1
A%	D00

The options **Multiple relationship** and **LIKE(%) relationship** can be enabled at the same time inside the same Lookup table.

- **Full relationship**: if this option is enabled, all data available in the *staging* table when the lookup table is applied by an **Enrichment: lookup table** type transformation task, must be recorded in the input combinations in the lookup table elements; this allows you to verify that each data record is effectively processed. This check is done by a **Lookup table diagnostic** type transformation task;
- **Matrix relationship**: when this option is enabled, for each combination of input elements / output elements, it is possible to manage some additional information that is required if an **Enrichment: matrix calculation** type task is included during the transformation step of a Routine.
- **Delegate management**: if this option is enabled, the Administrator user that usually manages the Lookup table and its elements, can delegate the management of the elements to the End User.

If a Lookup table diagnostic task has been defined among the transformation tasks of a Routine , the system will verify that the defined elements are consistent with the enabled options.

Detail

The **Elements** link provides access to the management of the Lookup table elements. This is enabled for all types accepted by the **Feeding type**.

Advanced

The fields described in this section allow defining a particular type of Lookup table that, instead of combining a set of output elements, link an amount with each combination of input elements. These particular types of lookup tables are used in the **Transfer Price from the rule** and **Allocation** transformation tasks. On the other hand, the standard **Enrichment: lookup table** tasks define the fields of the *staging* table related to the output dimensions with the values of the output elements at comparable values of the input elements available in the fields of the *staging* table related to the input dimensions. The tasks that use these type of lookup tables generate a result by using the amount defined in the elements at comparable values of the input elements available in the fields of the *staging* table related to the input dimensions.

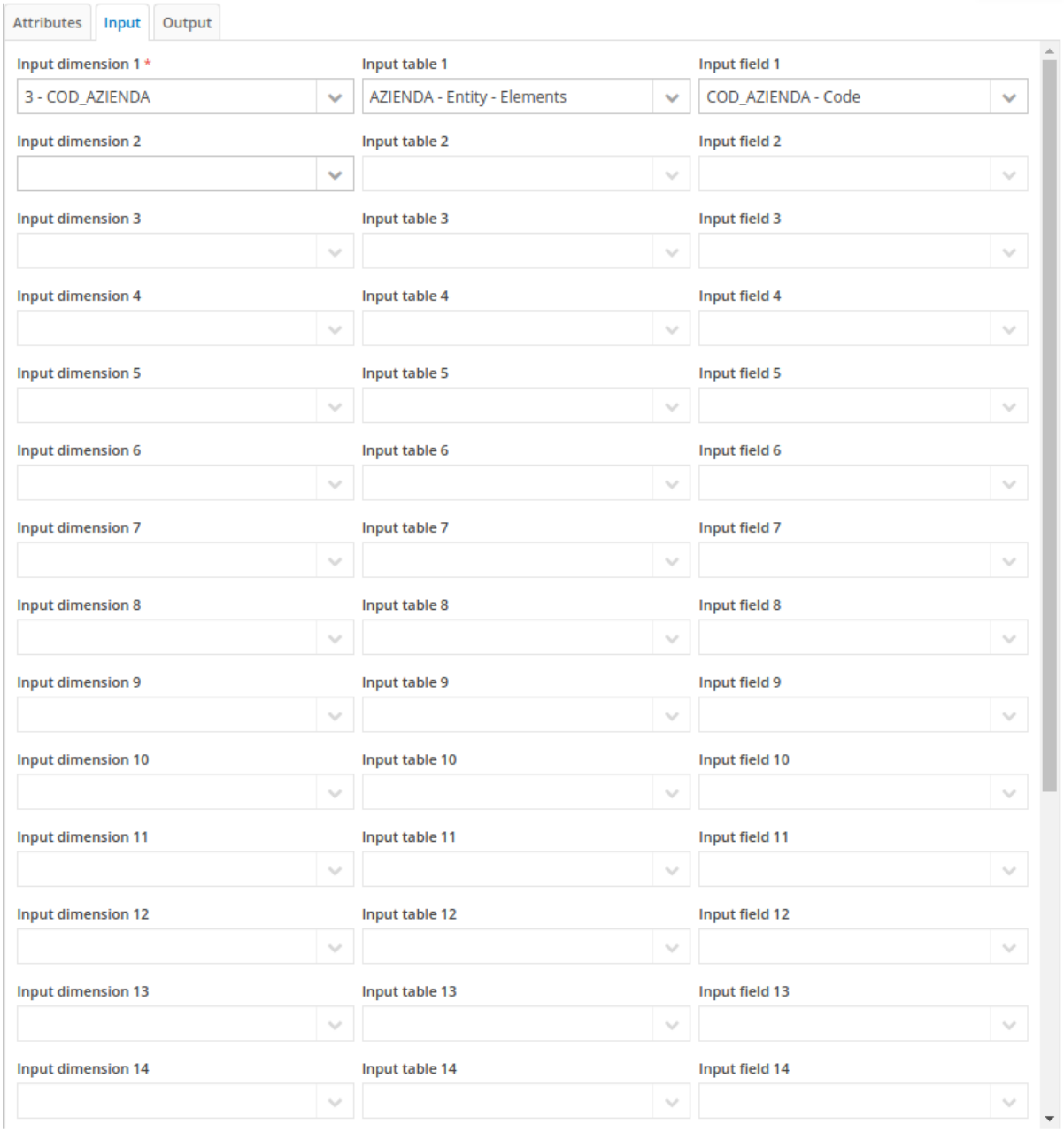
- **Set an Amount to input elements' combinations**: this option enabled the lookup table describe above;
- **Amount acronym**: this allows defining the name of the columns relayed to the amount that will be displayed in the Lookup Table elements in place of the output elements;
- **Amount feeding table**: table of the CCH Tagetik database from which the amount in the rule's elements is generated. The way this field behaves changes according to the value of the **Feeding type** field:
    - if it is *Deployed*, the field is visible and the default value is the **Feeding table** field's one;
    - if it is *Tagetik*, the field is hidden, the value is overridden and is that of the **Feeding table** field;
    - if is *Manual*, the field is hidden.
- **Amount feeding field**: fields related to the **Amount feeding table** from which the amount in the lookup table's elements is generated. The way this field behaves, as in the previous case, changes according to the value of the **Feeding type**field:
    - if it is *Deployed* or *Tagetik*, the field is visible and allows selecting a field of the table defined on **Amount feeding table**;
    - if it is *Manual*, the field is hidden.

Input

The detail tab related to the input dimensions includes 20 items composed of:

- **Input dimension**: ETL Domain dimension related to a field of the *staging* table and to the input element *n*;
- **Input table**: CCH Tagetik database table from which the input element *n* must be generated. The way this field behaves depend on the value of the **Feeding type** field:
    - if it is *Deployed*, the field is visible and the default value is that of the **Feeding table** field ;
    - if it is *Tagetik*, the field is not visible, the value is overridden and is that of the **Feeding table** field;
    - if it is *Manual*, the field is hidden.
- **Input field**: field related to the **Input table** from which the input elements *n* must be generated. The way this field behaves changes according to the values of the **Feeding type** field:
    - if it is *Deployed* or *Tagetik*, the field is visible and it is possible to select a field of the table defined on **Input table**;
    - if it is *Manual*, the field is hidden.

It is mandatory to define at least one input dimension; if more than dimension one is defined, they must be sequential.



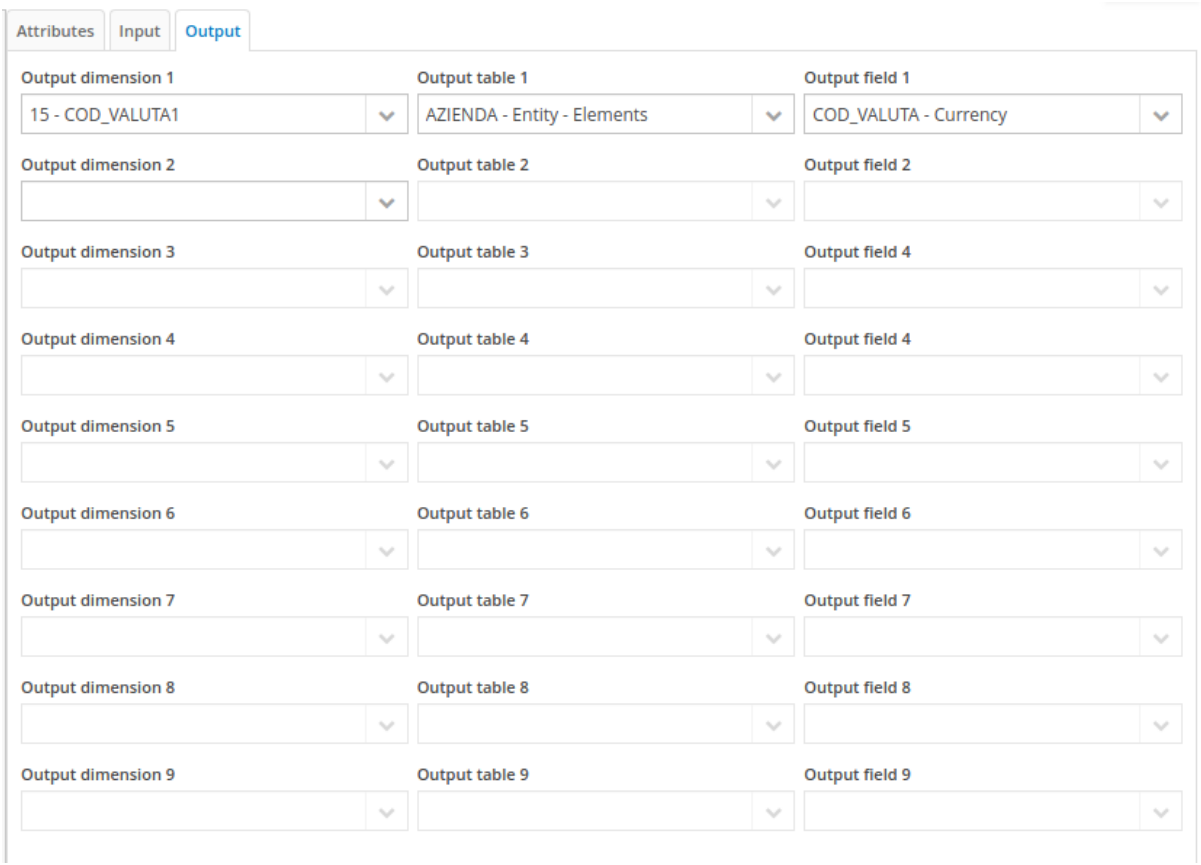
Output

The detail tab related to the output dimensions includes 9 items composed of:

- **Output dimension**: ETL Domain dimension related to a filed of the *staging* table and to the element of the output element *n*;
- **Output table**: table of the CCH Tagetik database from which the output element *n* must be generated. The way this field behaves depends on the value of the **Feeding type** field:
    - if it is *Deployed*, the field is visible and the default value is that of the **Feeding table** field;
    - if it is *Tagetik*, the field is hidden, the value is overridden and is that of the **Feeding table** field;
    - if it is *Manual*, the field is hidden.

- **Input field**: field related to the Output table from which the *n* output element must be generated. The way this field behaves depend on the value of the **Feeding type** field:
    - if it is *Deployed* or *Tagetik*, the field is visible and allows selecting a field of the table defined on **Output table**;
    - if it is *Manual*, the field is hidden.

It is mandatory to define at least one output dimension if the **Set an Amount to input elements' combinations** option is disabled; if more than one dimension is defined, they must be sequential.



Automatic feeding

The Actions menu of the window includes the **Automatic feeding** option which runs the data processing that generates the Lookup Table elements automatically. This menu is enabled only if one single record has been selected and if the value of the **Feeding type** field is *Tagetik* or *Deployed*.

Lookup Table elements definition

This window can be accessed from:

- *Elements* link of the Lookup table window.
- *Lookup table* workflow task of the Data Integration group
- *Lookup table - elements* tile in the ETL home page accessible from the End user home page, by selecting an ETL Domain and a Lookup table.

If the ETL Domain is locked, the window will be in read-only mode.

The system displays as many **Input 1..20** / **Output 1..9** fields as input/output Dimensions defined on the lookup table.

Each field has as suffix of the name the value of the **Acronym** field of the related dimension.

This window allows you to define - also synthetically - the rules for the creation of the actual elements that will be generated by the Automatic feeding data processing.

Lookup Table elements

This window can be accessed by the *Elements* link of the Lookup table window.

If the ETL Domain is locked, the window will be in read-only mode.

The system displays as many **Input 1..20** / **Output 1..9** fields as input/output Dimensions defined on the lookup table.

Each field has as suffix of the name the value of the **Acronym** field of the related dimension.

In this window it is possible to manage the records on which combinations of input elements and output elements are defined.

If the **Feeding type** is *Manual*, the definition of the elements must be performed manually by the user.

On the contrary, if the **Feeding type** is *Tagetik* or *Deployed*, the Automatic feeding data processing will pre-fill the lookup table elements on the basis of the lookup table definition; nevertheless, the user can access in order to inherit the records automatically generated by the data processing.

EXAMPLES

The following examples show how to define the Lookup table elements.

Example 1: Assume that we have a hierarchical structure that groups the entities by geographical area and for such hierarchy the flat structure correspondence with the two levels Country and Continent. We want to define a *Tagetik* lookup table that relates each entity to its Country and its Continent.

	Dimension	Table	Field
Input 1	ETL_AZI	AZIENDA_ GERARCHIA_ FLAT	COD_AZIENDA
Output 1	ETL_AZI_STATO	AZIENDA_ GERARCHIA_ FLAT	COD_ AZIENDA_ ELEGER3
Output 2	ETL_ AZI_ CONTINENTE	AZIENDA_ GERARCHIA_ FLAT	COD_ AZIENDA_ ELEGER2

Example 2: Let us assume we want to define a *Deployed* lookup table that relates the CO.GE. account to the CCH Tagetik Account / Custom dimension 1 pair.

Input Dimension 1 = ETL_CONTOCOGE

Output Dimension 1 = ETL_CONTO

Output table 1 = CONTO (CCH Tagetik table containing the accounts table)

Output field 1 = COD_CONTO (account code; in our example A01, A02, A11, A12, B01, B02, B11)

Output dimension 2 = ETL_DEST1

Output table 2 = DEST1 (CH Tagetik table containing the table of the custom dimension 1 elements)

Output field 2 = COD_DEST1 (custom dimension 1 element code; in our example DD1, DD2, DD3)

Synthetic definition:

Input1 (ETL_CONTOCOGE)	Output1 (ETL_CONTO)	Output2 (ETL_DEST1)
A	COD_CONTO LIKE 'A0%'	COD_DEST1 = 'DD1'
B	COD_CONTO LIKE 'B%'	COD_DEST1 <> 'DD1'
C	COD_CONTO = 'B01'	

By running the Automatic feeding, the system will generate the following combinations.

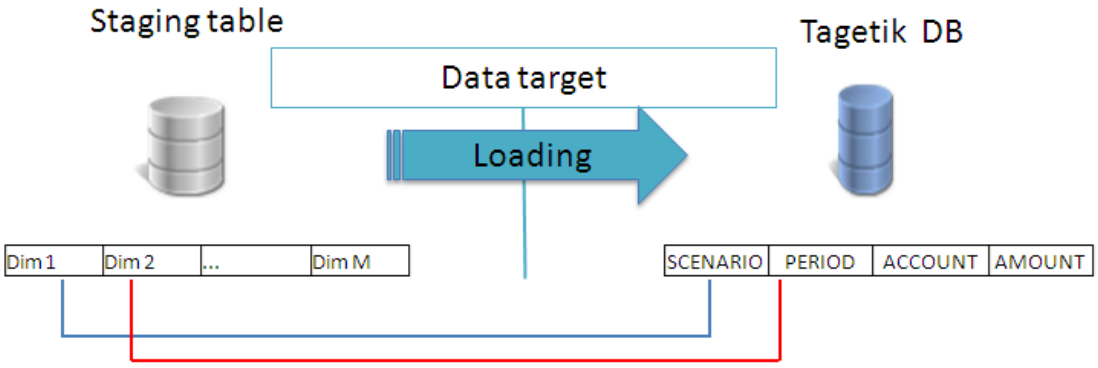
Input1 (ETL_CONTOCOGE)	Output1 (ETL_CONTO)	Output2 (ETL_DEST1)
A	A01	DD1

Input1 (ETL_CONTOCOGE)	Output1 (ETL_CONTO)	Output2 (ETL_DEST1)
A	A02	DD1
B	B01	DD2
B	B01	DD3
B	B02	DD2
B	B02	DD3
B	B11	DD2
B	B11	DD3
C	B01	DD1
C	B01	DD2
C	B01	DD3

Data Target: CCH Tagetik

Data targets: CCH Tagetik are used to define the tables on the CCH Tagetik database on which it is possible to load data extracted and possibly transformed during loading. Defining a Data target: CCH Tagetik is not enough to load data; it is also necessary to connect it to a loading task of a Routine.

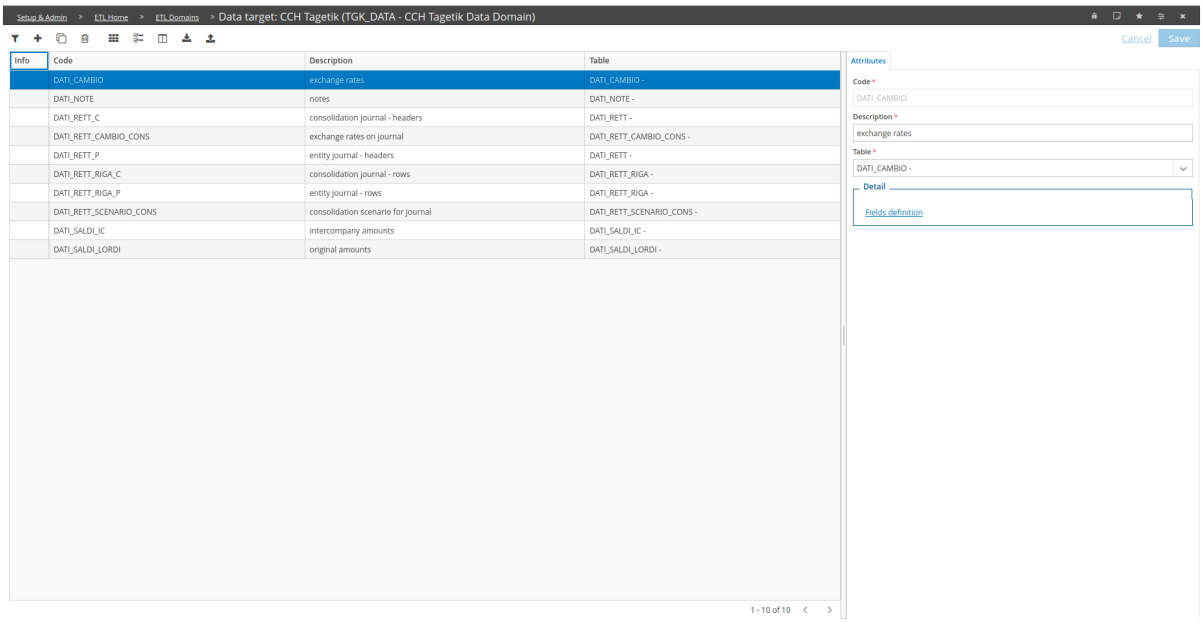
After defining the tables on which data can be loaded, it is necessary to access the Data target: CCH Tagetik fields definition in order to define the filling mode of each field of each table.



Definition

The Data target: CCH Tagetik definition can be accessed from the *Data target: CCH Tagetik* link of the ETL Domains window.

If the ETL Domain is locked, then the window will be in read-only mode.



Each Data target: CCH Tagetik is identified by:

- **Code**: code of the Data target: CCH Tagetik to use in the loading task;
- **Description**: description of the Data target: CCH Tagetik;
- **Table**: name of the CCH Tagetik application database's table on which data must be loaded.

Attributes

In addition to **Code**, **Description** and **Table**, the Attributes window provides a *Fields definition* link from which it is possible to access the Data target: CCH Tagetik fields definition window in which you can specify, for each field of the table defined in the Table field, if the field must be filled during loading with the values available in a field of the staging table related to a Dimension of the ETL Domain or with one single value.



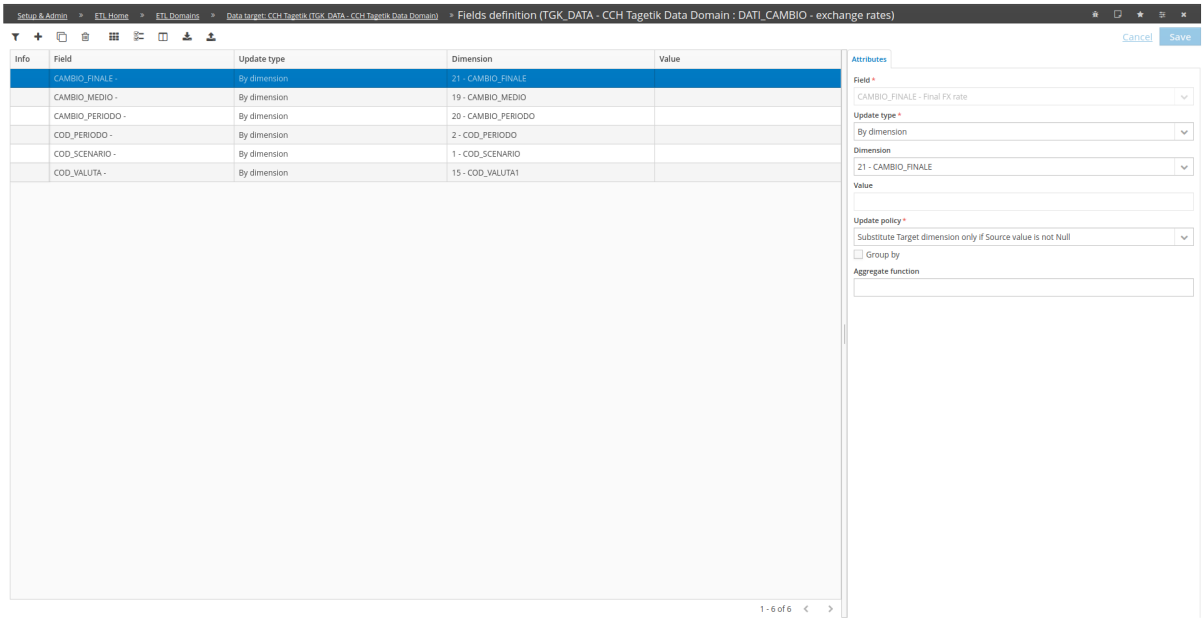
Data Target: CCH Tagetik - Fields definition

For each Data target: CCH Tagetik, it is necessary to specify the list of fields to process and how they shall be populated. For each field, it is possible to specify if it must be populated with the values present in a *staging* table's field related to a Dimension of the ETL Domain or with one single value. It is also possible to define a data grouping policy by editing the attributes of each field, in order to limit the number of records to process.

Definition

The Definition of the Data target: CCH Tagetik can be accessed from the *Fields definition* link in the Data target: CCH Tagetik window.

If the ETL Domain is locked, then the window will be in read-only mode.



Each record of the Field Definition of the Data target: CCH Tagetik is identified by:

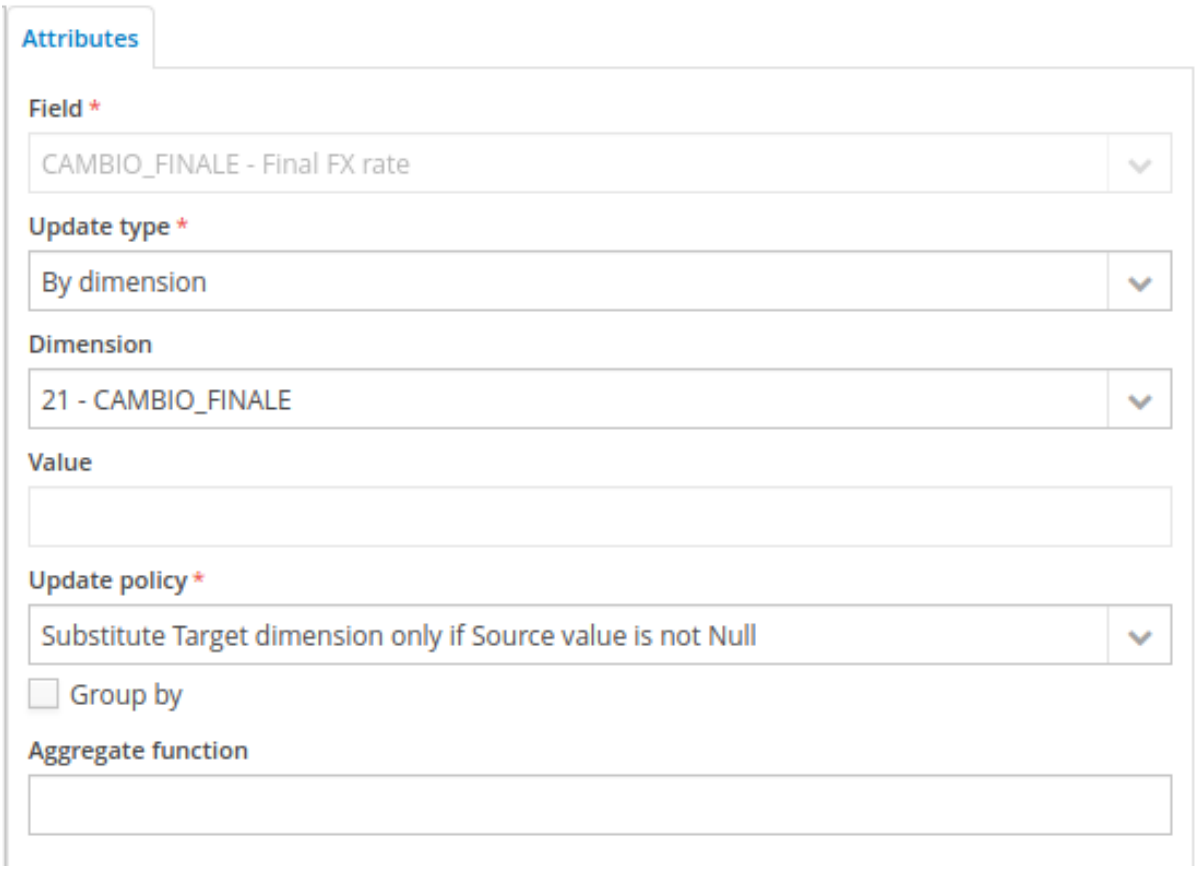
- **Field**: name of the field belonging to the Table defined in the Data target: CCH Tagetik to be handled;
- **Update type**: specifies how the field must be populated. Possible values:
  - *By dimension*: the field must be populated with the values present in the *staging* table's field related to a Dimension of the ETL Domain;
  - *By value*: the field must be populated with a fixed value;
- **Dimension**: if the **Update type** is *By dimension*, it specifies the Dimension of the ETL Domain related to the field of the *staging* table containing data to load on **Field**;
- **Value**: if the **Update type** is *By value*, it specifies the value to set up on **Field** for all uploaded records.

Attributes

In addition to **Field**, **Update type**, **Dimension** and **Values**, in the detail of the Attributes tab it is possible to define:

- **Update policy**: specifies how to populate the **Field** on the target table during update. Possible values:
  - *Always substitute Target dimension value with Source value.*: the field's value is always modified, even with a *NULL* value;
  - *Substitute Target dimension only if Source value is not Null*: the field's value is modified only if the new value is not *NULL*;
  - *Sum if loading value is not null (only for numeric field)*: the new value is added to the old value in the field.
- **Group by**: only if the **Update type** is *By dimension*, it means that before being uploaded, the *staging* table data must be grouped for the *staging* table's field related to the Dimension of the ETL Domain defined in the **Dimension** field;
- **Aggregate function**: only if the **Update type** is *By dimension*, it means that the aggregation function must be applied to the *staging* table's field related to the Dimension of the ETL Domain defined in the **Dimension** field.

After enabling the **Aggregate** option or specifying an **Aggregate function** for a field, it is necessary to set up one of the two fields for all other records for which the **Update type** is set to *By dimension*.



If the name of a field of a table of the CCH Tagetik application DB is deleted or modified, then it is necessary to modify the fields definition of any Data target: CCH Tagetik containing a reference to this field.

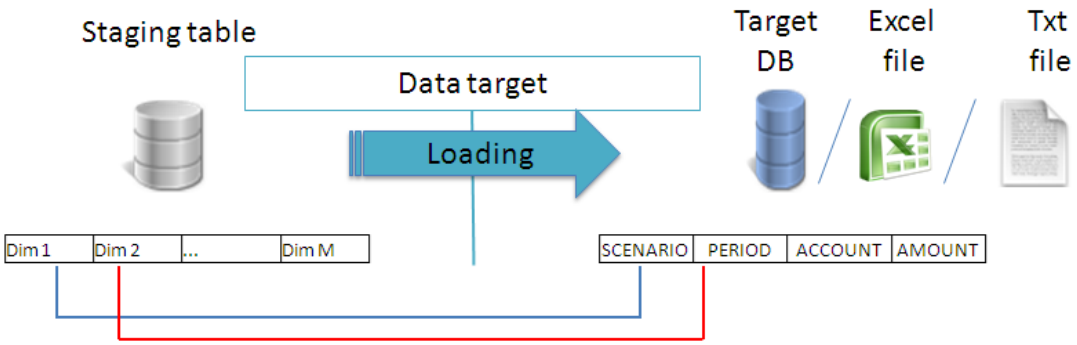
Data Target: Others

Data targets: Others are used to define the custom dimensions on which it is possible to load data that were extracted and possibly transformed during loading. Custom dimension types are:

- Excel (xls)
- Excel (xlsx)
- Xml
- Text fixed length
- Text delimited
- Text tab delimited
- Custom table of CCH Tagetik DB
- Custom table of a secondary DB

Defining a Data target: Others is not enough to load data but it is also necessary to link it to a loading task of a Routine.

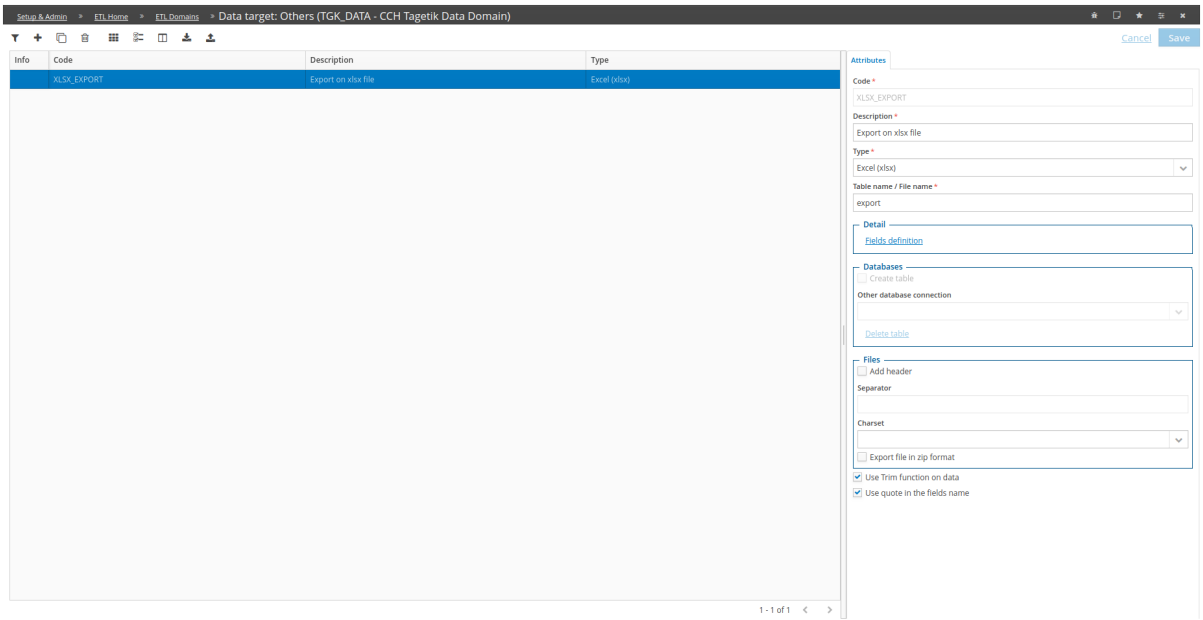
After defining the target tables, it is necessary to access the Data target: Others fields definition in order to define how each field of each custom dimension must be populated.



Definition

The Data target: Others Definition can be accessed from the *Data target: Others* link in the ETL Domain window.

If the ETL Domain is locked, the window will be in read-only mode.



Each Data target: Others is identified by:

- **Code**: code of the Data target: Others to use in the loading task;
- **Description**: description of the Data target: Others;
- **Type**: type of Data target: Others on which the data loading must be run.

Attributes

In addition to **Code**, **Description**and **Type**, the detail of the Attributes include:

- **Table name / File name**: depending on the selected type, it can specify:
  - the name of the custom table if the type is related to a database;
  - the name of a file if the type is related to a file.

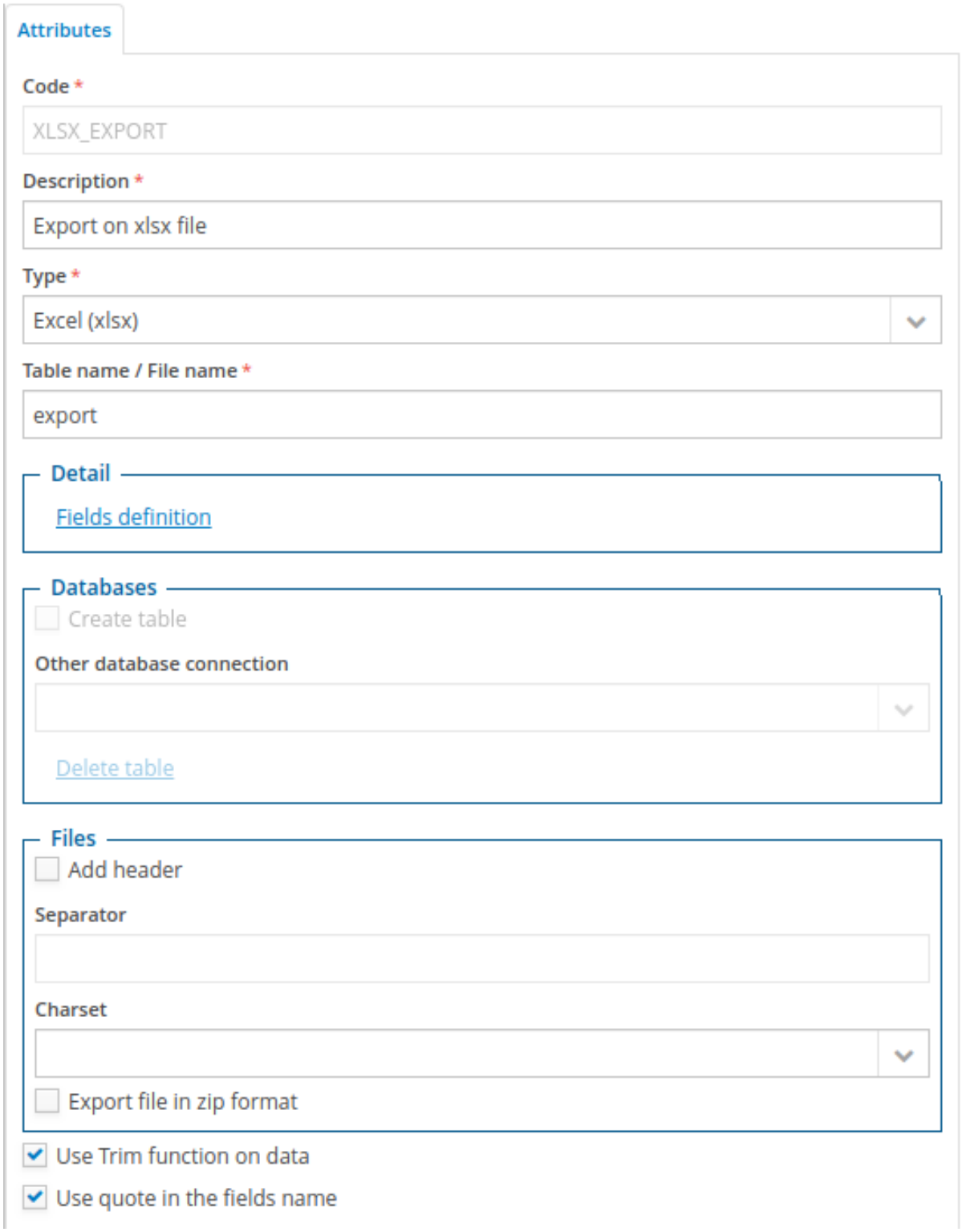
     In this case, when generating the file, the system will add a suffix to the name with a reference to the loading task and with the default extension in order to prevent file overwrite issues, in case the same Routine includes two loading tasks that use the same Data target: Others. For example, if the name defined for Data target: Others of *Text delimited* type is "FILE", and the sorting of the loading task to which the Data target: Others is linked is 10, the name of the generated file will be "FILE10.txt".

    - in order to tell the system to delete the sorting, you just have to add the character *#* at the end of the name. For example, if the defined name is "FILE#", the name of the generated file will be "FILE.txt".
    - in order to tell the system to delete the sorting and not to apply the default extension, you just have to add the character *** at the end of the name. For example, if the defined name is "FILE.csv*", the name of the generated file will be "FILE.csv".

  To define a parametric name, it is possible to use the ETL dynamic syntax (see Filter Strings Syntax or SQL Query).
- a *Fields definition* link to the Data target: Others fields definition window in which it is possible to specify, for each field of the custom dimension, if it must be populated with the values contained in a field of the *staging* table related to a Dimension of the ETL Domain or with one single value (defined manually or calculated automatically by the system).
- a *Databases* section, that can be used if the selected type is related to a database, containing:
  - **Create table**: if enabled, this option tells the system to create the table defined by the Data target: Others fields definition during the first run (if it does not exist);
  - **Other database connection**: if the type is related to a secondary DB, it specifies the JDBC connection to such database;
  - a *Delete table* link: this is used to delete the custom table defined in the **Table name / File name** field, directly from the application;
- a *Files* section that can be used if the selected type is related to a file; it includes:
- **Excel area name / Xml tag name**: when the type is related to an *Excel (.xls)*, *Excel (.xlsx)* or *Xml* files, it is possible to specify:
- the name of the sheet and of the data area, for the excel files;
- the name of a record's tag, for the xml files.

if the value is not specified, the system will use the value specified in the **Table name / File name** field.

- **Add header**: if enabled, this option tells the system to add as first row of the generated file, a record containing the columns' headers;
- **Separator**: if the type is related to a *Text delimited* file, it allows you to specify the separator character for the pieces of information contained in the file's columns;
- **Charset**: if defined, it specifies the character encoding with which the file must be created. If left empty, the system will use the default one;
- **Export file in zip format**: if enabled, this option tells the system to generate the file in zip format. This option is useful for avoiding "Out Of Memory" problems when a huge amount of data has to be loaded;
- **Use Trim function on data**: if enabled, this option tells the system to remove any space before or after the value to load;
- **Use quote in the fields name**: if enabled, this option tells the system to delimit the tables and fields' names by " when reading data as well as when creating and populating the custom table (if the type is related to a database), in order to keep the *case* of the names defined and to allow the use of names containing characters that are generally forbidden, such as the space.


Attributes
Code *
XLSX_EXPORT
Description *
Export on xlsx file
Type *
Excel (xlsx)
Table name / File name *
export
Detail
Fields definition
Databases
Create table
Other database connection
Delete table
Files
Add header
Separator
Charset
Export file in zip format
Use Trim function on data
Use quote in the fields name

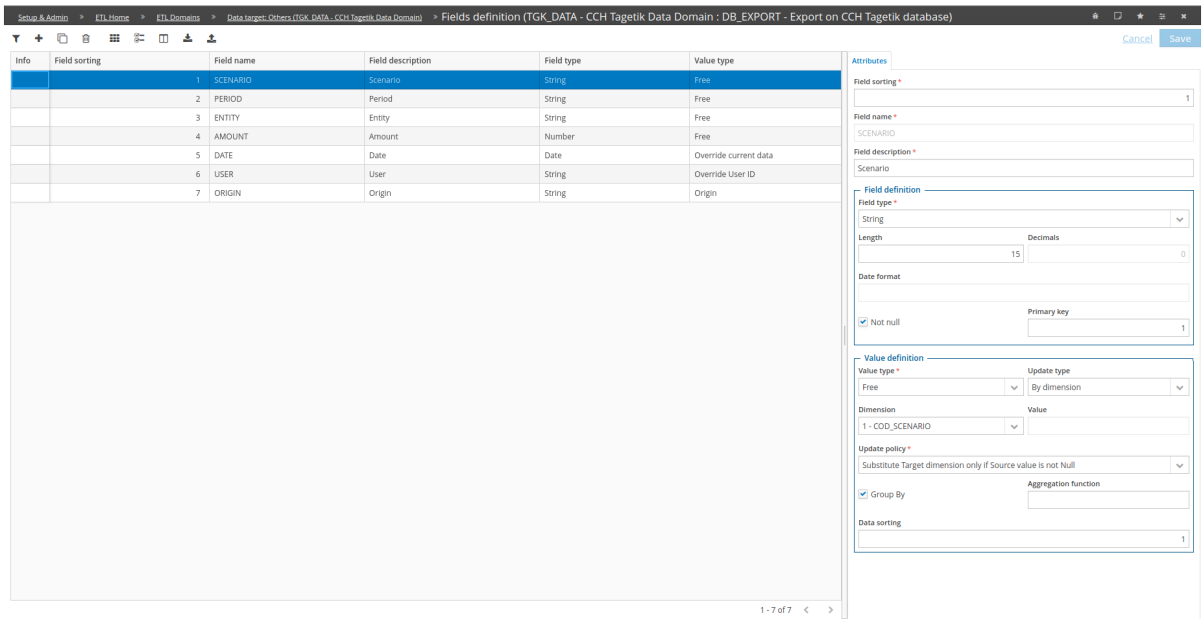
Loading - Data Target: Other

For each Data target: Others, it is necessary to list and describe the fields to process and how they shall be populated. For each field, it is possible to specify if it must be populated with the values present in a *staging* table's field related to a Dimension of the ETL Domain, with one single value or if it must be automatically calculated by the system (Oid, current date, current user or origin). It is also possible to define a data grouping policy by editing the attributes of each field, in order to limit the number of records to process.

Definition

The Definition of the Data target: Others can be accessed from the *Fields definition* link in the **Data target: Others** window.

If the ETL Domain is locked, the window will be in read-only mode.



Each record of the Fields Definition of the Data target: Others is identified by:

- **Field sorting**: specifies the order of the field in the custom table or in the file;
- **Field name**: specifies the name of the field; it is used for the creation of the custom tables if the Data target type is database;
- **Field description**: shows the description of the file; it is used in the files' headers if the Data target type is file;
- **Field type**: specifies the type of the field; possible values:
    - *String*: this is used for the varchar type fields and for those on which the **Value type** is *Override User ID* or *Origin*,
    - *Number*: this is used for the numeric fields,
    - *Date*: this is used for the date or time type fields for those on which the **Value type** is *Override current data*,
    - *Text*: this is used for the clob type fields,
    - *Oid*: this is used for those fields on which the **Value type** is *Automatic oid calculation*;

- **Value type**: specifies how the field must be defined; possible values:
  - *Free*: this is used to populate the field with the values of a dimension of the *staging* table or with a fix value,
  - *Override User ID*: this is used to automatically populate the field with the current user,
  - *Override current date*: this is used to automatically populate the field with the current date,
  - *Origin*: this is used to automatically populate the field with the origin of the current ETL,
  - *Automatic oid calculation*: this is used to automatically populate the field with a generated oid.

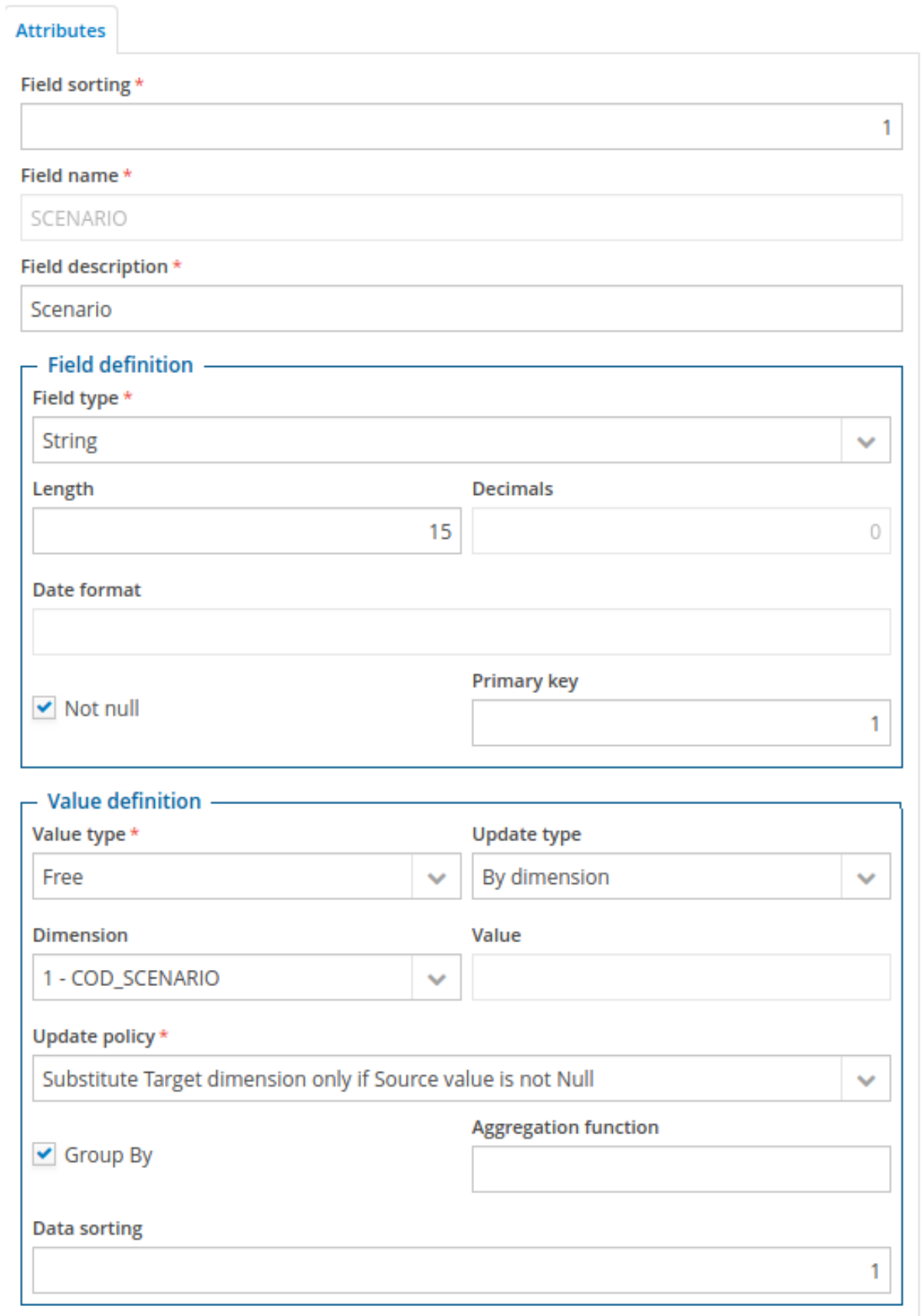
Attributes

In addition to **Field sorting**, **Field name**, **Field description**, **Field type** and **Value type**, in the detail of the Attributes tab, it is possible to define:

- **Length**: editable only if the **Field type** is *String*, *Number* or *Oid*, and specifies the maximum length of the value in the target field; this value is used to create the target table and to format data read from the *staging* table;
- **Decimals**: editable only if the **Field type** is *Number*, and specifies the maximum number of decimals of the value that will be loaded in the target field; this field is used to create the target table and to format data read from the *staging* table;
- **Date format**: editable only if the **Field type** is *Date*, and specifies the format of the value of date, time or datetime type that will be loaded in the target field; this value is used to format data read from the *staging* table;
- **Not null**: visible only if the Data target type is database, and specifies if the field of the target table accepts or not *NULL* values;
- **Primary key**: visible only if the Data target type is database and editable only if the **Not null** option is enabled; by setting a value that is not 0, this attribute allows including the field in the primary key of the target table; the defined values that are not 0 must be sequential (E.g. 1, 2, 3) in order to specify the order of the fields in the primary key univocally;
- **Text Fixed Length column length**: visible only if the Data target type is *Text Fixed Length*, and specifies the characters length for each column of the file related to the field; if the characters length of the value to insert is shorter than the one defined, then the column will be filled with "*spaces*" on the right or on the left, depending on whether or not the **Field type** is *Number*;
- **Update type**: only if the **Value type** is *free*, it specifies how the field must be populated. Possible values:
  - *By dimension*: the field must be populated with the values present in the *staging* table's field related to a Dimension of the ETL Domain;
  - *By value*: the field must be populated with a fixed value;
- **Dimension**: if the **Update type** is *By dimension*, it specifies the Dimension of the ETL Domain related to the field of the *staging* table containing data to load on **Field**;
- **Value**: if the **Update type** is *By value*, it specifies the value to set up on **Field** for all uploaded records.
- **Update policy**: specifies how to populate the **Field** on the target table during update. Possible values:

    - *Always substitute Target dimension value with Source value*: the field's value is always modified, even with a *NULL* value;
    - *Substitute Target dimension only if Source value is not Null*: the field's value is modified only if the new value is not *NULL*;
    - *Sum if loading value is not null (only for numeric field)*: the new value is added to the old value in the field.
- **Group by**: only if the **Update type** is *By dimension*, it means that before being uploaded, the *staging* table data must be grouped for the *staging* table's field related to the Dimension of the ETL Domain defined in the **Dimension** field;
- **Aggregate function**: only if the **Update type** is *By dimension*, it means that the aggregation function must be applied to the *staging* table's field related to the Dimension of the ETL Domain defined in the **Dimension** field.

   After enabling the **Aggregate** option or specifying an **Aggregate function** for a field, it is necessary to set up one of the two fields for all other records for which the **Update type** is set to *By dimension*.
- **Data sorting**: only if the **Update type** is *By dimension*; by specifying a value that is not 0, it allows adding the field of the *staging* table related to the ETL Domain's Dimension defined in the **Dimension** field in the list of fields on which data must be sorted; the defined values that are not 0 must be sequential (E.g. 1, 2, 3) in order to identify the order of the fields univocally;



If the Data target type is database and you want to modify the target table's trace, you have to remove the table from the database since the ETL run does not update its structure at runtime

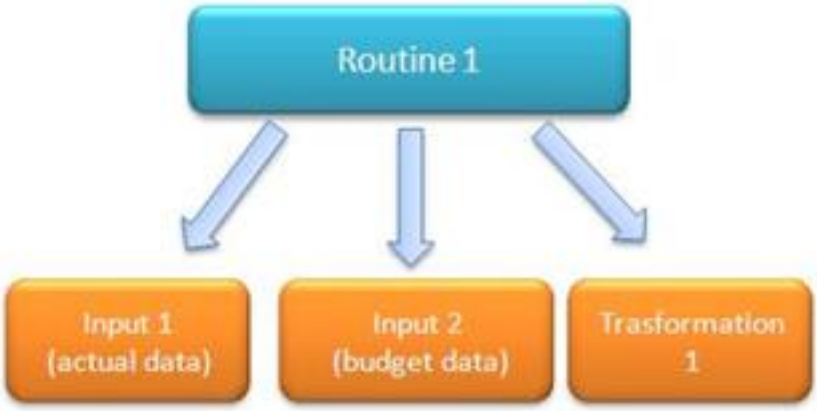
before loading data. If you cannot remove the table manually by directly accessing the database, you can do it by using the *Delete table* link in the **Data target: Others** window.

Routines

A routine is the data processing minimal unit in an ETL process and contains all data processing information of the routine and the link to the details for the data extraction, transformation and loading; Generally a routine is made up of:

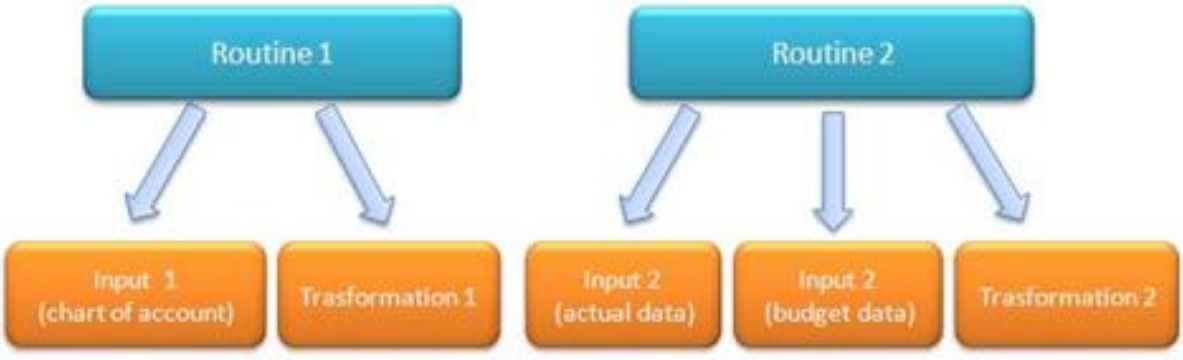
- one or more extraction tasks that allow extracting data from one or more data source types;
- one or more transformation tasks that allow processing and converting extracting data;
- one or more loading tasks that allow loading the extracted and/or transformed data on one or more target types.

When data to extracted and process are "homogeneous" (e.g. data come from various types of import or they are all structural elements), it is good practice to create one single Routine.



In this way data will be extracted all at the same time and transformed according to the same logics (Transformation 1).

On the contrary, if data are "heterogeneous" (e.g. imports and structural data) it is advisable to define one routine for each type of data to extract.



In this case data extracted from the Routine 1 will be processed by the Transformation 1, while those extracted from the Routine 2 (Input 1 and Input 2) will be processed by the Transformation 2.

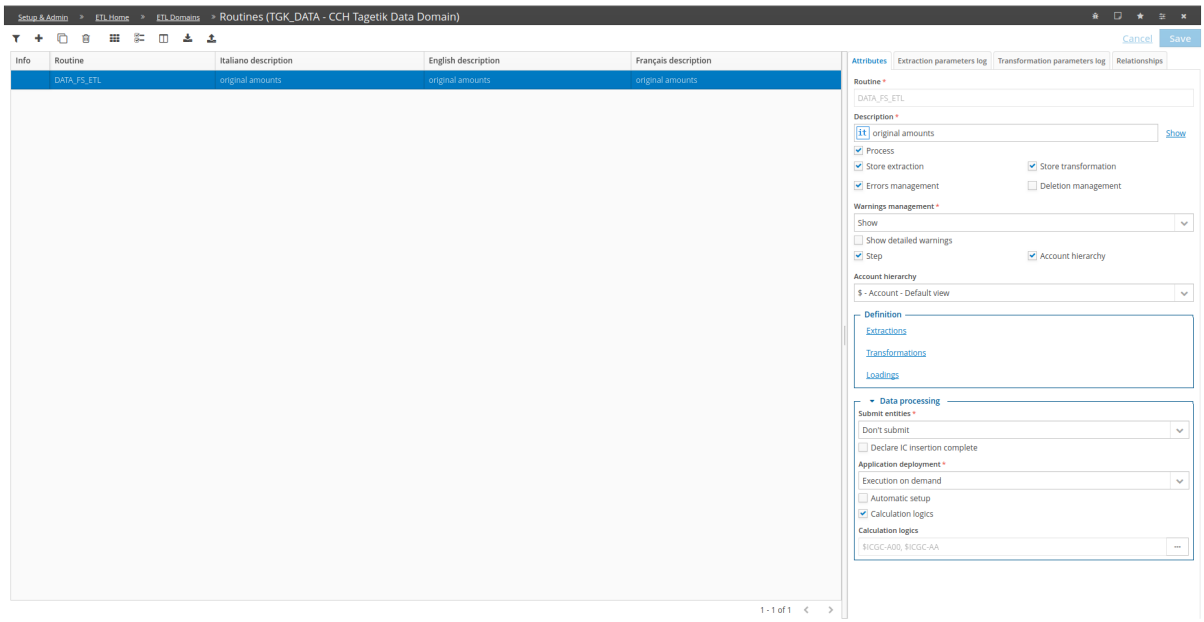
By default, CCH Tagetik provides the **Standard routines** of the **ETL Standard Domain** which allow loading both "structural" data (dimensions elements, hierarchies and groupings) and "values" (amounts, IC, adjustments and FX rates). These routines cannot be modified or deleted by the user.

It is possible to create new completely configurable **Custom Routines** (also as a copy of the standard ones), in order to meet specific business needs.

Definition

The definition of the Routines can be accessed by the *Routines* link of the **ETL Domain** window.

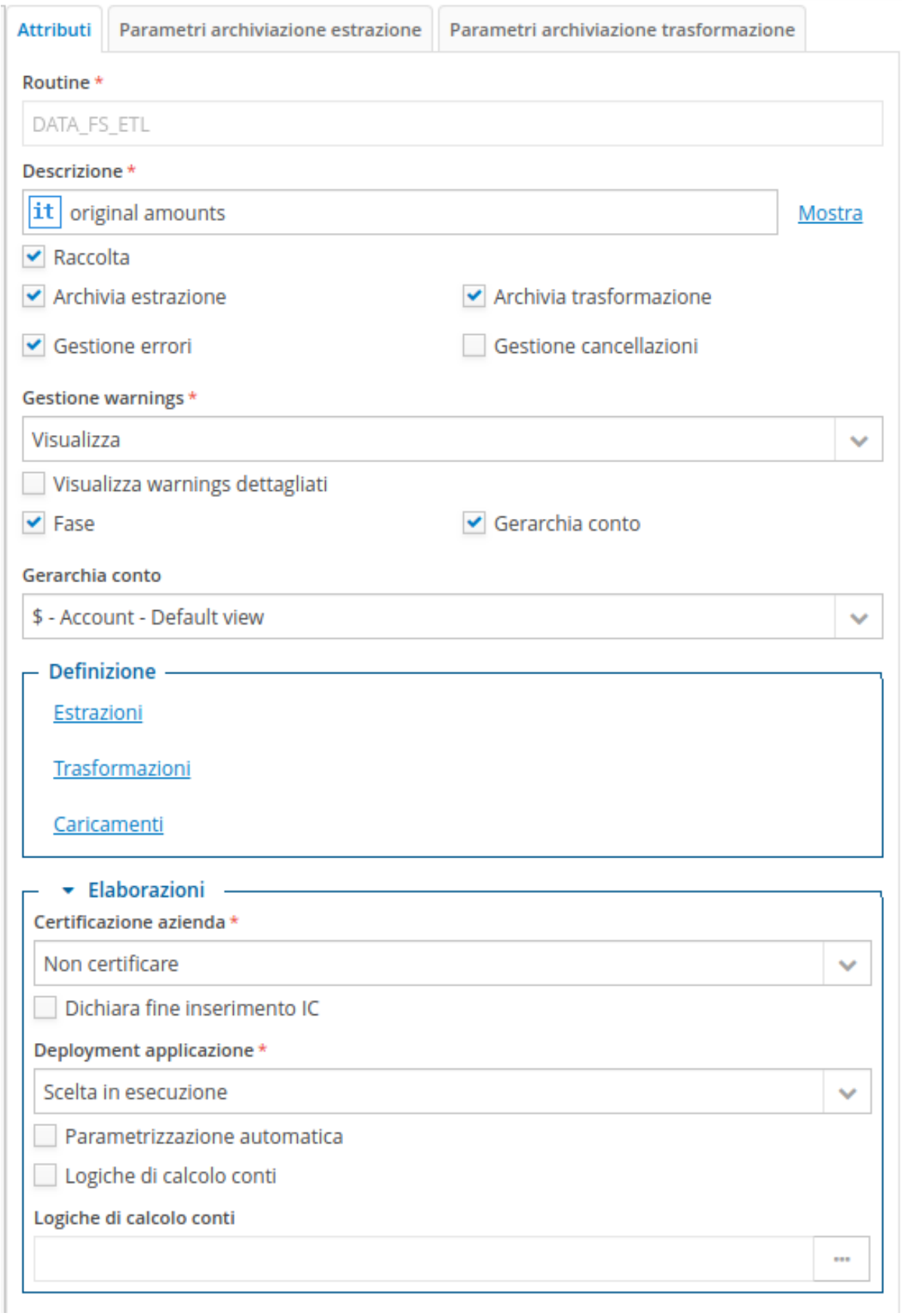
If the ETL Domain is locked, then the window will be in read-only mode.



Each Routine is identified by a **Code** and **Descriptions**.

Attributes

Besides the **Code** and the **Descriptions**, in the attributes detail it is possible to define other options as well.


Attributi
Parametri archiviazione estrazione
Parametri archiviazione trasformazione
Routine *
DATA_FS_ETL
Descrizione *
it original amounts
Mostra
Raccolta
Archivia estrazione
Archivia trasformazione
Gestione errori
Gestione cancellazioni
Gestione warnings *
Visualizza
Visualizza warnings dettagliati
Fase
Gerarchia conto
Gerarchia conto
$ - Account - Default view
Definizione
Estrazioni
Trasformazioni
Caricamenti
Elaborazioni
Certificazione azienda *
Non certificare
Dichiara fine inserimento IC
Deployment applicazione *
Scelta in esecuzione
Parametrizzazione automatica
Logiche di calcolo conti
Logiche di calcolo conti

- **Process**. If enabled, during runtime the Routine will need a filter on the Process. It must be enabled for those Routines that manage "values" data and have as target a CCH Tagetik table. In this way the system can perform all the editability checks required to understand if some data can be loaded or not;
- **Store Extraction**. If enabled, once the extraction is completed, the Routine must save data present in the "staging" table into the extracted data log table. Data are saved by ETL Domain / Routine / User and possibly by extraction log Parameters defined in the Routine. This is useful for:
  - running queries on extracted data and tracking over time which data have been extracted, by whom and when;
  - running a Routine in partial mode from the transformation step, namely by considering as data to extract those previously saved on the log table;
- **Stocker transformation**. If enabled, once the transformation process is completed, the Routine must save data contained in the "staging" table into the transformed data log table. Data are saved by ETL Domain / Routine and possibly by the transformation log Parameters defined in the Routine. This is useful for:
  - running queries on transformed data and tracking which data have been transformed over time, by whom and when;
  - running a drill - through query inside a form in order to understand in detail how a given data from the ETL has been generated. For example, assume that we extract from the subsystem the salaries and that in this subsystem, data are detailed by sector and that we load them into CCH Tagetik but on one single Salary account. Once the execution of the Routine on the Salary account is completed, we will have, for example, the amount 1000 but since we have already archived the extracted data, we can use a drill - through query inside a form and see how the number 1000 has been generated, namely how each sector has contributed to the generation of the total number 1000;
  - running a Routine in partial mode from the loading step, namely by considering as data to be transformed those previously saved on the log table;
- **Errors management**. If enabled, in case of errors the Routine will detect them, keep on running and, only at the end, display them by indicators (traffic lights) associated with each step of the Routine. On the contrary, if it is disabled, if errors occur while the Routine is running, the system displays the stack trace of the error as soon as it occurs and stops running;
- **Warnings management**. It defines how the Routine must behave if warnings (yellow traffic light) occur while it is running. It can be set to:
  - **None**: the routine keeps on running;
  - **Show**: the routine stops running and the user can decide whether to interrupt is or let it run ;
  - **Lock**: the routine stops running and does not allow the user to continue
- **Show detailed warnings**. If enabled, the Routine displays in detail (nor grouped pre-loading records) any warnings generated while loaded data on CCH Tagetik for the insertion and update of grouped data.
- when loading data on the CCH Tagetik tables of Amounts, IC and Adjustments, it is possible to partialize data to be inserted / deleted according to the Account. This can be achieved in two ways:
  - **Step**. If enabled, the Routine will load only data referring to accounts present in the steps (that will be specified at runtime) of the run process;

- **Account hierarchy**. If enabled, the Routine will load only data referring to accounts related to the nodes (that will be specified during runtime) of the **Account hierarchy** specified in the field of the same name (this value can be specified also during runtime).

- **Submit Entities**. If enabled, once the loading steps is completed, data will be submitted automatically. In particular, if this option is set to:
  - **Don't submit**, at the end of the loading step, the system does not run the data submission;
  - **Submit all steps**, the system submits automatically all the steps of the process for all the scenarios, periods and entities considered as parameters;
  - **Submit all previous steps to the IC matching step**, the system submits automatically all the steps below the one containing the first IC matching cockpit for all the scenarios, periods and entities considered as parameters
- **Declare IC insertion complete**. If enabled, at the end of the data loading step, the IC insertion is automatically declared complete for all IC matching cockpits belonging to the process run for all the scenarios, periods and entities considered as parameters;

   For these options to work properly once enabled, it is necessary that a Process is specified during runtime and that the Routine uses the parameters related to the Dimensions having as "Identity" respectively: Scenario, Period and Entity. This will specify the scenarios, periods and entities to submit.

- **Application deployment**. If enabled, before each loading task on the "values" table of CCH Tagetik, the Routine will run the deployments resulting from other loading tasks run by the same Routine on CCH Tagetik "structure" tables. If no loading task on "structure" tables exists within the Routine, the deployment will never run, independently of the value of this option. Possible values:
  - **Never**. The deployment must be run manually when the Routine execution is completed;
  - **Execution on demand**. The user will specify if the system must run the deployment during the Routine run;
  - **Always**. The system runs the necessary deployments automatically.
- **Automatic setup**. If enabled, the Routine will automatically create the user rights for the loaded "structural" data (entity, custom dimensions, categories,...). This is useful when the Routine includes the loading of data on "values" tables by using these elements, thus avoiding editability check errors;
- **Calculation logics**. If enabled, once the loading step is completed, the Routine will run the following data processing
  - "Entity data processing – Standard formulas"
  - "Entity data processing – Formulas".

  on the **Calculation logics** defined in the field of the same name.

   The data processing runs for the original scenarios, periods and entities selected during the execution of the Routine via ETL Parameters linked to dimensions having as identity Scenario, Period and Entity.

Routine tasks definition

By the links inside the *Definition* task it is possible to access the management of the following tasks:

- Extraction
- Transformation
- Loading

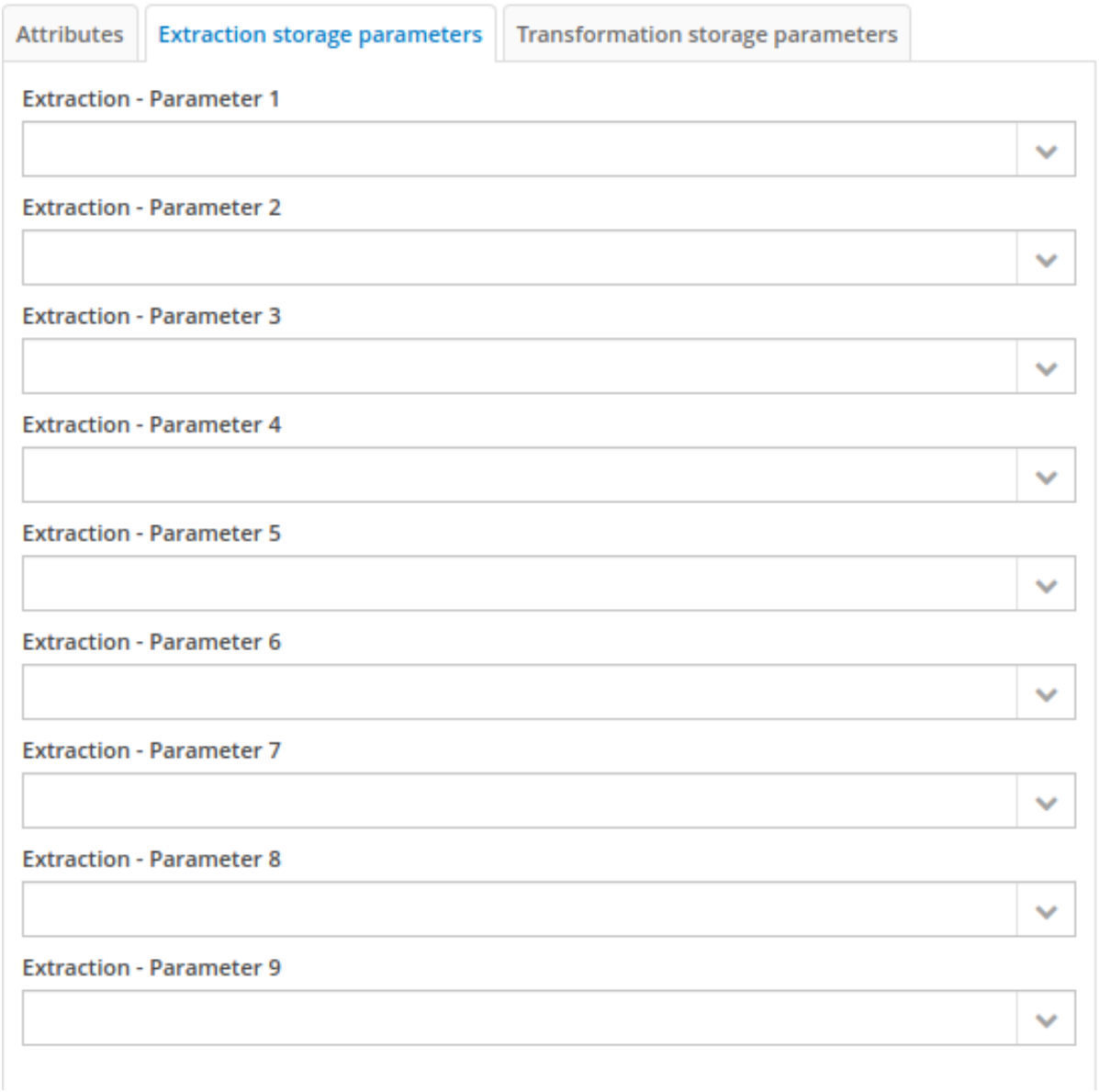
related to the selected Routine.

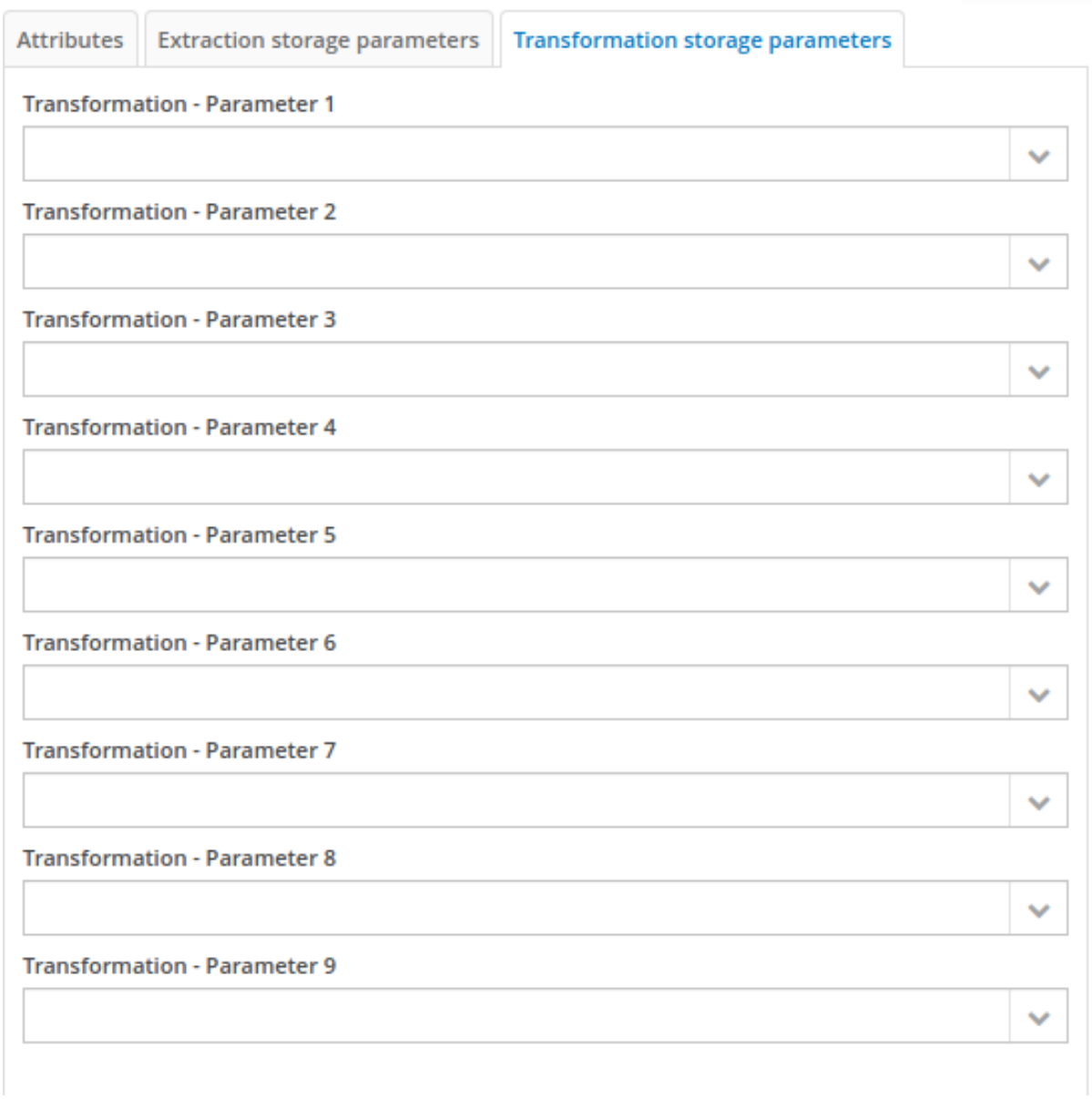
The Commands link provides access to the management of the *Custom Statements* and *Stored procedures* that can be run during a Routine. This link is visible only if the 0003874 function is included in the user's role and is enabled only if the Routine does not belong to a locked ETL Domain.

Extraction and transformation storage parameters

If the Routine manages the storage of extracted and/or transformed data, it is possible to define up to nine Parameters for each operation.

The parameters, along with the Routine ID, filter the deletion of the log table and the insertion of data from the "staging" table.

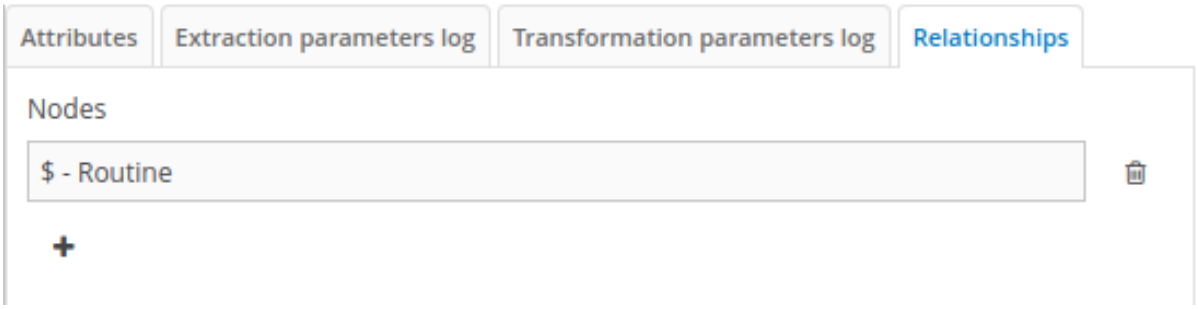

Attributes
Extraction storage parameters
Transformation storage parameters
Extraction - Parameter 1
Extraction - Parameter 2
Extraction - Parameter 3
Extraction - Parameter 4
Extraction - Parameter 5
Extraction - Parameter 6
Extraction - Parameter 7
Extraction - Parameter 8
Extraction - Parameter 9



Relationships

This tab shows and allows modifying the nodes of the grouping to which the selected Routine is linked.

Each new Routine will be automatically linked to the grouping nodes selected for the default relationship.



Run

The Routine can be run:

- individually (especially used as debug during development)
- within an ETL Job

from the specific window. (see **Run ETL** for details on the run modes and entry points).

Since the parametrization of a Routine can be very complex, it is *good practice* to proceed step by step in order to verify the effectiveness of the parametrization and detect any malfunctions. This will make it easier to find out where any possible error has been made. For example, if the user must extract data from an Excel sheet after defining an ETL Domain, the Dimensions, the Parameters and a Data Source, it is necessary to define a Routine with an extraction task that will be linked to the created Data Source in order to try and immediately extract data. If the extraction is successfully completed, then it is possible to proceed with the definition of the transformation tasks, otherwise it is possible to investigate the cases for failure to correctly extract data.

Utility

The Actions menu in the window includes some ETL Routine copy and deletion utilities. These utilities can be used only if one single record is selected.

Reset

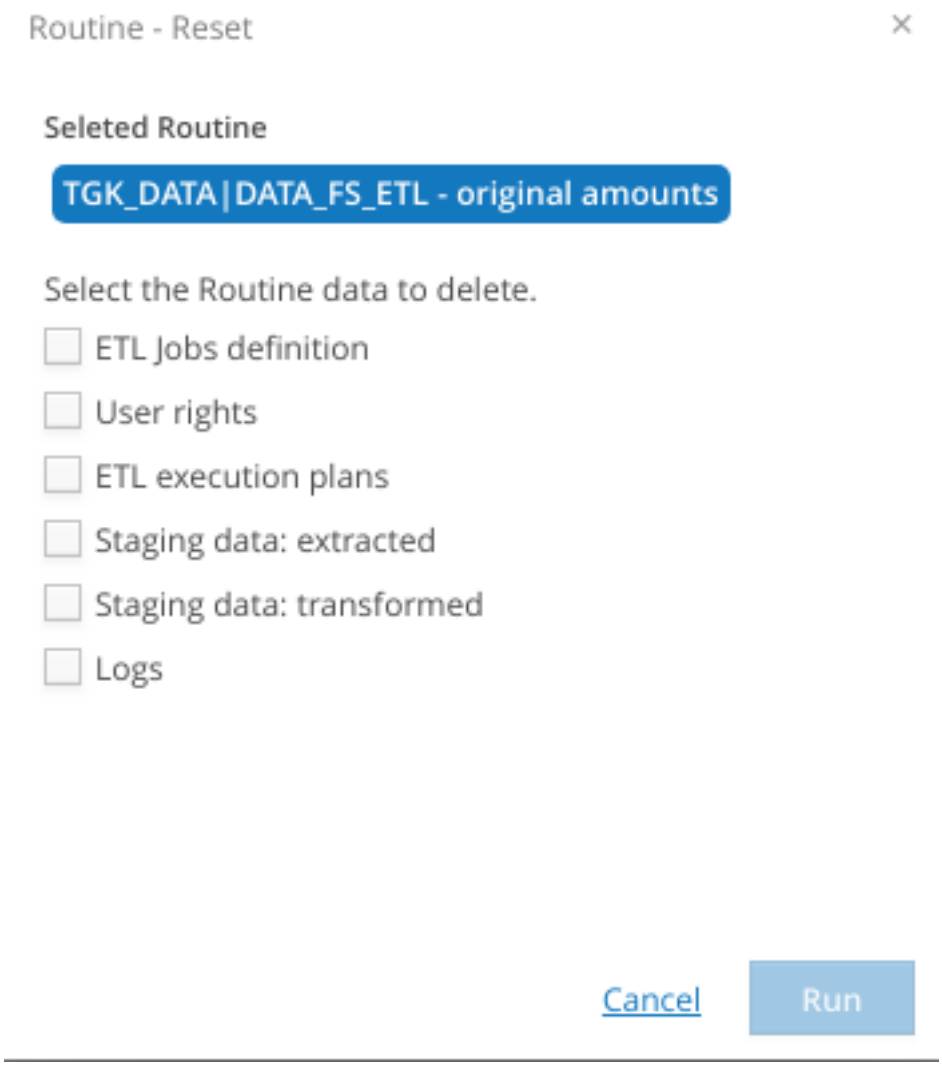
The *Reset* menu allows you to reset all uses of the Routine and of its parametrization:

- in the definition of all ETL Jobs,
- in the users rights,
- in the ETL execution plans of all processes

as well as the execution data, namely:

- the staging table of extracted data,
- the staging table of transformed data,
- the execution logs.

In order to run the operation, it is necessary to select at least one element to delete.

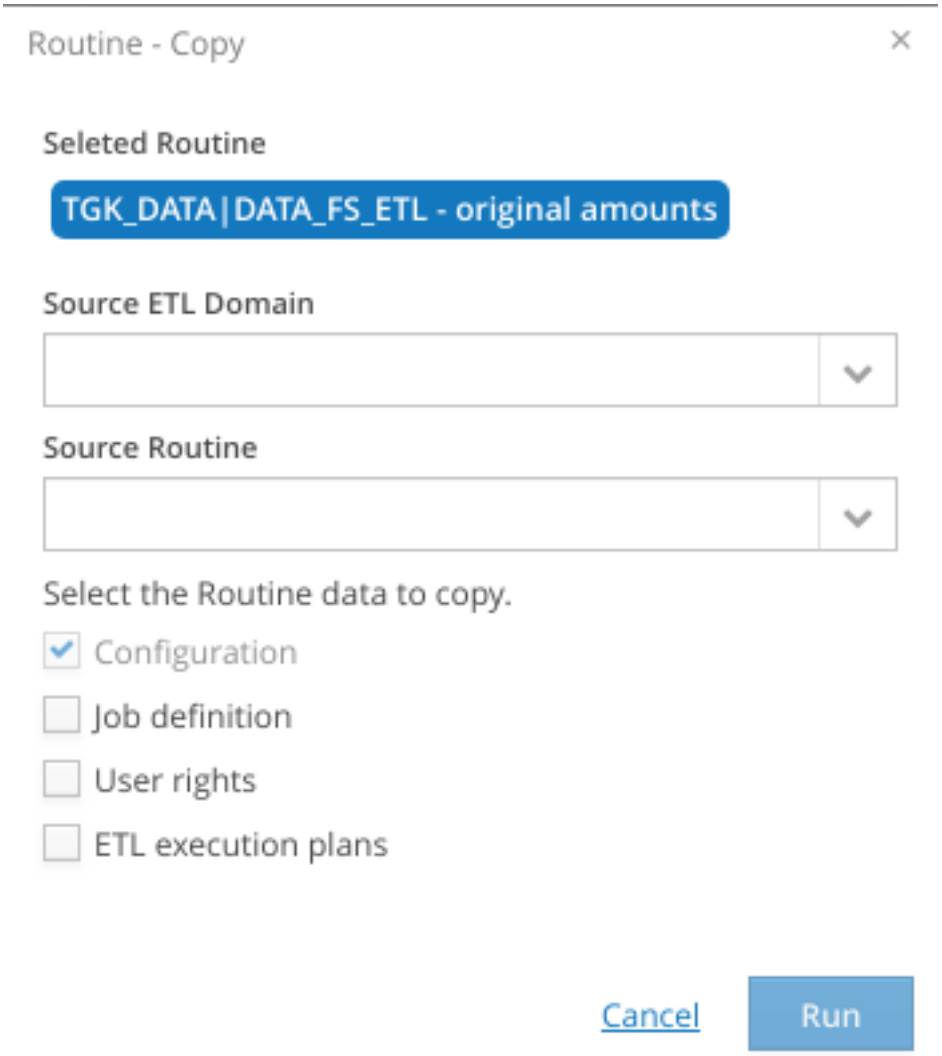


Copy from

The *Copy from* menu enables the copying (by overwriting) on the selected Routine of the entire default configuration and of the following uses of the Routine specified in the copy window:

- definition of ETL Jobs,
- user rights,
- ETL execution plans of all processes

In order to run this operation, it is necessary to specify the source ETL Domain.


Routine - Copy
Seleted Routine
TGK_DATA|DATA_FS_ETL - original amounts
Source ETL Domain
Source Routine
Select the Routine data to copy.
Configuration
Job definition
User rights
ETL execution plans
Cancel
Run

Routine - Commands

All the *Custom statements* and the *Stored procedures* that can be run during the execution of an ETL Routine are defined within the commands of a Routine.

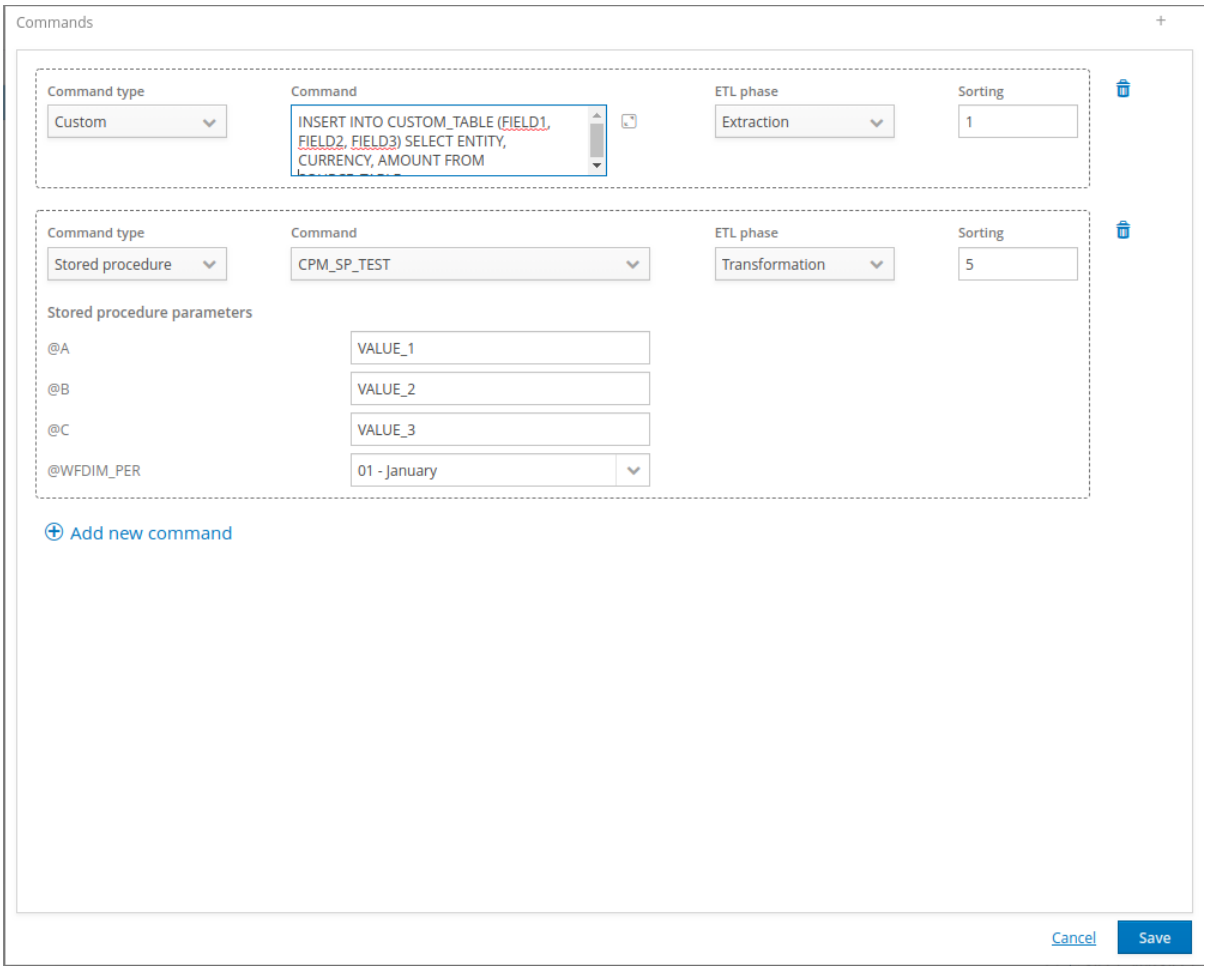
Configuration

The Commands' configuration can be accessed from the *Commands* link of the **Routines** window.

If the ETL Domain is locked, the link is not clickable.

To add a command, click on the button 

following the last inserted command. To remove a command, click on the button  displayed on the right hand side of each command.



Custom Statement

This type of command allows you to manually define a generic SQL statement.

Command type: *Custom*,

Command: contains the SQL text to be run; the ETL dynamic syntax is accepted (see Filter Strings Syntax or SQL Query

ETL phase: indicates the ETL Routine's step (Extraction, Transformation, Loading) in which the command has to be run;

Sorting: order in which the command has to be run in relation to the Extraction, Transformation or Loading tasks included in the step of the ETL Routine; with equal sorting, the task will be run first.

Stored procedure

This type of command allows you to define a Stored procedure.

Command type: *Stored procedure*,

Command: allows you to select one of the Stored procedures present on the database of CCH Tagetik having *CPM_SP* as a prefix;

ETL phase: indicates the ETL Routine's step (Extraction, Transformation, Loading) in which the command has to be run;

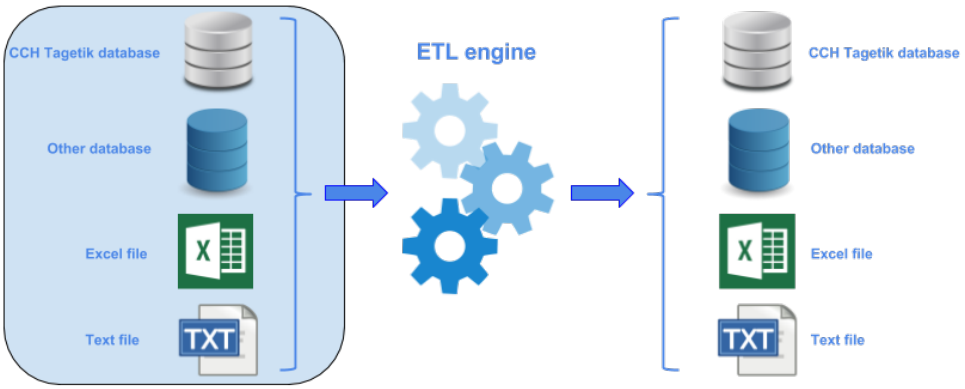
Sorting: order in which the command has to be run in relation to the Extraction, Transformation or Loading tasks included in the step of the ETL Routine; with equal sorting, the task will be run first.

Parameters

If the selected Stored procedure includes the selection of parameters, these will be shown in the **Stored procedure parameters** panel below. According to the parameter's name, the system displays different editors for the selection of the value (see Stored procedure). The dynamic syntax of the ETL (see Filter Strings Syntax or SQL Query) can be used as a parameter's value.

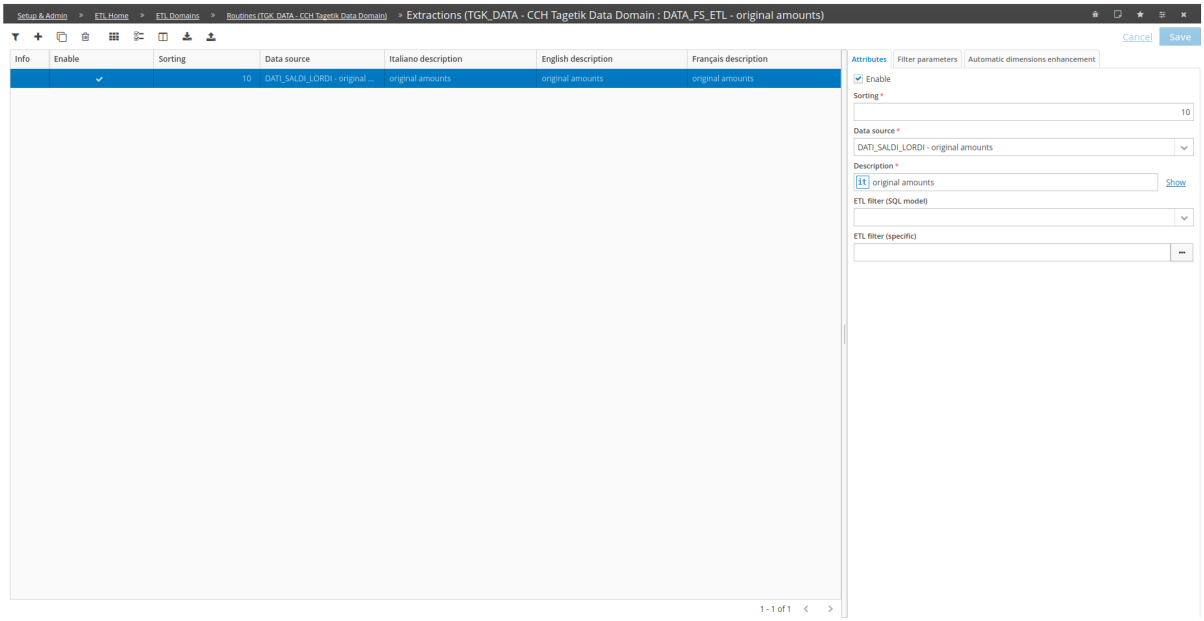
Routine - Extractions

Extractions are tasks of a Routine which allow for extracting data from sources defined in the Data sources and then saving them into the ETL *Staging* table.



Definition

The definition of the Extractions can be accessed from the *Extractions* link in the **Routines** window. If the ETL Domain is locked, then the window will be in read-only mode.

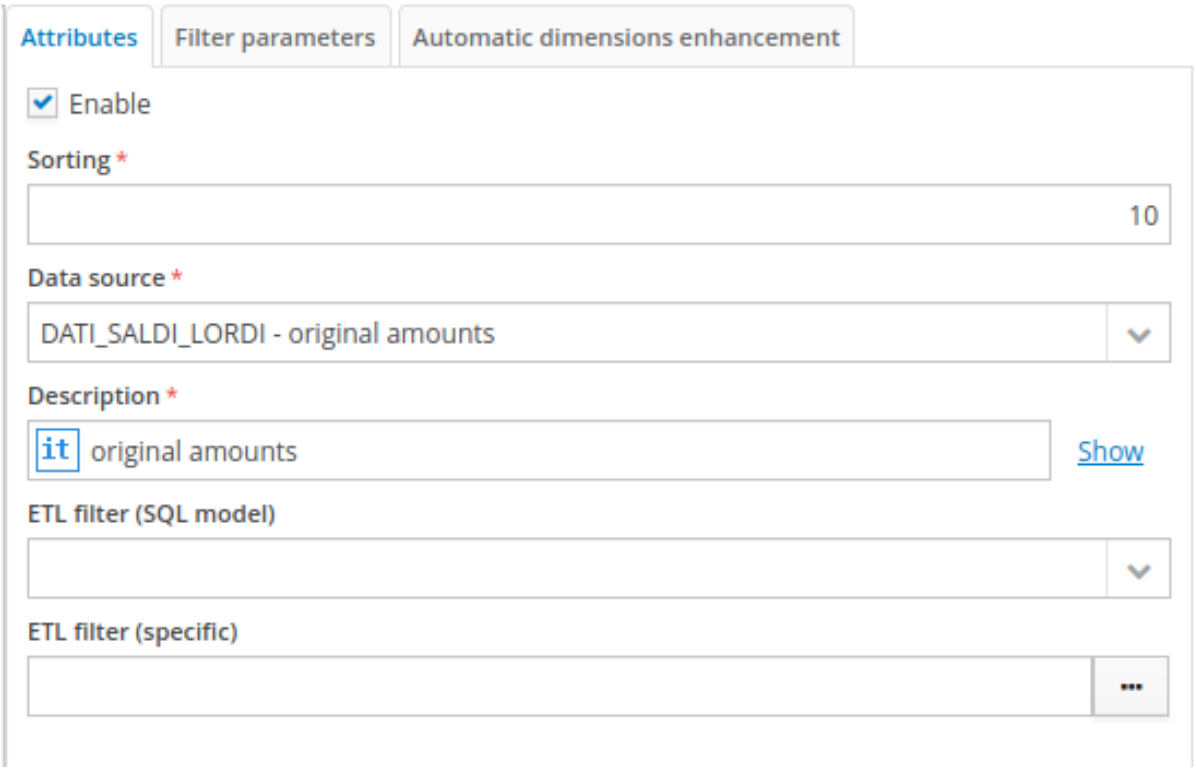


Each extraction is identified by:

Attributes

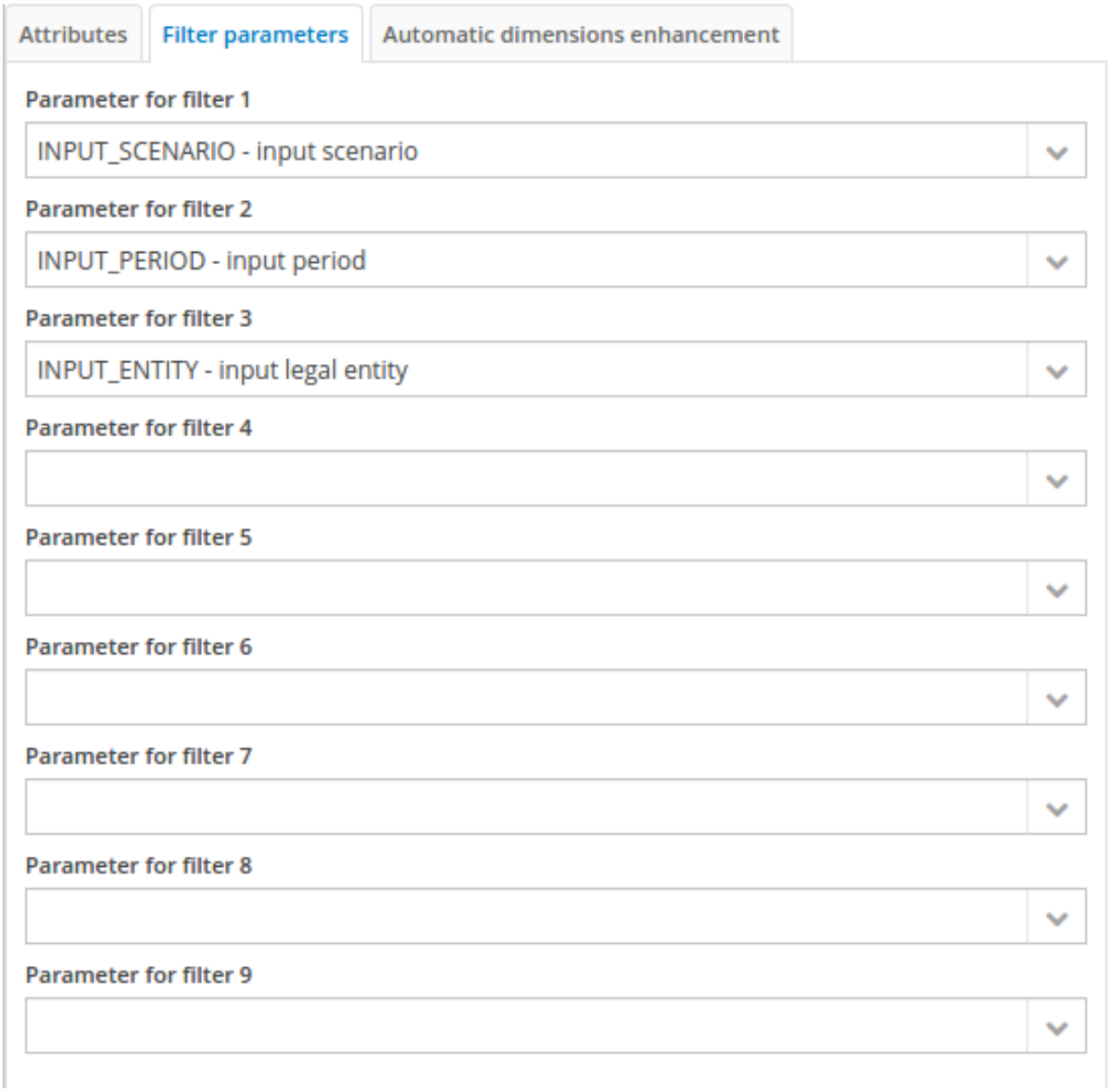
- an **Enable** option specifying if the task must be run by default;
- a **Sorting** specifying the order in which the task must be run; neither value 0 nor an existing sorting can be set up;
- a **Data source** specifying, through a Data source, the ways in which data that must be selected at runtime (mandatory) can be extracted;
- some **Descriptions** in the user language;
- an **ETL Filter (SQL Model)** specifying a filter previously defined in the SQL Models in order to limit the amount of data to extract;
- an **ETL Filter (specific)** specifying a filter with the SQL syntax in order to limit the amount of data to extract; it is possible to use the ETL dynamic syntax (see Filter Strings Syntax or SQL Query

  If a user specifies both the **ETL Filter (SQL Model)** and the **ETL Filter (specific)**, the system will link the conditions by the *AND* logical operator.



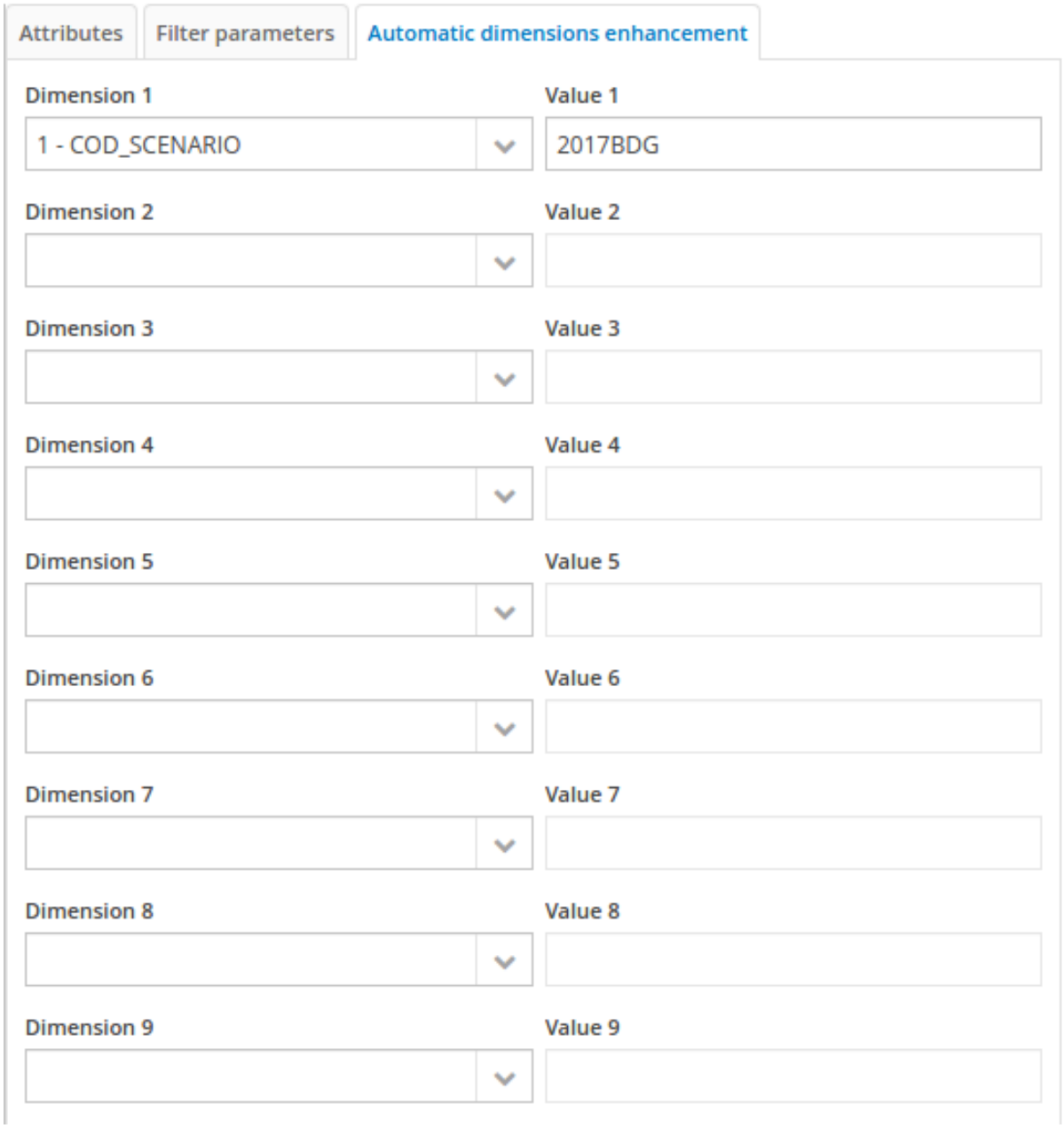
Filter parameters

- It is possible to define up to nine filter Parameters in order to limit the amount of data to extract dynamically; only the *From CCH Tagetik*, *From dictionary dimension* and *Free* parameters can be used. If the user specifies also one of the two ETL Filters, the system will link the conditions by the logical operator *AND*.



Automatic settings of dimensions

- At runtime, it is possible to define up to nine dimensions by setting **Dimension1...9** / **Value1...9**. These dimensions do not have to be specified also in the definition of Data source fields.

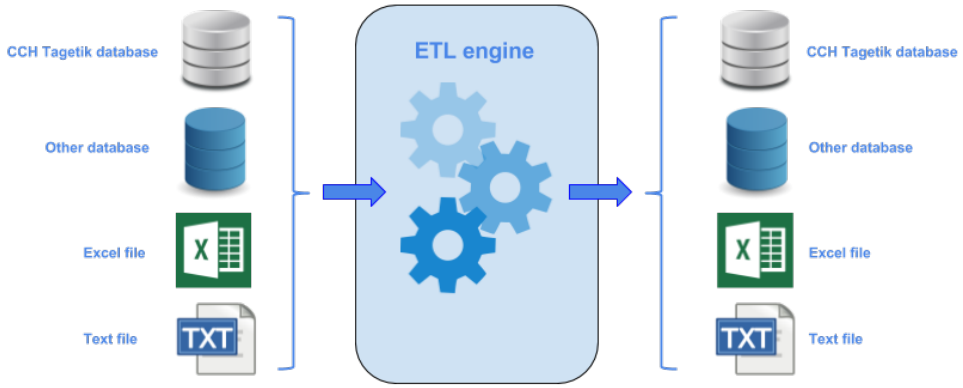


The data extraction will be performed only in the specific way, defined by the **Data source**, selected at runtime.

If several extraction tasks are required, for each task the system will record extracted data in the staging table after any previous loading tasks.

Routine - Transformation

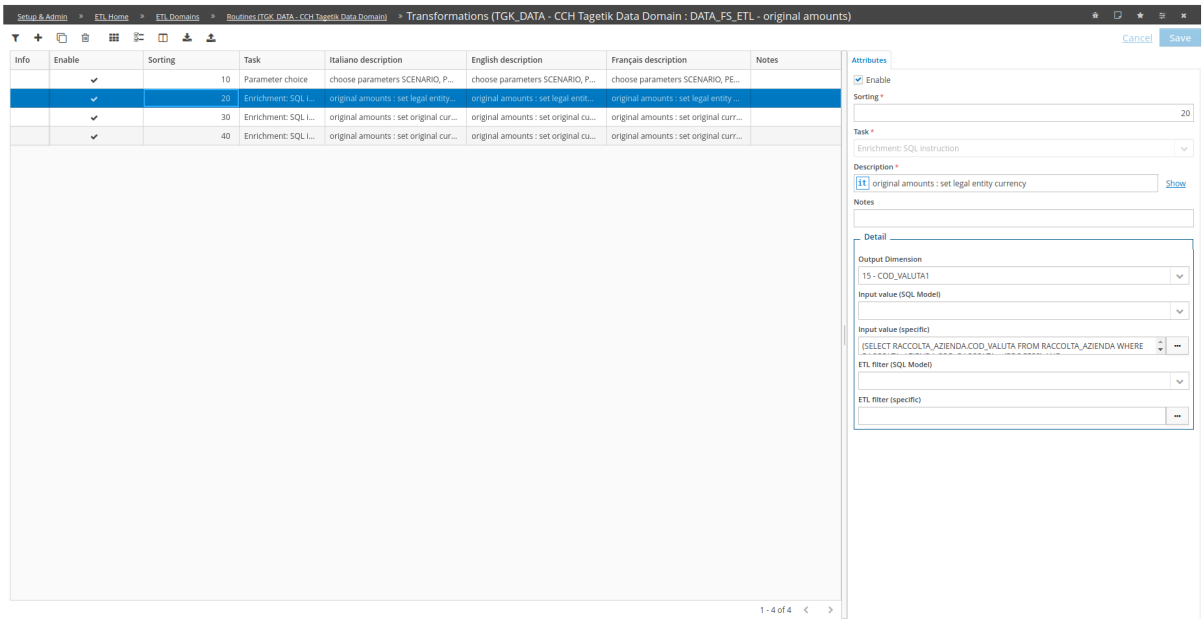
The transformations are tasks of a Routine that enable you to filter, check and enrich extracted data in order to make it ready for the next loading.



Definition

The definition of the Transformations can be accessed from the *Transformations* link of the **Routines** window.

If the ETL Domain is locked, then the window will be in read-only mode.



Each Transformation is identified by:

Attributes

- an **Enable** option that specifies if the task must be run by default;
- a **Sorting** that specifies the order in which the task must be run; neither 0 value nor an existing sorting can be set up;
- a **Task** specifying the task type. The tasks are grouped by macro type:
  - Filter Task:
    - Parameter choice
    - Delete by sql
  - Control Task:
    - Invalid characters check
    - Dictionary check
    - Completeness check
    - Completeness and dictionary check
    - SQL check
    - Lookup table diagnostic
  - Enrichment Task:
    - Lookup table feeding
    - Data grouping
    - Enrichment: lookup table
    - Enrichment: matrix calculation
    - Enrichment: SQL instruction
    - Enrichment: formula
    - Enrichment: amounts total
    - Enrichment: ranking
    - Year-to-date calculation
    - Verticalization
    - Counterparty Zone unbundling
    - Transfer Price from the rule
    - Allocation
- **Descriptions** ;
- any **Notes**

 Each **Task** may require additional information. A specific detail panel will appear in the window according to the selected **Task**. The detail of each panel is explained in the section related to the task type.

For most tasks it is possible to specify a filter to limit the amount of data on which the operation runs. This can be done by filling the following fields in each detail:

- **ETL Filter (SQL Model)** which allows you to specify a filter defined in the **SQL Models**;
- **ETL Filter (specific)** which allows you to specify a SQL syntax filter; the ETL dynamic syntax can be used (see **SQL Query or Filter Syntax**)

If the user specifies both the **ETL FIlter (SQL Model)** and the **ETL Filter (specific)**, the system will link the conditions by the *AND* logical operator.

The Transformation tasks are run in sequence according to the **Sorting** defined during the parametrization. Each task runs on the *Staging* table and processes data generated by the previous task (except the first one whose input data results from the extraction phase).



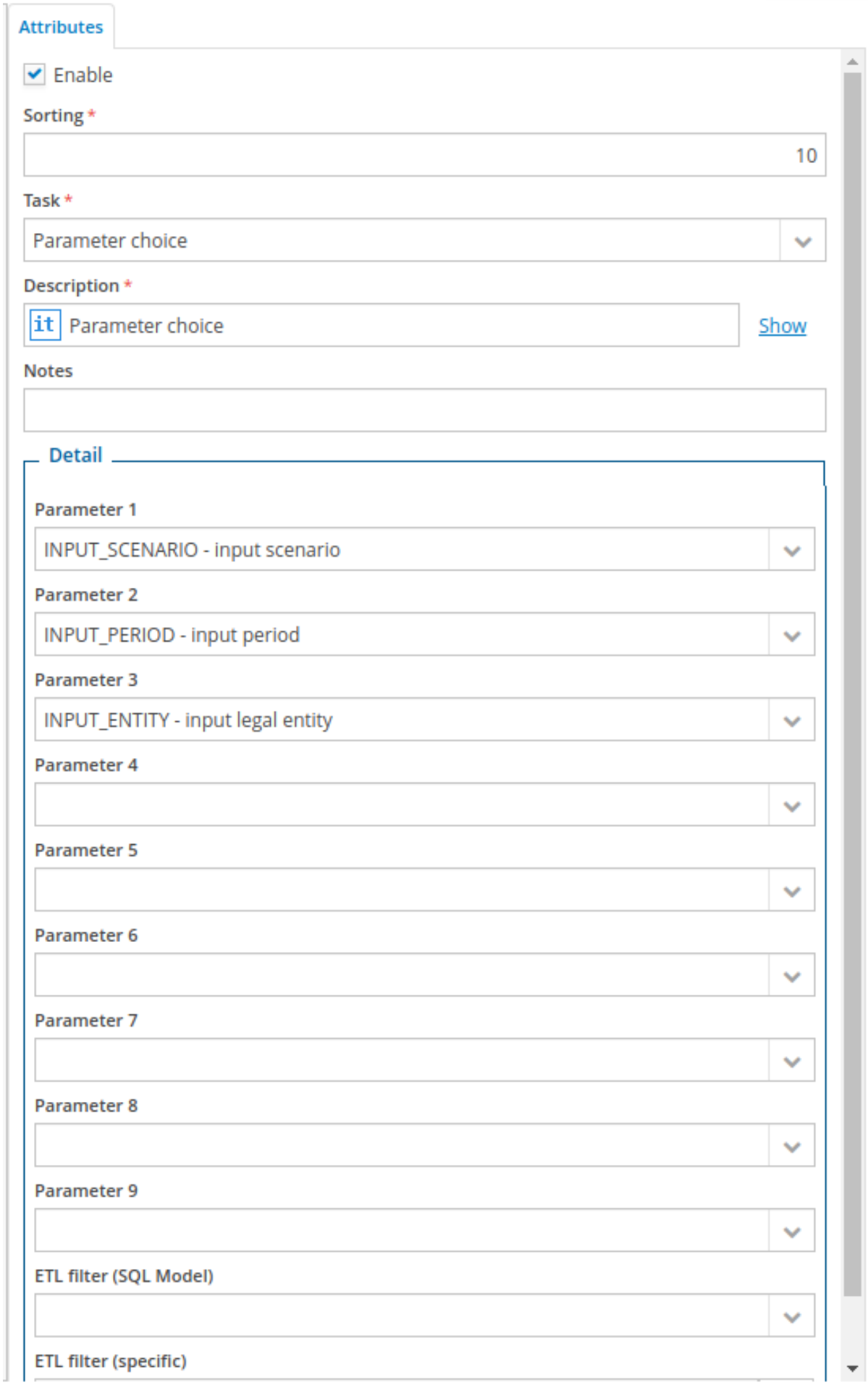
Filter Task

Parameters choice

This task allows you to define up to 9 **Parameters** that, depending on how they are defined, can:

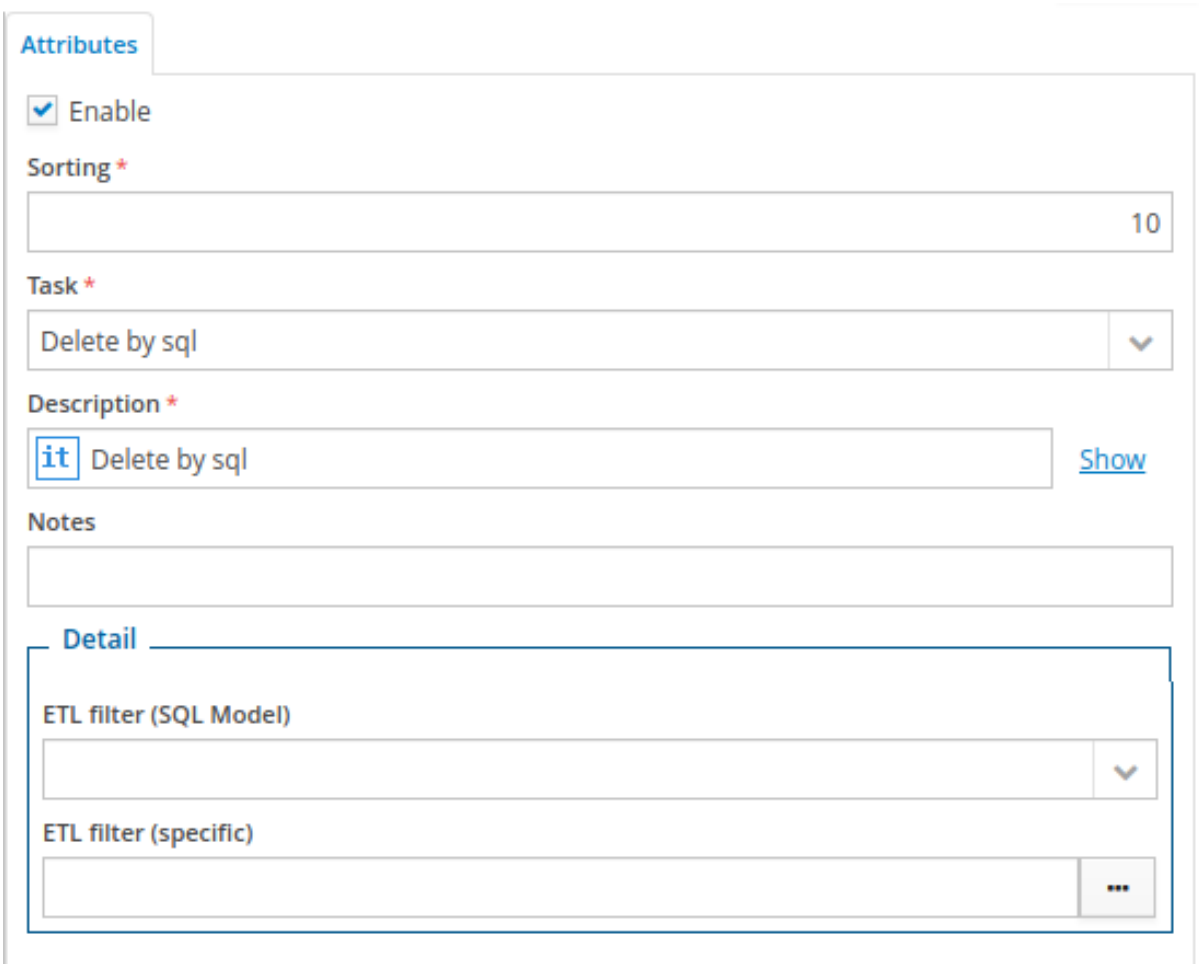
- define a dimension for all the records to be transformed (if the Parameter is *Free*, *From Tagetik* or *From dictionary dimension* and the **Multi-selection** option is disabled);
- filter data to be transformed by removing from the *Staging* table all the records for which the dimension related to the parameter does not contain a value selected for the parameter (if the Parameter configuration is different from that described before).

The selection of the values of a Parameter is done during execution and depends on the definition of the Parameter (see **Parameters**).


Attributes
Enable
Sorting *
10
Task *
Parameter choice
Description *
it Parameter choice
Show
Notes
Detail
Parameter 1
INPUT_SCENARIO - input scenario
Parameter 2
INPUT_PERIOD - input period
Parameter 3
INPUT_ENTITY - input legal entity
Parameter 4
Parameter 5
Parameter 6
Parameter 7
Parameter 8
Parameter 9
ETL filter (SQL Model)
ETL filter (specific)

Delete by SQL

This task allows you to delete from the *Staging* table all the records that do not meet the filter criterion defined in the **ETL Filer (SQL Model)** and/or **ETL Filter (specific)** field

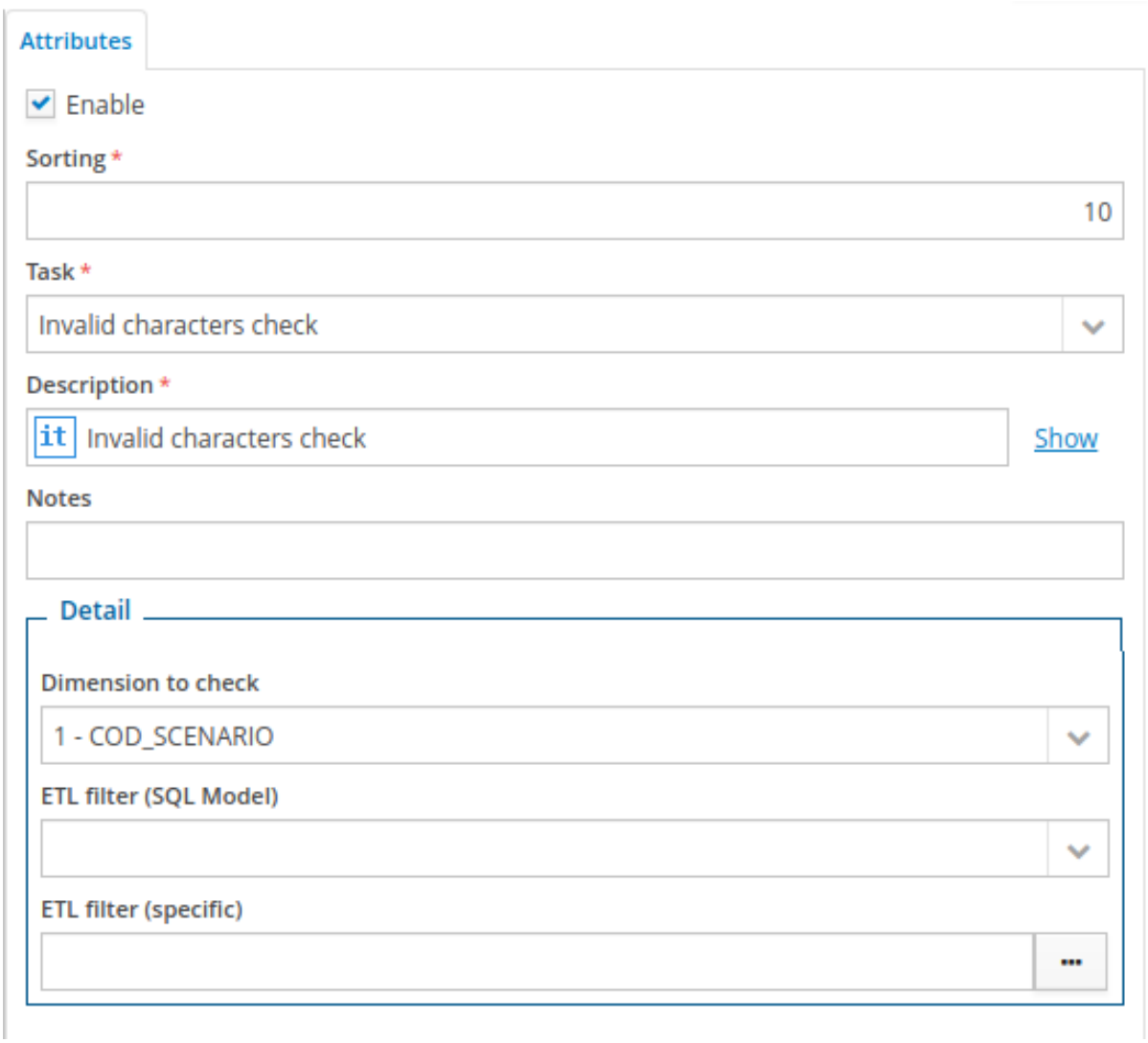


Control task

For those records of the *Staging* table on which the tasks are run and that do not meet the control requirements, these tasks generate warnings that are reported to the user.

Invalid characters check

This task allows you to check if the values of a given Dimension defined in the **Dimension to check** field contain invalid characters in the codes of CCH Tagetik.



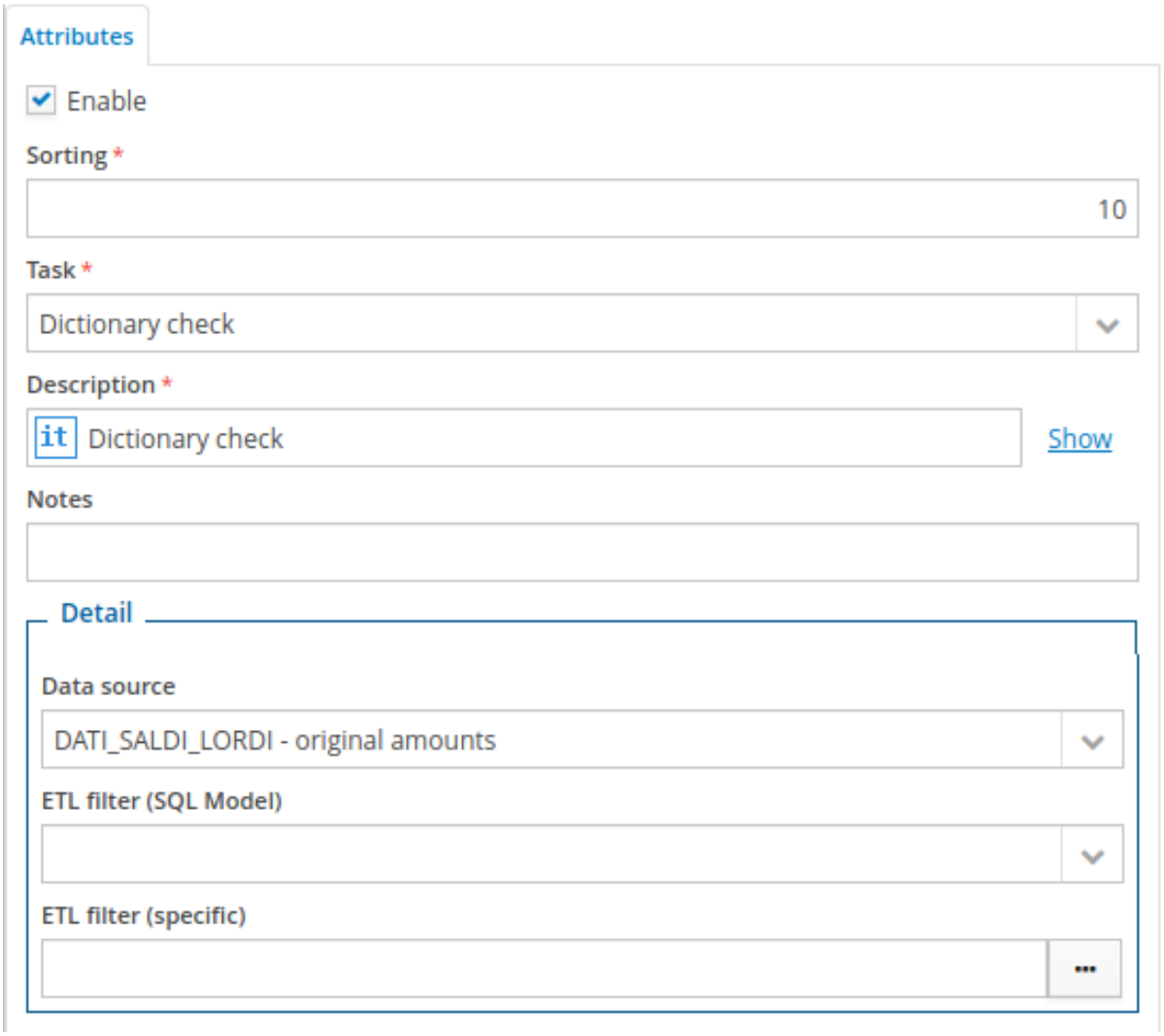
Dictionary check

This task allows you to check, for the dimensions of the ETL Domain of the Routine to which a Dictionary has been connected, if the values present in the run data belong to the set of valid values for the defined dictionary.

By default, the system checks all the dimensions present in the ETL Domain of the Routine for which a dictionary has been defined.

If the **Data Source** field is set up, it is possible to limit the control to the dimensions of the ETL Domain used in the Data source fields definition.

Moreover, by enabling the **Signal empty values in the check** option in the dictionary definition on the dimension, it is possible to make this task include as a cause of failure of the control also the lack of a value for the dimension.



Completeness check

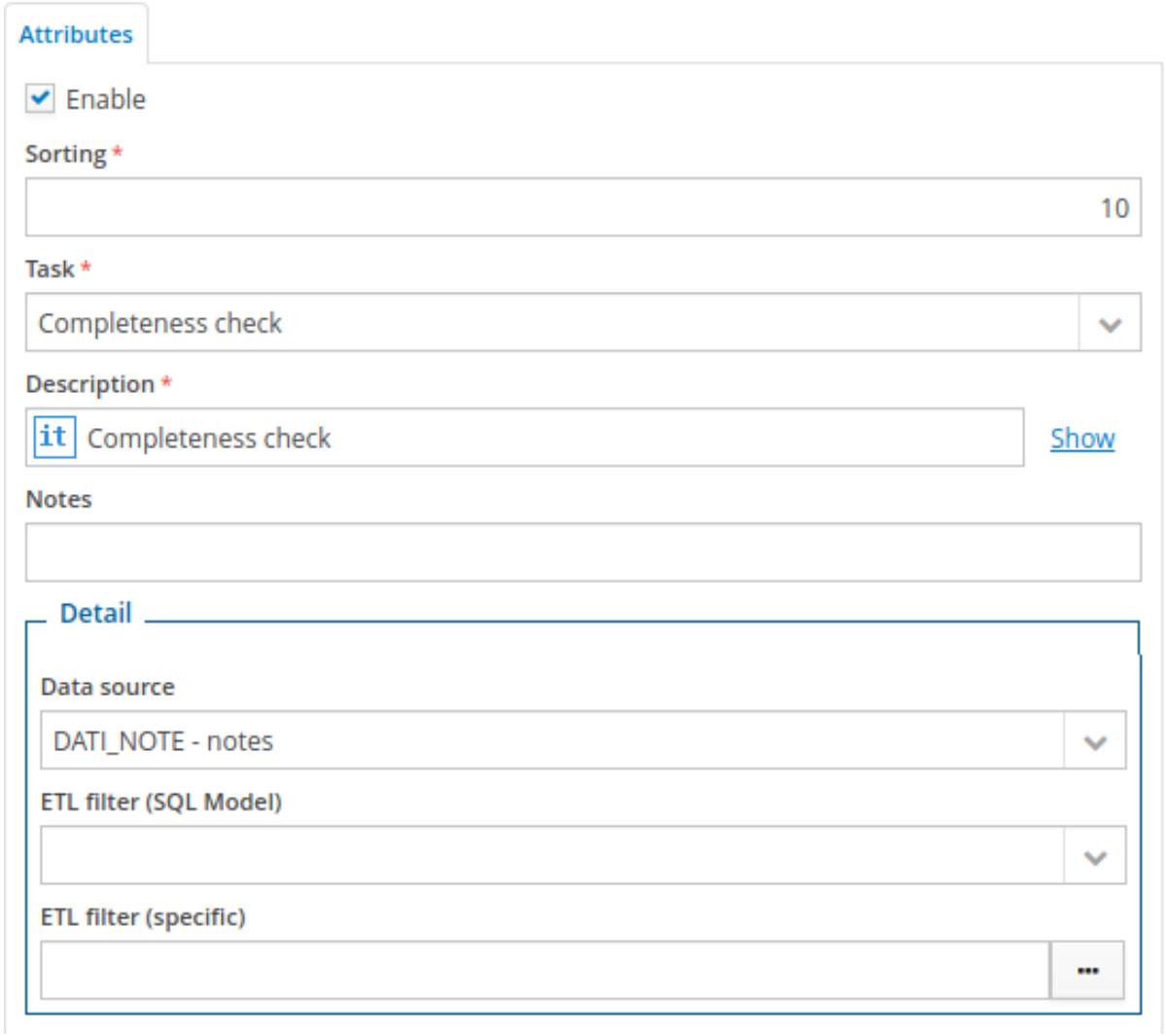
This task allows you to check, for the dimensions of the ETL Domain of the Routine to which a Completeness check has been linked, if the values present in the processed data meet the record completeness rules. This means that, for each record of the *Staging* table and for each value of a dimension included in the check, all the other mandatory dimensions must contain one value, while the forbidden dimensions must have no values (*NULL* value).

This check is useful to verify that:

- the data extraction has been correctly completed;
- the enrichment tasks following the extraction have been correctly completed.

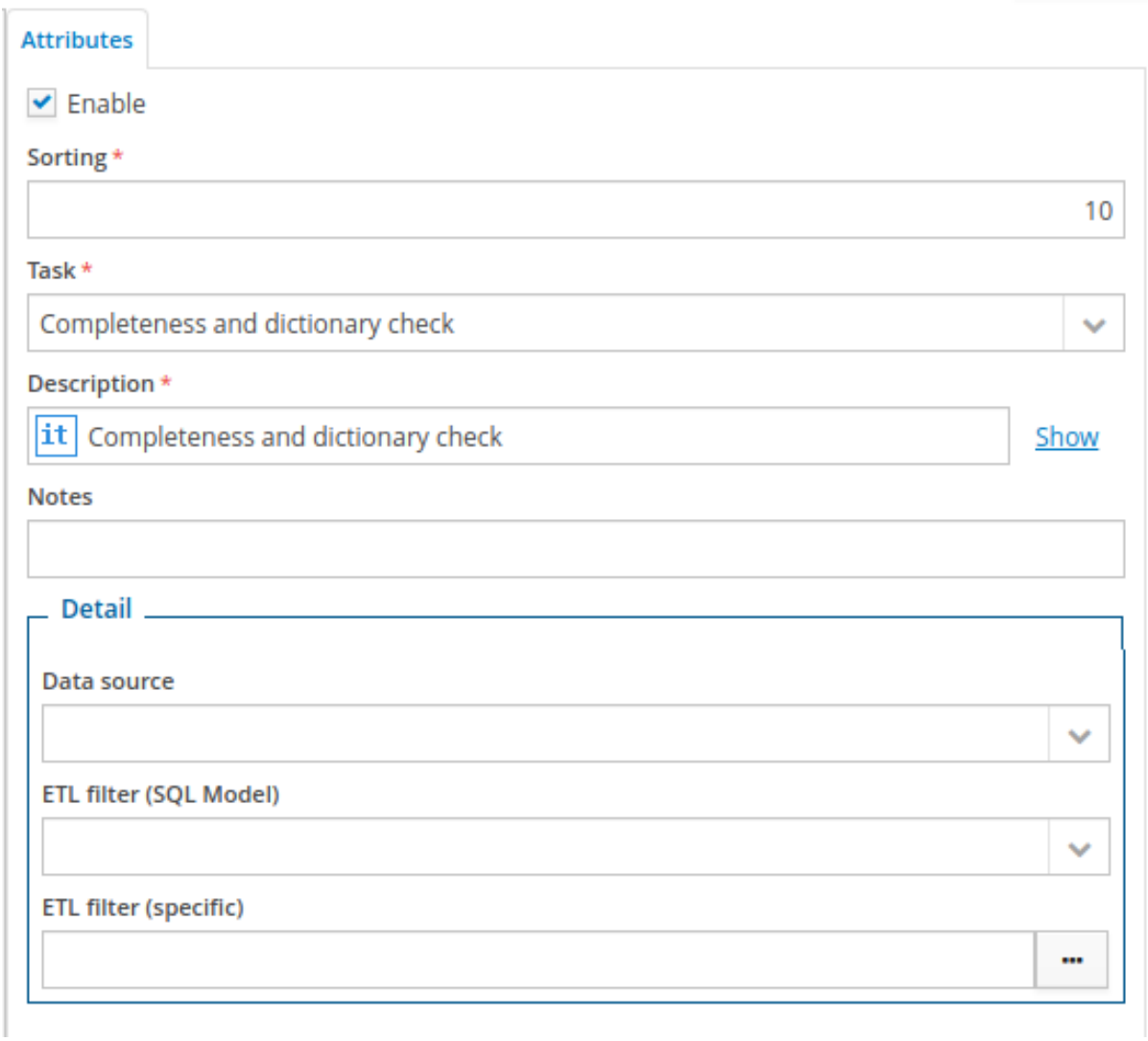
By default, the system checks all the dimensions present in the ETL Domain of the Routine for which a completeness check is defined.

If the **Data Source** field is defined, it is possible to limit the check to the dimensions of the ETL Domain used in the Data source fields definition.



Completeness and dictionary check

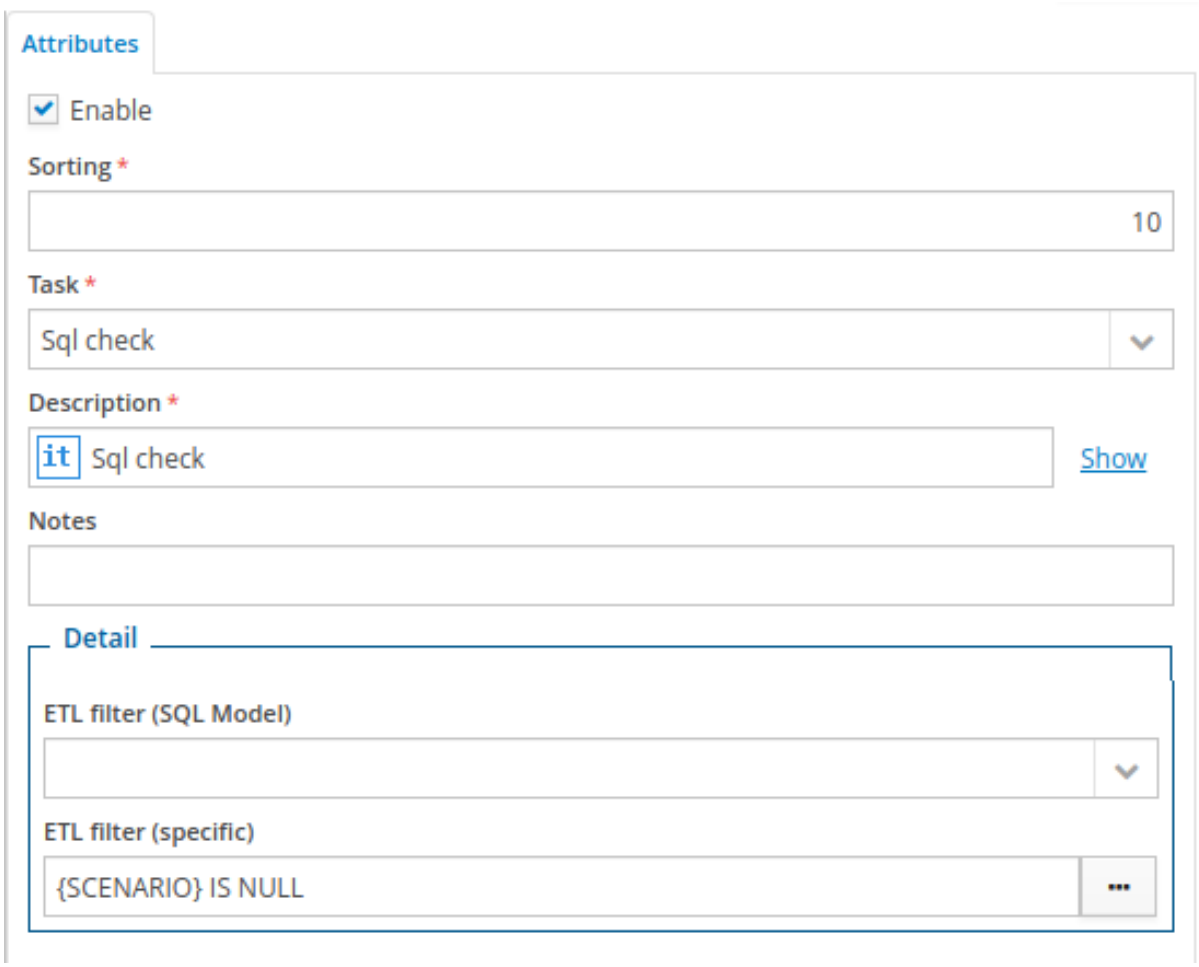
This task is used to run both the *Dictionary check* task and the *Completeness check* task.



SQL check

This check is used to identify all the records that do not meet the filter criterion defined in the **ETL Filter (SQL Model)** and/or **ETL Filter (specific)**field; generally, it is used to search for data containing wrong information.

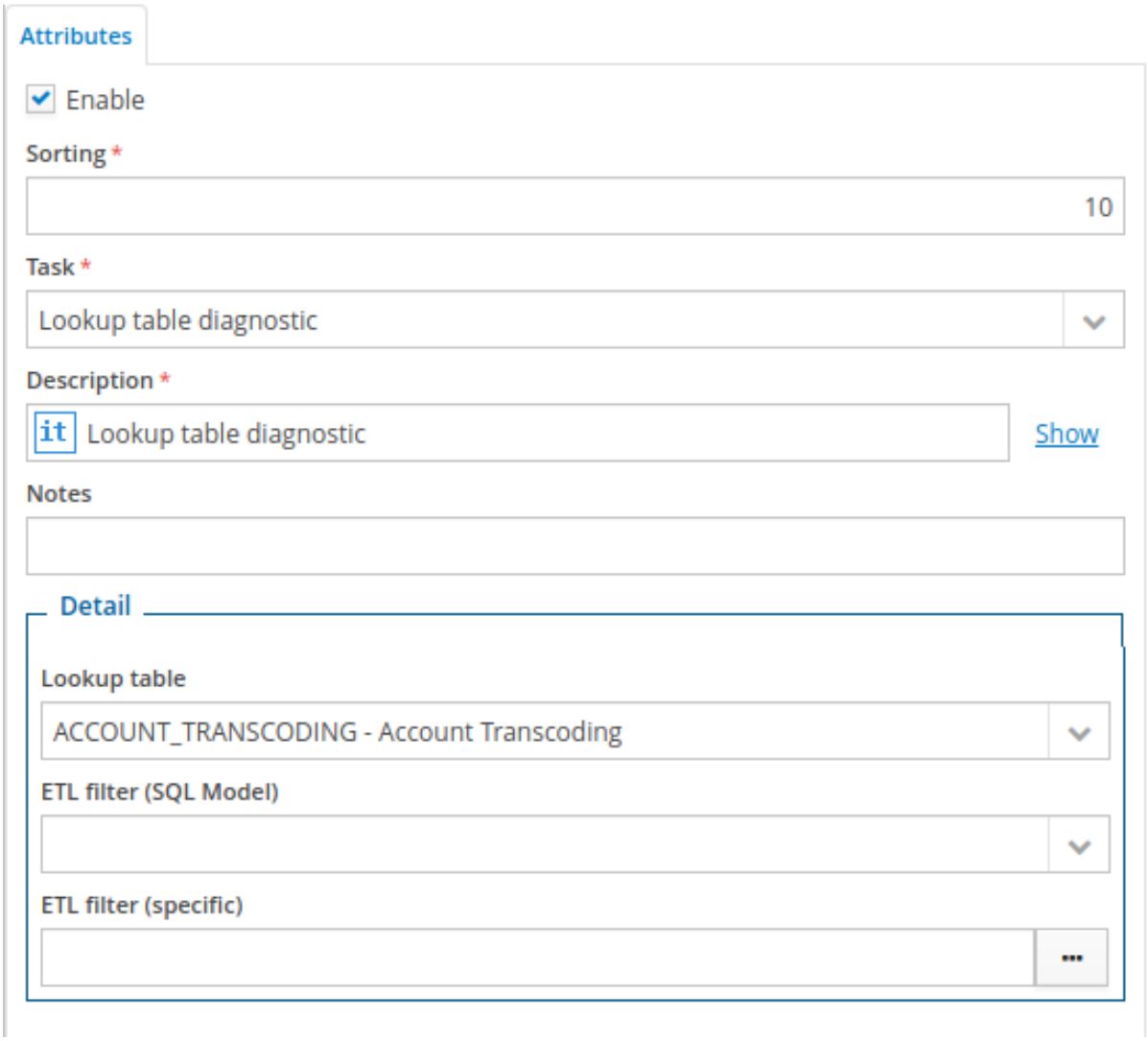
For example, it can be useful to run this task before a *Delete by sql* task in order to show the user what the system will delete in the next task and allow any corrections.



Lookup table diagnostic

This task verifies that the Lookup table defined in the **Lookup table** field is correctly set up. The task :

- if the **Full relationship** option is enabled, verifies that the *Staging* table contains all the values that compose the combination of input values defined in the elements of the lookup table;
- if the **Multiple relationship** option is not enabled, verifies that the combination of input values defined in the elements of the lookup table are univocal;
- if the **Multiple relationship** option is enabled, verifies that the combinations of input and output values defined in the elements of the lookup table are univocal.



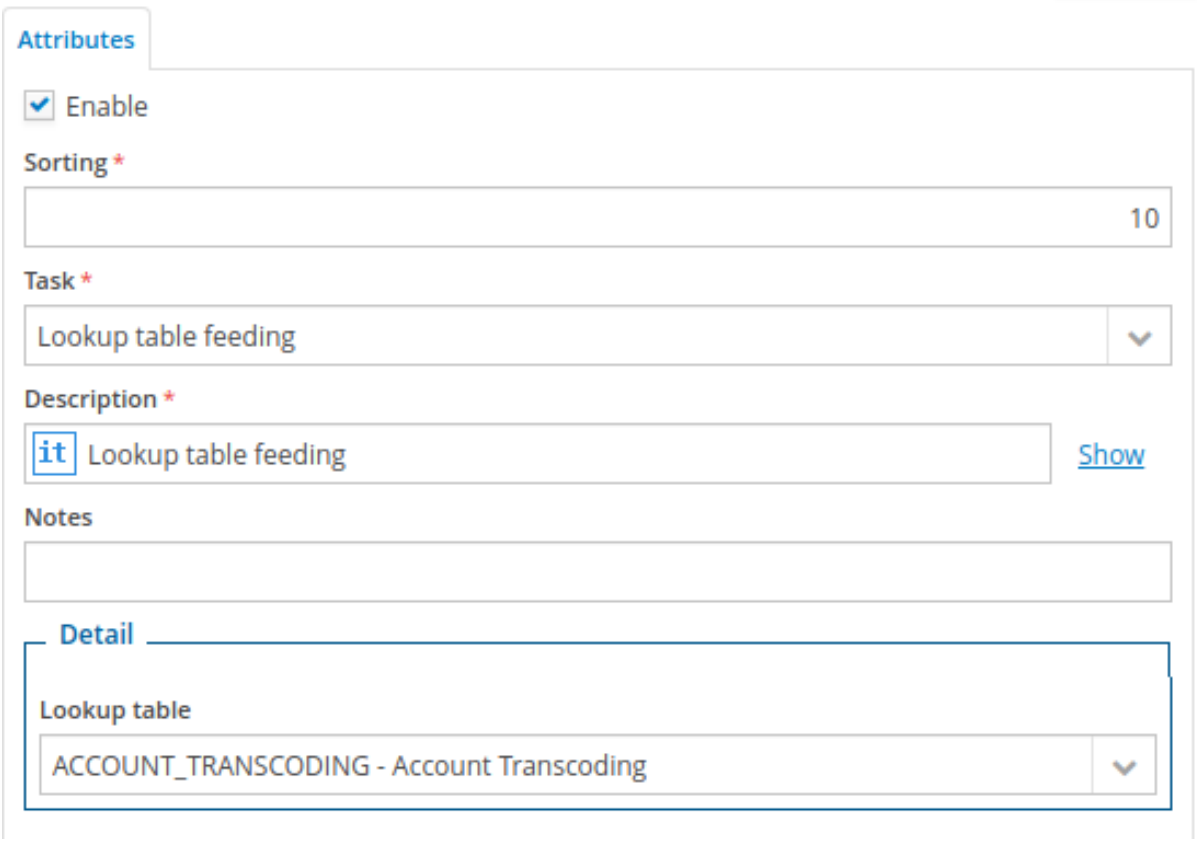
Enrichment task

Lookup table feeding

This task is used to feed the elements of the Lookup table defined in the **Lookup table** field.

Only the lookup tables whose feeding type is *Tagetik* or *Deployed* can be fed.

The system will delete all the existing elements (even if edited) and will insert the new elements from the data source defined by the lookup table.



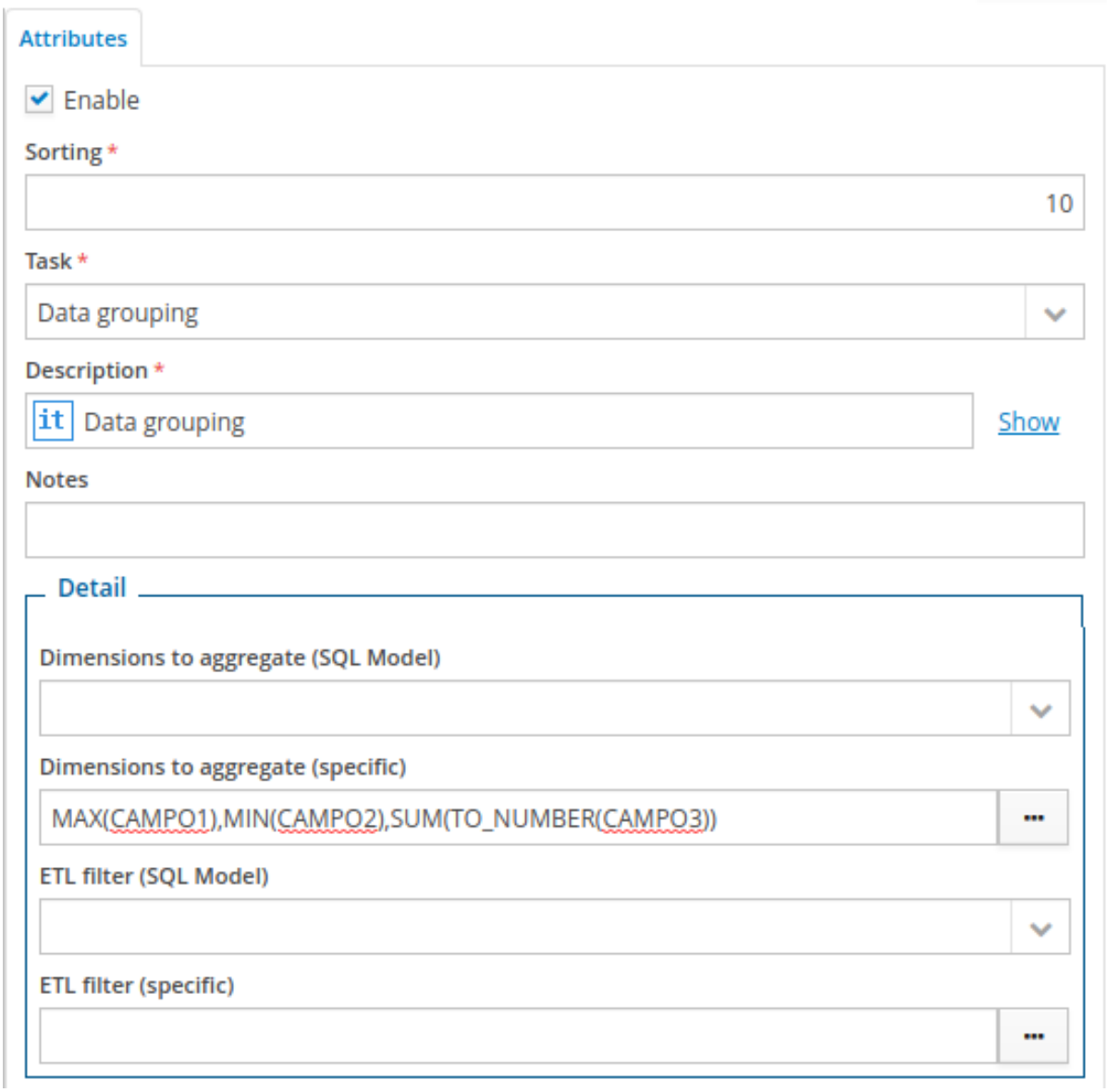
Data grouping

This task is used to aggregate data obtained during the extraction process or data enriched by a previous transformation task.

In the **Dimensions to aggregate (SQL model)** / **Dimensions to aggregate (specific)** fields, it is necessary to specify the list of fields (separated by comma) of the *Staging* table that must be aggregated and the aggregation function to be used. The text does not need to be written in upper case (E.g.: "MAX(CAMPO1), MIN(CAMPO2), SUM([number conversion function](CAMPO3))").

If a numeric aggregation function is used, it is necessary to set for the field of the *Staging* table, the conversion function provided for the database type. If you don't want to use these functions, you can replace the *CAMPOx* syntax with the value of the **Acronym** field, between *{ }*, of the dimension related to the field of the *Staging* table; so the system will manage the conversion.

A filter set on the task will aggregate only data that meet the filter criterion. Not aggregated data will be available for the next transformation task.

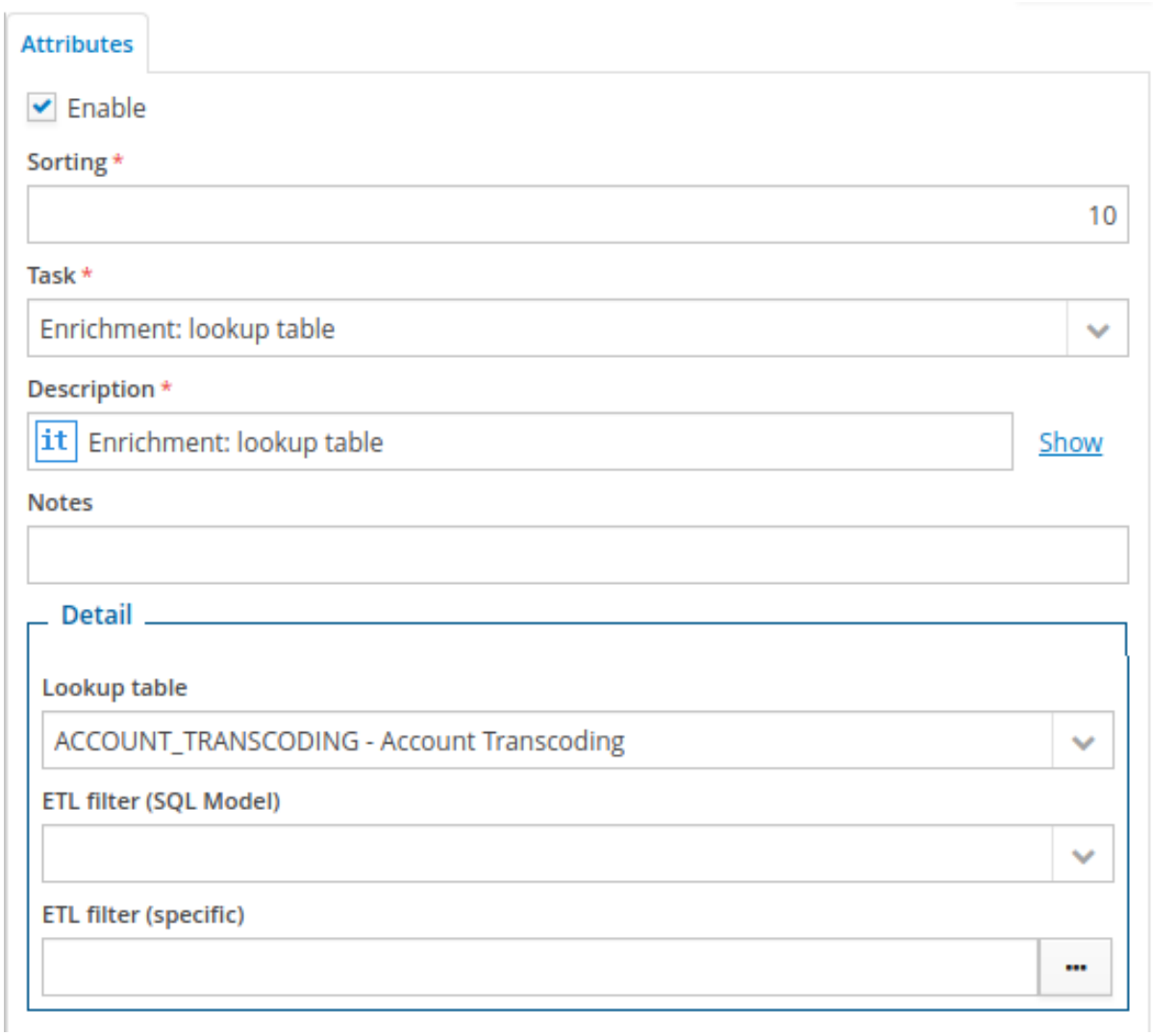


Enrichment: Lookup table

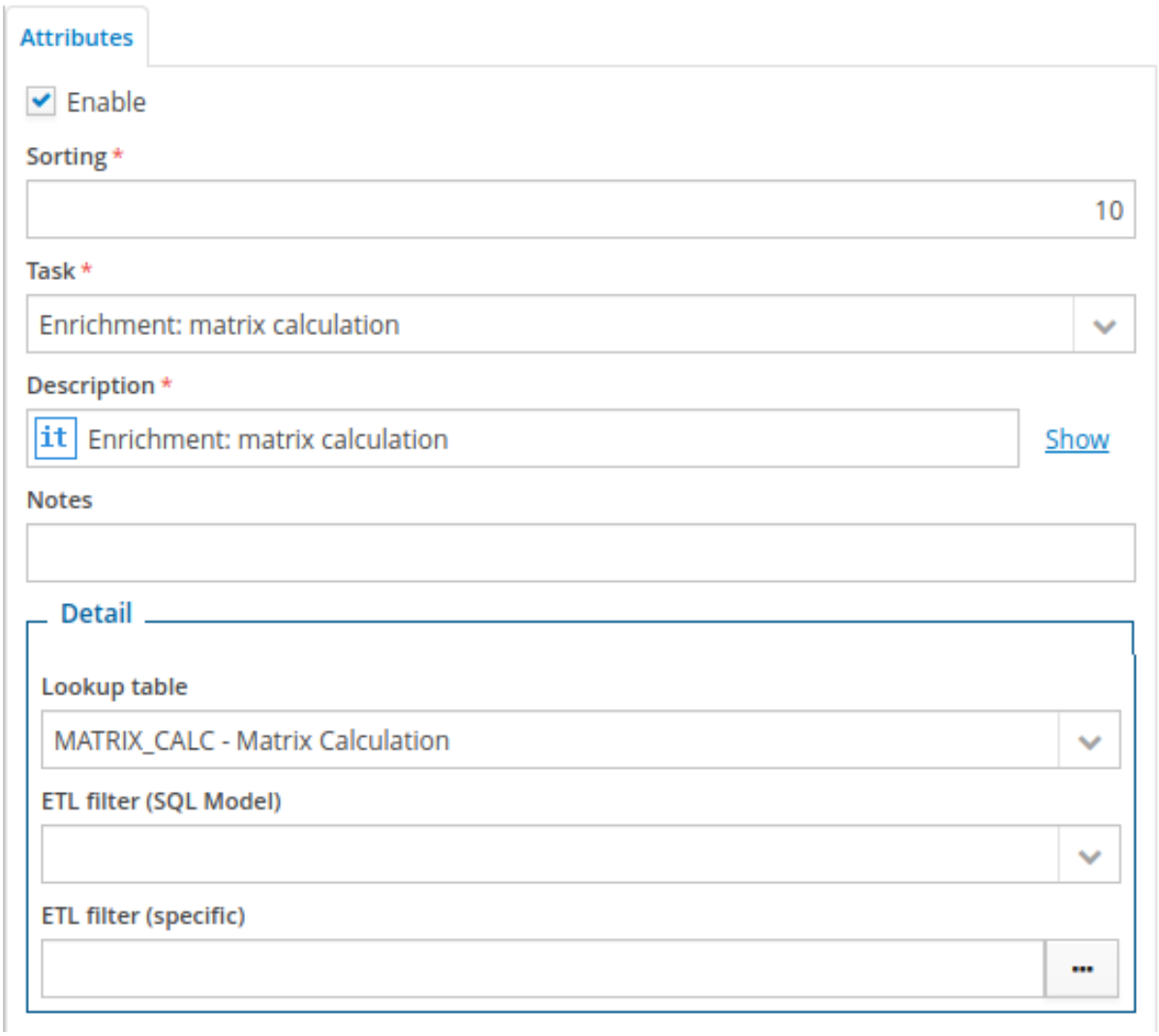
This task is used to apply the transcoding rules defined in the Lookup table specified in the **Lookup table** field.

The *output* dimensions are set in the *Staging* table with the values of the output elements for the records having a correspondence between the values of the *input* dimensions and the values defined in the input elements.

If the **Multiple relationship** option is enabled, for each combination of values found in the *input* dimensions, the system will open as many records as combinations of values defined in the output elements with an equal combination of input values.

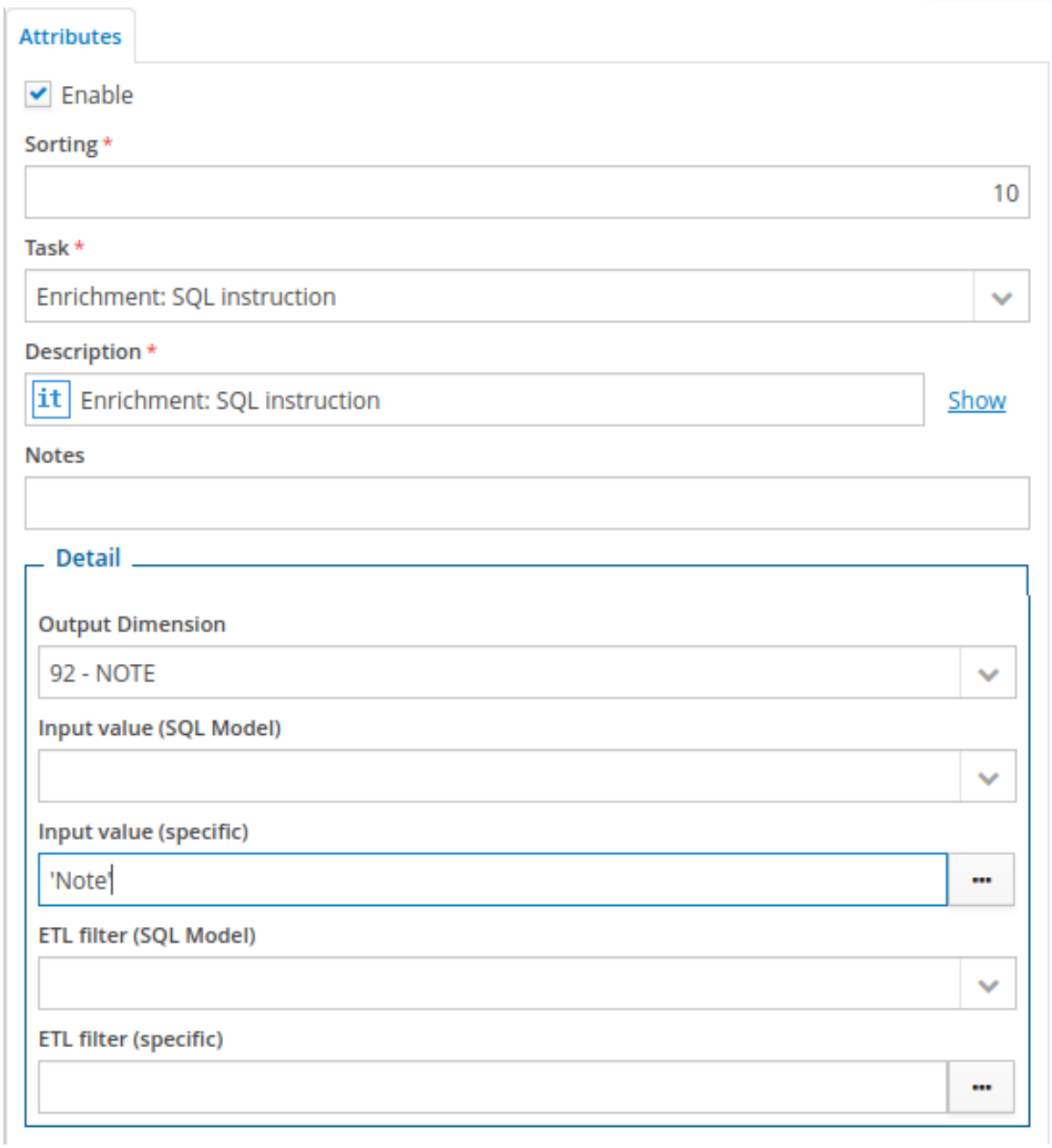

Attributes
Enable
Sorting *
10
Task *
Enrichment: lookup table
Description *
it
Enrichment: lookup table
Show
Notes
Detail
Lookup table
ACCOUNT_TRANSCODING - Account Transcoding
ETL filter (SQL Model)
ETL filter (specific)

Enrichment: matrix calculation



Enrichment: SQL instruction

This task is used to define the *Staging* table's field related to the Dimension defined in the **Output dimension** field by a SQL syntax defined in the **Input value (SQL Model)** / **Input value (specific)** fields where it is possible to use also the ETL dynamic syntax (see SQL Query or Filter Syntax). It is possible to specify a query or a fixed value put between quotation marks.



Example 1: Let us assume that we want to define the custom dimension field (6 - Custom dimension) with the last three characters of the value of the field related to the account (5 - Account) only for those accounts that starts with IC.

For this purpose, it is necessary to define:

Output dimension: 6 - Custom dimension

Input value: RIGHT(CAMPO5, 3)

ETL filter (specific): LEFT(campo5, 2) = 'IC'

The result will be:

5 - Account	6 - Custom dimension
IC_001	001
IC_002	002
003	
004	
IC_005	005

Example 2: Let us assume that we want to define the entity currency field (14 - Entity currency) with the currency defined for the entity (defined on the dimension 3 - Entity) on the process $PROCESS.

For this purpose, it is necessary to define:

Output dimension: 14 - Entity currency

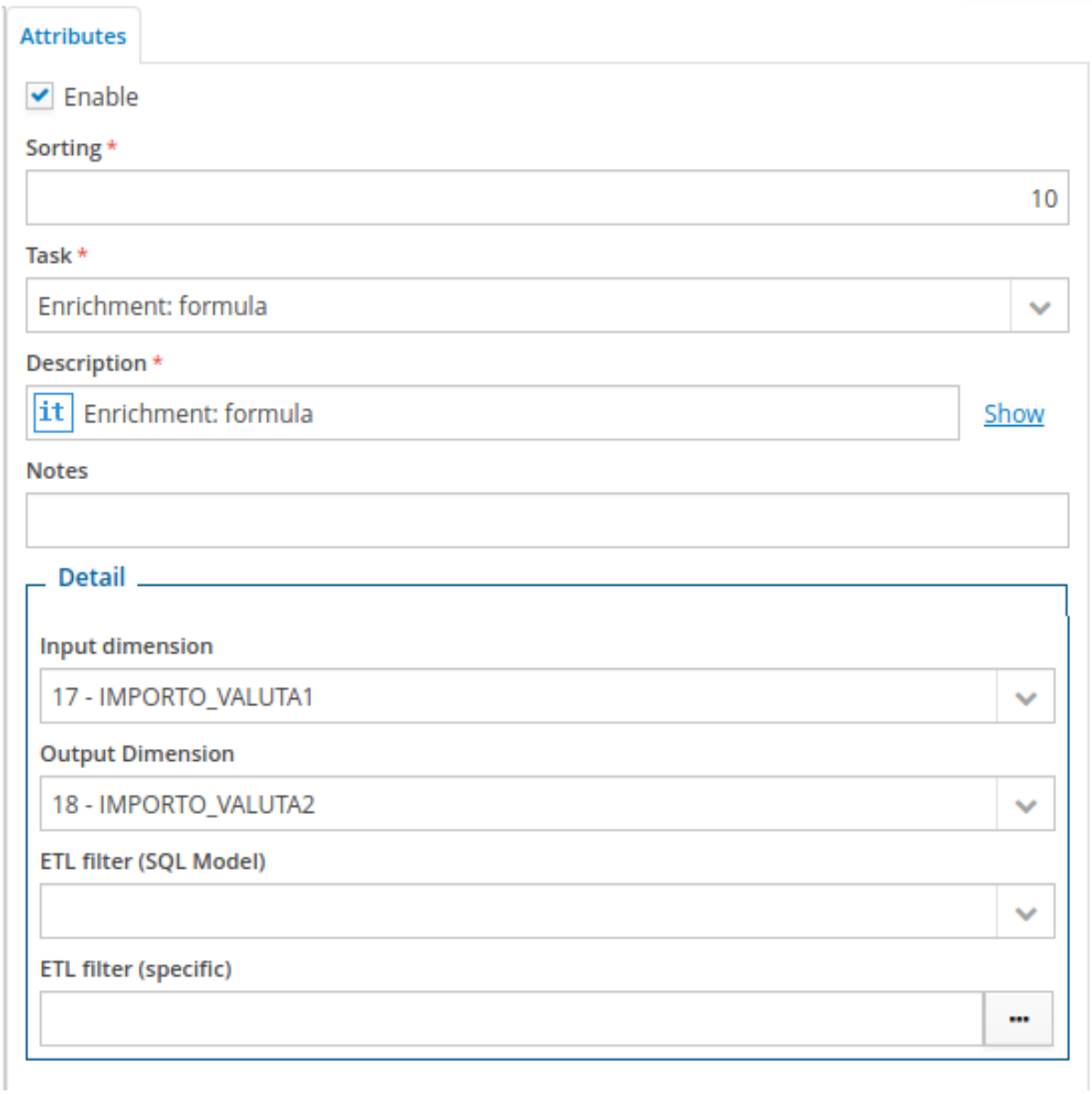
Input value: (SELECT COD_VALUTA FROM RACCOLTA_AZIENDA WHERE COD_RACCOLTA = '$PROCESS' AND COD_AZIENDA = CAMPO3)

The result will be:

3 - Entity	14 - Entity currency
ENTITY_001	EUR
NOT_AN_ENTITY	

Enrichment: formula

This task is used to define the *Staging* table's field related to the Dimension defined in the **Output dimension** field by the application of a numerical formula saved as value of the field related to the dimension defined in the **Input Dimension**field that uses in its syntax references to other fields using the identification number of the dimension placed between curly brackets (**EXAMPLE:** "({2}*{3} / 100)"). All the dimensions used in the definition of the task and in the syntax of the formula must be numerical and must contain a valid number.



Example 1: Let us assume that the field related to the amount to upload to CCH Tagetik (4 - Result) results from numerical operations defined as formula in the "1 - Formula" dimension field among the amounts contained in the fields related to the dimension "2 - Amount1" and "3 - Amount2" .

For this purpose, it is necessary to define:

Input dimension: 1 - Formula

Output dimension: 4 - Result

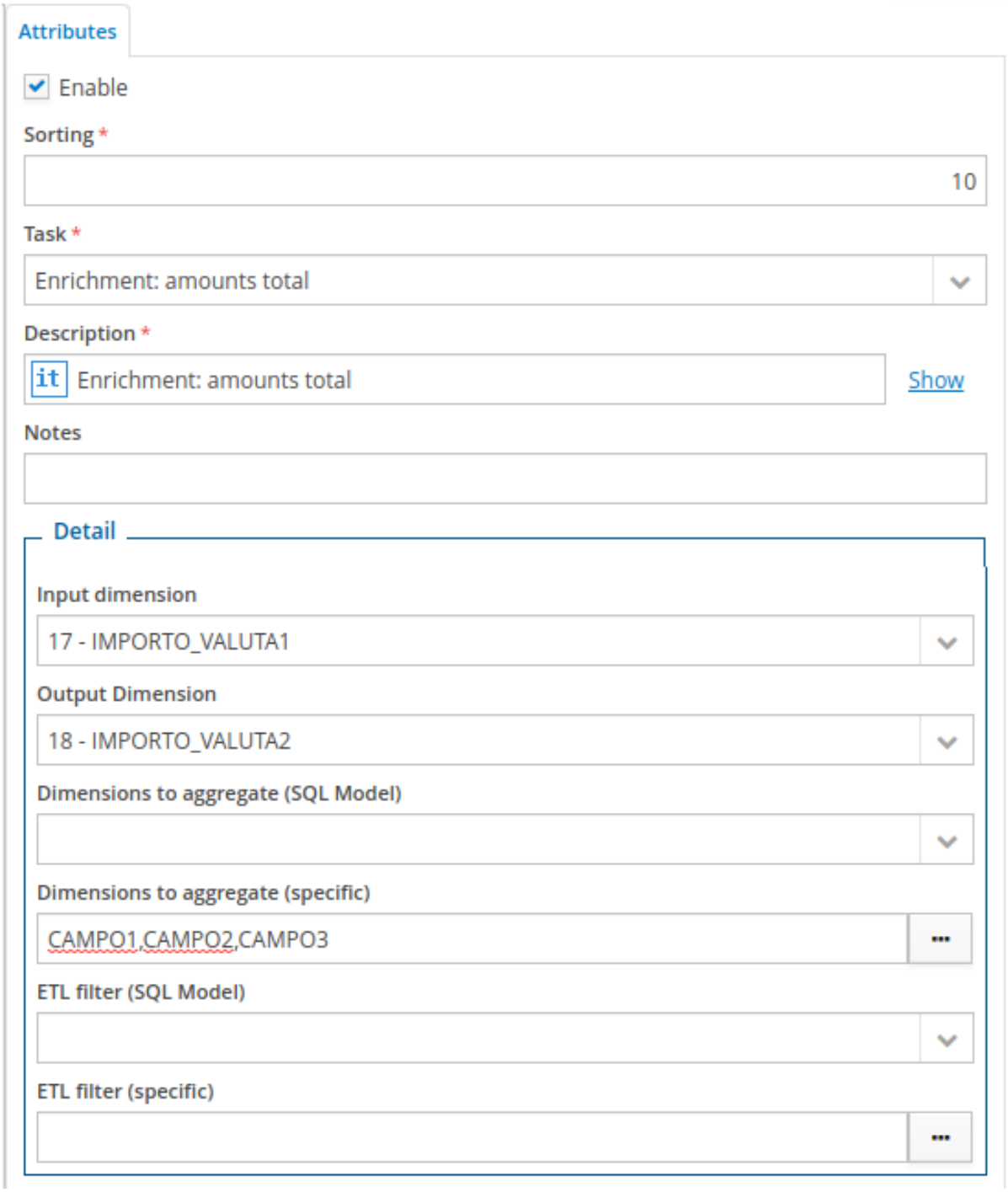
The result will be:

1 - Formula	2 - Amount1	3 - Amount2	4 - Result
({2} * {3}) / 100	7000	10	700
{3} - {2}	200	300	100

Enrichment: amounts total

This task is used to define the *Staging* table's field related to the Dimension defined in the **Output dimension** field with the sum of the amounts contained in the field related to the dimension defined in the **Input dimension** field, totalling at comparable values contained in other fields in the *Staging* table listed in the fields **Dimensions to aggregate (SQL model)** / **Dimensions to aggregate (specific)** (E.g.: "CAMPO1, CAMPO2, CAMPO3").

The input and output dimensions must be numerical and the related fields must contain numerical values.



Example 1: Let us assume that we want to add up the amount contained in the "4 - Amount" dimension field for the fields related to the dimensions "1 - Scenario", "2 - Period", "3 - Entity" and then save it in the "5 - Total amount" dimension field.

For this purpose, it is necessary to define:

Input dimension: 4 - Amount

Output dimension: 5 - Total amount

Dimensions to aggregate (specific): CAMPO1,CAMPO2,CAMPO3

The result will be:

1 - Scenario	2 - Period	3 - Entity	4 - Amount	5 - Total amount
2017ACT	01	001	100	300
2017ACT	01	001	200	300
2017ACT	01	002	200	500
2017ACT	01	002	300	500
2017ACT	02	001	100	150
2017ACT	02	001	50	150

Enrichment: ranking

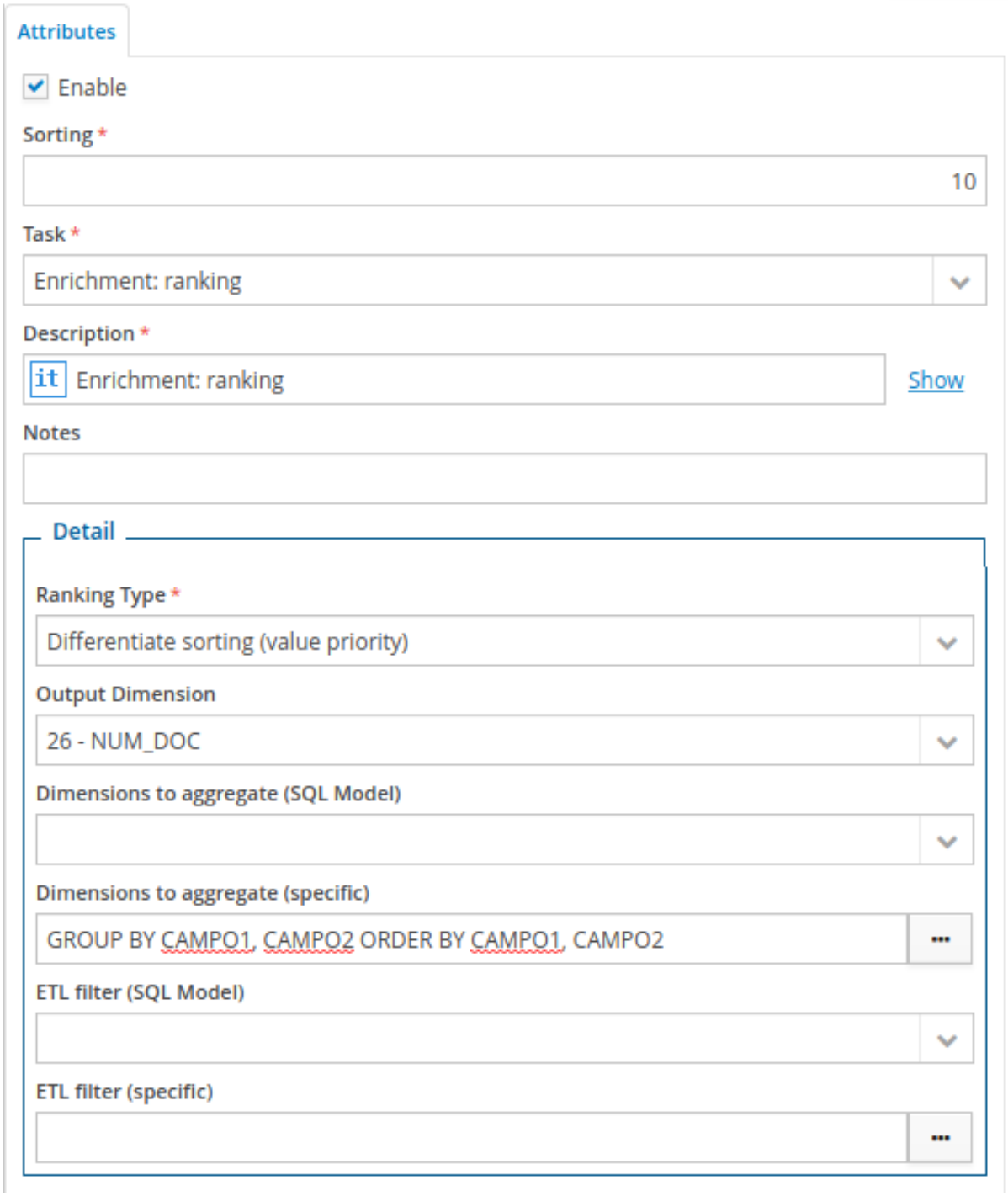
This task is used to define the *Staging* table's field related to the Dimension defined in the **Output dimension** with a sequential number at comparable values contained in other fields of the *Staging* table.

The criterion according to which the sequential number is assigned is specified in the **Ranking type** field:

- *Differentiate sorting (value priority)*: the sequential number is increased for each record contained in the range of records with the same values in the fields on which the ranking will run. On the other hand, it is reset for each new range of records;
- *Keep sorting (value priority)* : the same sequential number is assigned to the records contains in the same range of those records with the same values in the fields on which the ranking will run. It is increased for each new range of records.

The fields of the *Staging* table, on which the ranking must run and for which the sequential number is sorted and reset, are defined in the fields **Dimensions to aggregate (SQL model)** / **Dimensions to aggregate (specific)**(E.g..: GROUP BY CAMPO1, CAMPO2, ORDER BY CAMPO1, CAMPO2); these fields can contain:

- a GROUP BY clause that specifies the fields of the *Staging* table for which at comparable values some records to which the ranking applies as defined in the **Ranking type** field are inserted in the same range; therefore;
- an ORDER BY clause that specifies the fields of the *Staging* table for which the sorting is needed before assigning the range to each record;



Example 1: Let us assume that we want to sort the amounts contained in the "2 - Amount" dimension field at comparable value of the code of the account contained in the "1 - Account" dimension field sorted by amount.

For this purpose, it is necessary to define:

Ranking type: Differentiate sorting (value priority)

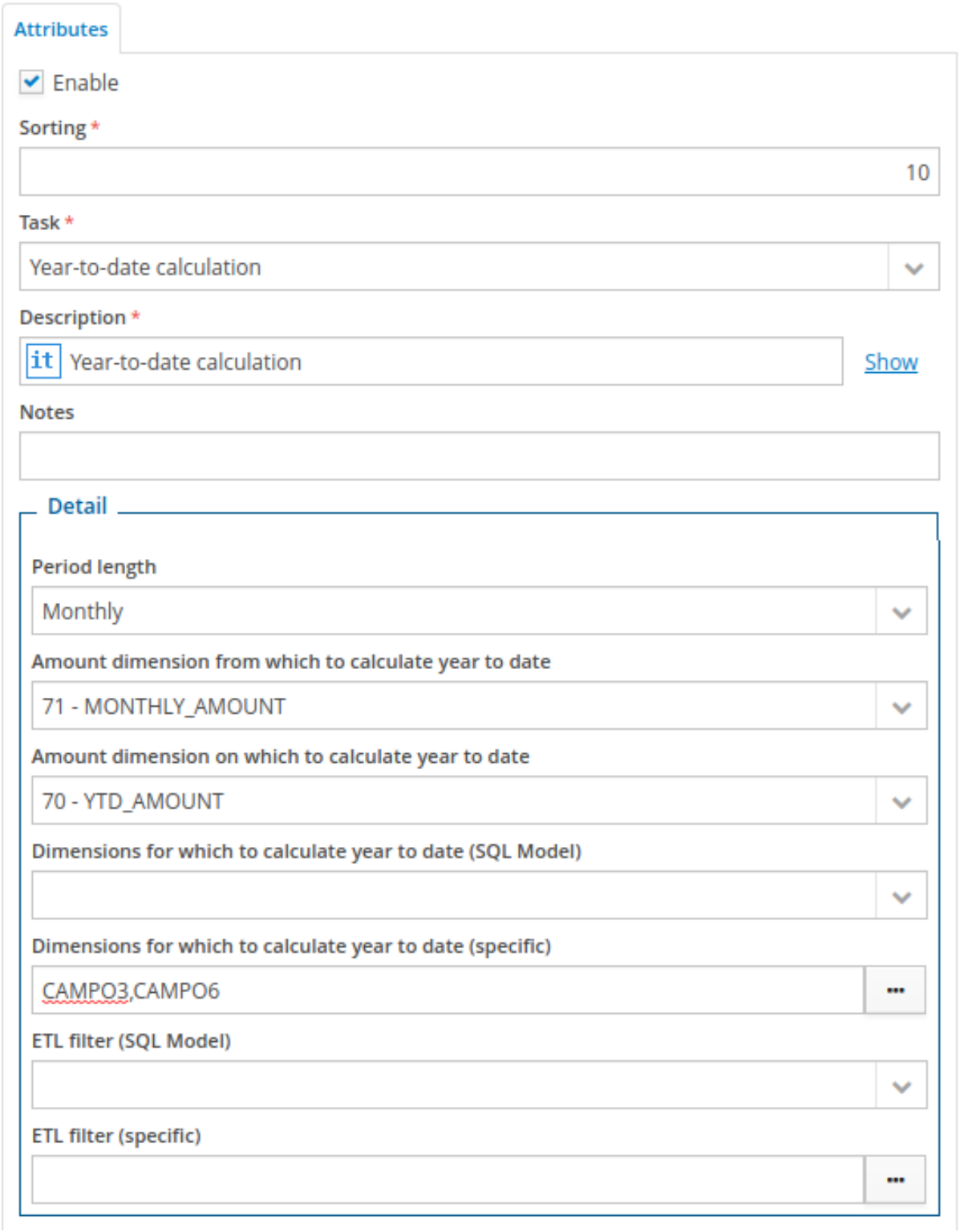
Output dimension: 3 - Output Ranking

Dimensions to aggregate: GROUP BY CAMPO1 ORDER BY CAMPO2 DESC

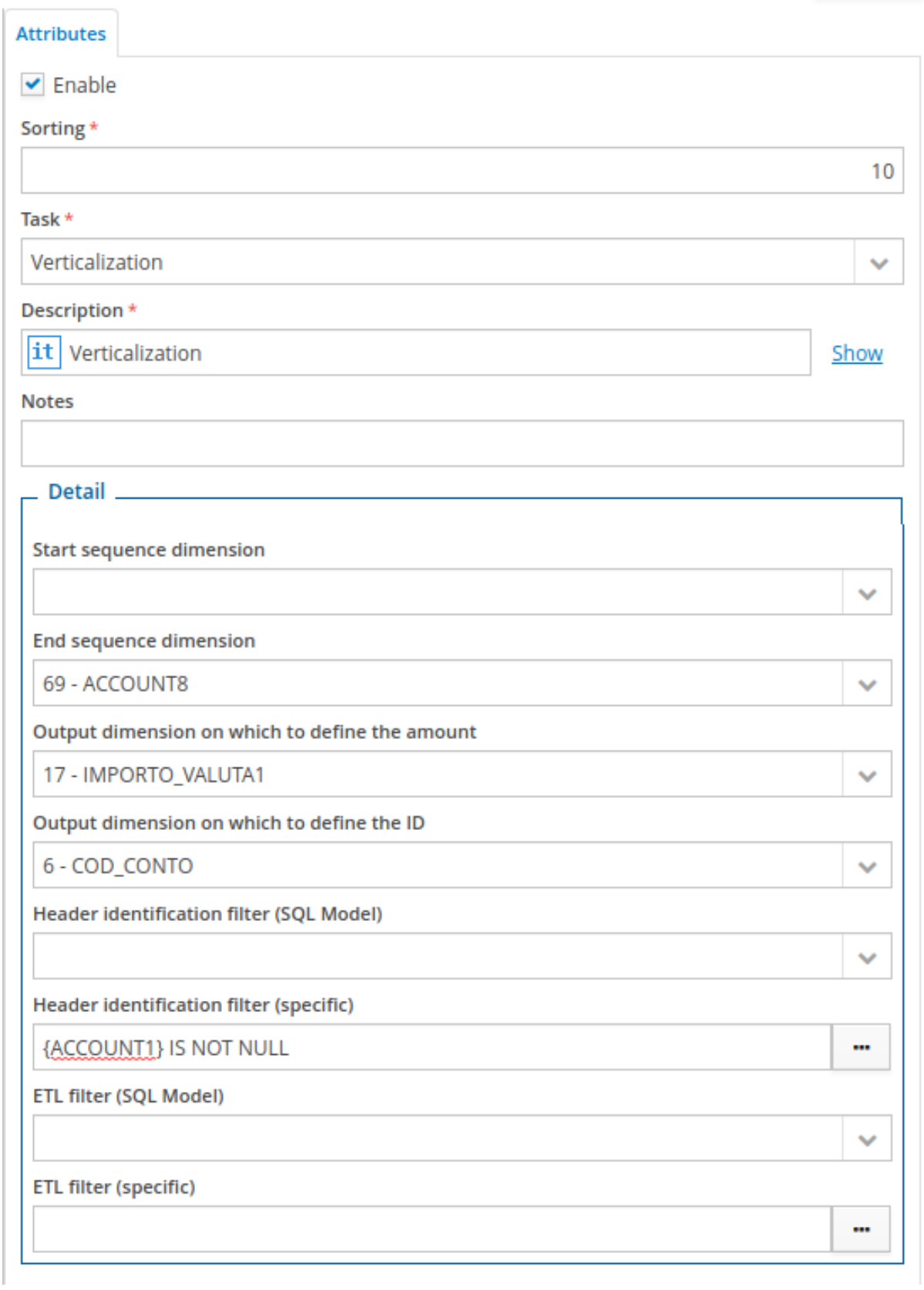
The result will be:

1 - Account	2 - Amount	3 - Rank-ing output
001	100	1
001	200	2
001	300	3
002	50	1
002	150	2
002	250	3

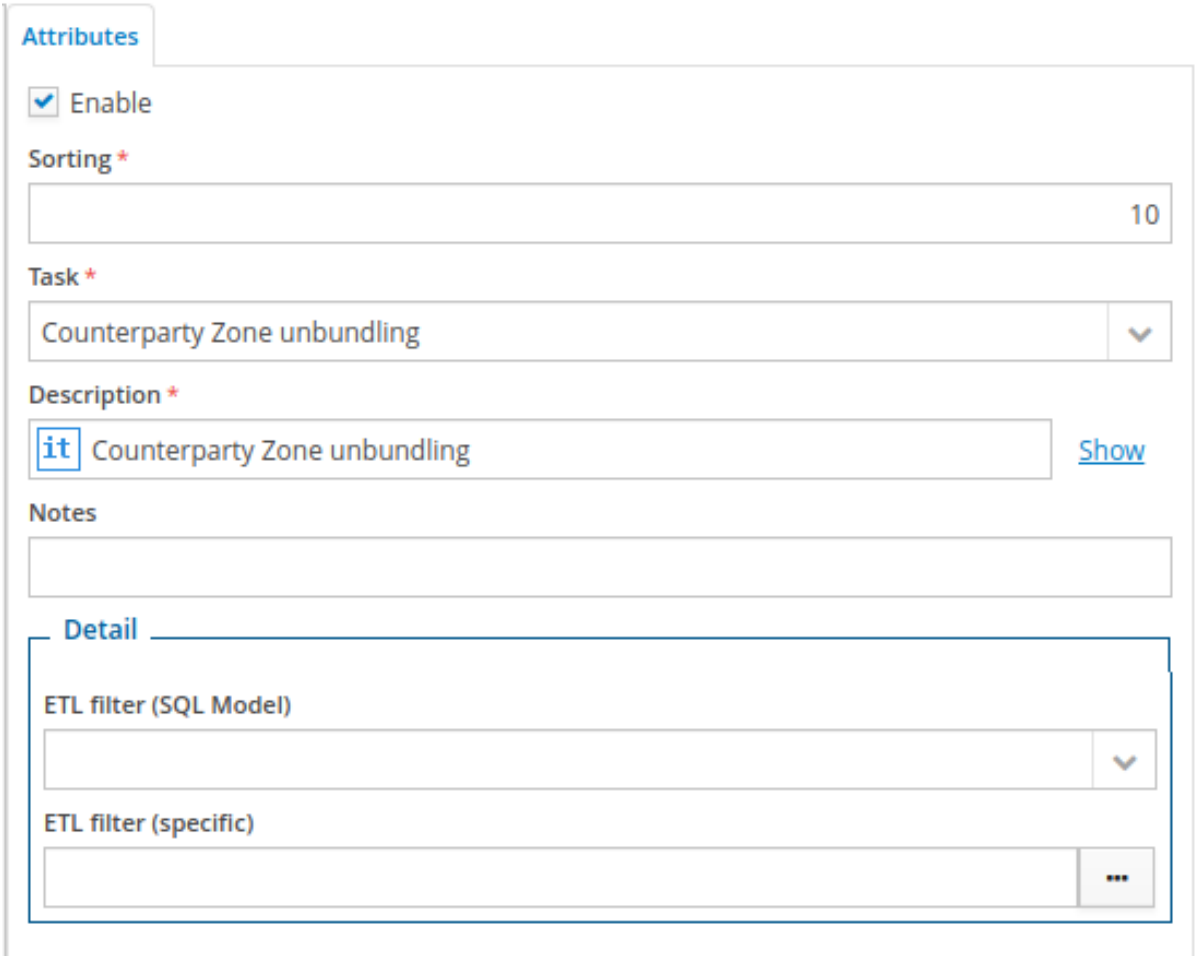
Year-to-date calculation



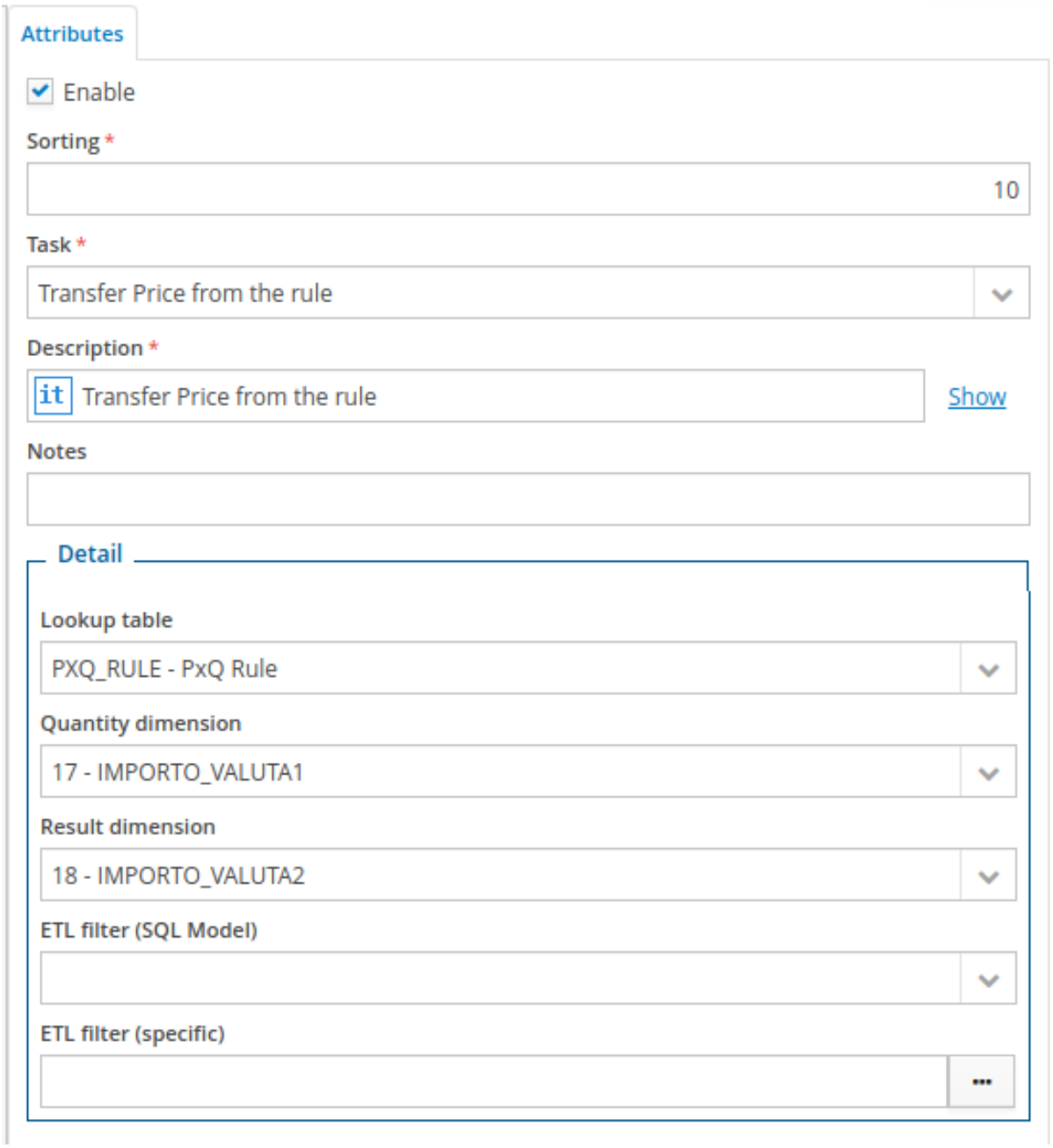
Verticalization



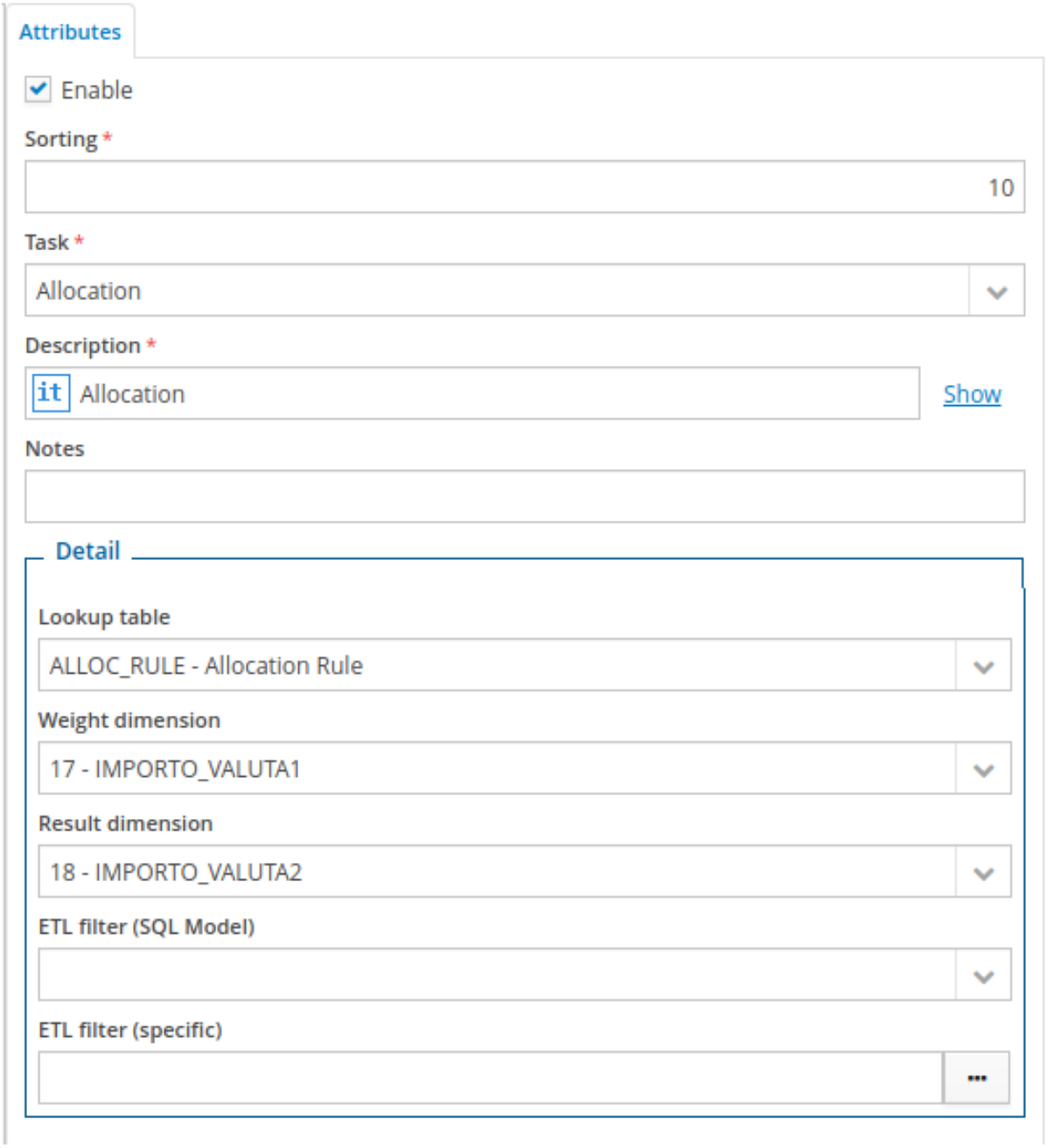
Counterparty Zone unbundling



Transfer price from the rule

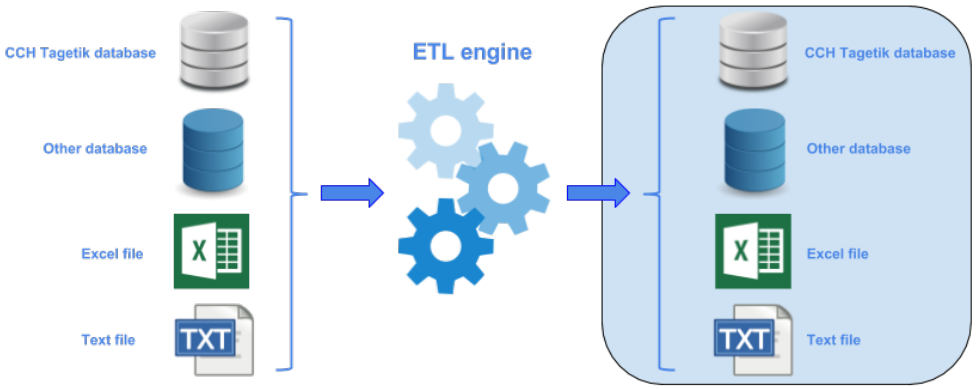


Allocation


Attributes
Enable
Sorting *
10
Task *
Allocation
Description *
it Allocation
Show
Notes
Detail
Lookup table
ALLOC_RULE - Allocation Rule
Weight dimension
17 - IMPORTO_VALUTA1
Result dimension
18 - IMPORTO_VALUTA2
ETL filter (SQL Model)
ETL filter (specific)

Routine - Loading

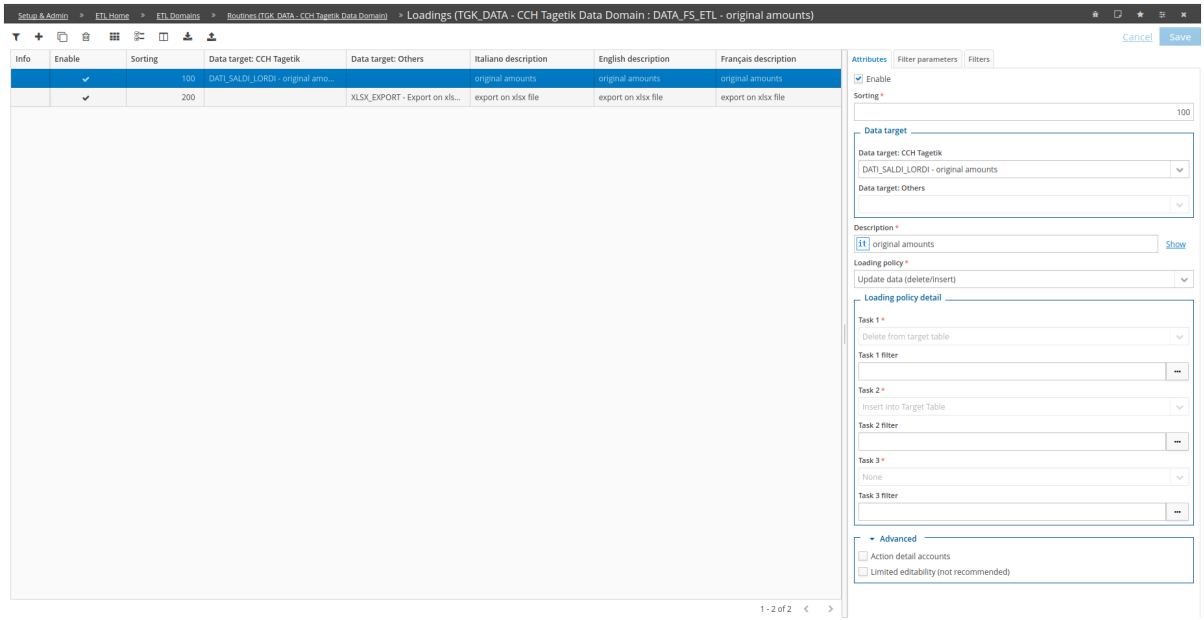
The Loading is a Routine's task that allows loading data from the *Staging* table to targets defined in the Data targets: CCH Tagetik and in the Data targets: Others. Editability checks are performed on the loadings run on CCH Tagetik in order to verify the correctness of the information to insert or to delete. Any wrong records will be displayed to the user in the form of warnings on completion of the ETL run.



Definition

The Loadings definition can be accessed by the *Loadings* link in the Routine window.

If the ETL Domain is locked, the window will be in read-only mode.



Each Loading is identified by:

Attributes

- the **Enable** option that specifies if the task must be run by default;
- a **Sorting** that specifies the task run order; neither value 0 nor a sorting that already exists for the loading type can be set;
- a **Data target: CCH Tagetik** or a **Data target: Others** that specifies, by a Data target: CCH Tagetik or a Data target: Others, the data loading mode on the CCH Tagetik database, on another database or on a file of one of the managed types;
- **Descriptions** in the selected languages;
- a **Loading policy** specifying how the system must load data into the target. This information is a template that allows specifying more quickly the detail tasks defined in the fields **Task1**, **Task2** and **Task3**. Available loading modes:
  - *Element (update/insert)*: available only if the data loading is run on a table for which a primary key is defined. Typically, this is used to insert *structural* data, namely those data related to the tables in which usually the deletion must not be run. In this case, the following detail tasks are run:
    - **Task1**: Update target table;
    - **Task2**: Append into target table;
    - **Task3**: None.
  - *Update data (delete/insert)*: available only if the data loading is run on a table for which a primary key is defined. Typically, this is used to update the *numeric* data. In this case, the following detail tasks are run:
    - **Task1**: Delete from target table;
    - **Task2**: Insert into Target Table;
    - **Task3**: None.
  - *Append data (insert)*: typically used to insert new *structural* or *numeric* data. In this case, the following detail tasks are inserted:
    - **Task1** : Insert into Target Table;
    - **Task2**: None;
    - **Task3**: None.
  - *Specific*: specifies that the detail tasks to run must be defined manually by the user.
- a **Task1** that specifies the first detail task to run;
- a **task1 filter** to specify a filter to apply to the operation run by the **Task1**; it is possible to use the ETL dynamic syntax (see SQL Query or Filter Syntax);
- a **Task2** that specifies the second detail task to run;
- a **task2 filter** to specify a filter to apply to the operation run by the **Task2**; it is possible to use the ETL dynamic syntax (see SQL Query or Filter Syntax);
- a **Task3** that specifies the third detail task to run;
- a **task3 filter** to specify a filter to apply to the operation run by the **Task2**; it is possible to use the ETL dynamic syntax (see SQL Query or Filter Syntax).

  The following task types can be selected for the fields **Task1**, **Task2** and **Task3** :

  - *Delete from target table*: deletes data from the target table defined in the data target;
  - *Delete ETL data from target table*; deletes data existing in the *staging* table from the target table defined in the data target (existence check by primary key);
  - *Delete non-ETL data from target table*; deletes data that do not exist in the *staging* table from the target table defined in the data target (existence check by primary key);

- *Insert into Target Table*: inserts into the data target all data available in the *staging* table.
- *Append into target table*: inserts into the target table defined in the data target any data available in the *staging* table that do not exist in the target table (existence check by primary key);
- *Update target table*: updates data present in the target table with values available in the *staging* table (existence check by primary key).

Moreover, only for **Data target: CCH Tagetik**, it is possible to define two options:

- **Action detail accounts**: for the loadings into the tables from Gross Amounts and Intercompany of CCH Tagetik with *Update data (delete/insert)* **Loading policy**, it enables the loading policy by the window containing details, for each account, on which amounts are available in the *staging* table, which amounts are available in the target table with origin related to the current Routine and which amounts are available on the target table with other origin. In this window it is possible to select the amounts that must be inserted and those that must be deleted.
- **Limited editability (not recommended)**: for the data loadings on CCH Tagetik tables, on which the editability check is run in order to verify the correctness of the inserted information, it disables the following checks:
  - Account restrictions - Custom dimensions
  - Entity rights - Custom dimensions
  - Counterparty rules

Attributes | Filter parameters | Filters

Enable

Sorting *

100

Data target

Data target: CCH Tagetik

DATI_SALDI_LORDI - original amounts

Data target: Others

Description *

it original amounts Show

Loading policy *

Update data (delete/insert)

Loading policy detail

Task 1 *

Delete from target table

Task 1 filter

Task 2 *

Insert into Target Table

Task 2 filter

Task 3 *

None

Task 3 filter

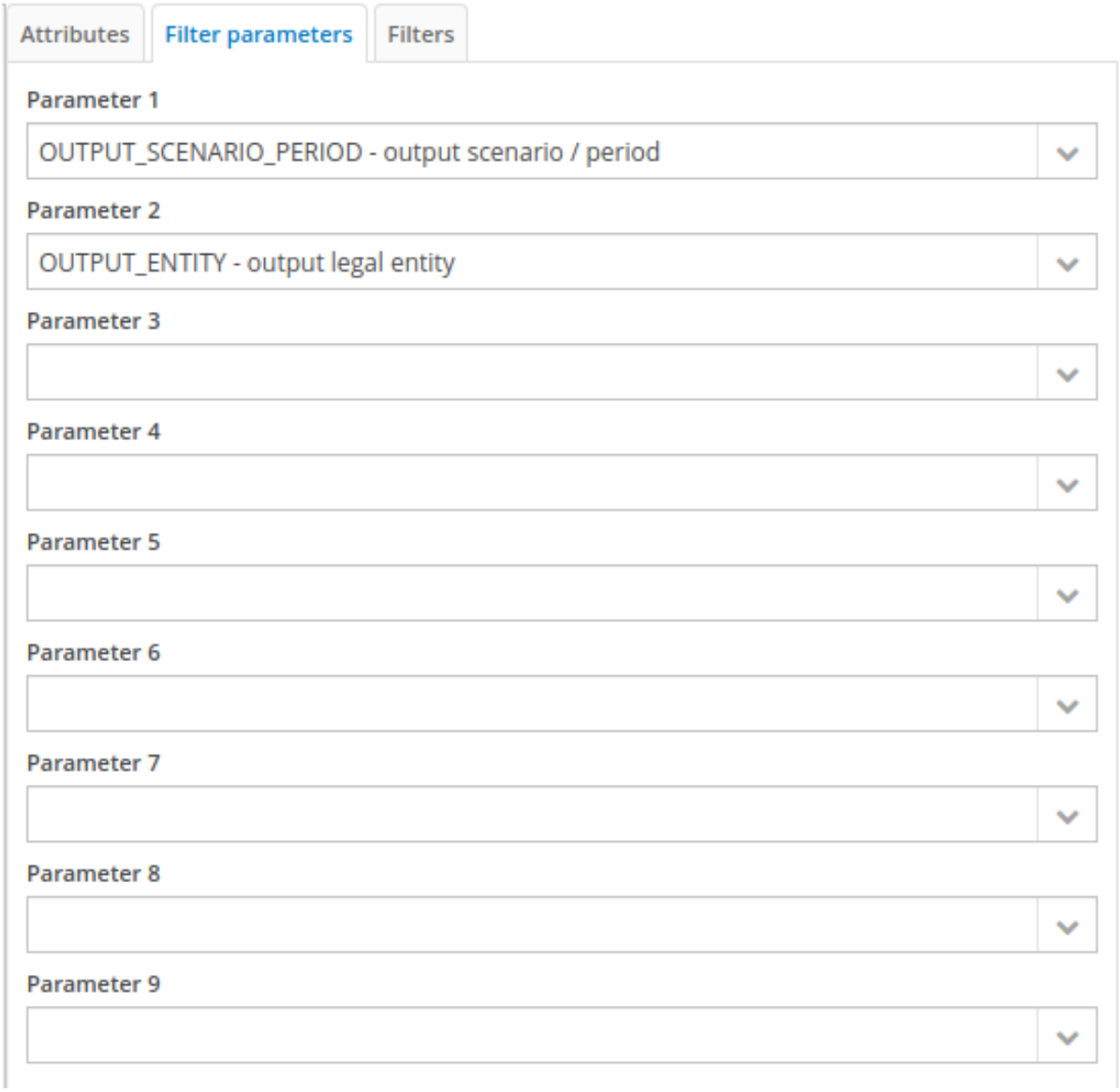
Advanced

Action detail accounts

Limited editability (not recommended)

Filter parameters

- It is possible to set up to nine Filter Parameters in order to limit the amount of data to load. If the user has specified also one of the other filters, then the filter will link the conditions by the logical operator *AND*.



Filters

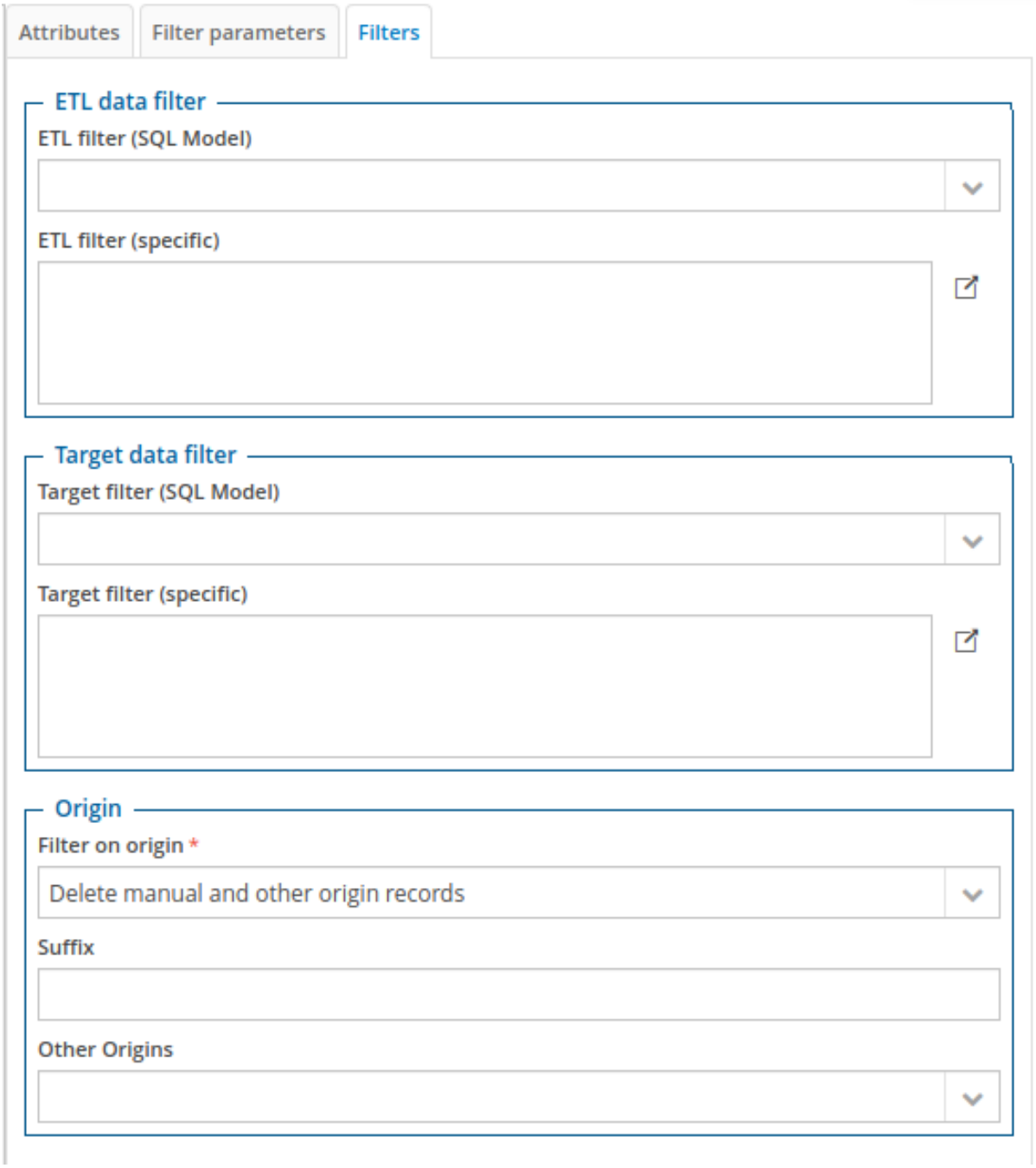
In order to limit the number of records processed by each loading detail task, it is possible to define:

- an **ETL Filter (SQL Model)** that specifies a filter previously define in the SQL Models; this filter applies to the *staging* tables when running the data entry and update operations.

- an **ETL Filter (specific)** which allows specifying a filter with SQL syntax manually; it is possible to use the ETL dynamic syntax (see SQL Query or Filter Syntax); this filter applies to the *staging* table when running the data entry and update.

  If the user specifies both filters, the system links the conditions by the logical operator *AND*.
- a **Target filter (SQL Model)** which allows specifying a filter previously defined in the SQL Models; this filter applies to the target table when running the deletion and update operations.
- an **ETL Filter (specific)** which allows specifying a filter with SQL syntax manually; it is possible to use the ETL dynamic syntax (see SQL Query or Filter Syntax); this filter applies to the target table when running the deletion and update operations.

  If the user specifies both filters, the system links the conditions by the logical operator *AND*.
- a **Filter on origin** which allows specifying the deletion policy to apply to the target table's field identified as origin of data. A predefined field (PROVENIENZA) is available for the CCH Tagetik tables, while for the custom tables of the CCH Tagetik database and for another database, the field identified as origin is specified by the **Parameters selection mode** attributes in the Data target: Others fields definition. Available options:
  - *Do not delete*: no data is deleted from the target table. Moreover, this option allows the user to specify a different value during runtime if the **Deletion management** option is enabled for the selected Routine;
  - *Delete all*: does not apply the filter to the origin field;
  - *Delete manual origin records*: deletes from the target table any data with origin *MAP**, *INPUT_DEFORM*, *INPUT_WEB* and *QDL*;
  - *Delete manual and other origin records* deletes from the target table any data with origin *MAP**, *INPUT_DEFORM*, *INPUT_WEB*, *QDL* and all other origins defined in the **Other Origins** field;
  - *Delete manual and calculation origin records*: deletes from the target table any data with origin *MAP**, *INPUT_DEFORM*, *INPUT_WEB*, *QDL*, *CONS_CALCULATED* and *PROC_CALCULATED*;
  - *Delete origin records from current Routine*: this is a default value and deletes from the target table those data whose origin is calculated by the current Routine (MAP_' ETL Domain code'_' Routine code').
- a **Suffix** aimed at differentiating the origins generated by the current Routine for each Loading task, when inserting and deleting data. If this filter is specified, the generated origin will be MAP_' ETL domain code'_'Routine code'_'Suffix';
- **Other origins** by which it is possible to specify a set of other origins to delete if the **Filter on origin** is set to *Delete manual and other origin records*.


Attributes
Filter parameters
Filters
ETL data filter
ETL filter (SQL Model)
ETL filter (specific)
Target data filter
Target filter (SQL Model)
Target filter (specific)
Origin
Filter on origin *
Delete manual and other origin records
Suffix
Other Origins

Jobs

A Job allows you to run several **ETL Routines** in sequence and following a pre-set sorting, all in one single data processing (also of different **ETL Domains**).

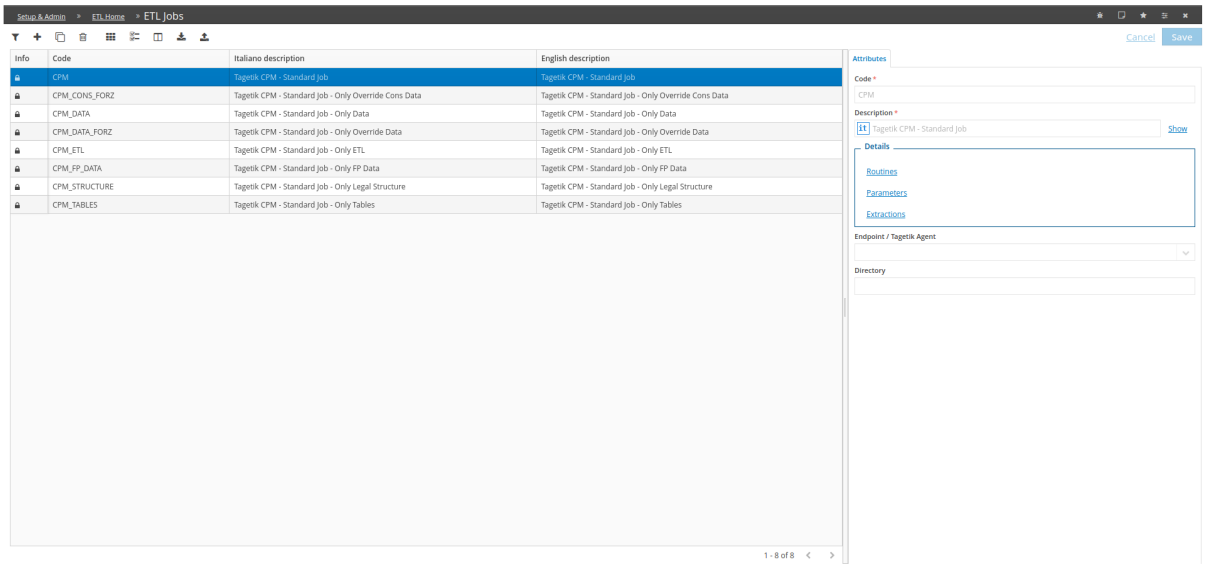
By default, CCH Tagetik provides the **Standard ETL Jobs** that are used to upload "structural" data (dimensions elements, hierarchies and groupings) and "values" (amounts, IC, journals and FX rates) through **Standard ETL Routines**.

It is possible to create new and completely configurable **Custom ETL Jobs** in order to meet specific business needs.

Definition

The Job definition can be accessed from the ETL tile on the admin user's home page.

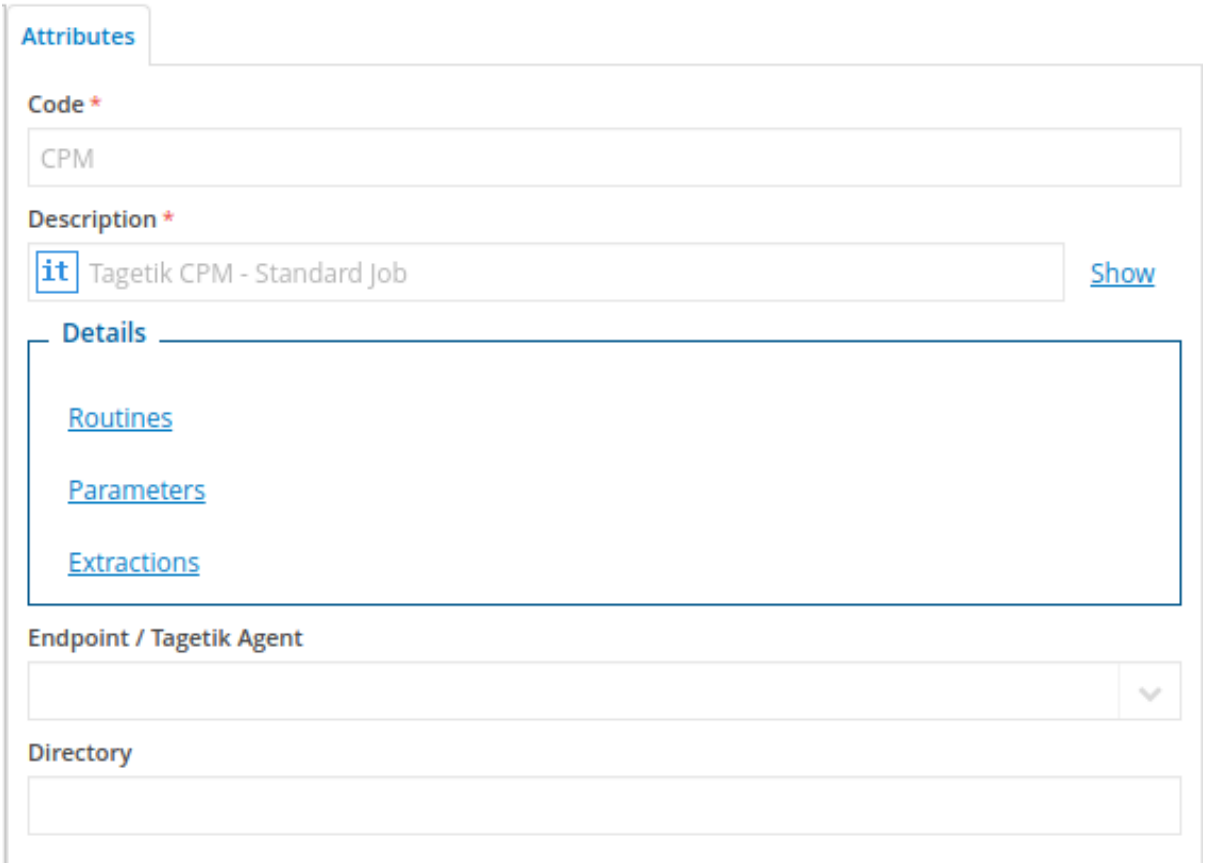
The definition of the standard jobs is locked. For these records, the icon is displayed in the first column of the grid.



Each Job is identified by a **Code** and **Descriptions**.

Attributes

Besides the **Code** and the **Descriptions**, further information can be defined in the Attributes details.



- **Endpoint / Tagetik Agent**: remote resource on which it is necessary to save the files created by a loading task that uses a Data target: Others. It is possible to specify either an Endpoint or a Tagetik Agent.
- **Directory**: directory on which it is necessary to save the files created by a loading task that uses a Data target: Others. If also an **Endpoint / Tagetik Agent** has been defined, the specified directory is considered as a sub-directory of the selected remote resource; on the other hand, if the user defines only the **Directory** field, it will be the absolute or relative path visible from the Application server.

  In order to meet the CCH Tagetik security requirements, this directory must coincide with or be a sub-folder of the *writable folder* (set up in the "it.-grupposervizi.easy.impexp.writableFolder" system property) or, if this property has not been defined, with the one defined in *java tmp path*. If the user, aware of the software security risks, wants to use a different directory, he will have to enable the **Allow ETL to read or write to directories other than the one that is configured as secure** option available in the *Advanced* section of the *Data model* general configuration.

Job details definition

Moreover, the *Definition* box allows you to manage:

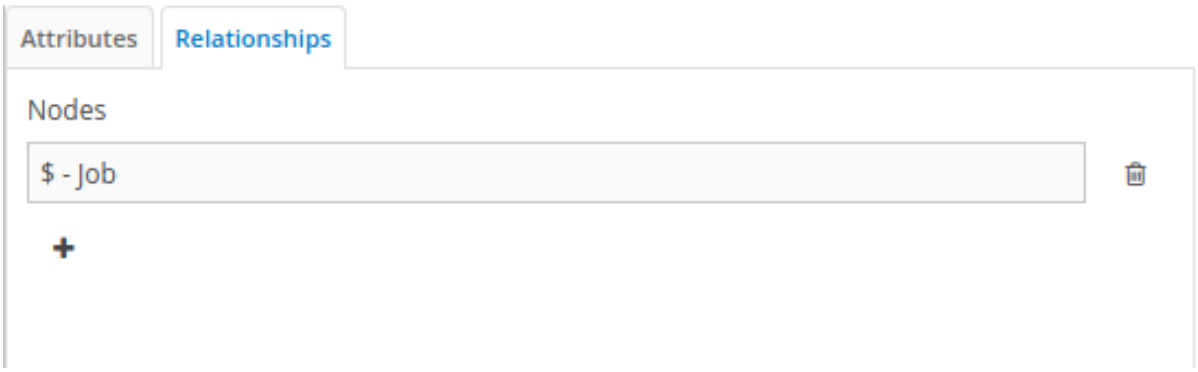
- Routines
- Parameters
- Extractions

related to the selected Job. These links are enabled only if one single Job is selected.

Relationships

In this tab, it is possible to display and modify the nodes of the grouping to which the selected Job is related.

Each new Job will be automatically associated with the grouping's nodes marked as default.



Run

The Job can be run:

- from the dedicated window; (for further details about the run modes and entry points, see Run ETL).
- by Scheduling;
- when saving a data entry form;
- from the corresponding task within an Application Job;
- via Web Service.

Utility

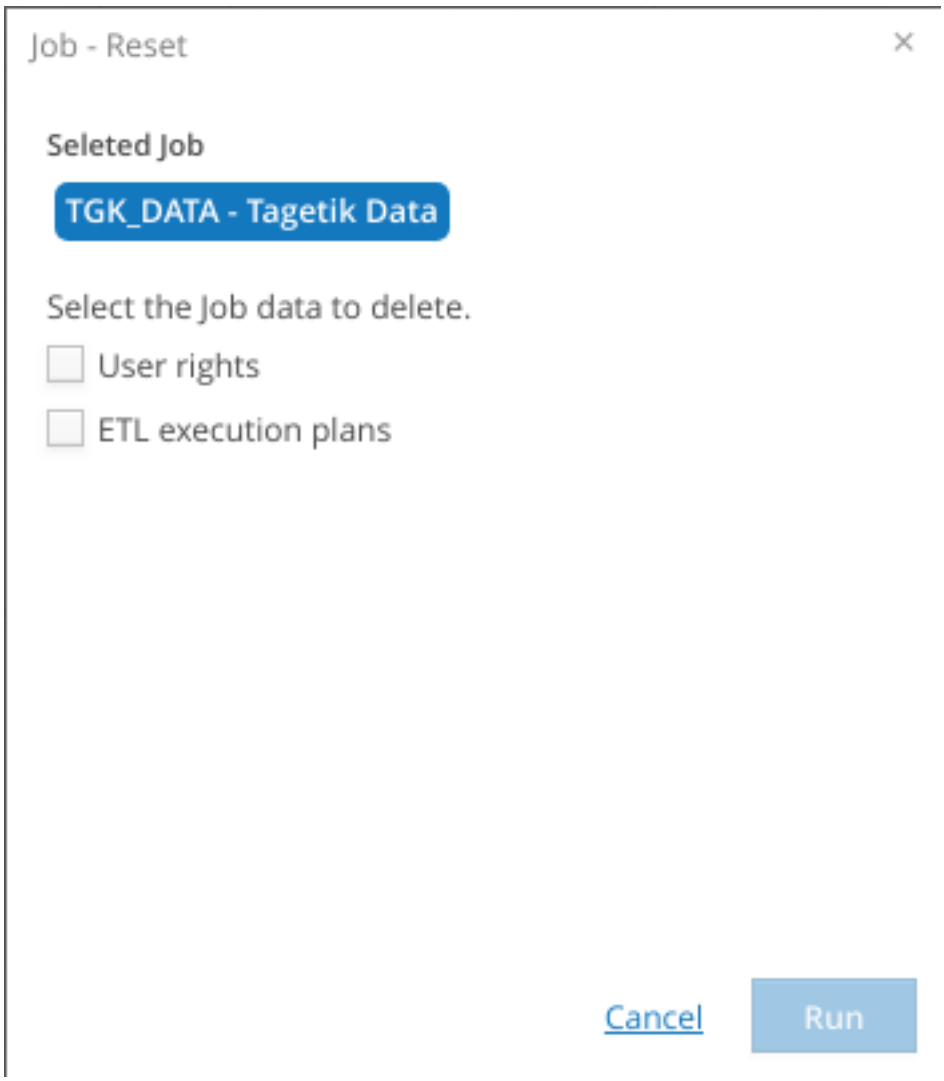
The Actions menu in the window includes some utilities for the copy and deletion of the ETL, as well as for the configuration of the SAP©, AX and NAV connectors. These utilities can be used only if one single record is selected.

Reset

The *Reset* menu allows you to reset all uses of the Job:

- in the users rights,
- in the ETL execution plans of all processes

In order to run the operation, it is necessary to select at least one element to delete.

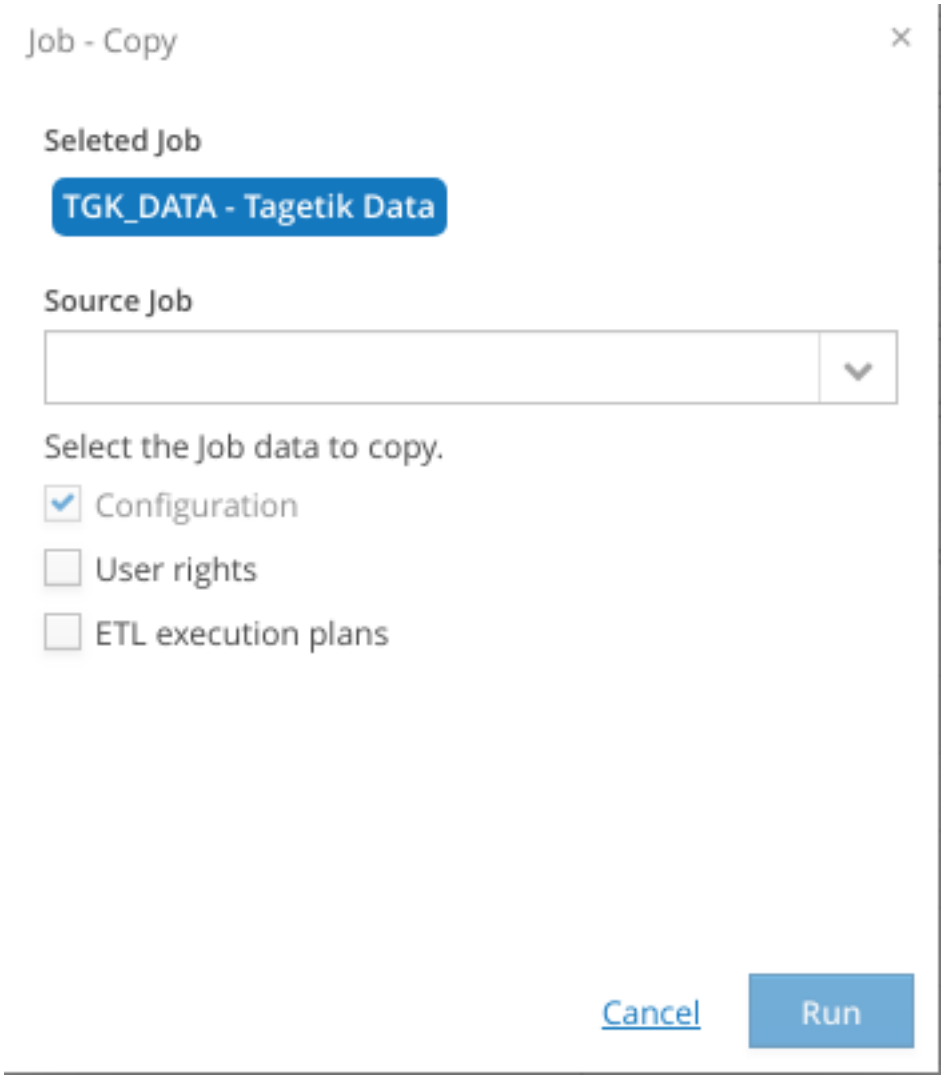


Copy from

The *Copy from* menu enables the copying (by overwriting) on the selected Job of the entire default configuration and of the following uses of the Job specified in the copy window:

- user rights,
- ETL execution plans of all processes

In order to run this operation, it is necessary to specify the source Job.



Connectors

From the actions menu of an ETL job, the user can automatically generate a parametrization including ETL domains and routines which allows for the upload of data into CCH Tagetik from the relevant source:

- SAP© connector wizard
- AX 2012 connector wizard

- AX 2012 R2 connector wizard
- NAV 2013 R2/2015/2016 connector wizard

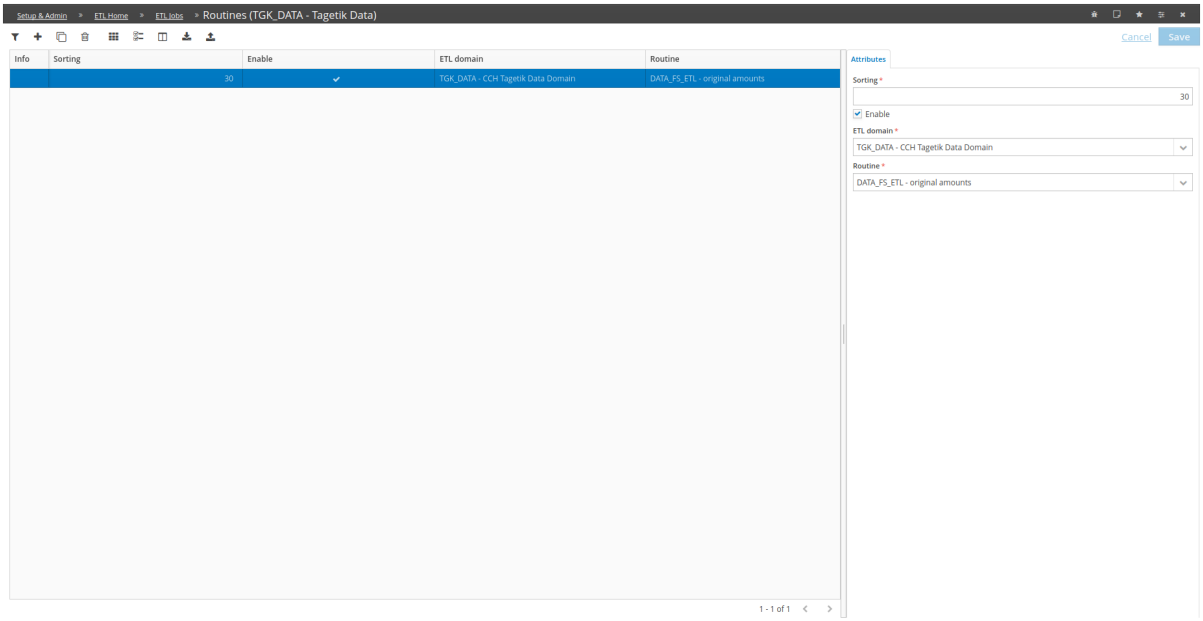
Job - Routines

For each Job it is necessary to specify which Routines must be run and in which order. Routines of different ETL Domains can be defined. Any Routine can be used more than once.

Definition

The Routines definition can be accessed from the *Routines* link in the **Jobs** window.

If the Job is locked, then the window will be in red-only mode.

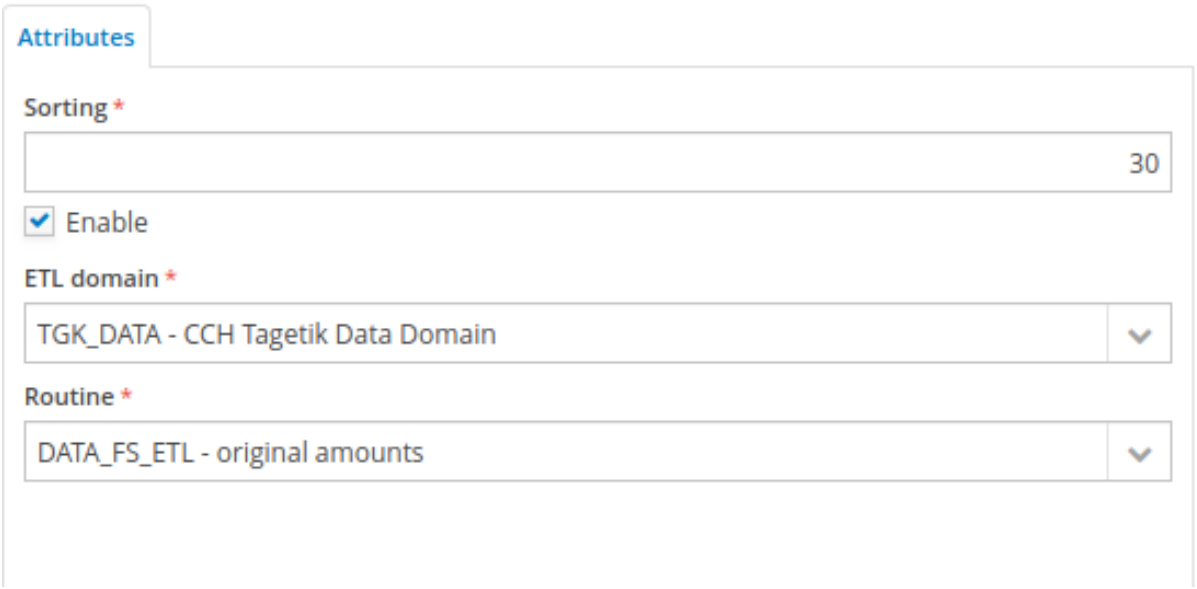


Each Routine of the Job is identified by:

- **Sorting**: the order in which the Routine will be run within the Job;
- **Enable**: specifies if the Routine must be run by default;
- **ETL Domain**: specifies the ETL Domain of the Routine;
- **Routine**: code of the Routine.

Attributes

The Attributes include only the fields described before. The values of the **Sorting** field must be univocal and not 0.



Open Job's Parameters

When saving, after any data entry, modification or deletion, the system automatically updates the Job's Parameters with all the parameters used by both the Routines defined and the extraction, transformation and loading tasks enabled.

Job - Extractions

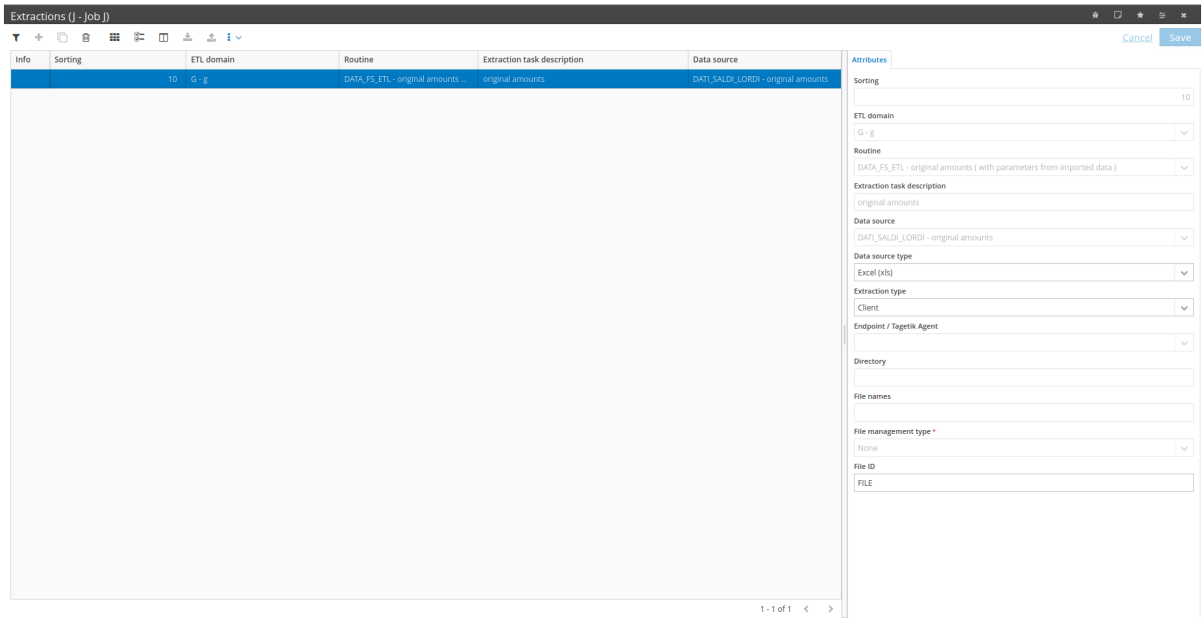
In order to run the active extraction tasks of the defined Job's Routines, it is necessary to define some information in the window of the Job's Extractions for those tasks.

New records cannot be inserted manually in the window. In order to define and edit the extraction tasks, it is necessary to run the *Update extractions* utility available in the window's actions menu. For each active Routine of the Job, the utility will insert as many records as extraction tasks enabled for that Routine. Nevertheless, you can delete records if you don't want to run any extraction task enabled in the Job's Routines that have been automatically inserted in the window by the *Update extractions* utility.

Definition

The definition of the Job's Extractions can be accessed from the *Extractions* link of the Jobs window.

If the Job is locked, then the window will be in read-only mode.

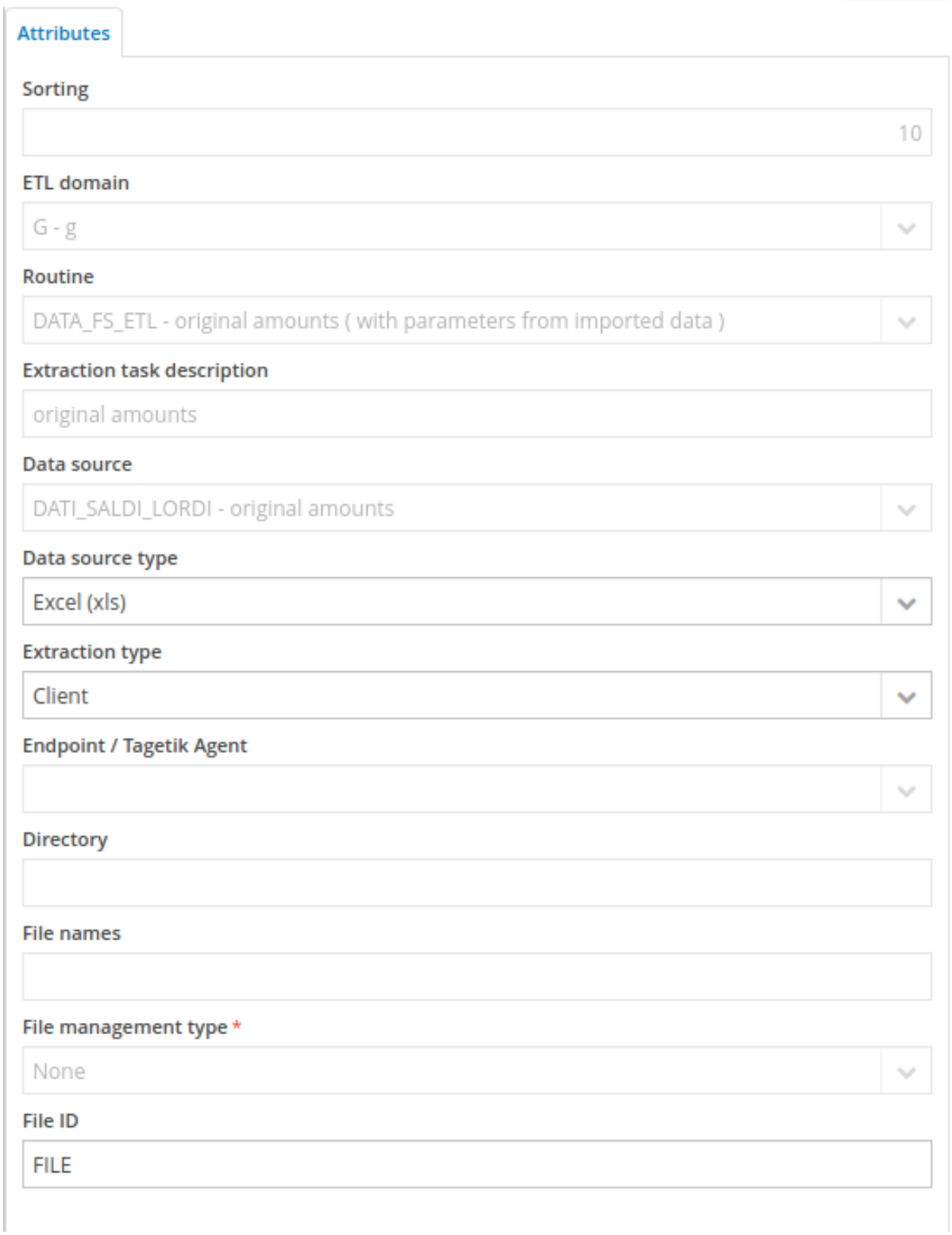


Each Job's Extraction is identified by:

- **Sorting**: order of the Routine inside the Job;
- **ETL Domain**: ETL Domain the extraction task belongs to;
- **Routine**: Routine the extraction task belongs to;
- **Extraction task description**: description of the extraction task;
- **Data source**: Data source defined for the extraction task.

Attributes

In addition to **Sorting**, **ETL Domain**, **Routine**, **Extraction task description** and **Data source**, further information can be defined in the Attributes window.



- **Data source type** : allows you to select the input type among those configured in the Data Source of the extraction task;
- **Extraction type**: allows you to select the extraction type among:
  - *Application server*: the loading is run from query or from files present on directories visible from the application server;
  - *Client*: the loading is performed by uploading files from client;

- *Endpoint / Tagetik Agent*: the loading is performed by using an Endpoint which enables also the use of the Tagetik Data Loader.

  If the external source is a database, the *Application server* and *Client* options are equivalent and tell the system that the query must be run on the database configured in the Data Source.

- **Endpoint / Tagetik Agent**: allows you to select an Endpoint that will be used if the **Extraction type** is *Endpoint / Tagetik Agent*;
- **Directory**: allows you to define a directory that will be used if the **Data source type** is from file and the **Extraction type** is not *Client*;
- **File names**: allows specifying the names of files (comma separated) that will be used if **data source type** is from file and the **Extraction type** is not from *Client*.
  - If you need to specify several files of the same type, you just have to separate the names by semicolon. Moreover, besides specifying more than one file, it is possible to use the wildcard "*" to extract those files that meet the specified filter criteria. For example, you can write:
    - [*.fbk] meaning all files ending with the extension ".fbk";
    - [nome*] meaning all files starting with "nome";
    - [nome*.fbk] meaning all files starting with "nome" and ending with ".fbk";
    - [nome{PARAM_AZIENDA}*] meaning all files starting with "nome" + entity parameter value.
  - The name of the file can be defined by using SQL strings that leverage the parameters defined in the Filter Strings Syntax or SQL Query
  - If no file name is specified, then the system extracts data of all files of the type related to the value of the **Data source type** field, present in the specified directory.
- **File management type**: allows you to select the action you want to run on the loaded file if the **data source type**is from file and the **Extraction type** is not from *Client*:
  - *None*: no action is run;
  - *Delete*: the file is removed if the upload is successfully completed;
  - *Store*: the file is moved to an "archive" folder if the upload is successfully completed.
- **File ID** : allows you to specify the ID of the file to upload if the **Data source type** is from file and the **Extraction type** is *Client*. This is used by the system to identify the extraction tasks that use the same data source in order to limit the number of uploads of files containing data to extract. If two or more tasks are labelled with the same **File ID**, the system will upload one file only, thus avoiding redundant upload of files that have already been uploaded.

When running the Job, it is possible to define or modify the attributes set up in this window for that specific run.

Job - Parameters

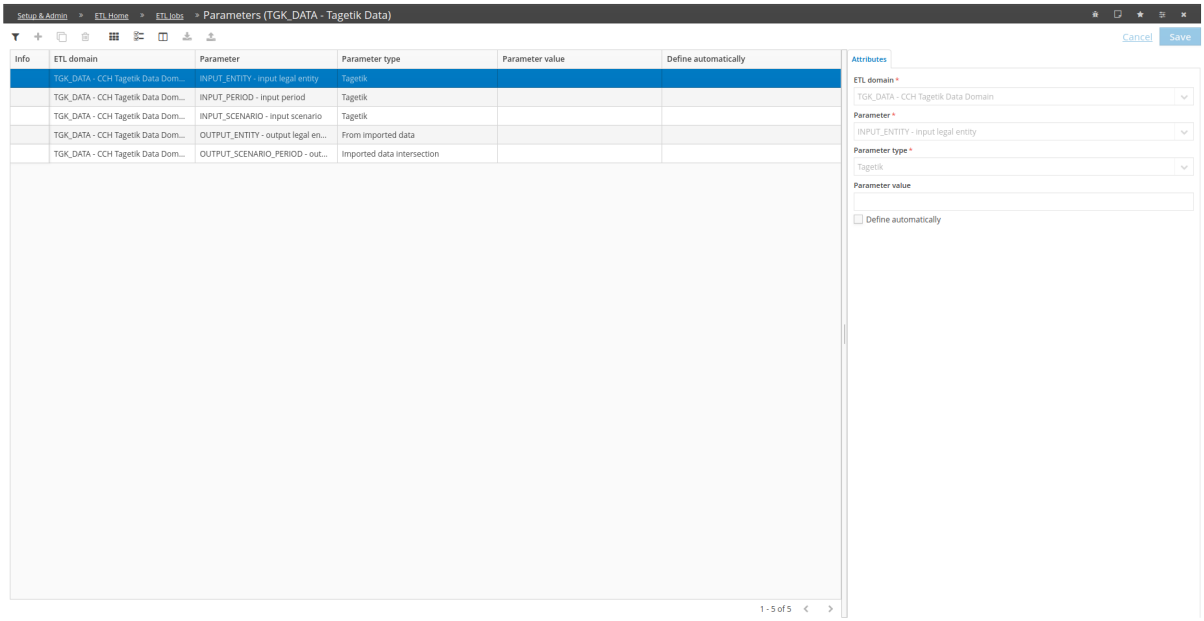
For any Parameter used by the Routines of the Job and by the extraction, transformation and loading tasks, it is possible to set up a default value or enable an option that makes the system auto-

matically set up the parameter when running the Job. If the same parameter is used more than one time, the system will consider it only once.

Definition

The definition of the Parameters of the Job can be accessed from the *Parameters* link of the **Jobs** window.

If the Job is locked, the window will be in read-only mode.

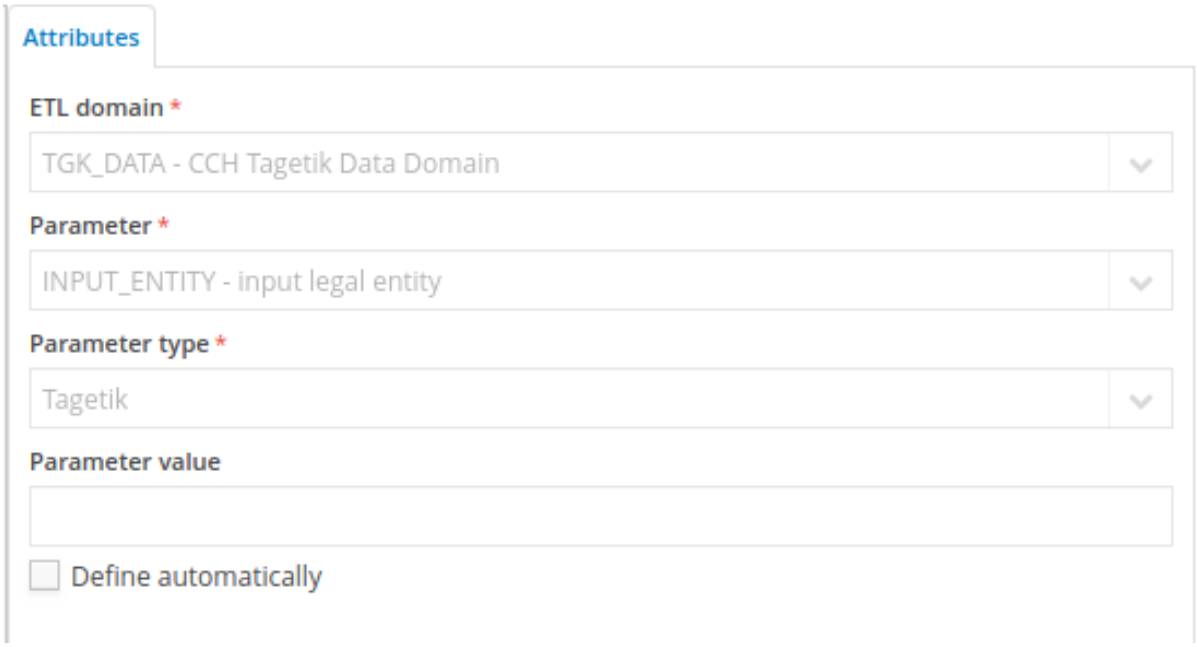


Each Parameter of the Job is identified by:

- **ETL Domain**: ETL Domain the parameter belongs to;
- **Parameter**: code of the parameter;
- **Parameter Type**: type of parameter;
- **Parameter value**: assigns a default value to the parameter;
- **Define automatically**: when enabled, this option means that the parameter will be defined automatically at runtime.

Attributes

The Attributes include only the fields described before. It is not possible to perform any insertion or deletion in the window, but it is only possible to edit the **Parameter value** and **Define automatically** fields according to the parameter type specified in the **Parameter type** field.



The user can:

- populate the **Value** field or enable the **Define Automatically** option for the parameters of *From Tagetik* and *From dimension dictionary* type;
- populate the **Parameter value** for the *Free* parameters;
- enable the **Define automatically** option for the *From imported data* or *Imported data intersection*type parameters.

The **Define automatically** option specifies that the parameter must be define automatically with all possible values in that context.

The *From imported data* or *Imported data intersection* type parameters are set to zero at the end of each Routine and then recalculated for the next Routine since they depend on the *staging* table's values that are generated by the Routine they belong to.

Open Job Parameters

When opening the window, the system automatically updates the Parameters of the Job with all the parameters used by both the Routines defined in the Job and all the extraction, transformation and loading tasks enabled.

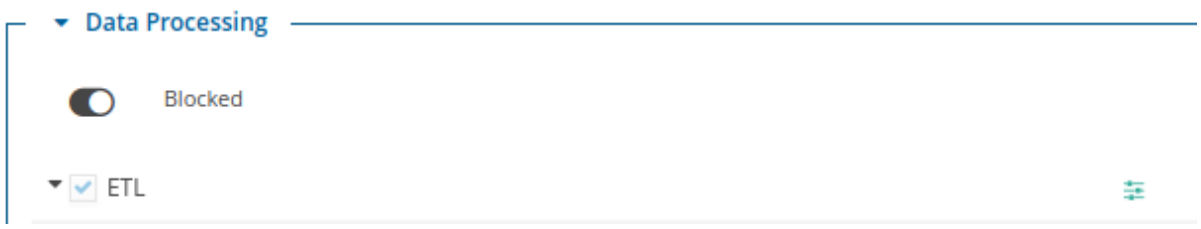
ETL Execution Plan

For each Entity of a given Process it is possible to specify which Jobs and/or ETL Routines can be run.

The restrictions defined in the Execution Plan are applied in the ETL Execution window only if it is opened from the *Process cockpit* and *Workflow* entry points described in the related manual.

Definition

The ETL execution plan definition window can be accessed by the dedicated button displayed in the process definition window.



Each definition record is identified by:

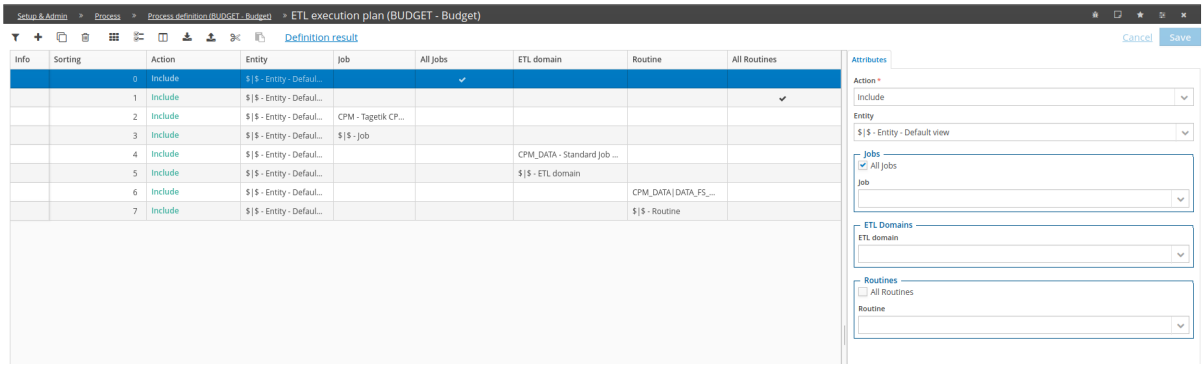
- **Sorting**: order in which the deploy data processing must run the defined records in order to correctly generate the execution plan;
- **Action**: specifies if the Jobs or the Routines defined in the record must be *Included* or *Excluded* from the execution plan;
- **Entity**: it is possible to select either one single Entity or a node of the Entity hierarchy;
- **Job**: it is possible to select either one single Job or a node of the ETL Jobs grouping;
- **All Jobs**: it is possible to select automatically all ETL Jobs of the table;
- **ETL Domain**: it is possible to select either one single ETL Domain or a node of the ETL Domains grouping;
- **Routine**: it is possible to select either one single ETL Routine or a node of the ETL Routines grouping;
- **All Routines**: it is possible to select automatically all the ETL Routines of the table.

The **Sorting**, **Action** and **Entity**fields are mandatory.

Then it is necessary to define one field between **Job** and **All Jobs** or one field among **ETL Domain**, **Routine** or **All Routines**.

The fields in which it is possible to specify one single entity or a node of the related hierarchy / grouping allows prompting the deploy data processing to consider the single entity or all entities of the selected node for the defined record.

The ETL Execution Plan generates restrictions only on Jobs and / or ETL Routines. Therefore, if an **ETL Domain** or a node of the ETL Domains grouping is selected, it means that all **Routines** of that specific ETL Domain or of the ETL Domains of the node must be considered.



Deploy

On completion of each saving, the system runs the *deploy* data processing that generates the result record automatically on the basis of the records inserted in the definition. The records will be displayed in the *Definition results* window.

Definition result

The Definition results window of the ETL execution plan can be accessed by the *Definition result* link in the button bar of the definition window.

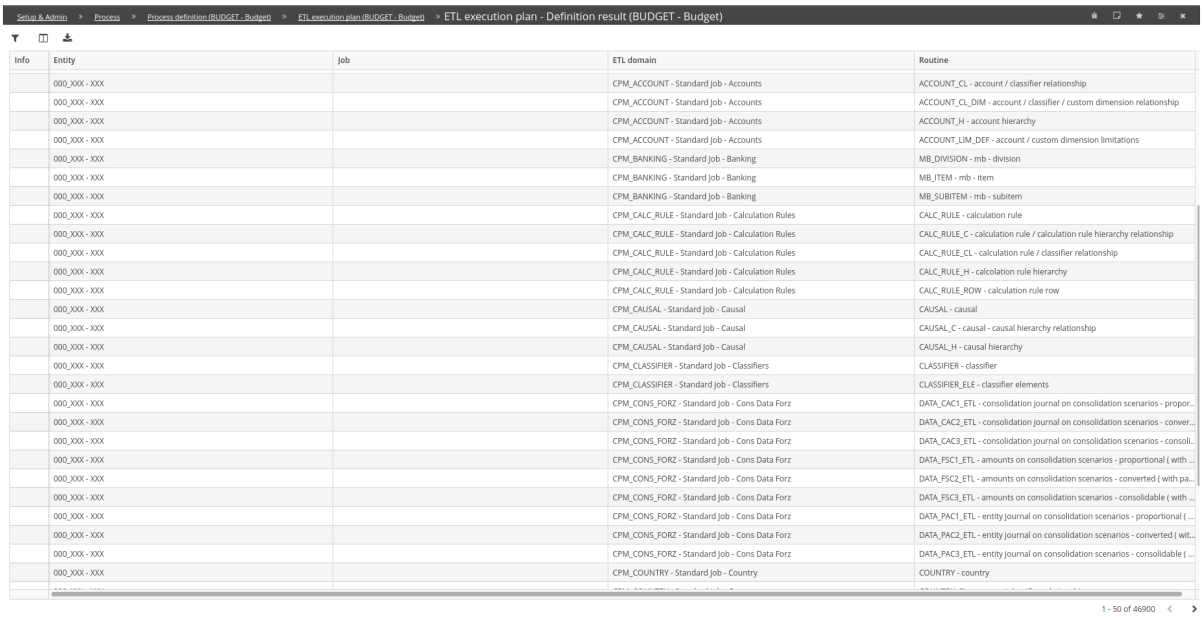


Each result record is identified by:

- **Entity**: Entity of the Process affected by the restriction concerning the execution of a Job or an ETL Routine;
- **Job**: ETL Job that can be run for the Entity;
- **ETL Domain** / **Routine**: ETL Routine that can be run for the Entity.

For each record of the result (Entity of the Process) a restriction is specified on a Job or on a Routine/ ETL Domain since one type excludes the other.

The ETL Routine restrictions apply only to the execution of the single Routine and not to the Routines of an ETL Job. If according to the Execution Plan it is possible to run an ETL Job, then it will be possible to run all its Routines even though they do not exist in the execution plan.


Setup & Admin » Process » Process definition (BUDGET - Budget) » ETL execution plan (BUDGET - Budget) » ETL execution plan - Definition result (BUDGET - Budget)
Info
Entity
Job
ETL domain
Routine
000_XXX - XXX
CPM_ACCOUNT - Standard Job - Accounts
ACCOUNT_CL - account / classifier relationship
ACCOUNT_CL_DIM - account / classifier / custom dimension relationship
ACCOUNT_H - account hierarchy
ACCOUNT_LIM_DEF - account / custom dimension limitations
CPM_BANKING - Standard Job - Banking
MB_DIVISION - mb - division
MB_ITEM - mb - item
MB_SUBITEM - mb - subitem
CPM_CALC_RULE - Standard Job - Calculation Rules
CALC_RULE - calculation rule
CALC_RULE_C - calculation rule / calculation rule hierarchy relationship
CALC_RULE_CL - calculation rule / classifier relationship
CALC_RULE_H - calcolation rule hierarchy
CALC_RULE_ROW - calculation rule row
CPM_CAUSAL - Standard Job - Causal
CAUSAL - causal
CAUSAL_C - causal - causal hierarchy relationship
CAUSAL_H - causal hierarchy
CPM_CLASSIFIER - Standard Job - Classifiers
CLASSIFIER - classifier
CLASSIFIER_ELE - classifier elements
CPM_CONS_FORZ - Standard Job - Cons Data Forz
DATA_CAC1_ETL - consolidation journal on consolidation scenarios - propor...
DATA_CAC2_ETL - consolidation journal on consolidation scenarios - conver...
DATA_CAC3_ETL - consolidation journal on consolidation scenarios - consoli...
DATA_FSC1_ETL - amounts on consolidation scenarios - proportional (with ...
DATA_FSC2_ETL - amounts on consolidation scenarios - converted (with pa...
DATA_FSC3_ETL - amounts on consolidation scenarios - consolidable (with ...
DATA_PAC1_ETL - entity journal on consolidation scenarios - proportional (...
DATA_PAC2_ETL - entity journal on consolidation scenarios - converted (wit...
DATA_PAC3_ETL - entity journal on consolidation scenarios - consolidable (...
CPM_COUNTRY - Standard Job - Country
COUNTRY - country
1 - 50 of 46900

Run ETL from HTML5 interface

Tagetik allows the user to run an ETL data processing from HTML5 interface.

Entry Points

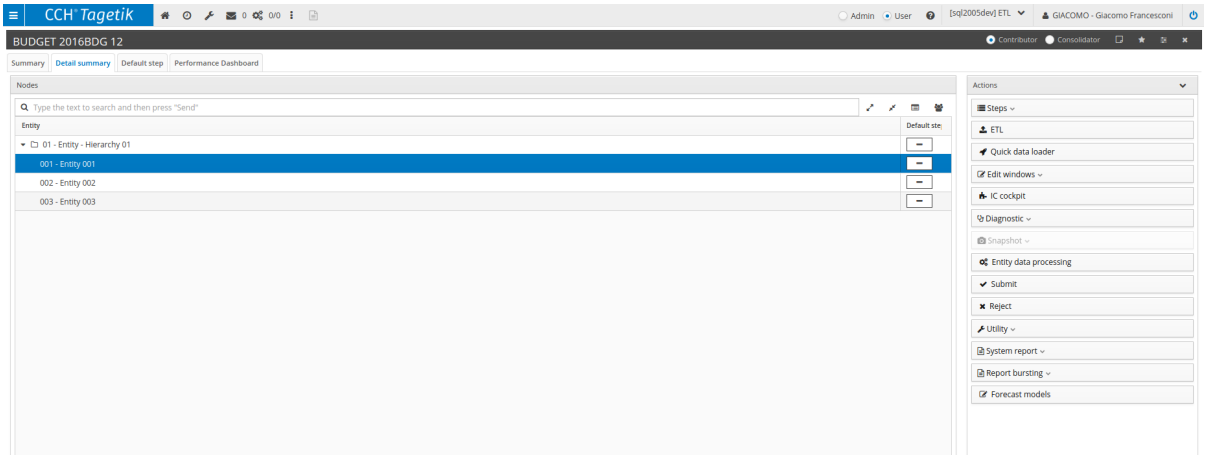
The ETL can be accessed and run from several entry points in the Home of the End User.

Process cockpit

By the cockpit - both Contributor and Consolidator - it is possible to access the ETL run window form the Detail summary and from each step by clicking the button



available in the **Action bar** on the right hand side. In both cases, it is necessary to select a node of the hierarchy or the single lowest level element. In the first case, the ETL will be run on all the entities of the node.



Workflow

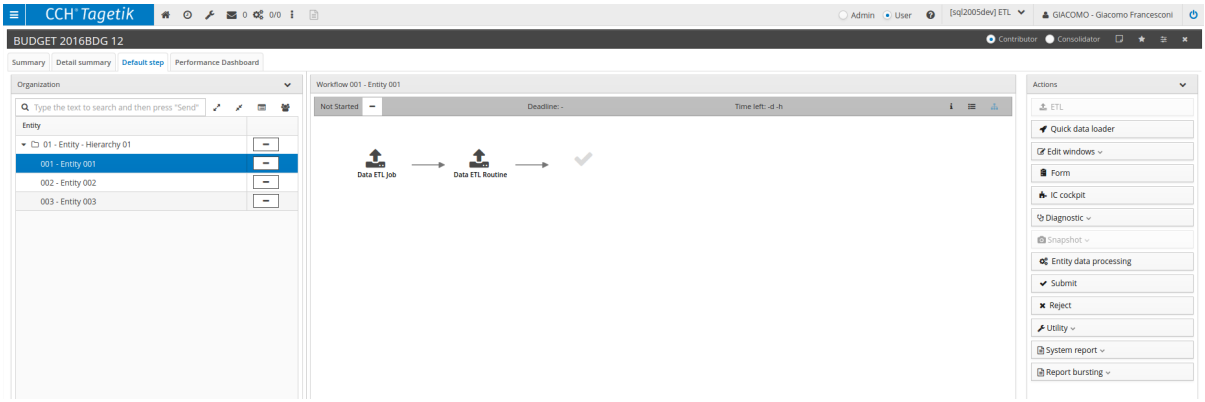
In the of each step, if the node or the entity are associated with a task flowchart containing a one of the following task types:

- ETL Job for values
- ETL Job for tables
- ETL Routine for values
- ETL Routine for tables

The button displayed in the **Action Bar** that enables the execution of the ETL is disabled.

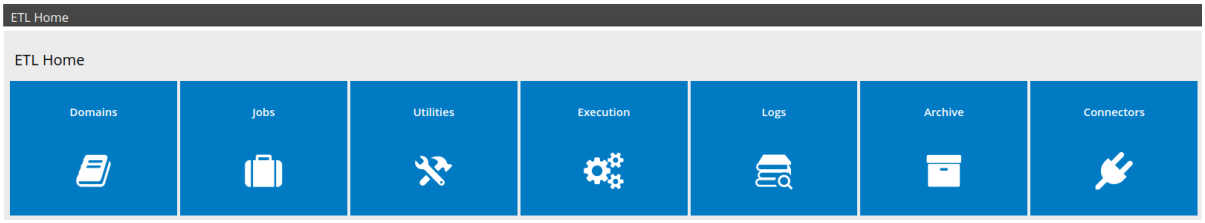
This happens because if one of these tasks exist, the ETL can be run only in the defined order.

In order to run the ETL from a task flowchart, it is necessary to click on the task and then select the **Run** menu.



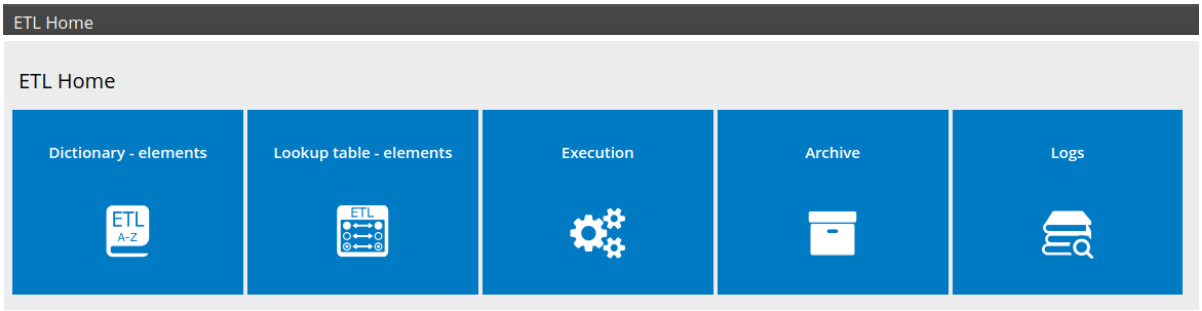
Tile ETL - Admin user

From the ETL tile in the Admin user home page, it is possible to access the window by clicking on the **Execution** tile.



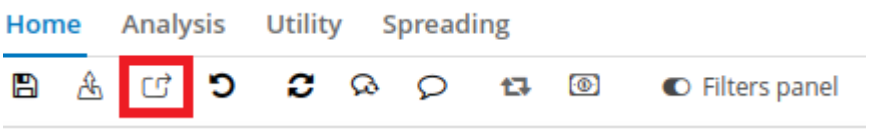
Tile ETL - End user

From the ETL tile in the End user home page, it is possible to access the window by clicking on the **Execution** tile.



Web form

From the Web Form button on the Home tab of the Web Form Data Entry Reporting .



The window that pops up from each of these entry points will be editable. The user can:

- select the ETL Job or a specific Routine to be run (except when accessing from the Workflow task)
- when an ETL Job is selected, it is possible to enable or disable one or more of its Routines
- enable or disable either the steps of a Routine or its extraction, transformation and loading tasks

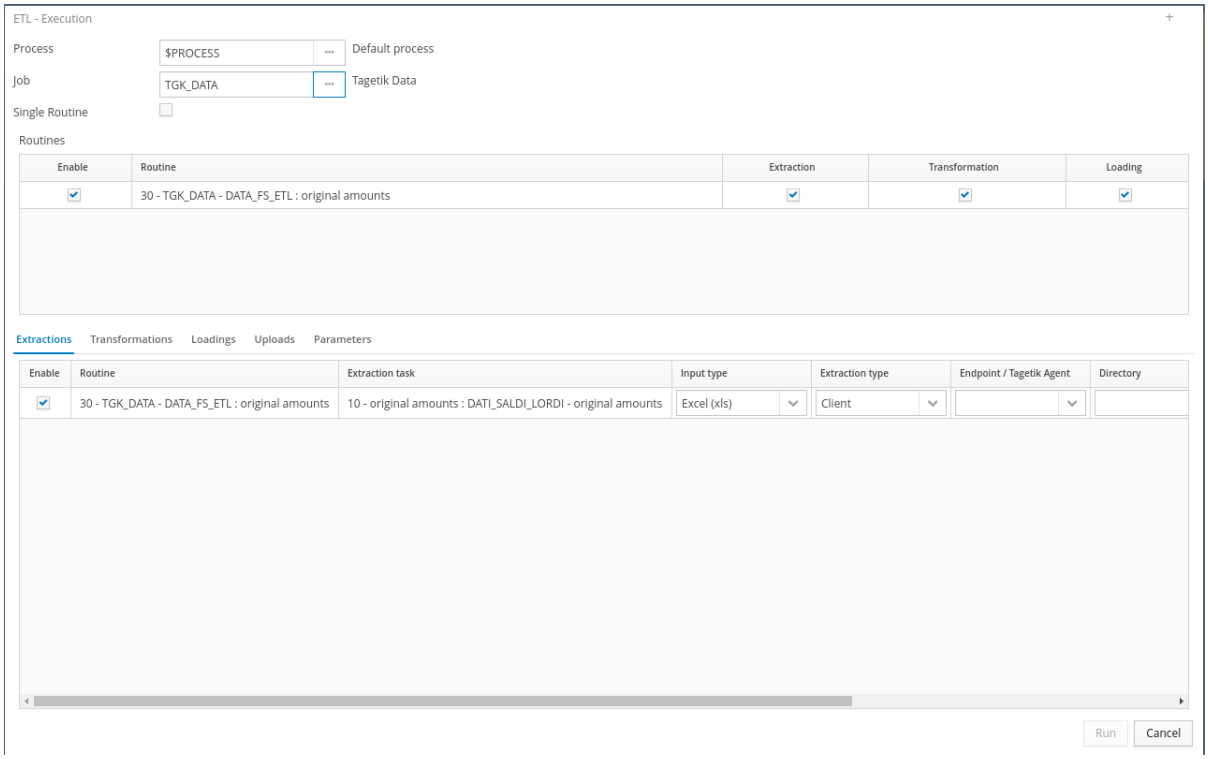
In order to prevent the window from being edited, it is necessary to enable the **0004108** function (that belongs to the ETL_ELAB_USER module but is disabled by default) in the role. In this way, the window's edit will be limited according to the specific entry point being used:

- Home Admin: always editable
- Home End user: always editable
- Process cockpit: the Job or single Routine selection is editable, whereas the steps selection and the selection / edit of the extraction, transformation, loading tasks is not
- Workflow: the Job or single Routine selection is not editable (values are pre-set from the task), the steps selection and the selection / edit of the extraction, transformation, loading tasks is not editable
- Web form: the Job or single Routine selection is editable, whereas the steps selection and the selection / edit of the extraction, transformation, loading tasks is not

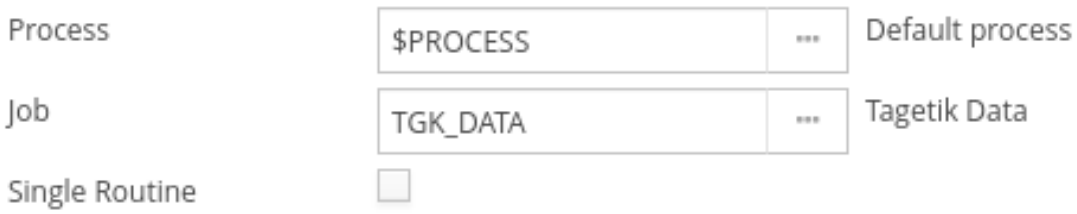
ETL run window

The window is made up of several panels, each of which manages a given part of the parametrization of the ETL Routine to be run. Such panels show the definition configured in the parametrization that can be modified for the current execution.

Any missing information preventing the execution of the ETL, is reported to the user by the red colour of the caption.



Job / Routine selection and process



After selecting - where necessary - the **Process**, it is possible to specify if a **Job** or a **Single Routine** has to be run.

Only the **Jobs** available for the current user can be selected.

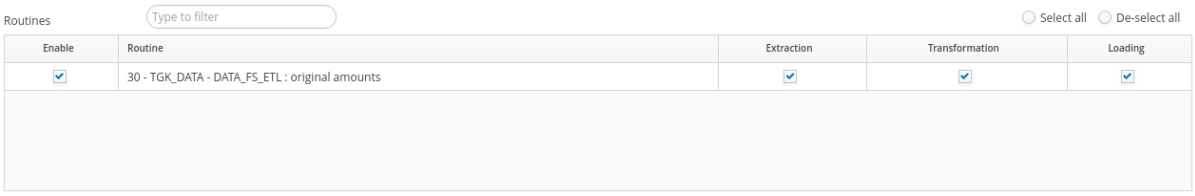
If it is run from the process cockpit, then the **Process** will be set automatically with that of the cockpit and it will not be editable; the selectable **Jobs** are those available in the execution plan of that process for the selected entities.

If it is run from a task flowchart:

- the **Process** will not be editable and will be set equal to that of the cockpit
- the **Job** selection lookup and the **Single Routine** option will not be editable
- if the tasks is related ti an ETL Job, the **Job** will be set automatically equal to the one set in the task definition.
- if the task is related to an ETL Routine, the **Single Routine** option will be enabled automatically.

After selecting the **Job** or enabling the **Single Routine** option, the system will update the content of the window.

Routines



This panel is filled out:

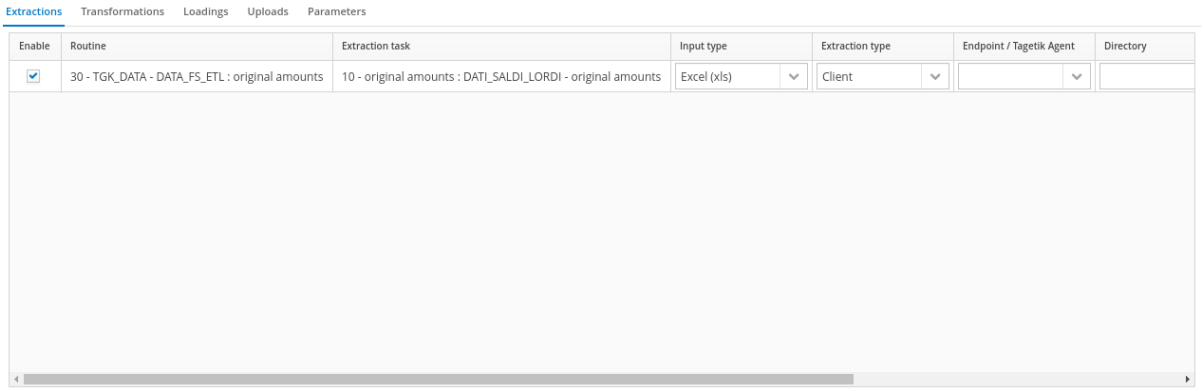
- with all the routines of the Job not restricted for the user nor for the Process execution plan, if a **Job** has been selected.
- with all the routines of all ETL Domains available to the current user, if the **Single Routine** option has been enabled and:
    - if the window has been opened from the process cockpit, it is filled out only with the routines of the execution plan of that specific process for the selected entities.
    - if the window has been opened from an ETL Routine task flowchart, it is filled out only with the routine set in the task definition.

The table is made up of the following columns:

- **Enable**: specifies if the Routine must be run. If the **Single Routine** option is enabled, only one Routine can be enabled. If the window has been run from an ETL Routine task flowchart, the option enabled by default on the Routine cannot be disabled.
- **Routine**: shows the order of the Routine in the Job (if a Job has been selected), the code of the ETL Domain, the code and description of the Routine.
- **Extraction**: specifies if the extraction must be run for the Routine. The option cannot be disabled if the storage of extracted data is not enabled in the definition and, therefore, it is not possible to run the transformation step
- **Transformation**: specifies if the transformation must be run for the Routine. The option cannot be disabled if the storage of transformed data is not enabled in the definition and, therefore, it is not possible to run the loading step.
- **Loading**: specifies if the loading step must be run for the Routine.

The activation / deactivation of the option causes the update of the underlying panels.

Extraction



This panel is filled out with all the extractions of all the Routines enabled in the **Routines** panel. If a **Job** has been selected, only the routines defined in the extractions of the selected Job will be displayed.

The table is made up of the following columns:

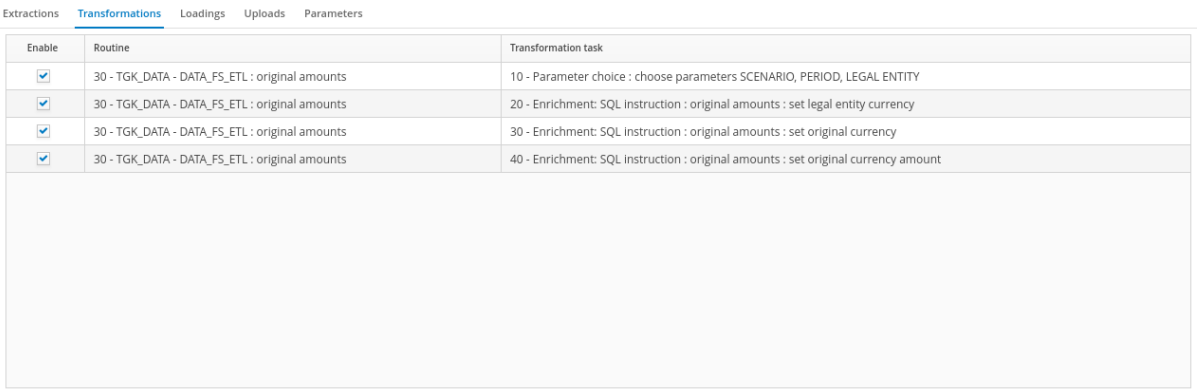
- **Enable**: specifies if the extraction task must be run.
- **Routine**: routine it belongs to.
- **Extraction task**: shows the sorting and the description of the task, the code and the description of the Data Source.
- **Input type**: allows you to select the input type among those configured in the Data Source of the task. The default value is defined in the Job extraction.
- **Extraction type**: allows you to select the extraction type:
  - *Application Server*: extraction run from query or files available on the directories visible from the application server.
  - *Client*: the loading is done by uploading files from client.
  - *Endpoint / Tagetik Agent*: the loading is run by using an **Endpoint** that enables also the use of **Tagetik Data Loader**.
- **Endpoint / Tagetik Agent**: allows you to select an Endpoint that will be used if **Extraction type** is *Endpoint / Tagetik Agent*.
- **Directory**: allows you to define a directory that will be used if the **Input type** is from file and the **Extraction type** is not from *Client*.
- **File names**: allows you to specify a list of files names (comma separated) that will be used if the **Input type** is from file and the **Extraction type** is not from *Client*
- **File ID**: allows you to specify the ID of the file to upload if the **Input type** is from file and the **Extraction type** is *Client*. It is used by the system to identify the extraction tasks that use the same data source in order to limit the uploading of files containing data to extract. If two or more tasks are labeled with the same **File ID**, the system will ask for uploading one single file in order to avoid uploading files that already exist.
- **File management type**: allows you to select the action to run on the uploaded file if the **Input type** is from file and the **Extraction type** is not from *Client* :
  - *None*: no operation is run.
  - *Delete*: the file is deleted if the upload is successfully completed.
  - *Store*: the file is moved to an *"archive"* folder if the upload is successfully completed.

The activation / deactivation of the tasks of the modification of the attributes cause the update of the **Upload** and **Parameters** panels.

 If the configuration of an extraction task is incomplete or if the same **File ID** has been defined for different **Data source types**, then the "*Extraction*" label of the panel turns red and the ETL cannot be run.

 If extraction tasks have been defined for the Routines but none of them is enabled, then the "*Extraction*" label of the panel turns yellow but the ETL can still be run.

Transformation



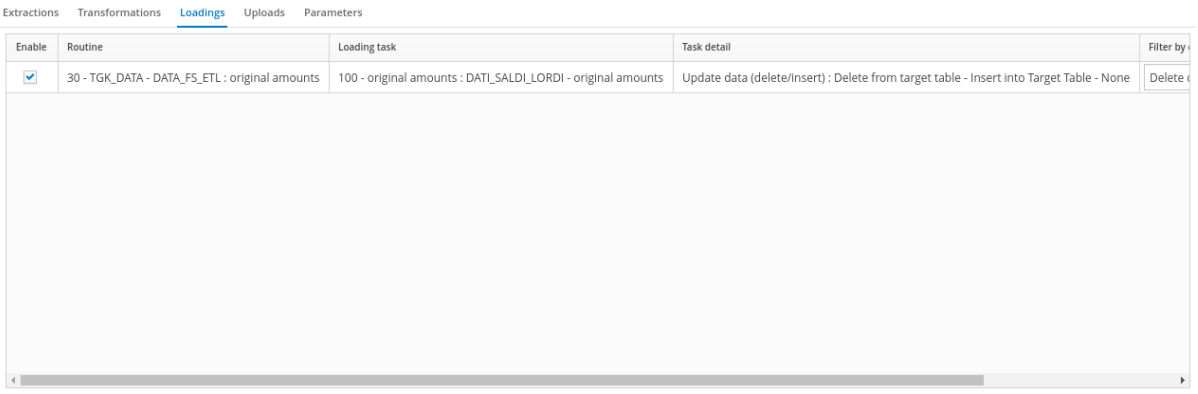
This panel is filled out with the information of all the Routines enabled in the **Routines** panel.

The table is made up of the following columns:

- **Enable**: specifies if the transformation task must be run.
- **Routine**: Routine it belongs to.
- **Transformation task**: task's sorting, type and description.

If transformation tasks have been defined for the Routines but none of them is enabled, then the "*Transformation*" label of the panel turns yellow but the ETL can still be run.

Loading



This panel is filled out with the uploads of all the Routines enabled in the **Routines** panel.

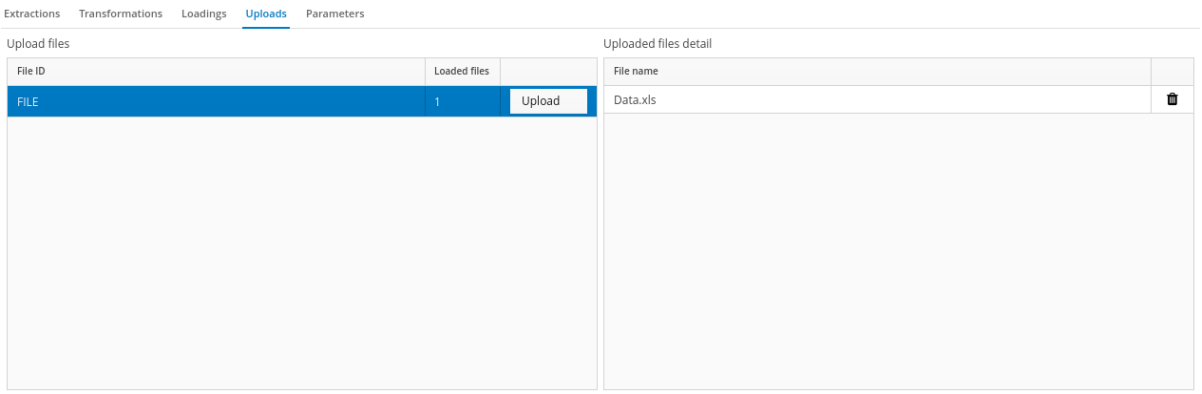
The table is made up of the following columns:

- **Enable**: specifies if the loading task must be run.
- **Routine**: Routine it belongs to.
- **Loading task**: sorting and description of the task, code and description of the Data Target.

- **Task detail**: description of the loading macro type and the description of the detail task.
- **Filter by origin**: type of filter to apply to the origin when running the deletion.

If loading tasks have been defined for the Routines but none of them is enabled, then the "*Loading*" label of the panel turns yellow but the ETL can still be run.

Uploads



This panel is made up of:

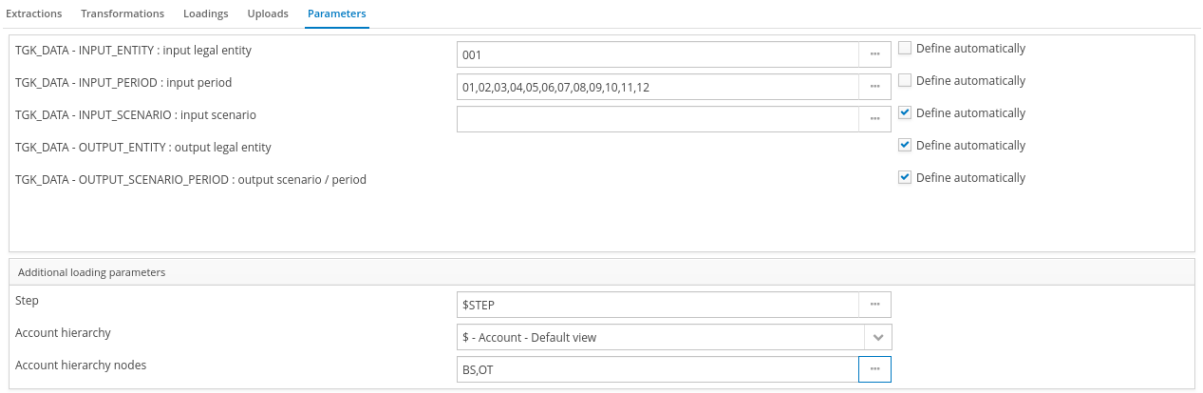
- **Uploaded files**: shows all the uploads to run, identified by the **File type** and the **File ID** if the **Extraction** panel contains tasks with **Input type** from file and **Extraction type** from *Client*.The tables contains the following columns:

- **File type**: shows the value of the **Data source type** field defined in the **Extraction** panel.

- **File ID**: value of the field of the same name defined in the **Extraction** panel.
- **Loaded files**: shows the number of files currently uploaded for that specific **File ID**.
- **Upload** button: allows uploading the files from client.
- **Uploaded files detail**: for each record selected in the **Uploaded files** panel, it shows the names of the uploaded files. The button next to each name allows cancelling the file upload.

If the loaded file has a different **Mime type** from that expected for the **File type**, the system returns an "*Invalid file type*" error and the upload fails.

Parameters



This panel is filled with all the parameters used in all active tasks in the **Extraction**, **Transformation** and **Loading** panels of all the active Routines in the **Routines** panel.

The following details are available for each parameter:

- a description including the code of the ETL Domain the code of the parameter and its description.
- if the parameter type is *Tagetik* or *From dictionary dimension*, a lookup to select the value and a check box to enable the automatic definition are available.
- if the parameter is *Free*, only a text input to define the value manually is available.
- if the parameter is *From imported data*, only a check box to enable the automatic definition option is available.

The **Additional loading parameters** panel shows, if defined for at least one Routine enabled in the **Routines** panel, the following parameters:

- **Step**: filter on the steps of the selected **Process** in order to load only data relating to the accounts available in these steps. This option is available only if the **Step** option has been enabled for at least one Routine;
- **Account hierarchy** and **Account hierarchy nodes**: filter on the nodes of the account hierarchy in order to load only data relating to the accounts associated with the specified nodes. These options are available only if the **Account hierarchy** option has been enabled for at least one Routine. If the account hierarchy has already been specified when defining the Routine, you just have to specify the nodes to use.

If it is run from the process cockpit or from a task flowchart, the ETL parameters associated with dimensions with **Identity** Scenario, Period and Entity and the additional parameter relating to the steps, will not be editable and will be automatically filled with the input parameters inherited from the process cockpit.

Run

To run the ETL Job, it is necessary to click on the **Run** button

This button is enabled only if:

- At least one Routine has been enabled in the **Routines** panel
- At least one among *Extraction*, *Transformation* and *Loading* has been selected for the enabled Routines

- If at least one Routine requires the process being selected, the **Process** filter must be defined.
- If active tasks are available in the **Extraction** panel, thy must be configured in a consistent manner (*see description of* **Extraction** panel's fields).
- If some extractions require loading the files via upload, this must be performed in the **Uploads** panel (at least one for each **File ID**).
- If parameters exist in the **Parameter** or **Additional loading parameters** panel, these must be set up.

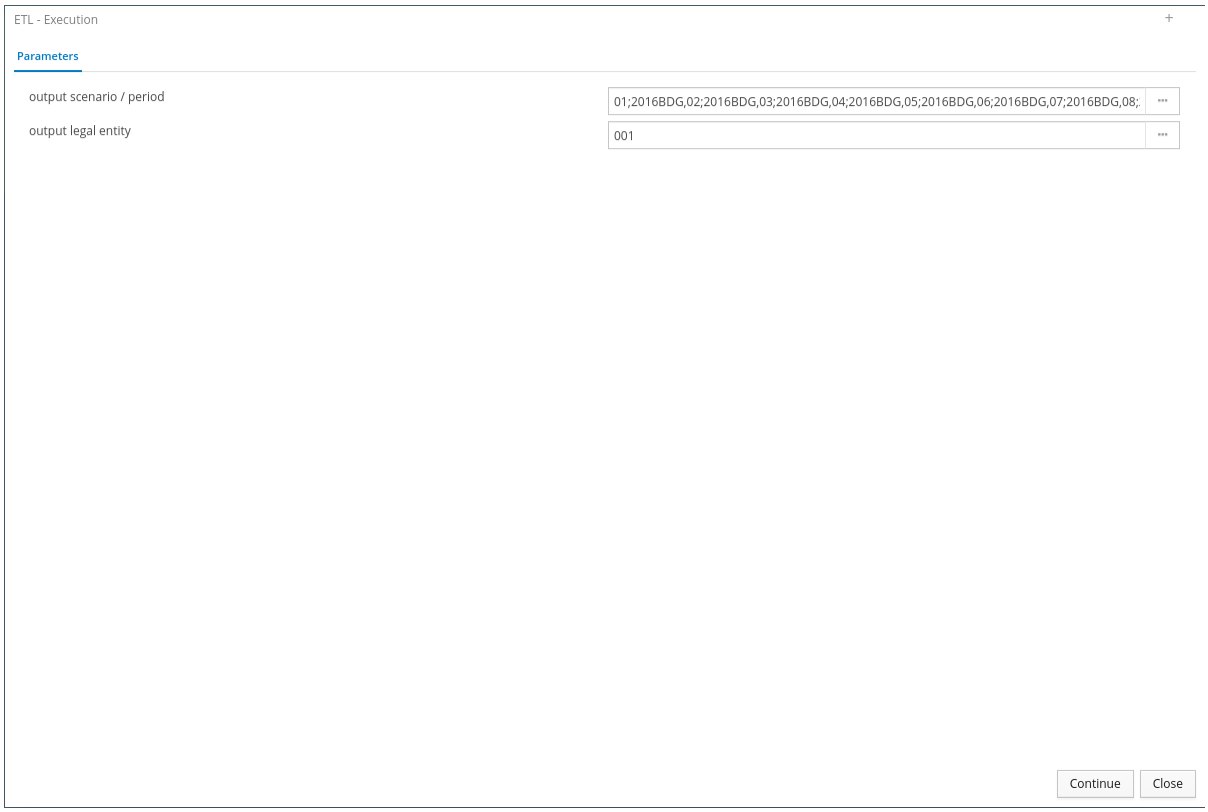
If a **Job** has been selected, the ETL data processing related to the Job is run (ETL_ESECUZIONE_JOB).

If the **Single Routine** option has been enabled, the ETL data processing related to the Routine is run (ETL_ESECUZIONE_ROUTINE).

Data processing interruption

The data processing may stop:

- in order to allow the user to select the values of the ETL parameters from *Imported Data* for which the Define automatically option is not enabled; for this purpose, the system displays a window containing as many records as parameters to be set up, made up of the description of the parameter and a lookup for the selection of the value. All the parameters must be defined before proceeding with the execution of the ETL.



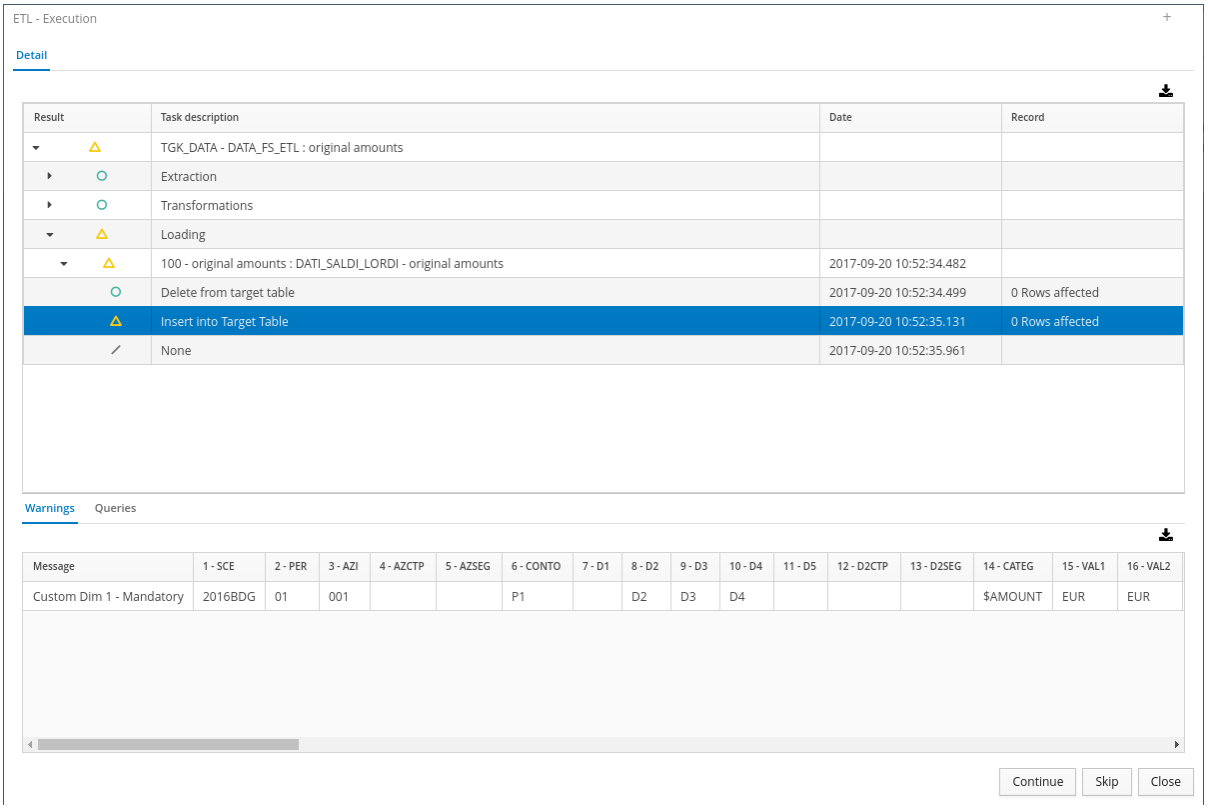
- in order to allow the user to decide whether to proceed with the execution or not if a task returned a warning. The following tasks may return warnings:

- extraction tasks that do not load any record.
- transformation tasks of *Control* type
- loading tasks that apply the editability checks

The ETL stops running only if the interruption option in case of warning is enabled on the Routine of the task.

Click **Continue** to proceed with the data processing.

When the interruption occurs after a loading task returning a warning, two buttons are available: **Continue** and **Skip**; the first one applies the loading rules to data not reported by the editability check and proceeds with the data processing, while the second one skips to the next loading task.



Result window

Once the execution of the ETL is completed, the result window appears. The window is divided into the following three panels:

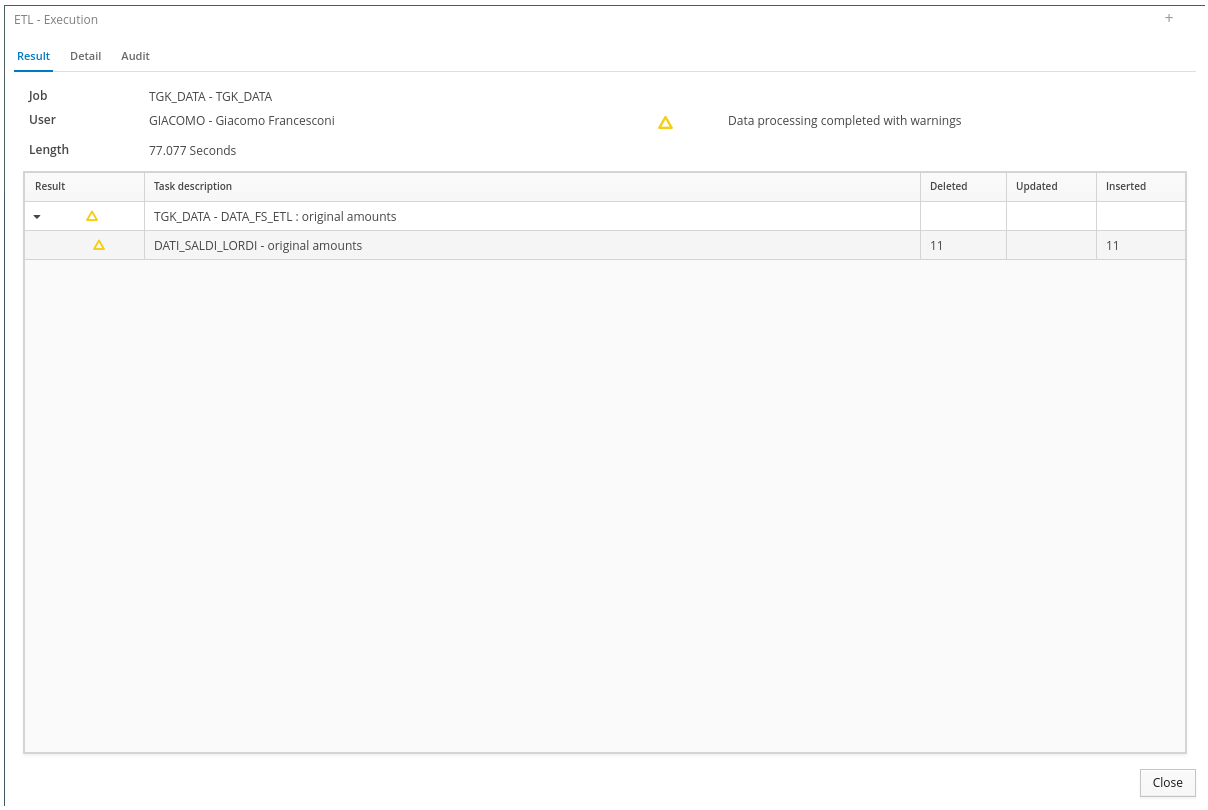
- **Result**, showing the overall result of the execution.

  The upper section displays the following:

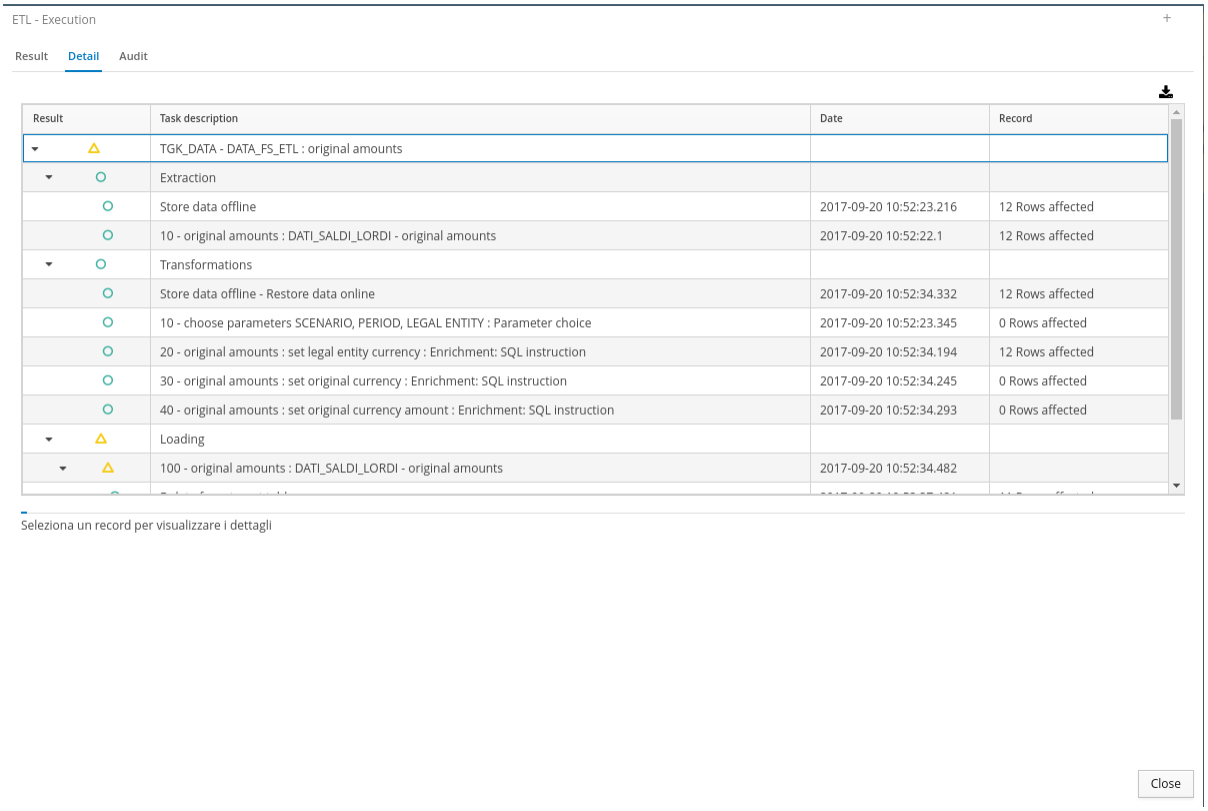
  - **Job/Routine** : the code and the description of the executed ETL Job or Routine
  - **User** : the code and the description of the user who run it
  - **Length** : data processing length expressed in seconds

- **Result** : total result displayed by a colored icon. Available colors:
  - GREEN: execution completed without error or warnings
  - YELLOW: a warning occurred during the execution of at least one task
  - RED: an error occurred during the execution of at least one task

while the lower section displays for each **Routine** / **Loading task** the total number of deleted, updated and inserted records

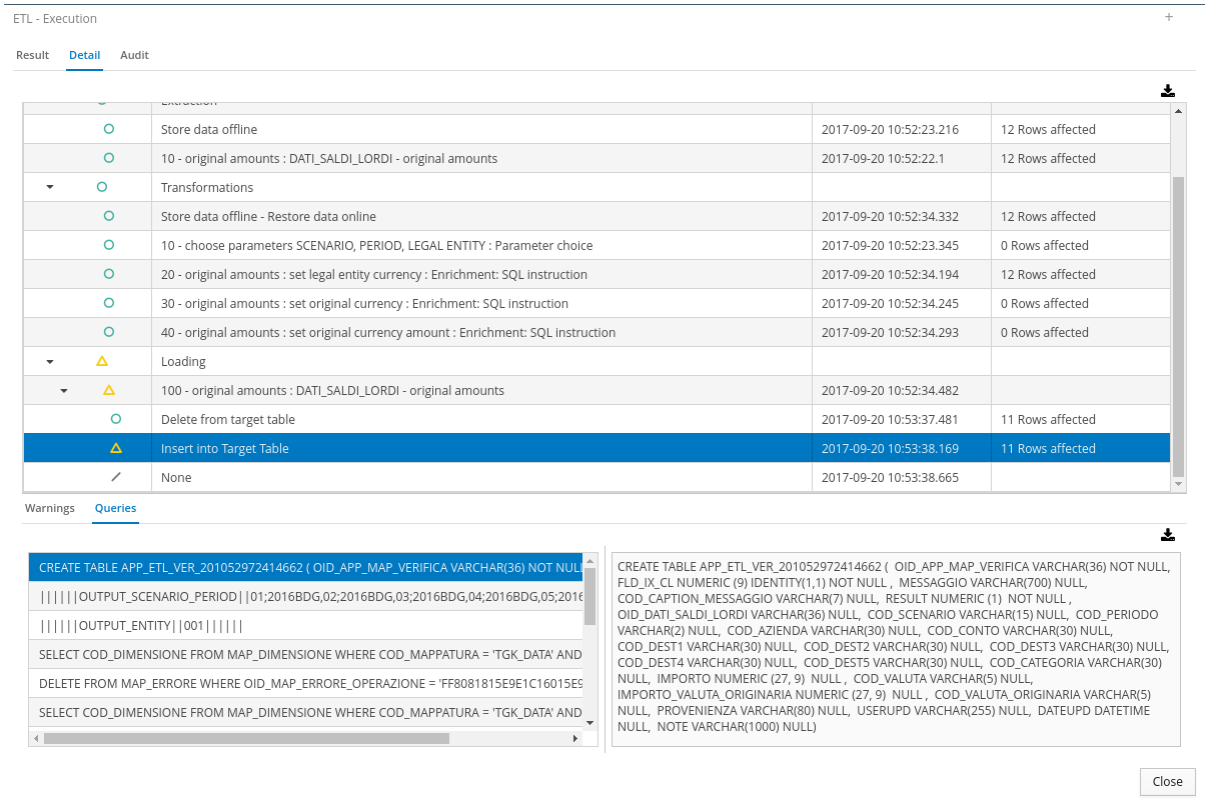


- **Detail** showing the details of the ETL execution's result. In particular, for each Routine run, the system displays the extraction, transformation and loading tasks (including micro tasks). The result is displayed by a colored icon. The result icon can be:
  - GREEN. Execution completed without errors
  - RED. An error occurred during execution
  - YELLOW. A warning occurred during execution
  - GREY. Lading micro task not parametrized



For each selected record further detail panels can be available according to the result of the task:

- **Query** : this panel appears only if the selected task if an extraction task, a transformation task or a micro loading task. It includes the list of queries run by the task.
- **Warning**: this panel appears only if the selected task is an extraction task, a transformation task or a micro loading task that generates warnings. It contains a table showing the warning message and:
  - the reported records and all the dimensions defined for the ETL domain, for the transformation tasks
  - the original records and all the dimensions defined for the ETL Domain, for the insertion/update operations if the data target has not been configured to group data to be loaded
  - if the data target is configured to group data to be loaded, for the insertion/update operations it shows:
    - the grouped records and only the dimensions defined in the data target if the *Show detailed warnings* option is not enabled in the routine definition
    - the source records and all the dimensions defined for the ETL Domain, if the *Show detailed warnings* option is enabled in the routine definition
  - it shows the records of the output table and only the dimensions defined in the data target, for the deletion operations.
- **Error**. This panel is available only if the selected task is an extraction task, a transformation task or a micro loading task that generates a blocking error. It shows the error message and the related stack trace.

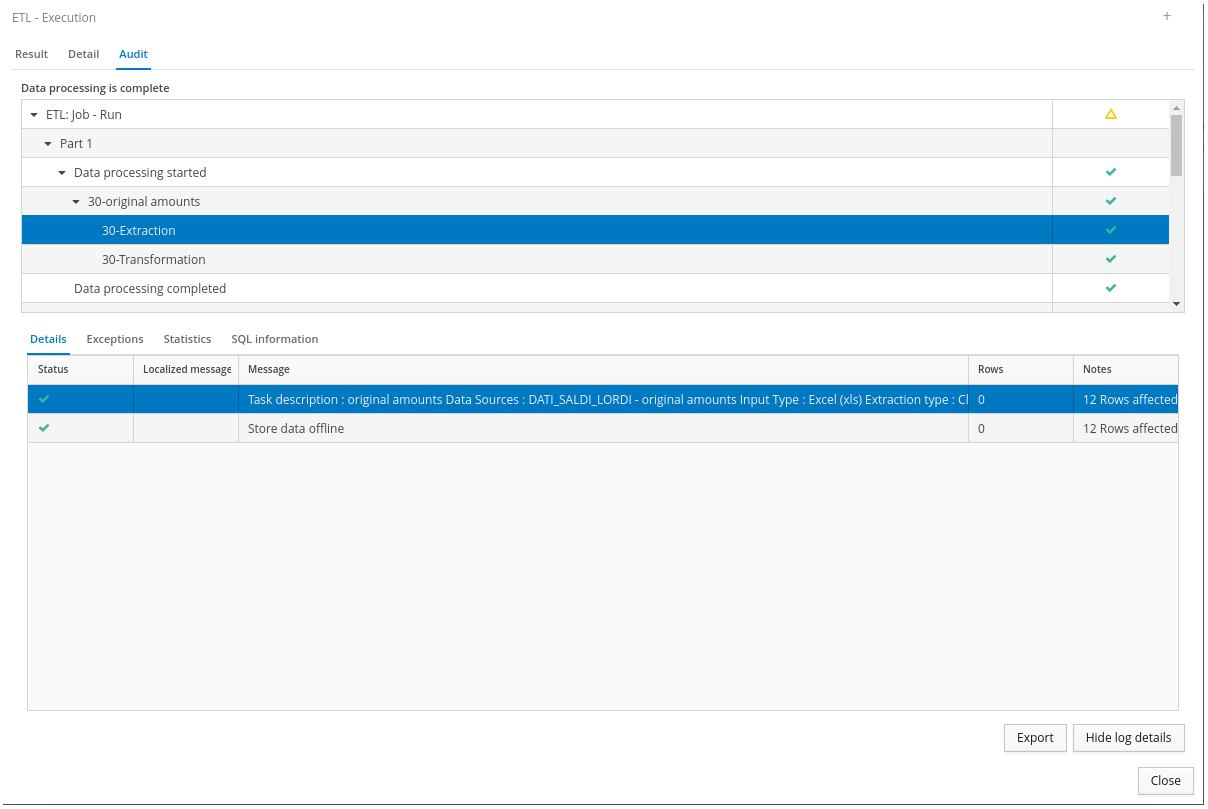


The content of each detail panel can be exported to Excel by the button  displayed at the top right of each panel

- **Audit** showing the standard track of the audit for the data processing of Tagetik.

  For each selected record it is possible to display the detail of the log in an underlying panel showing:

  - **Detail** : detail of the selected log
  - **Exceptions** : any stack traces generated by blocking errors
  - **Statistics** : statistics on the queries
  - **SQL information** : information on the queries


ETL - Execution
Result
Detail
Audit
Data processing is complete
ETL: Job - Run
Part 1
Data processing started
30-original amounts
30-Extraction
30-Transformation
Data processing completed
Details
Exceptions
Statistics
SQL information
Status
Localized message
Message
Rows
Notes
Task description : original amounts Data Sources : DATI_SALDI_LORDI - original amounts Input Type : Excel (xls) Extraction type : Cl
0
12 Rows affected
Store data offline
0
12 Rows affected
Export
Hide log details
Close

ETL Logs

This window shows the logs of the ETL Routines run individually or within an ETL Job.

The event / detail double-panel display shows for each event:

- the overall result of the data processing;
- the user who run it;
- the executed ETL Domain / Routine;
- the data processing start and end timestamps;

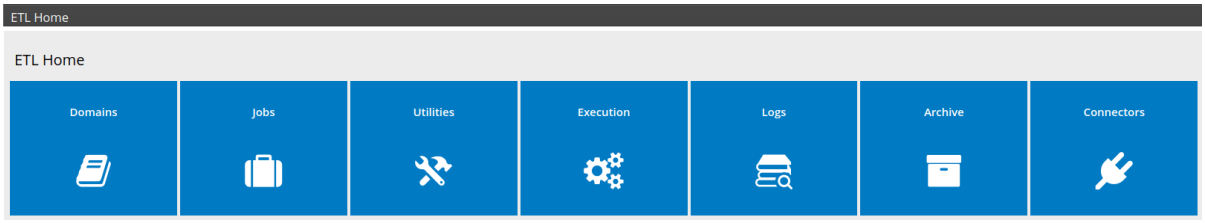
and the information about the results of each task, including:

- task result;
- data processing duration;
- the number of processed records;
- the blocking errors' detail;
- the warnings' details;
- the queries' details.

Entry Point

ETL Tile - Admin user

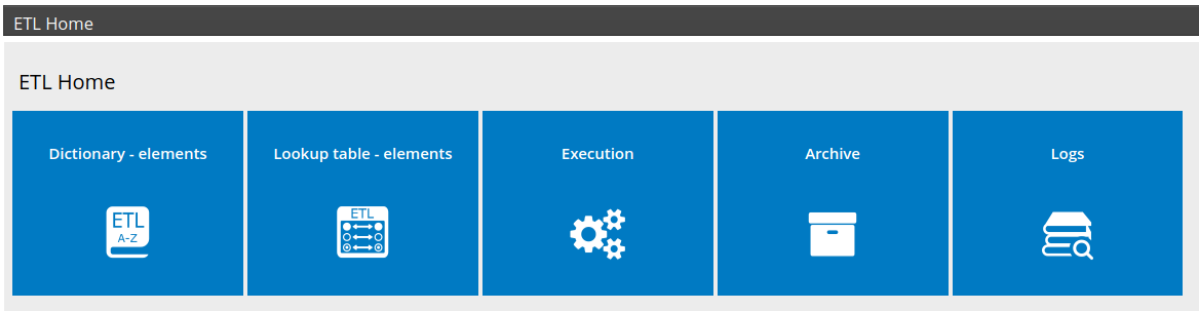
From the ETL tile in the Admin user home page, it is possible to access the window by clicking on **Logs**.



ETL Tile - End user

From the ETL tile in the End user home page, it is possible to access the window by clicking on **Logs**.

From this entry point it is not possible to display the logs of the executions run by users other than the current one.

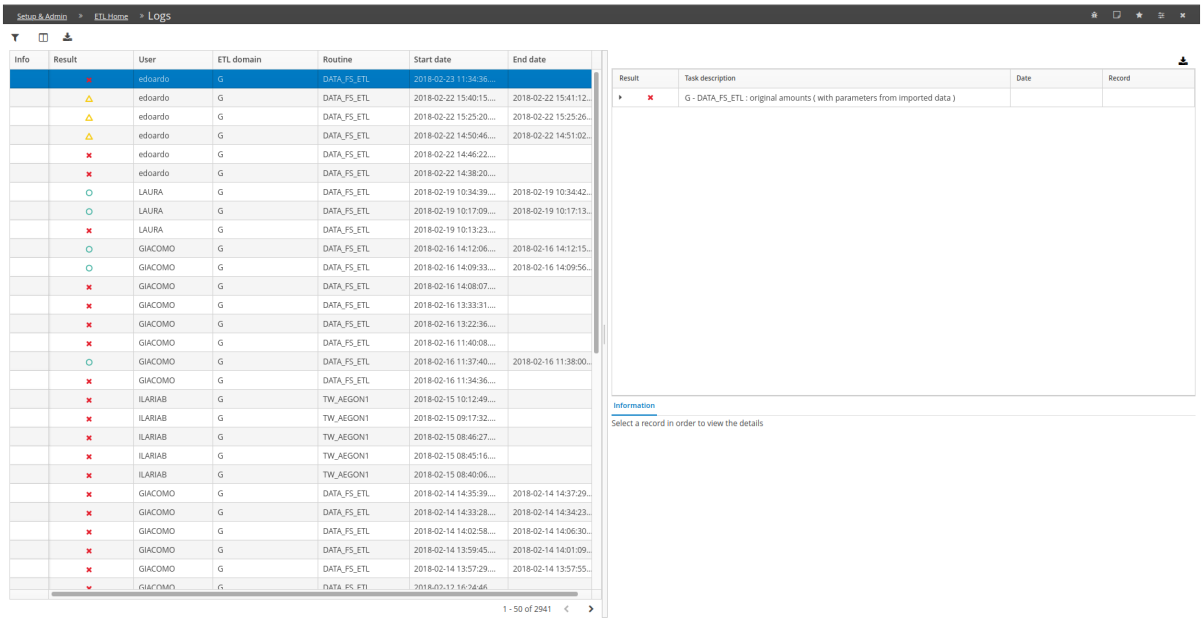


Window

The window is composed of two different panels:

- list of events
- the details of each event

the second panel is dynamic in relation to the record selected in the grid of the first panel.

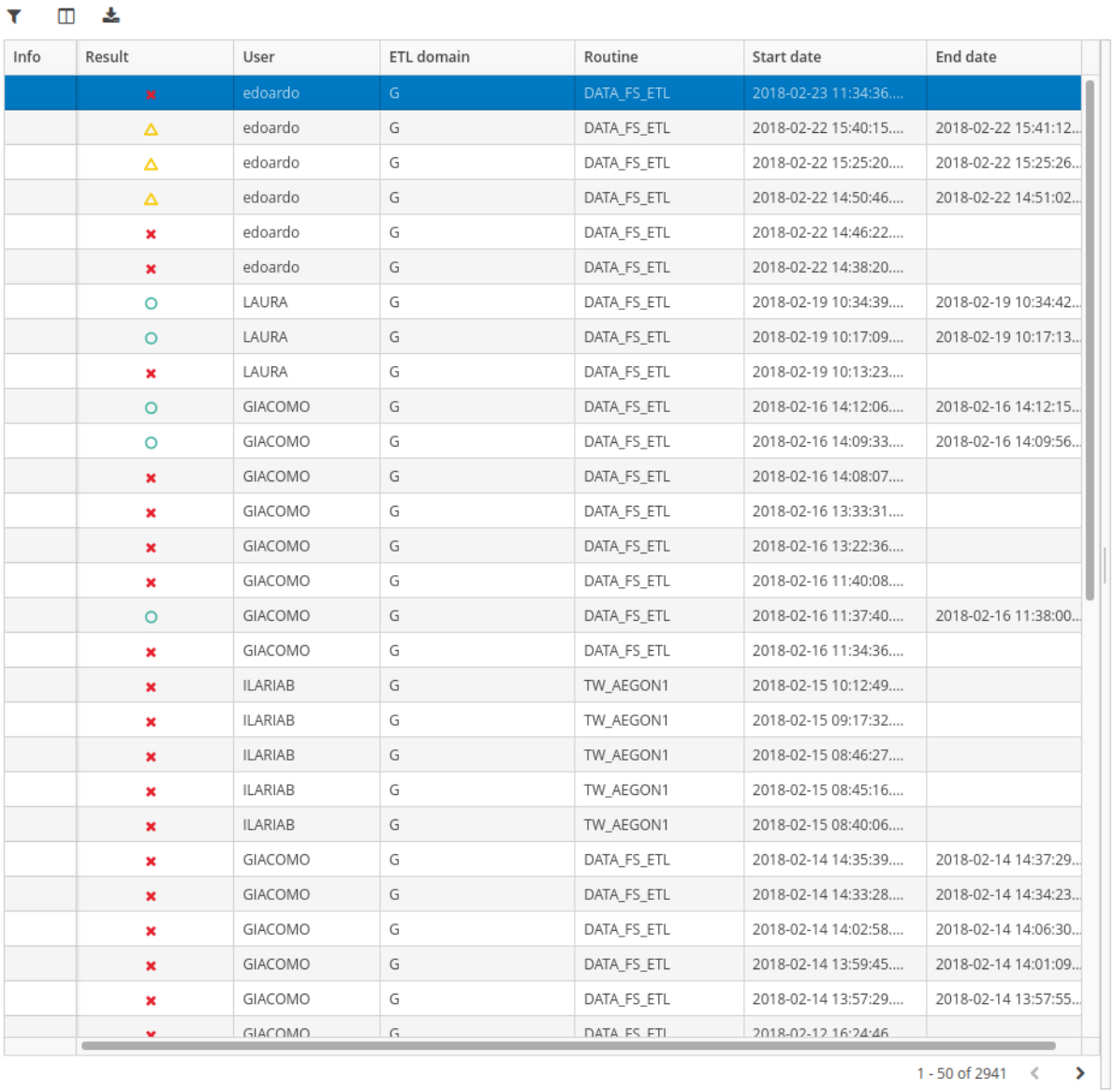


Events

The first panel shows the list of events related to the execution of the ETL Routines. For each event, the panel shows:

- **Result**: the overall result of the execution.
  - : Routine's tasks successfully completed;
  - : at least one task of the Routine ended with warnings;
  - : at least one task of the Routine ended with an error.
- **User**: code and description of the user who run the Routine (visible only if the window has been opened from an entry point showing the executions run by all users);

- **ETL Domain**: code and description of the ETL Domain related to the run Routine;
- **Routine**: code and description of the run Routine;
- **Start date**: timestamp related to the data processing start;
- **End date**: timestamp related to the data processing end. It can be omitted if the execution failed to end due to a blocking error.



Info	Result	User	ETL domain	Routine	Start date	End date
	✖	edoardo	G	DATA_FS_ETL	2018-02-23 11:34:36....	
	△	edoardo	G	DATA_FS_ETL	2018-02-22 15:40:15....	2018-02-22 15:41:12...
	△	edoardo	G	DATA_FS_ETL	2018-02-22 15:25:20....	2018-02-22 15:25:26...
	△	edoardo	G	DATA_FS_ETL	2018-02-22 14:50:46....	2018-02-22 14:51:02...
	✖	edoardo	G	DATA_FS_ETL	2018-02-22 14:46:22....	
	✖	edoardo	G	DATA_FS_ETL	2018-02-22 14:38:20....	
	○	LAURA	G	DATA_FS_ETL	2018-02-19 10:34:39....	2018-02-19 10:34:42...
	○	LAURA	G	DATA_FS_ETL	2018-02-19 10:17:09....	2018-02-19 10:17:13...
	✖	LAURA	G	DATA_FS_ETL	2018-02-19 10:13:23....	
	○	GIACOMO	G	DATA_FS_ETL	2018-02-16 14:12:06....	2018-02-16 14:12:15...
	○	GIACOMO	G	DATA_FS_ETL	2018-02-16 14:09:33....	2018-02-16 14:09:56...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-16 14:08:07....	
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-16 13:33:31....	
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-16 13:22:36....	
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-16 11:40:08....	
	○	GIACOMO	G	DATA_FS_ETL	2018-02-16 11:37:40....	2018-02-16 11:38:00...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-16 11:34:36....	
	✖	ILARIAB	G	TW_AEGON1	2018-02-15 10:12:49....	
	✖	ILARIAB	G	TW_AEGON1	2018-02-15 09:17:32....	
	✖	ILARIAB	G	TW_AEGON1	2018-02-15 08:46:27....	
	✖	ILARIAB	G	TW_AEGON1	2018-02-15 08:45:16....	
	✖	ILARIAB	G	TW_AEGON1	2018-02-15 08:40:06....	
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-14 14:35:39....	2018-02-14 14:37:29...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-14 14:33:28....	2018-02-14 14:34:23...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-14 14:02:58....	2018-02-14 14:06:30...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-14 13:59:45....	2018-02-14 14:01:09...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-14 13:57:29....	2018-02-14 13:57:55...
	✖	GIACOMO	G	DATA_FS_ETL	2018-02-12 16:24:46....	

1 - 50 of 2941

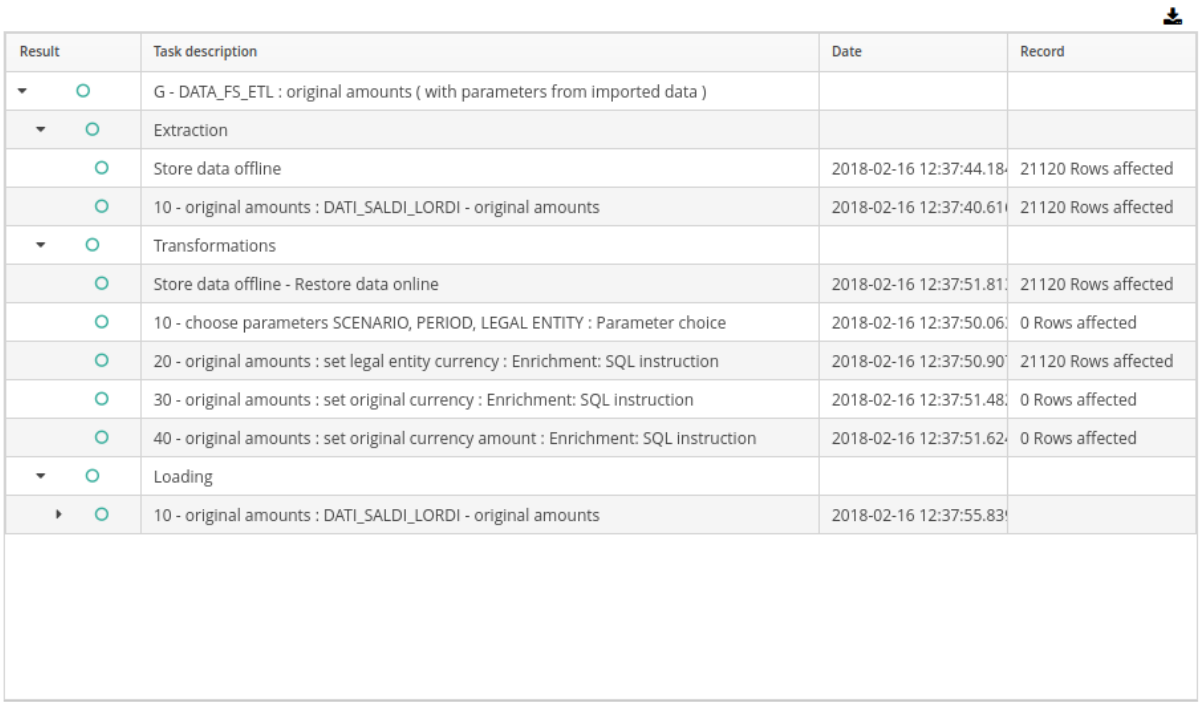
Detail

For each event selected in the first panel, a second panel is displayed, showing in a tree structure the details of the results generated by the execution of extraction, transformation and loading tasks (including micro tasks) of the Routine. For each task, the following details are displayed:

- **Result**: result of the task execution. Possible values:
  - ○: execution successfully completed;
  - △: warning occurred;

- : error occurred;
- : task not run.

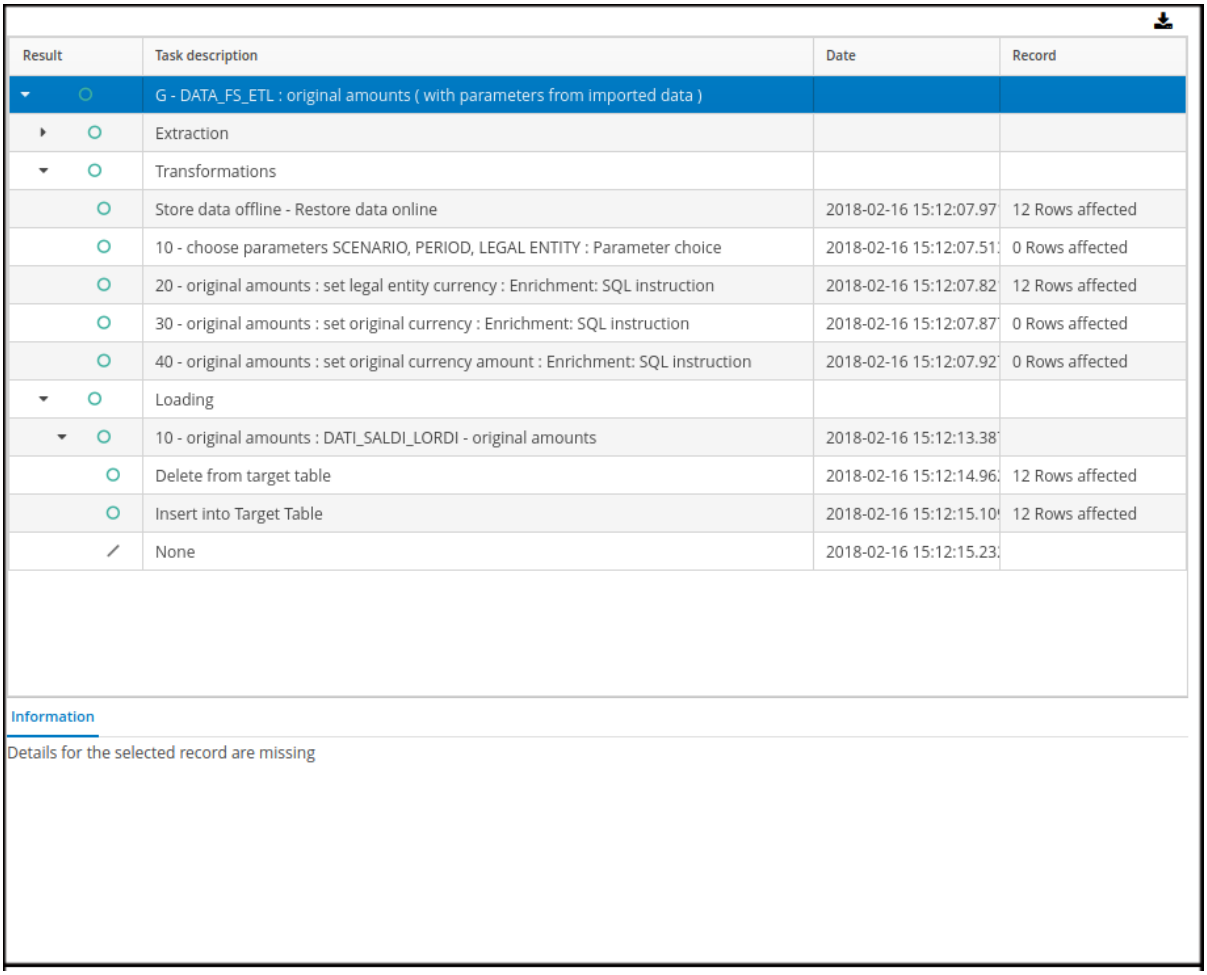
- **Task description**: description of the task; It may include:
  - the description of a step: *Extraction*, *Transformation*, *Loading*;
  - description of an activity related to the data storage: *Store data offline*, *Restore data online*;
  - description of a task containing the sorting and description defined by the user.
- **Date**: timestamp related to the task run start;
- **Record**: number of records processed by the task.



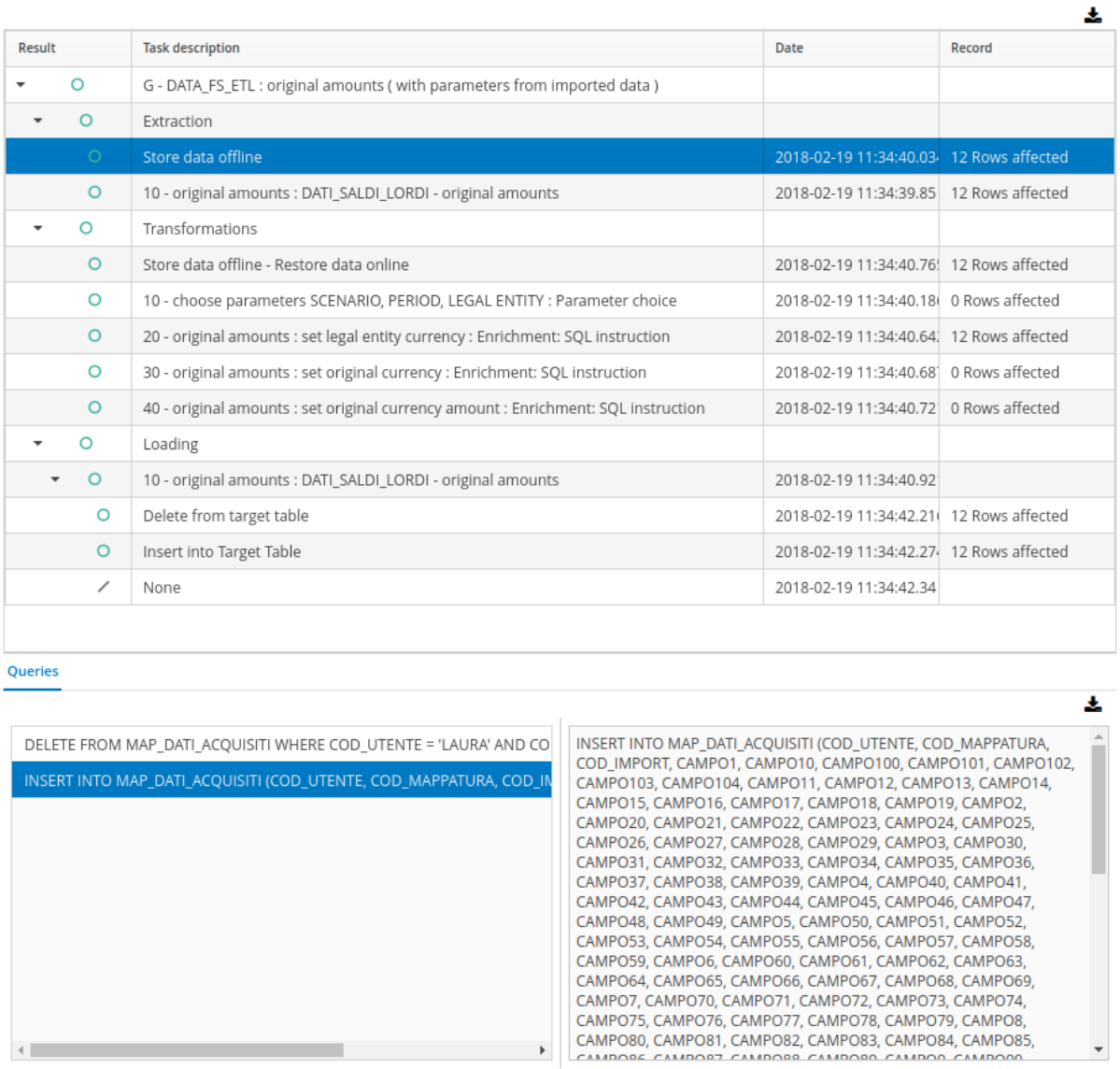
Result	Task description	Date	Record
	G - DATA_FS_ETL : original amounts (with parameters from imported data)		
	Extraction		
	Store data offline	2018-02-16 12:37:44.18	21120 Rows affected
	10 - original amounts : DATI_SALDI_LORDI - original amounts	2018-02-16 12:37:40.61	21120 Rows affected
	Transformations		
	Store data offline - Restore data online	2018-02-16 12:37:51.81	21120 Rows affected
	10 - choose parameters SCENARIO, PERIOD, LEGAL ENTITY : Parameter choice	2018-02-16 12:37:50.06	0 Rows affected
	20 - original amounts : set legal entity currency : Enrichment: SQL instruction	2018-02-16 12:37:50.90	21120 Rows affected
	30 - original amounts : set original currency : Enrichment: SQL instruction	2018-02-16 12:37:51.48	0 Rows affected
	40 - original amounts : set original currency amount : Enrichment: SQL instruction	2018-02-16 12:37:51.62	0 Rows affected
	Loading		
	10 - original amounts : DATI_SALDI_LORDI - original amounts	2018-02-16 12:37:55.83	

For each selected record, some sub-panels with additional details can be displayed according to the result of the task:

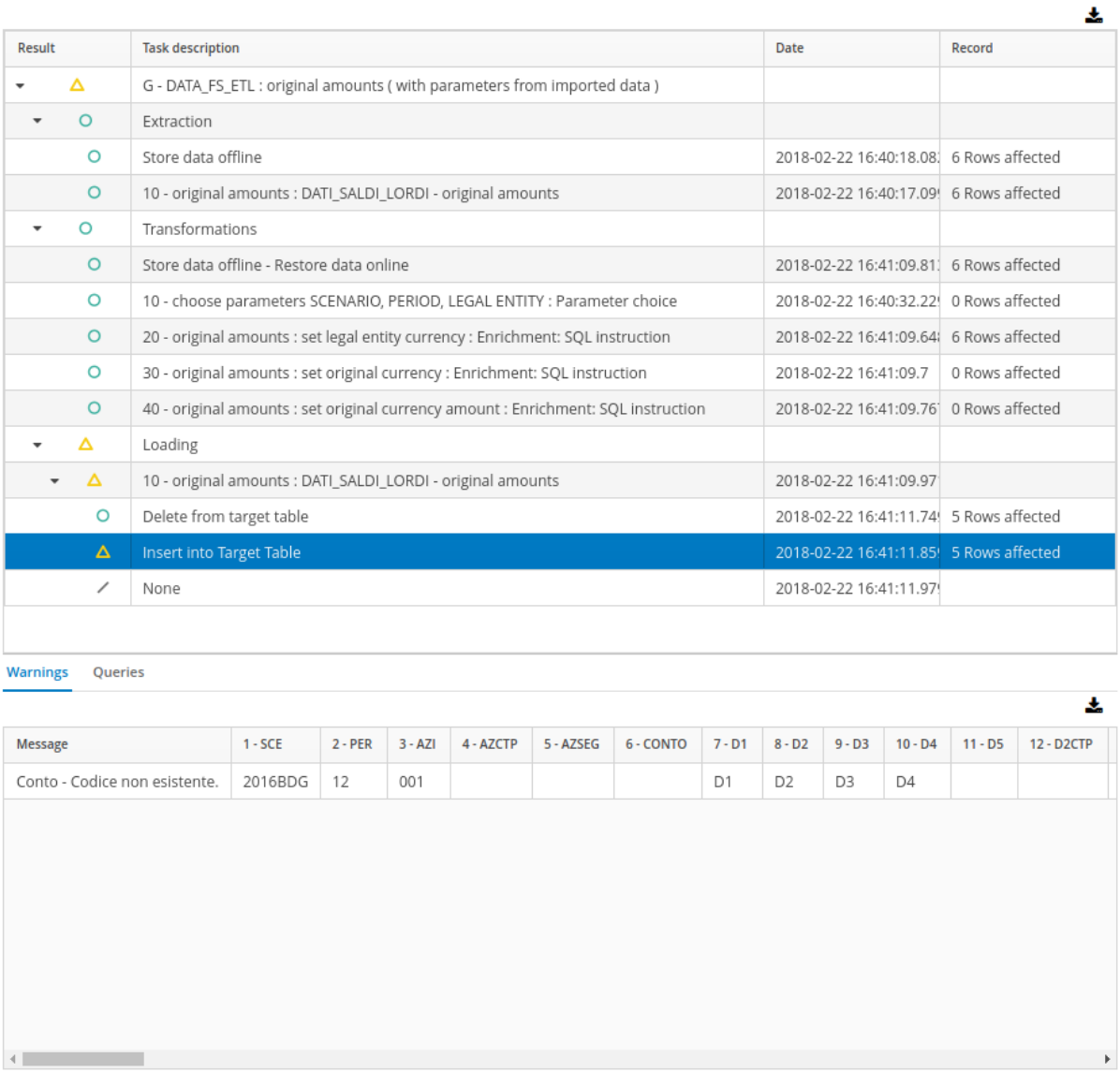
- **Information**: this panel is visible only if the selected record has not other details;



- **Query**: this panel is visible only if the selected record is an extraction or transformation task or a micro loading task. It contains the list of queries run by the task.



- **Warning**: this panel is visible only if the selected record is an extraction or transformation task or a micro loading task that generates warnings. IT contains a grid showing the warning message and:
  - for the transformation tasks, shows the reported records and all the dimensions defined for the ETL Domain
  - for the data entry / update operations, if the data target does not group the data to load, displays the original records and all the dimensions defined for the ETL Domain
  - for the data entry / update operations, if the data target groups the data to load, displays:
    - the grouped records and only the dimensions defined in the data target if the *Show detailed warnings* option is disabled in the routine's definition
    - the original records and all the dimensions defined for the ETL Domain if the *Show detailed warnings* option is enabled in the routine's definition
  - for the deletion operations, displays the records of the output table and only the dimensions defined in the data target.

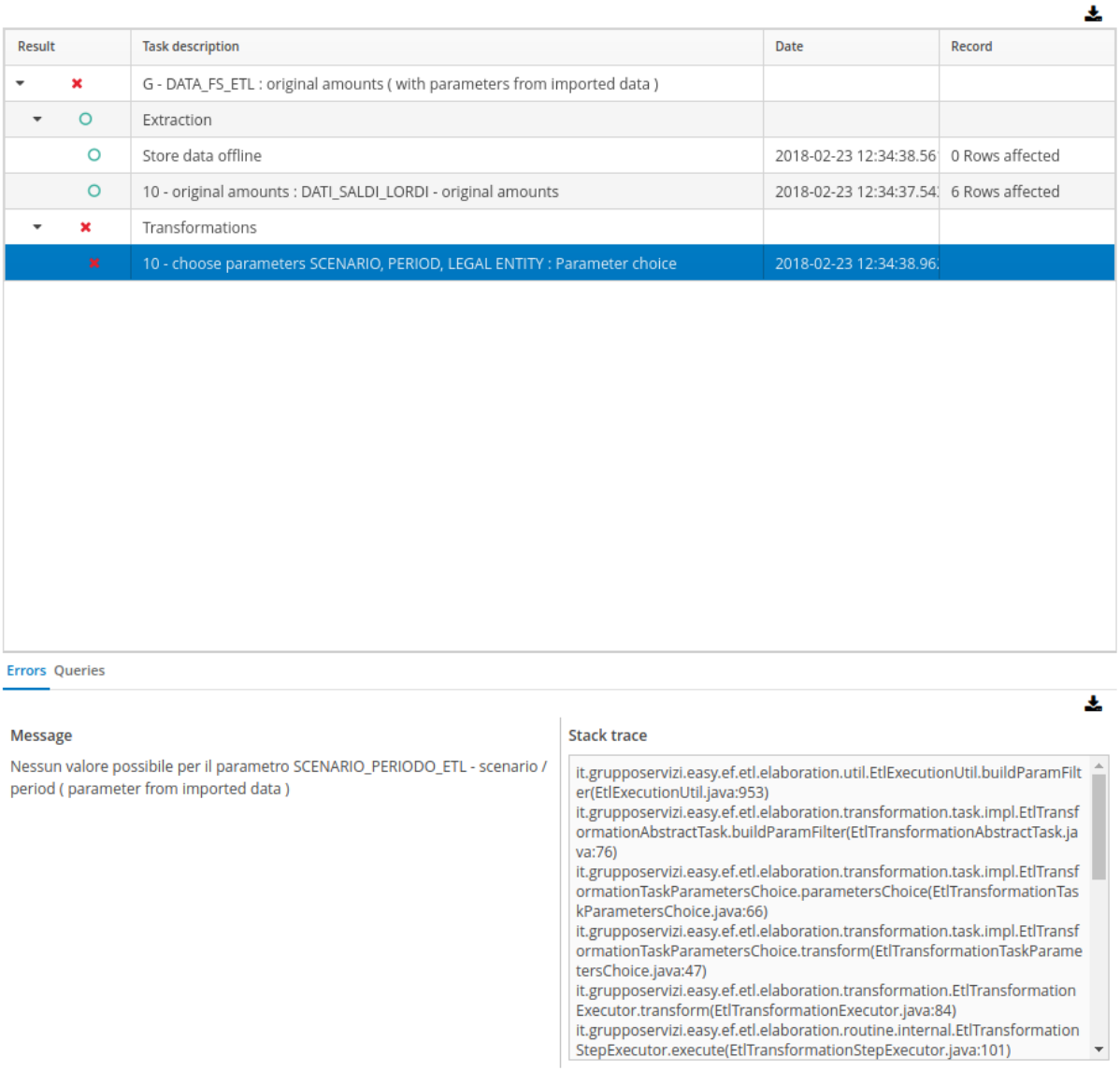


Result	Task description	Date	Record
△	G - DATA_FS_ETL : original amounts (with parameters from imported data)		
○	Extraction		
○	Store data offline	2018-02-22 16:40:18.08.	6 Rows affected
○	10 - original amounts : DATI_SALDI_LORDI - original amounts	2018-02-22 16:40:17.09	6 Rows affected
○	Transformations		
○	Store data offline - Restore data online	2018-02-22 16:41:09.81	6 Rows affected
○	10 - choose parameters SCENARIO, PERIOD, LEGAL ENTITY : Parameter choice	2018-02-22 16:40:32.22	0 Rows affected
○	20 - original amounts : set legal entity currency : Enrichment: SQL instruction	2018-02-22 16:41:09.64	6 Rows affected
○	30 - original amounts : set original currency : Enrichment: SQL instruction	2018-02-22 16:41:09.7	0 Rows affected
○	40 - original amounts : set original currency amount : Enrichment: SQL instruction	2018-02-22 16:41:09.76	0 Rows affected
△	Loading		
△	10 - original amounts : DATI_SALDI_LORDI - original amounts	2018-02-22 16:41:09.97	
○	Delete from target table	2018-02-22 16:41:11.74	5 Rows affected
△	Insert into Target Table	2018-02-22 16:41:11.85	5 Rows affected
/	None	2018-02-22 16:41:11.97	

Warnings Queries

Message	1 - SCE	2 - PER	3 - AZI	4 - AZCTP	5 - AZSEG	6 - CONTO	7 - D1	8 - D2	9 - D3	10 - D4	11 - D5	12 - D2CTP
Conto - Codice non esistente.	2016BDG	12	001				D1	D2	D3	D4		

- **Errors**: this panel is visible only if the selected record is an extraction, transformation task or a micro loading task that generates a blocking error. It contains the error message and the related stack trace.



The content of each detail panel can be exported to Excel by the button on the right hand corner of each panel.

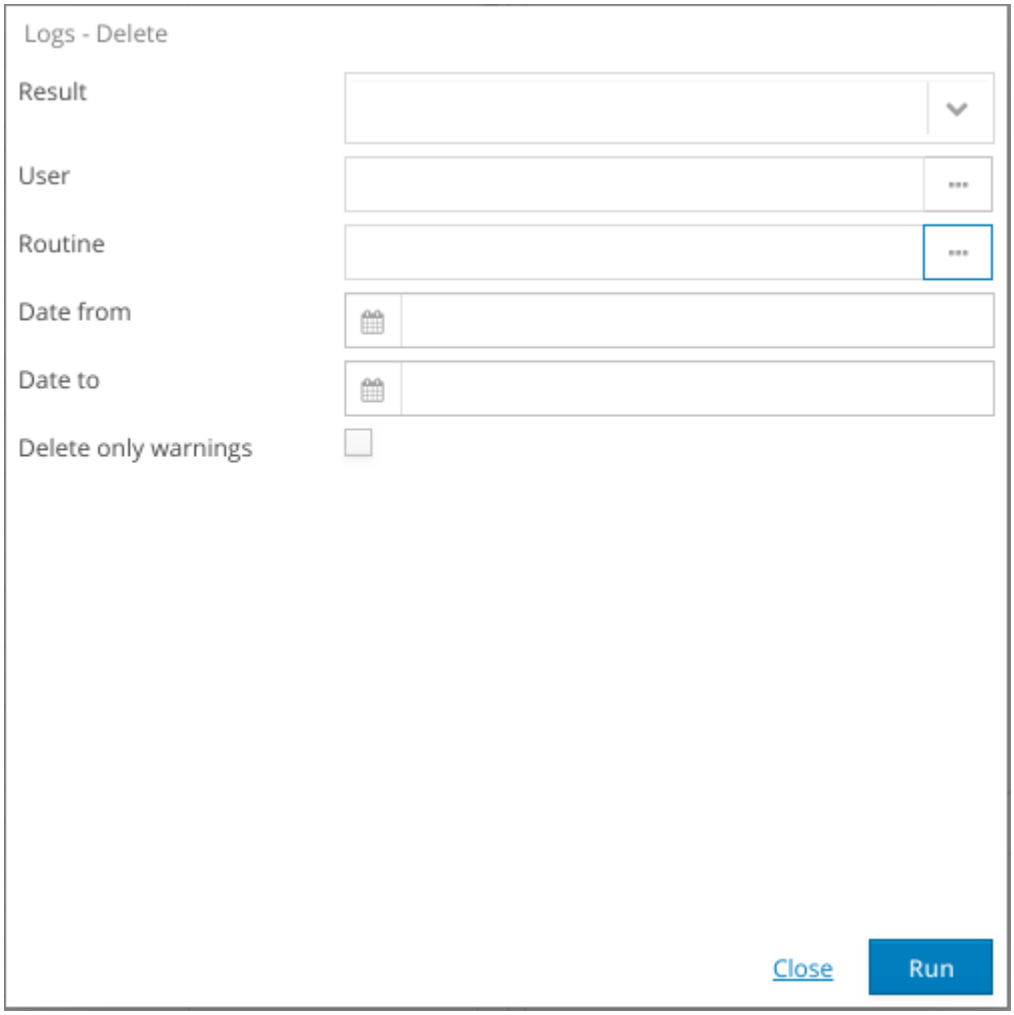
Utility

The window's actions menu includes the ETL Logs advanced deletion utility.

Logs deletion

The *Logs deletion* menu opens a window in which you can select the logs related to the execution of the ETL routines. The selection of the logs can be filtered by:

- **Result**: the result of the execution of the task. Possible values:
  - : Successfully executed;
  - : Executed with warnings;
  - : Executed with errors;
  - : Not executed.
- **User**: the code of one or more users
- **Routine**: the code of one or more ETL routines
- **Date from / Date to**: two date fields identifying a time interval in which the execution runs
- **Delete only warnings**: if selected, the system deletes only the warnings contained in the warnings tab of the logs with result "Executed with warnings". Otherwise, the system will entirely delete the selected logs.


Logs - Delete
Result
User
Routine
Date from
Date to
Delete only warnings
Close
Run

ETL data archive

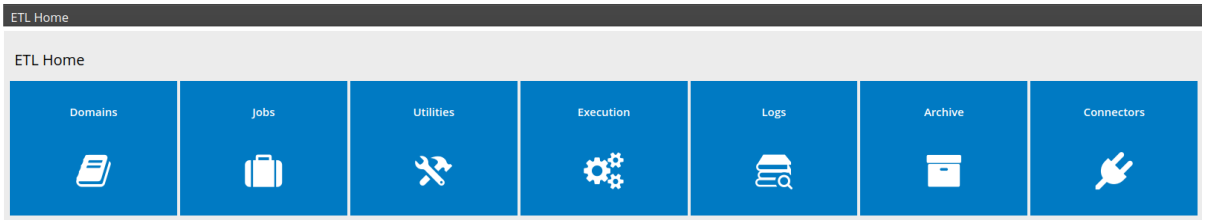
This window shows data, recorded during extraction and transformation, related to the ETL Routines run individually or inside an ETL Job.

Stored data are available only if the options that enabled the storage of data are enabled for an ETL Routine.

Entry Point

ETL Tile - Admin user

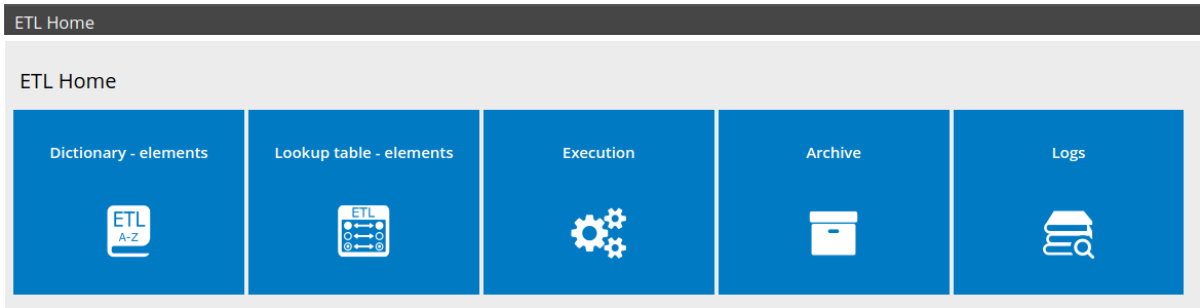
From the ETL tile in the Admin user home page, it is possible to access the window by clicking on the **Archive** tile.



ETL Tile - End user

From the ETL tile in the End user home page, it is possible to access the window by clicking on the **Archive** tile.

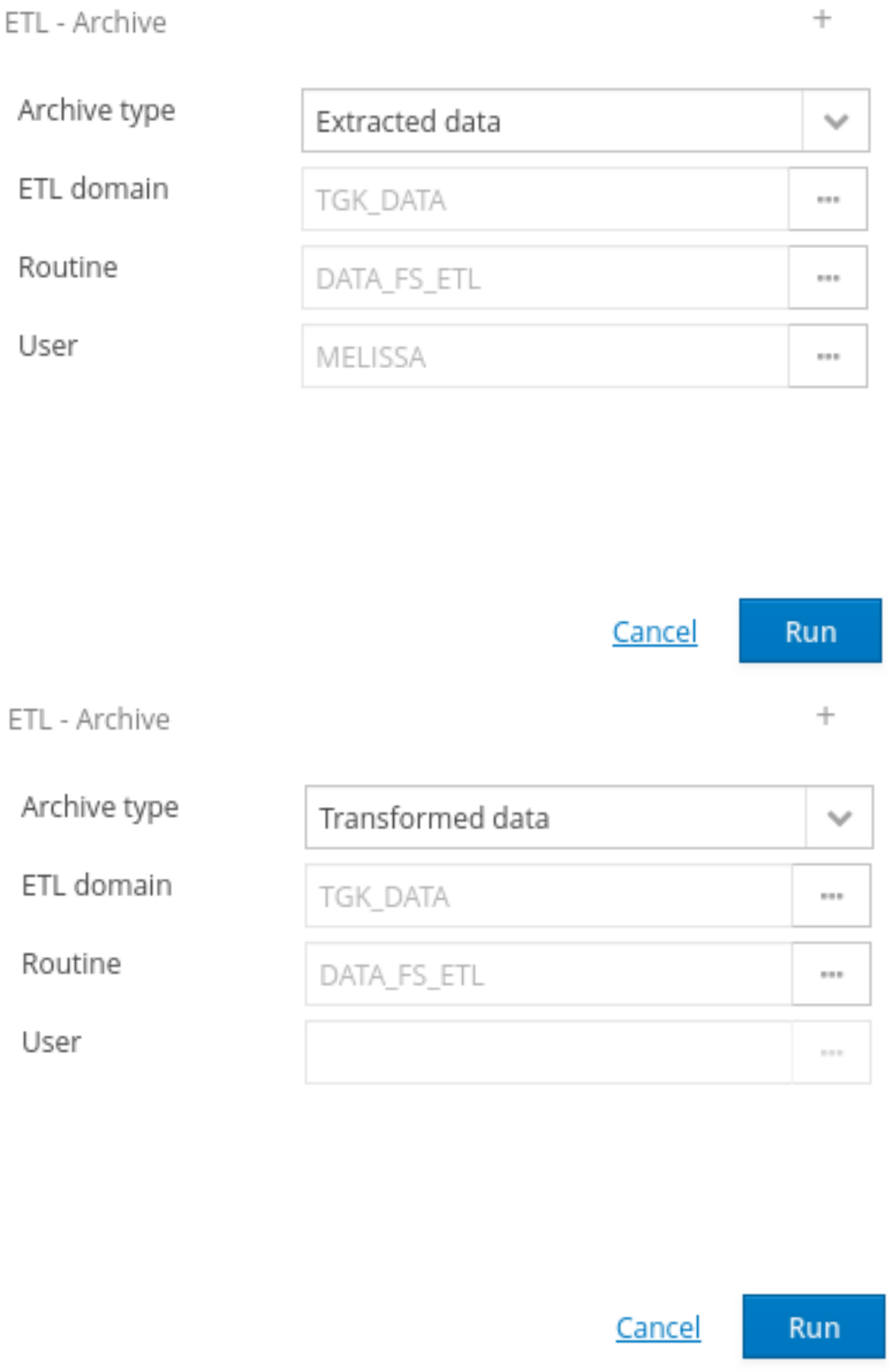
From this entry point, it is not possible to display data stored during extraction by users other than the current one.



Filter

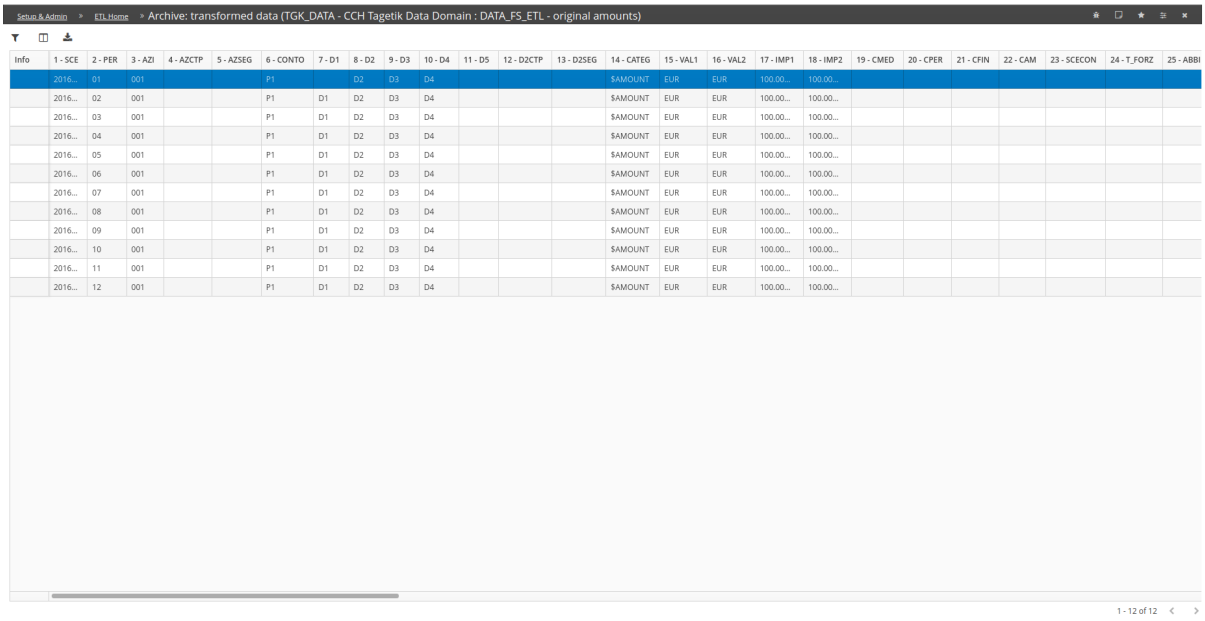
In the window containing stored data, it is necessary to specify the following initial filters:

- **Archive type**: it specifies if the archive to display is that of *Extracted data* or of *Transformed data*;
- **ETL Domain** / **ETL Routine**: it filters the archive by ETL Routine;
- **User**: it filters the archive by user but only for the **Admin user** entry point and only for the *Extracted data* **Archive type**



Window

The window displays a read-only table containing only the columns of the ETL Dimensions belonging to the ETL Domain selected when setting the filters, showing data stored during extraction or transformation for the Archive type and for the ETL Routine (and, if need be, for the User).



The content of each archive can be exported to Excel by the button on the toolbar on the left hand side of the screen.

Appendix

Filter Strings Syntax or SQL Query

In all management windows related to an ETL process where it is possible to insert an ETL filter or a Query in SQL language, text parametric parts can be used in order to report the value of run filters of a routine or of an ETL job.

The parametric elements that can be inserted are:

Parameter	Value replaced during the run phase
{TABLE}	Staging table
{MAP}	ETL Domains Code
{IMP}	Routine Code
{JOB}	ETL Job Code
{USER}	User code that follows the Routine or the ETL Job
{PROCESS}	Application process code
{DATETIME}	Date and time in which the Routine or the ETL Job is run
{DATE}	Date in which the Routine or the ETL Job is run
{TIME}	Time in which the Routine or the ETL Job is run
{DATE_YYYY}	Year (4 numbers) in which the Routine or the ETL job is run
{DATE_YY}	Year (2 numbers) in which the Routine or the ETL job is run
{DATE_MM}	Month (2 numbers) in which the Routine or the ETL job is run
{DATE_DD}	Day (2 numbers) in which the Routine or the ETL job is run
{TIME_HH}	Time (2 numbers) in which the Routine or the ETL job is run
{TIME_MI}	Minute (2 numbers) in which the Routine or the ETL job is run
{TIME_SS}	Second (2 numbers) in which the Routine or the ETL job is run
{TIME_ML}	Millisecond (2 numbers) in which the Routine or the ETL job is run

Parameter	Value replaced during the run phase
{codice_ parametro_ETL}	Parameter value. EXAMPLE: if we assume to have defined a "PARAM_AZIENDA" Parameter of "imported data" type, that gets the value of the ETL Dimension and contains the entities for which data are extracted (e.g. A01, A02, A03) it is enough to write COD_AZIENDA IN ({PARAM_AZIENDA} to let the system translate this instruction into COD_AZIENDA IN ('A01', 'A02', 'A03')
{sigla_dimensione_ ETL}	From the ETL Dimensions the system determines the dimension type having the specified acronym and that is able to convert automatically the value contained in the dimension element into the dimension type. EXAMPLE: if we have a number type ETL Dimension having {IMP1} as acronym and assuming that we want to multiply the value of this dimension by -1 and considering that in the staging table the columns are all string types, it is enough to write {IMP1}*-1 instead of TO_NUMBER(IMP1) *-1
{AW|[workspace code]|[dataset code]}	Used to retrieve the physical name of the partition of the specified Analytical Workspace if the Dataset is not partitioned or is partitioned by Business Cycle and the ETL is run from a single submission Process.
{AW| [workspace code]| [dataset code]| [scenario code]| [period code]}	Used to retrieve the physical name of the partition of the specified Analytical Workspace if the Dataset is partitioned by Business Cycle and the ETL is run from a submission for scenario/period Process
{AW| [workspace code]| [dataset code]| [entitiy code]}	Used to retrieve the physical name of the partition of the specified Analytical Workspace if the Dataset is partitioned by Business Cycle / Entity and the ETL is run from a single submission Process.
{AW| [workspace code]| [dataset code]| [scenario code]| [period code]| [entitiy code]}	Used to retrieve the physical name of the partition of the specified Analytical Workspace if the Dataset is partitioned by Business Cycle / Entity and the ETL is run from a submission for scenario/period Process
{AW| [workspace code]| [dataset code]| [typology code]| [process	Used to retrieve the physical name of the partition of the specified Analytical Workspace whatever are the Dataset partition and the execution context. If one of the parameters of the syntax is not relevant, it is necessary to set up the **$** character

Parameter	Value replaced during the run phase
code]| [scenario code]| [period code]| [entitiy code]}	E.g.: • {AW|001|00001|$\|$|$\|$|$} • {AW\|001\|00001\|$|BUDGET|$\|$|$} • {AW\|001\|00001\|$|ACTUAL|2020ACT|12|$} • {AW\|001\|00001\|$|ACTUAL|2020ACT|12|001} • {AW|001|00001|CALC|BUDGET|$\|$|001} • {AW|001|00001|CALC|ACTUAL|2020ACT|12|$} • {AW|001|00001|CALC|ACTUAL|2020ACT|12|001}

 It is possible to use the syntax **#{}#**to prevent the system from delimiting the value between single quotes, when converting some key words.

For example, the syntax **{PROCESS}** is converted with *'BUGDET'*; while the syntax **#{PROCESS}#** is converted with *BUDGET*.

This is useful, for example, when you want to use a parametric syntax as a part of a table's name or as a parameter of a stored procedure.

 Moreover, it is possible to use the Unified SQL syntax.



- Every time an ETL Dimension must be reported in an ETL filter or in a Query in SQL language, it is necessary to use not the personal names but the names that the staging tables columns fields have in the application database of Tagetik, namely CAMPO1, CAMPO2 etc.
- Every time it is required to write a filter to apply to the staging table in order to report its fields, it is mandatory to use the names they have in the application database of Tagetik, namely CAMPO1, CAMPO2 etc.
- Every time it is required to write a filter to apply to the table where the data are uploaded to report its field, it is mandatory to use the names they have in the database where the table is.

Glossary

A

account

The Account represents the basic nature of each information stored in Tagetik. In the account dimension the user can open any measure or metric, both financial and statistical.

actual

The Actual represents a set of information related to a past period For example, the costs incurred in January are considered actual costs in February

application database

An application database is a database where all data and the data model's custom dimensions related to the application are stored. A typical installation of Tagetik consists of a Repository database containing all the information about the system management (users, roles, licenses etc.) and an application database for each application to manage.

application language

The Application Language is the interface language, namely the language in which the application is displayed (menu items, headers, fields names in management windows, error messages)

audit

The Audit is the means by which Tagetik tracks all the activities performed and all errors occurred in the course of time. The audit allows the user to track all the steps of the activities (directly run by the system or with the intervention of the user) carried out during a given time interval.

B

budget

The Budget represents a goal the Entity intends to achieve in relation to a specific period of time. Usually the Budget is related to a one-year period (for example budget 2003); with periods longer than one year, we talk about plan. In our case, the budget is a forecast of revenues, costs, actual balance sheet and cash flow that the Entity is engaged to achieve in a given period of time

C

Carrying Value

The Carrying Value is the value of the financial asset on the financial statement of the owner entity

category

The Category dimension of Tagetik allows you to classify and segment data (in particular the structures) and track the way data is generated

cell field

A cell field is used to recover and display in a form the attributes of the dimensions used as filters

consolidation currency

The Consolidation Currency is the currency in which the consolidation is generated. It usually corresponds to the currency of the Holding (but not necessarily)

consolidation scenario

The Consolidation Scenario is the scenario that specifies the entity that defines the characteristics of the consolidation scope: the Holding, the entities to consolidate, the consolidation methods, the consolidation currency, the source scenario etc. This scenario contains the "consolidated" data that cannot be modified manually. The Consolidation Scenario is also the rule that explains Tagetik how to consolidate the entities of a given consolidation scope.

currency

The Currency is the national currency in which a transaction or a financial statement is prepared.

custom dimension

A Custom Dimension is a dimension that allows you to meet the specific needs of the business model (products, clients, cost centres ...). Tagetik allows you to define and manage up to five custom dimensions

D

data entry mode

The “Data entry” mode allows the user to insert new data and to modify data present in a data entry form

data processing

A Data Processing is a process that runs on data. It reads input data and generates output data according to built-in rules (namely, encoded inside the system) or rules defined by the User.

database

A database is a collection of information set to be easily accessed for consultation, modifications and updates

design mode

The “Design” mode enables the modification of a form structure

dictionary

A Dictionary is a list of admitted values for an ETL dimension

Document activity

A Document activity represents the single part (chapter, section, paragraph ...) that composes the document type and can enclose user-entered content (texts, images, tables), or simply a content-free organizational element.

Document part

A Document part is a document activity that represents a content-free organizational element (reading of a document, a report, ...)

Document type

A Document Type is the type of document that you want to prepare such as the Annual Report, Board Book, 10Q-10K etc. The document type can be considered as a container or more precisely a template that defines the basic information that serves to create the word document used by end users to enter information (comments, tables, graphs ...). It is entirely designed in the application's web interface

drill down

The Drill down utility allows you to analyze the parts that compose an aggregated data

E

ETL

The ETL (Extraction, Transformation & Loading) is a completely configurable engine able to extract data from external sources such as Excel,

text files, warehouse data, ERP systems (SAP, Oracle or Microsoft) or from a Tagetik database; it is able to change and load data into several custom dimensions, both internal (Tagetik database) and external (generic database, Excel or text file) to Tagetik

ETL dimension

An ETL dimension is a column of the staging table where the ETL engine copies data extracted from one or more external sources

ETL domains

An ETL Domain represents a macro container that includes all attributes and information concerning an ETL rule (configurations, extraction formats, transformation and load rules)

ETL routine

An ETL Routine is the smallest data processing unit which can be run in an ETL process and contains all the information about the execution of the Routine and the link to the details for the data extraction, transformation and loading

Event

An Event is a fact that causes a change in the financial investment relationship between two companies of a group and that Tagetik must detect because it affects the consolidation journals.

event scenario

The Event Scenario is a fictitious scenario used by Tagetik to manage events by date, particularly the events of the group structure

F

financial statement template

A Financial Statement Template is a particular aggregation structure of accounts or accounts / custom dimensions which is generally used to create standard financial statement templates (P&L and IAS balance sheet, IV directive, direct / indirect cash flow) but also KPI and other needed indexes for the reporting

fiscal year

A Fiscal Year represents the financial year to which the entity management refers. Usually the fiscal year goes from the 1st of January to the 31st of December; but different dates can be also chosen. Tagetik does not

manages an out-and-out fiscal years table; rather, their definition is implicit in the definition of the scenarios.

forecast

The forecast is the budget data update. The forecast is elaborated during the course of the year when the Entity closes the final statements. When the final statement is compared to the budget, differences can occur; these generate new forecasts. The Forecast refers to a specific month: for example in March the forecast comes from the actual data of January - March and a new forecast for April -December

form

a Form is a special Excel spreadsheet used by Tagetik inside its reporting and data entry "unified" system based on Microsoft Excel to interact directly with the application database of Tagetik so as to display / extract database information and save data inserted by the user into the database.

form filter

A Form Filter is a filter type that affects the whole form

function

A Function is a basic operation the user can perform. Typically the functions are the management of tables, data processing, reports. In most cases a function corresponds to a menu key or to a window menu; in some cases to more simple operations. The functions definition is aimed at enabling the definition of the roles and from version 4.2.2 also the control of the validity of the licence, both of "Adim" type or "Basic" type. The “Admin” functions distinguish an administrator user from a basic user.

G

grouping

A grouping is a multi-level structure with unlimited depth and multi-parent (each node or element of the dimension can be related to several parent nodes) which allows you to group / organize a set of elements in order to meet the various business needs

H

hierarchy

A Hierarchy is a closely hierarchical multilevel structure that allows to aggregate / sort a group of elements in order to meet the various business needs

holding

The Holding is the Entity that controls all the other entities of the group thanks to the ownership of shareholdings (control or anyway relevant shareholdings). It is the entity that presents the consolidation financial statement

I

Invoice currency

The Invoice Currency is the currency in which a given operation is performed (regardless of the currency of the entity that performed it)

M

Management Consolidated Financial Statement

The Management Consolidated Financial Statement contains information about the financial position and allows the holding to analyse the group's performance. The management consolidation, unlike the statutory consolidation, has two main goals: 1) to respond to control needs by analysing accounting and management information. Therefore, it is necessary to have non-accounting information (such as information about sales quantity, production, purchase or kpi) and financial information with management details in order to create economic reports for a specific business dimension (products, sales channels, business units or others); 2) to analyse more often the balance sheet items (monthly or quarterly) and before the accounting closing of each company. For this reason, it is necessary to consolidate the balance sheet items with faster methods and integrate data with manual adjustments (since data are incomplete upon preparation).

matrix

The Matrix identifies the form's area in which data present on the application database are displayed. Technically, the matrix represents the combination of several filters to apply to the Tagetik dimensions so as to have the set of data to display, like a SQL query

matrix filter

A Matrix Filter is a filter type which affects the whole area of a form matrix and thus not only the rows or the columns

module

The Module represents a set of functions. Each function belongs to one module only. The modules definition has two goals: - to simplify the roles definition. Indeed, when assigning the functions to a role, it is possible to use the module to assign to the role all the functions belonging to the module. – to check the system's functions accessibility according to license purchased by the customer. The list of functions of a module is pre-set, and the users cannot modify it

N

navigation mode

The "Navigation" mode allows the user to display data of a report or of a data entry form. When the "Navigation" mode is used it is not possible to edit data

O

original scenario

The Original Scenario is the scenario used by Tagetik for all the data entries: Amount, Preconsolidation and Consolidation Adjustments. It includes the concepts of fiscal year and data version (for example 2008ACT to describe the fiscal year 2008 and the actual data version, or 2008BDG to describe the fiscal year 2008 and the budget data version, or 2008F6 to describe the fiscal year 2008 and the forecast version edited at the end of June)

Owned Entity

An Owned Entity is an Entity whose shares are held by another entity

Owner Entity

An Owner Entity is an entity that holds the shares of another entity

P

parameter

A Parameter is a particular type of filter that can be used inside a form whose value remains unknown until the form is run

period

The Period represents the basic unit by which the fiscal year is divided. Typically, the Period represents the month (it manages 12 Periods) but a manual management is also possible in order to meet the user's business needs.

period length

The Period length is a specific dimension of the Reporting that enables the management of aggregated data, for example in quarters, semesters, etc... from cumulated data present in the Tagetik application database

process

The Process represents a business cycle (budget, actual, forecast...) having a start date and an end date. Each Process: - can have a single expiration date or several intermediate expiration dates; - is repeatable with different expiration dates (monthly, quarterly, half-year, annual); - can run on the entire data model or on a part of it

R

Reporting language

The Reporting Language is the language in which the descriptions of the metadata elements are displayed in Tagetik reporting. In order to enable the reporting language, it is necessary to define at least one extended reporting language in the Reporsitory Configuration (Actions menu in the Localization tab)

Repository

The Repository is a centralized database where the users who can access Tagetik and the roles (both system and custom) assigned to the users that define the activities a user can do are managed

role

A Role defines the activities that a User can do in Tagetik. A role is described by a list of functions, that is a list of simple operations of the application that a user can choose and perform. Typically the functions are: management of tables, data processing, report; usually a function corresponds to a menu key or to a window menu. The Role defines if the User can use each function fully, only in display mode (as far as tables are concerned)or if he cannot use it at all.

S

scenario

The Scenario represents, together with the period, the time dimension of Tagetik and includes also the concept of data version

scheduling

The Scheduling is the delayed run of a data processing or a job; it defines parameters and runtime

SharePoint

SharePoint is the Microsoft cooperative platform, based on .NET Framework, that provides a complete environment for the management of files, contents, business processes, etc

snapshot scenario

The Snapshot Scenario is a scenario where to "Freeze" data before further modifications

SQL model

A SQL Model is a SQL language query that can be simply used in the management windows where an ETL filter is employed

Statutory consolidated Financial Statement

The Statutory consolidated Financial Statement is an actual document drawn up according to the EEC IV directive. The IV directive arises from the need to ensure that financial statements of the limited companies of the various EEC Countries are prepared with comparable assessment criteria, methods and terminologies. This directive is intended to harmonize the legislations of the member states in terms of: 1) information contained in the annual financial statement and in the document and attached reports; 2) publication modes; 3) accounting standards to apply

T

tab filter

A Tab Filter is a filter type which affects the whole area of a template present in the form

tables language

The Tables Language is the language in which the metadata elements' descriptions are displayed

template
: A Template identifies each Excel sheet present in a form

U

user
: The Users are those who can access Tagetik